American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005-4070
202-682-8000

Date: December 1998
To: Purchasers of API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction,* Second Edition
Re: Addendum 3

This package contains Addendum 3 of API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction,* Second Edition. The addendum consists of the pages that have changed since the December 1997 printing of Addendum 2.

To update your copy of API Standard 653, replace the following pages as indicated:

Part of Book Changed	Old Pages to be Replaced	New Pages
Cover	Front and back covers	Front and back covers
Front matter	Title page to Addendum 2 v-ix	Title Page to Addendum 3 v-ix (+ blank)
Section 1	1-1–1-2	1-1–1-2
Section 2	2-5–2-8	2-5–2-8
Section 3	3-1–3-3	3-1–3-3 (+ blank)
Section 4	4-1–4-4	4-1–4-4
Section 6	6-1	6-1 (+ blank)
Section 7	7-1–7-2, 7-5–7-10	7-1–7-2, 7-5–7-11 (+ blank)
Section 8	8-1–8-2	8-1–8-2
Section 10	10-1–10-4	10-1–10-4
Section 11	11-1–11-2	11-1–11-2
Appendix A	A-1	A-1 (+ blank)
Appendix B	B-1–B-11	B-1–B-11 (+ blank)
Appendix D	D-1	D-1 (+ blank)
Appendix E	E-1	E-1 (+ blank)
Appendix F (new)	None	F-1–F-2

The parts of the text, tables, and figures that contain changes are indicated by a vertical bar and a small "98" in the margin.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0616719 207 ■

Tank Inspection, Repair, Alteration, and Reconstruction

API STANDARD 653
SECOND EDITION, DECEMBER 1995
ADDENDUM 1, DECEMBER 1996
ADDENDUM 2, DECEMBER 1997
ADDENDUM 3, DECEMBER 1998



American Petroleum Institute

Helping You Get The Job Done Right.℠

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34
Information Handling Services, 2000

Tank Inspection, Repair, Alteration, and Reconstruction

Downstream Segment

API STANDARD 653
SECOND EDITION, DECEMBER 1995
ADDENDUM 1, DECEMBER 1996
ADDENDUM 2, DECEMBER 1997
ADDENDUM 3, DECEMBER 1998



American Petroleum Institute

Helping You Get The Job Done Right.℠

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34
Information Handling Services, 2000

SPECIAL NOTES

API publications necessarily address problems of a general nature. With respect to particular circumstances, local, state, and federal laws and regulations should be reviewed.

API is not undertaking to meet the duties of employers, manufacturers, or suppliers to warn and properly train and equip their employees, and others exposed, concerning health and safety risks and precautions, nor undertaking their obligations under local, state, or federal laws.

Information concerning safety and health risks and proper precautions with respect to particular materials and conditions should be obtained from the employer, the manufacturer or supplier of that material, or the material safety data sheet.

Nothing contained in any API publication is to be construed as granting any right, by implication or otherwise, for the manufacture, sale, or use of any method, apparatus, or product covered by letters patent. Neither should anything contained in the publication be construed as insuring anyone against liability for infringement of letters patent.

Generally, API standards are reviewed and revised, reaffirmed, or withdrawn at least every five years. Sometimes a one-time extension of up to two years will be added to this review cycle. This publication will no longer be in effect five years after its publication date as an operative API standard or, where an extension has been granted, upon republication. Status of the publication can be ascertained from the API Downstream Segment [telephone (202) 682-8000]. A catalog of API publications and materials is published annually and updated quarterly by API, 1220 L Street, N.W., Washington, D.C. 20005.

This document was produced under API standardization procedures that ensure appropriate notification and participation in the developmental process and is designated as an API standard. Questions concerning the interpretation of the content of this standard or comments and questions concerning the procedures under which this standard was developed should be directed in writing to the general manager of the Downstream Segment, American Petroleum Institute, 1220 L Street, N.W., Washington, D.C. 20005. Requests for permission to reproduce or translate all or any part of the material published herein should also be addressed to the general manager.

API standards are published to facilitate the broad availability of proven, sound engineering and operating practices. These standards are not intended to obviate the need for applying sound engineering judgment regarding when and where these standards should be utilized. The formulation and publication of API standards is not intended in any way to inhibit anyone from using any other practices.

Any manufacturer marking equipment or materials in conformance with the marking requirements of an API standard is solely responsible for complying with all the applicable requirements of that standard. API does not represent, warrant, or guarantee that such products do in fact conform to the applicable API standard.

All rights reserved. No part of this work may be reproduced, stored in a retrieval system, or transmitted by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior written permission from the publisher. Contact the Publisher, API Publishing Services, 1220 L Street, N.W., Washington, D.C. 20005.

Copyright © 1995, 1996, 1997, 1998 American Petroleum Institute

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

FOREWORD

This standard is based on the accumulated knowledge and experience of owners, operators, manufacturers, and repairers of steel storage tanks. The object of this publication is to provide guidance in the inspection, repair, alteration, and reconstruction of steel storage tanks used in the petroleum and chemical industries.

If tanks are inspected, repaired, altered, or reconstructed in accordance with this standard, the owner/operator may elect to modify, delete, or amplify sections of this standard. It is strongly recommended that such modifications, deletions, or amplifications be made by supplementing this standard rather than by rewriting or incorporating sections into another complete standard.

The rules given in this standard are minimum requirements. This standard is not to be interpreted as approving, recommending, or endorsing any specific design nor limiting the methods of inspection, repair, alteration, or reconstruction.

Each edition, revision, or addenda to this API standard may be used beginning with the date of issuance shown on the cover page for that edition, revision, or addenda. Each edition, revision, or addenda to this API standard becomes effective six months after the date of issuance for equipment that is certified as being rerated, reconstructed, relocated, repaired, modified (altered), inspected, and tested per this standard. During the six-month time between the date of issuance of the edition, revision, or addenda and the effective date, the purchaser and manufacturer shall specify to which edition, revision, or addenda the equipment is to be rated, reconstructed, relocated, repaired, modified (altered), inspected, and tested.

API standards are published as an aid to procurement of standardized equipment and materials. API standards are not intended to inhibit purchasers or producers from purchasing or producing products made to specifications other than those of API.

API standards may be used by anyone desiring to do so. Every effort has been made by the Institute to assure the accuracy and reliability of the data contained in them; however, the Institute makes no representation, warranty, or guarantee in connection with the publication of API standards and hereby expressly disclaims any liability or responsibility for loss or damage resulting from their use; for any violation of any federal, state, or municipal regulation with which an API standard may conflict; or for the infringement of any patent resulting from the use of an API standard.

Suggested revisions are invited and should be submitted to the general manager of the Downstream Segment, American Petroleum Institute, 1220 L Street, N.W., Washington, D.C. 20005.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

CONTENTS

Page

SECTION 1—INTRODUCTION
1.1 General 1-1
1.2 Compliance With This Standard 1-1
1.3 Jurisdiction 1-1
1.4 Safe Working Practices 1-1
98 | 1.5 Definitions 1-1
97 | 1.6 Referenced Publications 1-2

SECTION 2—SUITABILITY FOR SERVICE
2.1 General 2-1
2.2 Tank Roof Evaluation 2-1
2.2.1 General 2-1
97 | 2.2.2 Fixed Roofs 2-1
2.2.3 Floating Roofs 2-1
2.2.4 Change of Service 2-1
2.3 Tank Shell Evaluation 2-2
2.3.1 General 2-2
2.3.2 Actual Thickness Determination 2-2
97 | 2.3.3 Minimum Thickness Calculation for Welded Tank Shell 2-3
2.3.4 Minimum Thickness Calculation for Riveted Tank Shell 2-4
2.3.5 Distortions 2-4
2.3.6 Flaws 2-5
2.3.7 Wind Girders and Shell Stiffeners 2-5
2.3.8 Shell Welds 2-5
2.3.9 Shell Penetrations 2-5
2.4 Tank Bottom Evaluation 2-5
98 | 2.4.1 General 2-5
2.4.2 Causes of Bottom Failure 2-5
2.4.3 Cathodic Protection of Tank Bottoms 2-5
2.4.4 Internal Lining Protection of Tank Bottoms 2-6
2.4.5 Bottom Leak Detection 2-6
2.4.6 Bottom Plate Thickness Measurements 2-6
98 | 2.4.7 Minimum Thickness for Tank Bottom Plate 2-6
96 | 2.4.8 Minimum Thickness for Annular Plate Ring 2-7
2.5 Tank Foundation Evaluation 2-7
2.5.1 General 2-7
2.5.2 Foundation Repair or Replacement 2-7
2.5.3 Anchor Bolts 2-8

SECTION 3—BRITTLE FRACTURE CONSIDERATIONS
3.1 General 3-1
3.2 Basic Considerations 3-1
3.3 Assessment Procedure 3-1

SECTION 4—INSPECTION
96 | 4.1 General 4-1
4.2 Inspection Frequency Considerations 4-1
96 | 4.3 Inspections from the Outside of the Tank 4-1

Previous page is blank

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

		Page
	4.3.1 Routine In-Service Inspections	4-1
96	4.3.2 External Inspection	4-1
98	4.3.3 Ultrasonic Thickness Inspection	4-1
96	4.3.4 Cathodic Protection Surveys	4-2
	4.4 Internal Inspection	4-2
98	4.4.1 General	4-2
	4.4.2 Inspection Intervals	4-2
98	4.4.3 Alternative Internal Inspection Interval	4-2
	4.5 Alternative to Internal Inspection to Determine Bottom Thickness	4-3
	4.6 Preparatory Work for Internal Inspection	4-3
	4.7 Inspection Checklists	4-3
	4.8 Records	4-3
	4.8.1 General	4-3
	4.8.2 Construction Records	4-3
	4.8.3 Inspection History	4-3
	4.8.4 Repair/Alteration History	4-3
	4.9 Reports	4-3
	4.10 Inspector Qualifications	4-4

SECTION 5—MATERIALS

	5.1 General	5-1
	5.2 New Materials	5-1
	5.3 Original Materials for Reconstructed Tanks	5-1
	5.3.1 Shell and Bottom Plates Welded to the Shell	5-1
	5.3.2 Structural	5-1
	5.3.3 Flanges and Fasteners	5-1
	5.3.4 Roof, Bottom, and Plate Wind Girders	5-1
	5.4 Welding Consumables	5-1

SECTION 6—DESIGN CONSIDERATIONS FOR RECONSTRUCTED TANKS

	6.1 General	6-1
	6.2 New Weld Joints	6-1
	6.3 Existing Weld Joints	6-1
	6.4 Shell Design	6-1
	6.5 Shell Penetrations	6-1
	6.6 Wind Girders and Shell Stability	6-1
	6.7 Roofs	6-1
	6.8 Seismic Design	6-1

SECTION 7—TANK REPAIR AND ALTERATION

98	7.1 General	7-1
	7.2 Removal and Replacement of Shell Plate Material	7-1
	7.2.1 Minimum Thickness of Replacement Shell Plate	7-1
	7.2.2 Minimum Dimensions of Replacement Shell Plate	7-1
	7.2.3 Weld Joint Design	7-1
	7.3 Shell Repairs Using Lap-Welded Patch Plates	7-1
	7.4 Repair of Defects in Shell Plate Material	7-4
96	7.5 Alteration of Tank Shells to Change Shell Height	7-4
	7.6 Repair of Defective Welds	7-4
	7.7 Repair of Shell Penetrations	7-4
97	7.8 Addition or Replacement of Shell Penetrations	7-4

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

		Page
97	7.9 Alteration of Existing Shell Penetrations	7-5
97	7.10 Repair of Tank Bottoms	7-5
98	7.10.1 General Repair Requirements	7-5
98	7.10.2 Replacement of Entire Tank Bottom	7-6
	7.11 Repair of Fixed Roofs	7-7
	7.11.1 Supported Cone Roofs	7-7
	7.11.2 Self-Supporting Roofs	7-7
	7.12 Floating Roofs	7-7
	7.12.1 External Floating Roofs	7-7
	7.12.2 Internal Floating Roofs	7-7
	7.12.3 Repair of Leaks in Pontoons	7-7
	7.13 Repair or Replacement of Floating Roof Perimeter Seals	7-7
96	7.13.1 Primary Seals	7-7
	7.13.2 Secondary Seals	7-7
	7.13.3 Seal-to-Shell Gap	7-9
	7.13.4 Mechanical Damage	7-9
	7.13.5 Deterioration of Seal Material	7-9
	7.13.6 Installation of Primary and Secondary Seals	7-9
	7.14 Hot Taps	7-9
	7.14.1 General	7-9
	7.14.2 Hot Tap Procedures	7-10
	7.14.3 Preparatory Work	7-10
	7.14.4 Material Limitations	7-10
	7.14.5 Installation Procedure	7-10

SECTION 8—DISMANTLING AND RECONSTRUCTION

	8.1 General	8-1
	8.2 Cleaning and Gas Freeing	8-1
	8.3 Dismantling Methods	8-1
	8.3.1 General	8-1
97	8.3.2 Bottoms	8-1
98	8.3.3 Shells	8-1
	8.3.4 Roofs	8-1
	8.3.5 Piece Marking	8-1
	8.4 Reconstruction	8-2
	8.4.1 General	8-2
98	8.4.2 Welding	8-2
	8.4.3 Bottoms	8-2
	8.4.4 Shells	8-2
	8.4.5 Roofs	8-3
	8.5 Dimensional Tolerances	8-3
	8.5.1 General	8-3
97	8.5.2 Plumbness	8-3
	8.5.3 Roundness	8-3
	8.5.4 Peaking	8-3
	8.5.5 Banding	8-3
	8.5.6 Foundations	8-3

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

SECTION 9—WELDING
9.1 Welding Qualifications 9-1
9.2 Identification and Records 9-1

SECTION 10—EXAMINATION AND TESTING
10.1 Nondestructive Examinations 10-1
10.1.1 General 10-1
10.1.2 Shell Penetrations 10-1
10.1.3 Repaired Weld Flaws 10-1
96 10.1.4 Temporary and Permanent Attachments to Shell Plates 10-1
10.1.5 Shell Plate to Shell Plate Welds 10-1
98 10.1.6 Shell-to-Bottom Weld 10-1
10.1.7 Bottoms 10-1
10.2 Radiographs 10-2
97 10.2.1 Number and Location of Radiographs 10-2
10.2.2 Acceptance Criteria for Existing Shell Plate to Shell Plate Welds 10-2
96 10.2.3 Marking and Identification of Radiographs 10-2
10.3 Hydrostatic Testing 10-2
98 10.3.1 When Hydrostatic Testing Required 10-2
10.3.2 When Hydrostatic Testing Not Required 10-3
10.4 Leak Tests 10-4
10.5 Measured Settlement During Hydrostatic Testing 10-4
97 10.5.1 Initial Survey 10-4
10.5.2 Survey During Hydrostatic Testing 10-4

SECTION 11—MARKING AND RECORDKEEPING
11.1 Nameplates 11-1
98 11.2 Recordkeeping 11-1
11.3 Certification 11-1

98 APPENDIX A—BACKGROUND ON PAST EDITIONS OF API WELDED STORAGE TANK STANDARDS A-1
98 APPENDIX B—EVALUATION OF TANK BOTTOM SETTLEMENT B-1
APPENDIX C—CHECKLISTS FOR TANK INSPECTION C-1
APPENDIX D—AUTHORIZED INSPECTOR CERTIFICATION D-1
98 APPENDIX E—TECHNICAL INQUIRIES E-1
APPENDIX F—NDE REQUIREMENTS SUMMARY F-1

Figures
2-1—Inspection of Corrosion Areas 2-2
2-2—Pit Measurement 2-3
3-1—Brittle Fracture Considerations 3-2
98 3-2—Exemption Curve for Tanks Constructed From Carbon Steel of Unknown Toughness 3-3
7-1—Acceptable Details for Replacement of Shell Plate Material 7-2
7-2—Lapped Patch Repair Plates at the External Shell-to-Bottom Joint 7-3
96 7-2a—Typical Details for Addition of Reinforcing Plate to Existing Shell Penetration 7-4
7-3—Typical Details for Addition of "Tombstone" Shape Reinforcing Plate to Existing Shell Penetration 7-5
7-4—Method for Raising Shell Nozzles 7-6

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

Figures

98 7-5—Typical Welded-on Patch Plates on Tank Bottom Plates 7-8
96 7-6—Curve for Hot Tapping Tanks With Steels of Unknown Toughness 7-9
98 7-7—Hot Tap for Tanks 7-11
97 8-1—Tank Shell and Bottom Cut Locations 8-1
98 10-1—DELETED 10-3
11-1—Nameplate 11-2
11-2—Certification Forms 11-3
B-1—Measurements of Shell Settlement (External) B-1
97 B-2—Measurements of Bottom Settlement (Internal) Tank Out-of-Service B-2
B-3—Graphical Representation of Tank Shell Settlement B-3
B-4— Out-of-Plane Settlement Measurement Points B-4
B-5—Edge Settlement B-4
B-6—Correction for Measured Edge Settlement B-5
B-7—Bottom Settlement Near Shell B-6
98 B-8—Localized Bottom Depressions or Bulges Remote From Shell B-6
B-9—Localized Tank Bottom Settlement Limits for Single Pass Welds B-8
B-10—Maximum Allowable Edge Settlement for Areas with Bottom Lap Welds Approximately Parallel to the Shell B-9
B-11—Maximum Allowable Edge Settlement for Areas with Bottom Lap Welds Approximately Perpendicular to the Shell B-10
B-12— Edge Settlement with a Lap Weld at an Arbitrary Angle to the Shell B-11

Tables

2-1—Joint Efficiencies for Welded Joints 2-4
2-2—Joint Efficiencies for Riveted Joints 2-4
2-3—Annular Bottom Plate Thicknesses (Inches) (Product Specific Gravity <1.0) 2-7
4-1—Bottom Plate Minimum Thickness 4-3
98 7-1—Hot Tap Connection Sizes and Shell Plate Thicknesses 7-9
8-1—Maximum Thicknesses on New Welds 8-2
8-2—Radii Tolerances 8-3
98 10-1—DELETED 10-3
A-1—Editions of API Standard 650 and Its Precursor, Standard 12C A-1

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Tank Inspection, Repair, Alteration, and Reconstruction

SECTION 1—INTRODUCTION

1.1 General

1.1.1 This standard covers carbon and low alloy steel tanks built to API Standard 650 and its predecessor 12C. This standard provides minimum requirements for maintaining the integrity of welded or riveted, nonrefrigerated, atmospheric pressure, aboveground storage tanks after they have been placed in service. It covers the maintenance inspection, repair, alteration, relocation, and reconstruction of such tanks.

1.1.2 The scope is limited to the tank foundation, bottom, shell, structure, roof, attached appurtenances, and nozzles to the face of the first flange, first threaded joint, or first welding-end connection. Many of the design, welding, examination, and material requirements of API Standard 650 can be applied in the maintenance inspection, rating, repair, and alteration of in-service tanks. In the case of apparent conflicts between the requirements of this standard and API Standard 650 or its predecessor API 12C, this standard shall govern for tanks that have been placed in service.

1.1.3 This standard employs the principles of API Standard 650; however, storage tank owner/operators, based on consideration of specific construction and operating details, may apply this standard to any steel tank constructed in accordance with a tank specification.

1.1.4 This standard is intended for use by organizations that maintain or have access to engineering and inspection personnel technically trained and experienced in tank design, fabrication, repair, construction, and inspection.

1.1.5 This standard does not contain rules or guidelines to cover all the varied conditions which may occur in an existing tank. When design and construction details are not given, and are not available in the standard to which the tank was originally constructed, then details that will provide a level of integrity equal to the level provided by the current edition of API Standard 650 must be used.

1.2 Compliance With This Standard

The owner/operator has ultimate responsibility for complying with the provisions of this standard. Should a party other than the owner/operator be assigned certain tasks, such as relocating and reconstructing a tank, the limits of responsibility for each party shall be defined by the owner/operator prior to commencing work.

1.3 Jurisdiction

If any provision of this standards presents a direct or implied conflict with any statutory regulation, the regulation shall govern. However, if the requirements of this standard are more stringent than the requirements of the regulation, then the requirements of this standard shall govern.

1.4 Safe Working Practices

An assessment shall be made of the potential hazards to which personnel may be exposed when conducting internal tank inspections, making repairs, or dismantling tanks. Procedures shall be developed according to the guidelines given in API Publications 2015 and 2217 that will include safeguard for personnel health and safety, prevention of accidental fires and explosions, and the prevention of property damage.

Special procedures may need to be developed for certain activities described in this standard that are not fully covered by the referenced API publications; for example, safety precautions for personnel accessing floating roof tanks that are in service, or gas freeing the bottom side of a tank. Finally, procedures must comply with any federal or state safety regulations pertaining to "confined spaces" or any other relevant provisions.

1.5 Definitions

1.5.1 *Alteration* means any work on a tank involving cutting, burning, welding, or heating operations that changes the physical dimensions and/or configuration of a tank. Typical examples of alterations include:

a. The addition of manways and nozzles greater than 12-inch nominal pipe size (NPS).
b. An increase or decrease in tank shell height.

1.5.2 *Applicable standard* refers to the original standard of construction, such as API standards or specifications or Underwriter Laboratories (UL) standards, unless the original standard of construction has been superseded or withdrawn from publication; in this event, applicable standard means the current edition of the appropriate standard. See Appendix A for background on editions of API welded storage tank standards.

1.5.3 *Atmospheric pressure* is used to describe tanks designed to withstand an internal pressure up to but not exceeding $2^1/_2$ pounds per square inch gauge.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0616729 156 ■

1.5.4 An *Authorized Inspection Agency* is one of the following organizations or individuals.

a. The inspection organization of the jurisdiction in which the aboveground storage tank is operated.

b. The inspection organization of an insurance company which is licensed or registered to and does write aboveground storage tank insurance.

c. An owner or operator of one or more aboveground storage tank(s) who maintains an inspection organization for activities relating only to his equipment, and not for aboveground storage tanks intended for sale or resale.

d. An independent organization or individual under contract to and under the direction of an owner or operator and recognized or otherwise not prohibited by the jurisdiction in which the aboveground storage tank is operated. The owner or operator's inspection program shall provide the controls necessary for use by Authorized Inspectors contracted to inspect aboveground storage tanks.

1.5.5 *Change in service* means a change from previous operating conditions involving different properties of the stored product such as specific gravity or corrosivity and/or different service conditions of temperature and/or pressure.

1.5.6 *Corrosion rate* is defined as the total metal loss divided by the period of time over which the metal loss occurred.

1.5.7 *Critical zone* for repairs to tank bottoms is that portion of the bottom or annular plate within 3 inches of the inside edge of the shell, measured radially inward.

1.5.8 *Hot tap* identifies a procedure for installing a nozzle in the shell of a tank that is in service.

1.5.9 An *Inspector* as used in this document is an Authorized Inspector and is an employee of an Authorized Inspection Agency who is qualified and certified to perform tank inspections under this standard.

1.5.10 *Owner/operator* refers to the legal entity having both control of and/or responsibility for the operation and maintenance of an existing storage tank. 98

1.5.11 *Reconstruction* means the work necessary to reassemble a tank that has been dismantled and relocated to a new site.

1.5.12 *Reconstruction organization* refers to that organization having assigned responsibility by the owner/operator to design and/or reconstruct a tank.

1.5.13 *Repair* means any work necessary to maintain or restore a tank to a condition suitable for safe operation. Typical examples of repairs include:

a. Removal and replacement of material (such as roof, shell, or bottom material, including weld metal) to maintain tank integrity.

b. Re-leveling and/or jacking of a tank shell, bottom, or roof.

c. Addition of reinforcing plates to existing shell penetrations.

d. Repair of flaws, such as tears or gouges, by grinding and/or gouging followed by welding.

1.6 Referenced Publications

The following standards, codes, publications, and specifications are cited in this standard. The latest edition or revision shall be used unless otherwise noted.

API

Std 620	*Design and Construction of Large, Welded Low-Pressure Storage Tanks*
Std 650	*Welded Steel Tanks for Oil Storage*
RP 651	*Cathodic Protection of Aboveground Petroleum Storage Tanks*
RP 652	*Lining of Aboveground Petroleum Storage Tank Bottoms*
Std 2000	*Venting Atmospheric and Low-Pressure Storage Tanks (Non-refrigerated and Refrigerated)*
RP 2003	*Protection Against Ignitions Arising Out of Static, Lightning, and Stray Currents*
Publ 2015	*Cleaning Petroleum Storage Tanks*
Publ 2201	*Procedure for Welding or Hot Tapping on Equipment Containing Flammables*
Publ 2207	*Preparing Tank Bottoms for Hot Work*
Publ 2217A	*Guidelines for Work in Inert Confined Spaces in the Petroleum Industry* 97

ASME[1]

Boiler and Pressure Vessel Code, Section V, "Nondestructive Examination"; Section VIII, "Pressure Vessels" Alternative Rules, Division 2; Section IX, "Welding and Brazing Qualifications"

ASTM[2]

A6	*General Requirements for Rolled Steel Plates, Shapes, Sheet Piling, and Bars for Structural Use*
A20	*General Requirements for Steel Plates for Pressure Vessels*
A36	*Structural Steel*
A370	*Standard Test Methods and Definitions for Mechanical Testing of Steel Products*

[1] American Society for Mechanical Engineers, 345 East 47th Street, New York, New York 10017.

[2] American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959. 97

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

2.3.5.2 Shell distortions can be caused by many conditions such as foundation settlement, over- or under-pressuring, high wind, poor shell fabrication, or repair techniques, and so forth.

2.3.5.3 Shell distortions shall be evaluated on an individual basis to determine if specific conditions are considered acceptable for continuing tank service and/or the extent of corrective action.

2.3.6 FLAWS

Flaws such as cracks or laminations shall be thoroughly examined and evaluated to determine their nature and extent and need for repair. If a repair is needed, a repair procedure shall be developed and implemented. The requirement for repairing scars such as arc strikes, gouges, or tears from temporary attachment welds must be evaluated on a case-by-case basis. Cracks in the shell-to-bottom weld are critical and shall be removed and the weld repaired.

2.3.7 WIND GIRDERS AND SHELL STIFFENERS

The evaluation of an existing tank shell for suitability for service must also consider the details and condition of any wind girders or shell stiffeners. Degradation by corrosion of these structural elements or their attachment welds to the shell may render these elements inadequate for the design conditions.

2.3.8 SHELL WELDS

The condition of the tank shell welds shall be evaluated for suitability for service. Any deterioration of the existing welds that results from corrosion or pitting must be evaluated and appropriate repair procedures established or the tank rerated as necessary. Some typical shell butt-weld flaws and recommended procedures for repairs are given in 7.6. 96

2.3.9 SHELL PENETRATIONS

2.3.9.1 The condition and details of existing shell penetrations (nozzles, manways, cleanout openings, etc.) shall be reviewed when assessing the integrity of an existing tank shell. Details such as type and extent of reinforcement, weld spacing, and thickness of components (reinforcing plate, nozzle neck, bolting flange, and cover plate), are important considerations and shall be reviewed for structural adequacy and compliance with the applicable standard. Any noncompliance, or deterioration due to corrosion, must be assessed and repair procedures established where appropriate or the tank rerated, as necessary.

2.3.9.2 Nozzle wall thickness shall be evaluated for pressure and all other loads.

2.4 Tank Bottom Evaluation

2.4.1 GENERAL

Tank bottom inspection strategies shall provide suitable data which when used with the procedures in this standard will determine the tank bottom integrity necessary to prevent leakage of fluids that may cause environmental damage. Each aspect of corrosion phenomena, and other potential leak or failure mechanism must be examined. Periodic assessment of tank bottom integrity shall be performed in addition to the internal inspections specified in 4.4. The assessment period shall be less than or equal to the appropriate internal inspection interval given in 4.4.2 or 4.4.3. The use of leak detection tests or monitoring systems (such as double bottoms or liners under tank bottoms with leak detection pipes) will satisfy the requirement for periodic assessment between internal inspections. 98 98

Excessive foundation settlement of storage tanks can affect the integrity of tank shells and bottoms. Therefore, monitoring the settlement behavior of tanks is a recognized practice to assess the integrity of tank bottoms. Refer to Appendix B for techniques for evaluating tank bottom settlement.

2.4.2 CAUSES OF BOTTOM FAILURE

The following list gives some historical causes of tank bottom leakage or failure that shall be considered in the decision to line, repair, or replace a tank bottom:

a. Internal pitting and pitting rates in the anticipated service.
b. Corrosion of weld joints (weld and heat affected zone).
c. Weld joint cracking history.
d. Stresses placed on the bottom plates by roof support loads and shell settlement.
e. Underside corrosion (normally in the form of pitting).
f. Inadequate drainage resulting in surface water flowing under the tank bottom.
g. The lack of an annular plate ring when required.
h. Uneven settlement that results in high localized stresses in the bottom plates.
i. Roof support columns or other supports welded to the tank bottom where adequate allowance for movement was not made.
j. Rock or gravel foundation pads with inadequately filled-in surface voids.
k. Nonhomogeneous fill under the tank bottom (for example, a lump of clay in a sand foundation pad).
l. Inadequately supported sumps.

2.4.3 CATHODIC PROTECTION OF TANK BOTTOMS

A selection basis for cathodic protection systems for the underside of tank bottoms is covered by API RP 651.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

2.4.4 INTERNAL LINING PROTECTION OF TANK BOTTOMS

Applied linings for internal surfaces of tank bottoms are covered by API RP 652.

2.4.5 BOTTOM LEAK DETECTION

If a tank bottom is to be replaced, consideration should be given to installing a leak detection (tell-tale) system that will channel any leak in the bottom to a location where it can be readily observed from the outside of the tank.[1]

2.4.6 BOTTOM PLATE THICKNESS MEASUREMENTS

Various methods for determining tank bottom plate underside corrosion are available. The methods vary by the extent that they can reliably measure general corrosion and pitting. A combination of these methods may be required along with extrapolation techniques and analysis to establish the probable conditions of the entire tank bottom.

2.4.7 MINIMUM THICKNESS FOR TANK BOTTOM PLATE

Quantifying the remaining thickness of tank bottoms based on the results of measurement (see 2.4.6) can be done by either a deterministic method (see 2.4.7.1) or a probabilistic method (see 2.4.7.2) as follows:

2.4.7.1 An acceptable deterministic method for calculating the minimum remaining bottom thicknesses is as follows:

$$MRT_1 = T_o - GC_a - StP_a - UP_m - (StP_r + UP_r + GC_r)O_r$$

$$MRT_2 = T_o - GC_a - StP_m - UP_a - (StP_r + UP_r + GC_r)O_r$$

Where:

MRT_1, MRT_2 = minimum remaining thickness at the end of the in-service period of operation, in inches. MRT_1 represents minimum remaining thickness due to average internal pitting and maximum external pitting. MRT_2 represents minimum remaining thickness due to maximum internal pitting and average external pitting.

T_o = original plate thickness, in inches.

StP_a = average depth of internal pitting prior to repairs (in accordance with 2.4.7.5), in inches, measured from the original thickness. 96

StP_m = maximum depth of internal pitting remaining in the bottom plates after repairs in accordance with 2.4.7.5 are completed, in inches, measured from the original thickness. 96

UP_a = average depth of underside pitting, in inches.

UP_m = maximum depth of underside pitting, in inches.

StP_r = maximum internal pitting rate, in inches per year; $StP_r = 0$ if the tank bottom is internally lined (see API RP 652).

UP_r = maximum underside pitting rate, in inches per year; $UP_r = 0$ if tank bottom is cathodically protected (see API RP 651).

O_r = anticipated in-service period of operation (normally 10 years).

GC_a = average depth of generally corroded area, in inches.

GC_r = maximum rate of general corrosion, in inches per year.

2.4.7.2 For the probabilistic method, a statistical analysis is made of thickness data from measurements (see 2.4.6) projecting remaining thickness, based on sample scanning of the bottom.

2.4.7.3 If the minimum bottom thicknesses, at the end of the in-service period of operation, are calculated to be less than the minimum bottom renewal thicknesses given in Table 4-1, or less than the minimum bottom renewal thicknesses providing acceptable risk as determined by a risk-based inspection methodology, the bottom shall be lined, repaired, replaced, or the interval to the next internal inspection shortened. 98

2.4.7.4 Unless a stress analysis is performed, the minimum bottom plate thickness in the critical zone of the tank bottom defined by paragraph 7.10.1.2 shall be the smaller of $^1/_2$ the original bottom plate thickness (not including the original corrosion allowance) or 50 percent of t_{min} of the lower shell course per paragraph 2.3.3.1 but not less than 0.1 inch. Isolated pitting will not appreciably affect the strength of the plate. 98

[1] For existing tanks, API supports the use of a Release Prevention System (RPS). The term RPS refers to the suite of API standards that are designed to maintain aboveground storage tank integrity and thus protect the environment. These are: the frequency of internal inspections; lining the bottom of the tank interior; fitting the tank with Release Prevention Barriers (RPBs); installing cathodic protections; or some combination of these measures, depending on the operating environment and service of the tank.

If a decision is made to replace an existing bottom, API supports the evaluation of installing an RPB or continued use of an RPS. The evaluation should consider the effectiveness of RPS controls, the product stored, the location of the tank, and environmental sensitivities. An RPB includes steel bottoms, synthetic materials, clay liners, and all other barriers or combination of barriers placed in the bottom of or under an aboveground storage tank, which have the functions of: 1) preventing the escape of contaminated material; and 2) containing or channeling released material for leak detection (covered in detail in the non-mandatory Appendix I of Standard 650).

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

2.4.7.5 The repair of internal pitting, when performed to extend the in-service period of operation, shall be by pit welding, overlay welding, or lap patching, followed by inspection and testing. The extent of weld repairs is limited in 98 the critical zone in accordance with 7.10.1.2.

2.4.7.6 The treatment of bottom pitting by the use of non-welded repairs (for example, coatings, caulking) is not considered a repair in the context of how StP_m affects the remaining service life of a tank bottom. 96

2.4.7.7 The thickness of the projection of the bottom plate beyond the shell as measured at the toe of the outside bottom-to-shell fillet weld shall not be less than 0.1 inch. The projection of the bottom plate beyond the outside toe of the shell-to-bottom weld shell shall be at least $^3/_8$ inch.

2.4.8 MINIMUM THICKNESS FOR ANNULAR PLATE RING

2.4.8.1 Due to strength requirements, the minimum thickness of annular plate ring is usually greater than 0.10 inch. Isolated pitting will not appreciably affect the strength of the plate. Unless a stress analysis is performed, the annular plate thickness shall be in accordance with 2.4.8.2 or 2.4.8.3, as applicable.

2.4.8.2 For tanks in service with a product specific gravity less than 1.0, which require annular plates for other than seismic loading considerations, the thickness of the annular plates shall be not less than the thicknesses given in Table 2-3, plus any specified corrosion allowance.

2.4.8.3 For tanks in service with a product specific gravity of 1.0 or greater, which require annular plates for other than seismic loading considerations, the thickness of the annular plates shall be in accordance with Table 3-1 of API Standard 650 plus any specified corrosion allowance.

2.4.8.4 For tanks that utilize thickened annular plates for seismic considerations, a seismic evaluation shall be performed in accordance with the requirements of the applicable standard, using the actual thickness of the existing annular plate.

2.4.8.5 For the thickness and projection of the annular 96 plate beyond the shell refer to 2.4.7.7.

2.5 Tank Foundation Evaluation

2.5.1 GENERAL

2.5.1.1 The principal causes of foundation deterioration are settlement, erosion, cracking, and deterioration of concrete initiated by: calcining, attack by underground water, attack by frost, and attack by alkalies and acids. To ensure suitability for service, all tank foundations shall be inspected periodically (see 4.3).

Table 2-3—Annular Bottom Plate Thicknesses (Inches) (Product Specific Gravity < 1.0)

Plate Thickness[a] of First Shell Course (Inches)	Stress[b] in First Shell Course (pounds per square inch) <24,300	<27,000	<29,700	<32,400
$t \leq 0.75$	0.17	0.20	0.23	0.30
$0.75 < t \leq 1.00$	0.17	0.22	0.31	0.38
$1.00 < t \leq 1.25$	0.17	0.26	0.38	0.48
$1.25 < t \leq 1.50$	0.22	0.34	0.47	0.59
$t > 1.50$	0.27	0.40	0.53	0.68

Note: The thicknesses specified in the table are based on the foundation providing a uniform support under the full width of the annular plate. Unless the foundation is properly compacted, particularly at the inside of a concrete ringwall, settlement will produce additional stresses in the annular plate.
[a]Plate thickness refers to the tank shell as constructed.
[b]Stresses are calculated from $[2.34\ D\ (H\text{-}1)]/t$.

2.5.1.2 Some mechanisms of concrete deterioration are briefly described below:

a. Calcining (loss of water of hydration) can occur when concrete has been exposed to sufficiently high temperature for a period of time. During intermediate cooling periods, the concrete can absorb moisture, swell, lose its strength, and crack.

b. Deterioration of concrete exposed to underground water can be caused by chemical attack, by cyclic changes in temperature, and by freezing moisture.

c. Expansion of freezing moisture in porous concrete, or in concrete with minor settlement cracks or temperature cracks, can result in spalling and/or the development of serious structural cracks.

d. Sulfate-type alkalies, and to a lesser extent, chlorides, can act corrosively to destroy the bond of the concrete.

e. Temperature cracks (hairline cracks of uniform width) do not seriously affect the strength of the concrete foundation structure; however, these cracks can be potential access points for moisture or water seepage that could eventually result in corrosion of the reinforcing steel.

2.5.2 FOUNDATION REPAIR OR REPLACEMENT

2.5.2.1 If there is a need for foundation repair or replacement, foundations shall be restored to the tolerance limits of 8.5.6.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

2.5.2.2 Concrete pads, ringwalls, and piers, showing evidence of spalling, structural cracks, or general deterioration, shall be repaired to prevent water from entering the concrete structure and corroding the reinforcing steel.

2.5.3 ANCHOR BOLTS

Distortion of anchor bolts and excessive cracking of the concrete structures in which they are embedded may be indications of either serious foundation settlement or a tank overpressure uplift condition.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 3—BRITTLE FRACTURE CONSIDERATIONS

3.1 General

This section provides a procedure for the assessment of existing tanks for suitability for continued operation or change of service with respect to the risk of brittle fracture and does not supplement or replace the requirements of Section 10 for the hydrotesting of repaired, modified, or reconstructed tanks. The procedure applies to both welded and riveted tanks; however, the procedure is based primarily on experience and data obtained from welded tanks.

3.2 Basic Considerations

A decision tree, Figure 3-1, is used to present the assessment procedure for failure due to brittle fracture. The decision tree is based on the following principles:

3.2.1 In all reported incidents of tank failure due to brittle fracture, failure occurred either shortly after erection during hydrostatic testing or on the first filling in cold weather, after a change to lower temperature service, or after a repair/alteration. This experience shows that once a tank has demonstrated the ability to withstand the combined effects of maximum liquid level (highest stresses) and lowest operating temperature without failing, the risk of failure due to brittle fracture with continued service is minimal.

3.2.2 Any change in service must be evaluated to determine if it increases the risk of failure due to brittle fracture. In the event of a change to a more severe service (such as operating at a lower temperature or handling product at a higher specific gravity) it is necessary to consider the need for a hydrostatic test to demonstrate fitness for a new more severe service. The following aspects should be considered:

a. The likelihood of repairs/alterations since the original hydrostatic test not meeting requirements of this standard.
b. Deterioration of the tank since original hydrostatic test.

3.3 Assessment Procedure

The assessment procedure as illustrated in Figure 3-1 shall be used for evaluating aboveground atmospheric storage tanks in petroleum and chemical services. Each of the key steps, numbered 1 through 11 on the decision tree, correspond sequentially to the explanations provided below.

3.3.1 These tanks meet the API Standard 650 (seventh edition or later) requirements to minimize the risk of failure due to brittle fracture. Tanks may also be shown to meet the toughness requirements of API Standard 650 (seventh edition or later) by impact testing coupon samples from a representative number of shell plates.

3.3.2 Many tanks that continue to operate successfully in the same service were not built to the requirements of API Standard 650 (seventh edition or later). These tanks are potentially susceptible to failure due to brittle fracture and require an assessment as illustrated by the decision tree.

3.3.3 For the purpose of this assessment, hydrostatic testing demonstrates that an aboveground atmospheric storage tank in a petroleum or chemical service is fit for continued service and at minimal risk of failure due to brittle fracture, provided that all governing requirements for repairs, alterations, reconstruction, or change in service are in accordance with this standard (including a need for hydrostatic testing after major repairs, modifications or reconstruction). The effectiveness of the hydrostatic test in demonstrating fitness for continued service is shown by industry experience.

3.3.4 If a tank shell thickness is no greater than 0.5 inch, the risk of failure due to brittle fracture is minimal, provided that an evaluation for suitability of service in accordance with Section 2 has been performed. The original nominal thickness for the thickest tank shell plate shall be used for this assessment.

3.3.5 No known tank failures due to brittle fracture have occurred at shell metal temperatures of 60°F or above. Similar assurance against brittle fracture can be gained by increasing the metal temperature by heating the tank contents.

3.3.6 Industry experience and laboratory tests have shown that a membrane stress in tank shell plates of at least 7 ksi is required to cause failure due to brittle fracture.

3.3.7 Tanks constructed from steels listed in Figure 2-1 of API Standard 650 can be used in accordance with their exemption curves, provided that an evaluation for suitability of service in conformance with Section 2 of this standard has been performed. Tanks fabricated from steels of unknown toughness thicker than $^1/_2$ inch and operating at a shell metal temperature below 60°F can be used if the tank meets the requirements of Figure 3-2. The original nominal thickness for thickest tank shell plate shall be used for the assessment. For unheated tanks, the shell metal temperature shall be the design metal temperature as defined in 2.2.9.3 of API Standard 650. | 98

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

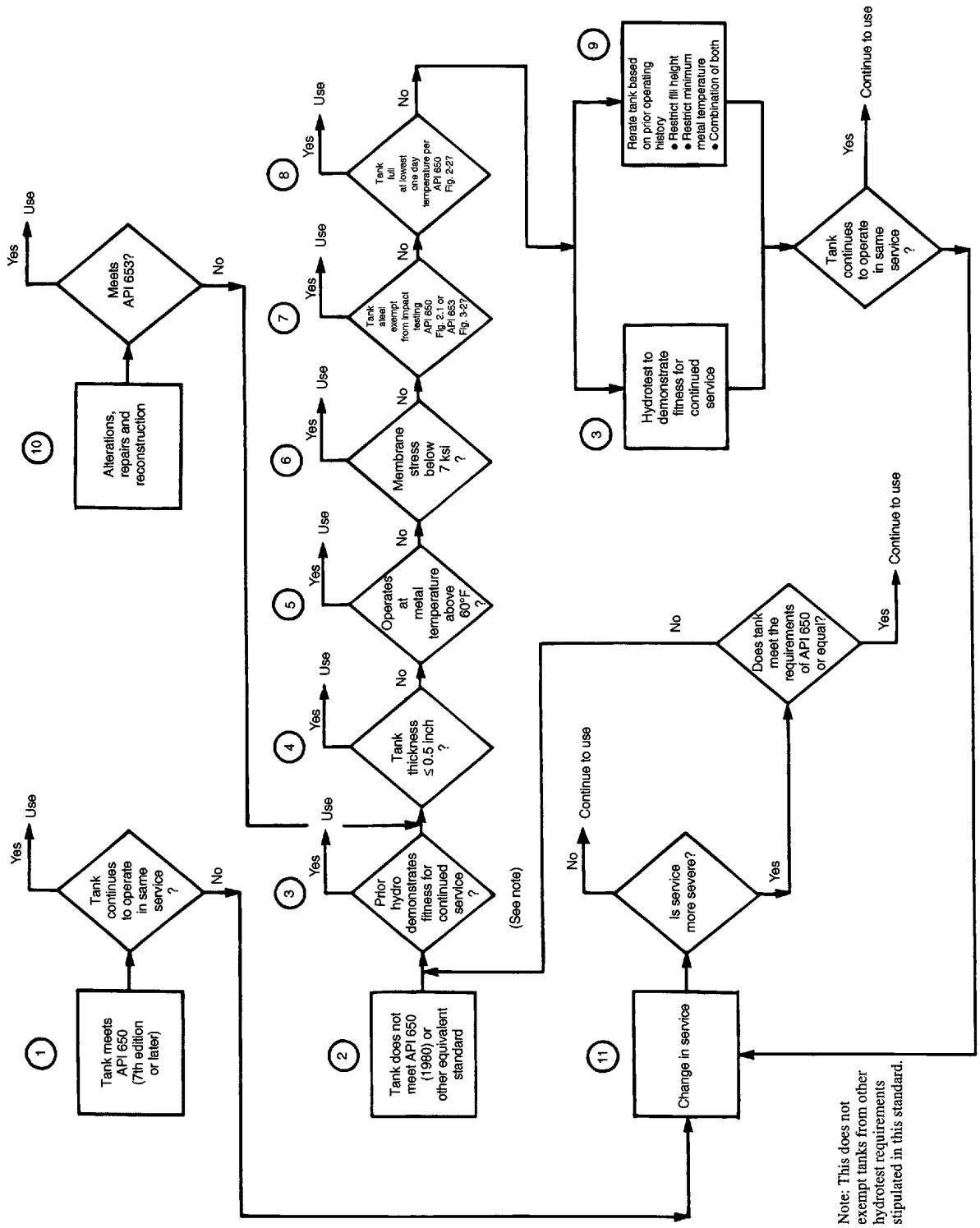


Figure 3-1—Brittle Fracture Considerations

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

3.3.8 The risk of failure due to brittle fracture is minimal once a tank has demonstrated that it can operate at a specified maximum liquid level at the lowest expected temperature without failing. For the purpose of this assessment, the lowest expected temperature is defined as the lowest one-day mean temperature as shown in Figure 2-2 of API Standard 650 for the continental United States. It is necessary to check tank log records and meteorological records to ensure that the tank had operated at the specified maximum liquid level when the one-day mean temperature was as low as shown in Figure 2-2 of API Standard 650.

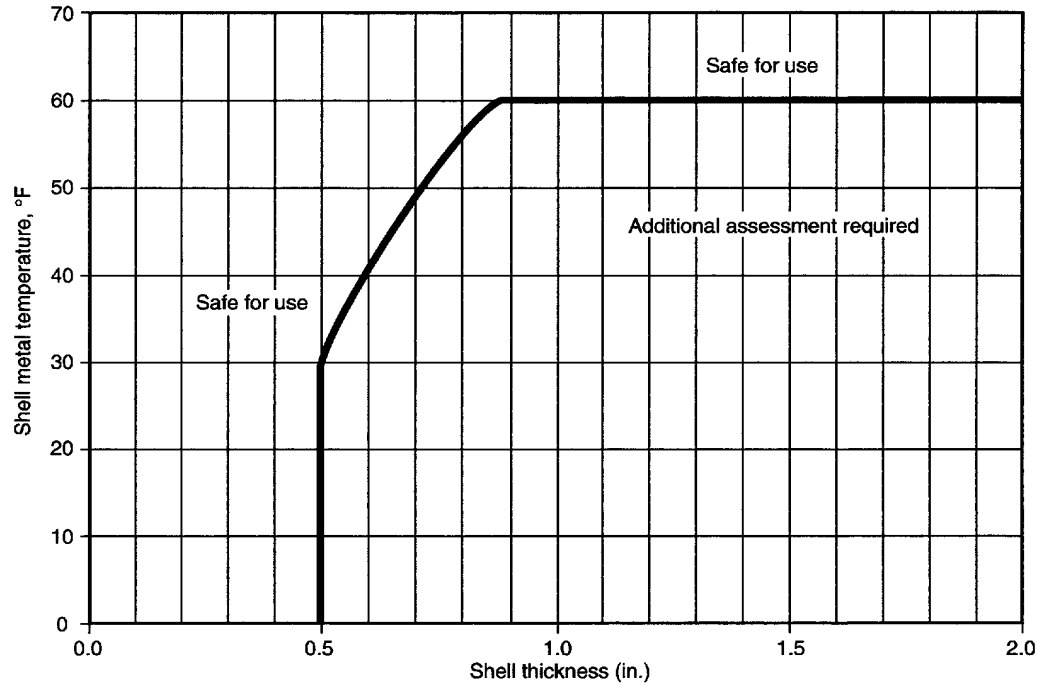
3.3.9 An evaluation can be performed to establish a safe operating envelope for a tank based on the past operating history. This evaluation shall be based on the most severe combination of temperature and liquid level experienced by the tank during its life. The evaluation may show that the tank needs to be rerated or operated differently; several options exist:

a. Restrict the liquid level.
b. Restrict the minimum metal temperature.
c. Change the service to a stored product with a lower specific gravity.
d. Combinations of a., b., and c., above.

The owner/operator can also make a more rigorous analysis to determine the risk of failure due to brittle fracture by performing a fracture mechanics analysis based upon established principles and practices. The procedures and acceptance criteria for conducting an alternative analysis are not included in this standard.

3.3.10 All repairs, alterations, and relocations shall be made in compliance with this standard.

3.3.11 An assessment shall be made to determine if the change in service places the tank at greater risk of failure due to brittle fracture. The service can be considered more severe and create a greater risk of brittle fracture if the service temperature is reduced (for example, changing from heated oil service to ambient temperature product), or the product is changed to one with a greater specific gravity and thus increased stresses.



Note: For unheated tanks, the shell metal temperature shall equal the design metal temperature as defined in 2.2.9.3 of API Standard 650. The original nominal thickness for the thickest tank shell plate shall be used for the assessment.

Figure 3-2—Exemption Curve for Tanks Constructed From Carbon Steel of Unknown Toughness

98

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 4—INSPECTION

4.1 General

Periodic in-service inspection of tanks shall be performed as defined herein. The purpose of this inspection is to assure continued tank integrity. 96

4.2 Inspection Frequency Considerations

4.2.1 Several factors must be considered to determine inspection intervals for storage tanks. These include, but are not limited to, the following:

a. The nature of the product stored.
b. The results of visual maintenance checks.
c. Corrosion allowances and corrosion rates.
d. Corrosion prevention systems.
e. Conditions at previous inspections.
f. The methods and materials of construction and repair.
g. The location of tanks, such as those in isolated or high risk areas.
h. The potential risk of air or water pollution.
i. Leak detection systems.
j. Change in operating mode (for example: frequency of fill cycling, frequent grounding of floating roof support legs).
k. Jurisdictional requirements.
l. Changes in service (including changes in water bottoms).

4.2.2 The interval between inspections of a tank (both internal and external) should be determined by its service history unless special reasons indicate that an earlier inspection must be made. A history of the service of a given tank or a tank in similar service (preferably at the same site) should be available so that complete inspections can be scheduled with a frequency commensurate with the corrosion rate of the tank. On-stream, nondestructive methods of inspection shall be considered when establishing inspection frequencies.

4.2.3 Jurisdictional regulations, in some cases, control the frequency and interval of the inspections. These regulations may include vapor loss requirements, seal condition, leakage, proper diking, and repair procedures. Knowledge of such regulations is necessary to insure compliance with scheduling and inspection requirements.

4.3 Inspections from the Outside of the Tank 96

4.3.1 ROUTINE IN-SERVICE INSPECTIONS

4.3.1.1 The external condition of the tank shall be monitored by close visual inspection from the ground on a routine basis. This inspection may be done by owner/operator personnel, and can be done by other than inspectors described in 4.10. Personnel performing this inspection should be knowledgeable of the storage facility operations, the tank, and the characteristics of the product stored.

4.3.1.2 The interval of such inspections shall be consistent with conditions at the particular site, but shall not exceed one month.

4.3.1.3 This routine in-service inspection shall include a visual inspection of the tank's exterior surface checking for: leaks; shell distortions; signs of settlement; corrosion; and condition of the foundation, paint coatings, insulation systems, and appurtenances.

4.3.2 EXTERNAL INSPECTION

4.3.2.1 All tanks shall be given a visual external inspection by an Authorized Inspector. This inspection shall be called the external inspection and must be conducted at least every 5 years or at the quarter corrosion rate life of the shell, whichever is less. Tanks may be in operation during this inspection. 96

4.3.2.2 Insulated tanks need to have insulation removed only to the extent necessary to determine the condition of the exterior wall of the tank or the roof.

4.3.2.3 Tank grounding system components such as shunts or mechanical connections of cables shall be visually checked. Recommended practices dealing with the prevention of hydrocarbon ignition are covered by API Recommended Practice 2003. 96

4.3.3 ULTRASONIC THICKNESS INSPECTION 96

4.3.3.1 External, ultrasonic thickness measurements of the shell can be a means of determining a rate of uniform general corrosion while the tank is in service, and can provide an indication of the integrity of the shell. The extent of such measurements shall be determined by the owner/operator.

4.3.3.2 When used, the ultrasonic thickness measurements shall be made at intervals not to exceed the following:

a. When the corrosion rate is not known, the maximum interval shall be five years.

Note: Corrosion rates may be estimated from tanks in similar service based on thickness measurements taken at an interval not exceeding five years.

b. When the corrosion rate is known, the maximum interval shall be the smaller of $RCA/2N$ years (where RCA is the remaining corrosion allowance in mils and N is the shell corrosion rate in mils per year) or 15 years. 98

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

4.3.3.3 Internal inspection of the tank shell, when the tank is out of service, can be substituted for a program of external ultrasonic thickness measurement if the internal inspection interval is equal to or less than the interval required in 4.3.3.2.b.

4.3.4 CATHODIC PROTECTION SURVEYS

4.3.4.1 Where exterior tank bottom corrosion is controlled by a cathodic protection system, periodic surveys of the system shall be conducted in accordance with API RP 651. The owner/operator shall review the survey results. 96

4.3.4.2 The owner/operator shall assure competency of personnel performing surveys.

4.4 Internal Inspection

4.4.1 GENERAL

4.4.1.1 Internal inspection is primarily required to:

a. Ensure that the bottom is not severely corroded and leaking.

b. Gather the data necessary for the minimum bottom and shell thickness assessments detailed in Section 2. As applicable, these data shall also take into account external ultrasonic thickness measurements made during in-service inspections (see 4.3.3).

c. Identify and evaluate any tank bottom settlement.

4.4.1.2 All tanks shall have a formal internal inspection conducted at the intervals defined by 4.4.2 or 4.4.3 below. The Authorized Inspector who is responsible for evaluation of a tank must conduct a visual inspection and assure the quality and completeness of the NDE results. If the internal inspection is required solely for the purpose of determining the condition and integrity of the tank bottom, the internal inspection may be accomplished with the tank in-service utilizing various ultrasonic robotic thickness measurement and other on-stream inspection methods capable of assessing the thickness of the tank bottom, in combination with methods capable of assessing tank bottom integrity as described in 2.4.1. Electromagnetic methods may be used to supplement the on-stream ultrasonic inspection. If an in-service inspection is selected, the data and information collected shall be sufficient to evaluate the thickness, corrosion rate, and integrity of the tank bottom and establish the internal inspection interval, based on tank bottom thickness, corrosion rate, and integrity, utilizing the methods included in this standard. An individual, knowledgeable and experienced in relevant inspection methodologies, and the Authorized Inspector who is responsible for evaluation of a tank must assure the quality and completeness of the in-service NDE results. 98

4.4.2 INSPECTION INTERVALS

4.4.2.1 Intervals between internal inspections shall be determined by the corrosion rates measured during previous inspections or anticipated based on experience with tanks in similar service. Normally, bottom corrosion rates will control and the inspection interval will be governed by the measured or anticipated corrosion rates and the calculations for minimum required thickness of tank bottoms (see 2.4.7). The actual inspection interval shall be set to ensure that the bottom plate minimum thicknesses at the next inspection are not less than the values listed in Table 4-1. In no case, however, shall the internal inspection interval exceed 20 years.

4.4.2.2 When corrosion rates are not known and similar service experience is not available to determine the bottom plate minimum thickness at the next inspection, the actual bottom thickness shall be determined by inspection(s) within the next 10 years of tank operation to establish corrosion rates.

4.4.3 ALTERNATIVE INTERNAL INSPECTION INTERVAL

As an alternative to the procedures in 4.4.2, an owner-operator may establish the internal inspection interval using risk-based inspection (RBI) procedures. Combining the assessment of the likelihood of tank leakage or failure and the consequence of tank leakage or failure is the essential element of RBI. A RBI assessment may increase or decrease the internal inspection intervals obtained using the procedures of 4.4.2.1. The RBI process may be used to establish as acceptable the risk of a minimum bottom plate thickness at the next inspection interval independent of the values in Table 4-1. The RBI assessment may also increase or decrease the 20-year inspection interval 4.4.2.1. The initial RBI assessment shall be reviewed and approved by an authorized inspector and an engineer(s), knowledgeable and experienced in tank design (including tank foundations) and corrosion. The RBI assessment shall be subsequently reviewed and approved by an authorized inspector and an engineer(s), knowledgeable and experienced in tank design (including tank foundations) and corrosion, at intervals not to exceed 10 years, or more often if warranted by changes in service. 98

Some of the factors that should be considered in a RBI assessment of a tank include the following:

a. The material of construction, including liners and coatings, relative to the product temperature and ambient conditions.

b. The design codes or standards utilized in the tank construction and repair (including tank bottoms).

c. The methods used for determination of the shell and bottom plate thickness.

d. The availability and effectiveness of the inspection methods and quality of the data collected.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

e. The analysis methods used to determine the product side, soil side, and external corrosion rates and the accuracy of these methods and corrosion rates.

f. The availability, accuracy, and need for leak detection methods and procedures.

g. The effectiveness of corrosion mitigation methods, such as cathodic protection systems, liners, and coatings.

h. The quality of the maintenance, including previous repairs.

i. The probability of and type of failure, i.e., slow leak to the environment, tank bottom rupture or tank shell brittle fracture.

j. The environmental consequence and likelihood of a tank
98 leak or failure.

Historic tank leakage and failure data and information will also be important for this assessment.

It is essential that all RBI assessments be conducted by trained, qualified individuals knowledgeable in RBI methodology, and knowledgeable and experienced in tank foundation design, construction, and corrosion. RBI assessments shall be thoroughly documented, clearly defining all the factors contributing to both the likelihood and consequence of tank leakage or failure.

After an effective RBI assessment is conducted, the results can be used to establish a tank inspection strategy and better define the most appropriate inspection methods, appropriate frequency for internal, external and on-stream inspections, and prevention and mitigation steps to reduce the likelihood and consequence of a tank leak or failure.

4.5 Alternative to Internal Inspection to Determine Bottom Thickness

In cases where construction, size, or other aspects allow external access to the tank bottom to determine bottom thickness, an external inspection in lieu of an internal inspection is allowed to meet the data requirements of Table 4-1. However, in these cases, consideration of other maintenance items may dictate internal inspection intervals. This alternative approach shall be documented and made part of the permanent record of the tank.

4.6 Preparatory Work for Internal Inspection

Specific work procedures shall be prepared and followed when conducting inspections that will assure personnel safety and health and prevent property damage in the workplace (see 1.4).

4.7 Inspection Checklists

Appendix C provides sample checklists of items for consideration when conducting in-service and out-of-service inspections.

Table 4-1—Bottom Plate Minimum Thickness

Minimum Bottom Plate Thickness[a] at Next Inspection (inches)	Tank Bottom/Foundation Design
0.10	Tank bottom/foundation design with no means for detection and containment of a bottom leak.
0.05	Tank bottom/foundation design with means to provide detection and containment of a bottom leak.
0.05	Applied tank bottom reinforced lining, >0.05 inch thick, in accordance with API RP 652.

[a] See 2.4.7

4.8 Records

4.8.1 GENERAL

Inspection records form the basis of a scheduled inspection/maintenance program. (It is recognized that records may not exist for older tanks, and judgments must be based on experience with tanks in similar services.) The owner/operator shall maintain a complete record file consisting of three types of records, namely: construction records, inspection history, and repair/alteration history.

4.8.2 CONSTRUCTION RECORDS

Construction records may include nameplate information, drawings, specifications, construction completion report, and any results of material tests and analyses.

4.8.3 INSPECTION HISTORY

The inspection history includes all measurements taken, the condition of all parts inspected, and a record of all examinations and tests. A complete description of any unusual conditions with recommendations for correction of details which caused the conditions shall also be included. This file will also contain corrosion rate and inspection interval calculations.

4.8.4 REPAIR/ALTERATION HISTORY

The repair/alteration history includes all data accumulated on a tank from the time of its construction with regard to repairs, alterations, replacements, and service changes (recorded with service conditions such as stored product temperature and pressure). These records should include the results of any experiences with coatings and linings.

4.9 Reports

4.9.1 Reports recommending repairs shall include reasons for the repairs, and sketches showing location and extent.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

4.9.2 General inspection reports shall include metal thickness measurement, conditions found, repairs, any settlement measurements, and recommendations.

4.10 Inspector Qualifications

4.10.1 Qualified inspectors shall have education and experience equal to at least one of the following:

a. A degree in engineering plus one year of experience in inspection of tanks, pressure vessels, or piping.

b. A two-year certificate in engineering or technology from a technical college, and two years of experience in construction, repair, operation, or inspection, of which one year must be in inspection of tanks, pressure vessels, or piping.

c. The equivalent of a high school education, and three years of experience in construction, repair, operation, or inspection, of which one year must be in inspection of tanks, pressure vessels, or piping.

d. Five years of experience in the inspection of aboveground storage tanks in the petroleum or chemical industries.

4.10.2 An owner/operator of tanks may designate tank inspectors qualified in accordance with 4.10.1. Such inspectors shall have the necessary authority and organizational freedom to perform their duties. Authorized Inspectors shall be certified by an agency as provided in this standard, in accordance with Appendix D. This requirement will become effective eighteen (18) months after the date of issuance of the requirement.

4.10.3 Personnel performing nondestructive examinations shall meet the qualifications identified in 10.1.1.2, but need not be certified in accordance with 4.10.2. The results of their work, however, must be considered in the evaluation of the tank by an inspector who is so certified.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 6—DESIGN CONSIDERATIONS FOR RECONSTRUCTED TANKS

6.1 General

Any specific design considerations other than normal product loading shall be specified by the owner/operator. Refer to 2.4.5, footnote 1, for release prevention systems and release prevention barrier definition.

6.2 New Weld Joints

6.2.1 Weld joint details shall meet the welding requirements of the applicable standard.

6.2.2 All new shell joints shall be butt-welded joints with complete penetration and complete fusion.

6.3 Existing Weld Joints

Existing weld joints shall meet the requirements of the original standard of construction for the tank.

6.4 Shell Design

6.4.1 Thickness to be used for each shell course when checking tank design shall be based on measurements taken within 180 days prior to relocation. (See 2.3.2 for measuring procedure, number, and locations of measured thicknesses.)

6.4.2 The maximum design liquid level for product shall be determined by calculating the maximum design liquid level for each shell course based on the specific gravity of the product, the actual thickness measured for each shell course, the allowable stress for the material in each course, and the design method to be used. The allowable stress for the material shall be determined using Table 3-2 of API Standard 650. For material not listed in Table 3-2, an allowable stress value of the lesser of $^{2}/_{3}$ yield strength or $^{3}/_{8}$ tensile strength shall be used.

6.4.3 The maximum liquid level for hydrostatic test shall be determined by using the actual thickness measured for each shell course, the allowable stress for the material in each course, and the design method to be used. The allowable stress for the material shall be determined using Table 3-2 of API Standard 650. For material not listed in Table 3-2, an allowable stress value of the lesser of $^{3}/_{4}$ yield strength or $^{2}/_{5}$ tensile strength shall be used.

6.4.4 If a corrosion allowance is required for the reconstructed tank, the required corrosion allowance shall be deducted from the actual thickness before calculating the maximum liquid level. If the actual thickness is greater than that necessary to allow the liquid level required, the extra thickness can be considered as corrosion allowance.

6.4.5 The joint efficiency and allowable stress levels used for the design liquid level calculations shall be consistent with the design method used and with the degree and type of inspection made on welded joints. The joint efficiency and allowable stress levels for existing welded joints that are not to be removed and replaced shall be based on the original degree and type of inspection.

6.5 Shell Penetrations

6.5.1 Replacement and new penetrations shall be designed, detailed, welded, and examined to meet the requirements of the current applicable standard.

6.5.2 Existing penetrations shall be evaluated for compliance with the original standard of construction.

6.6 Windgirders and Shell Stability

6.6.1 Top and intermediate windgirders for open top tanks shall meet the requirements of the applicable standard.

6.6.2 Tanks to be reconstructed shall be checked for wind-induced buckling in accordance with the procedures of the applicable standard, using the wind requirements for the location where the tank will be reconstructed.

6.7 Roofs

6.7.1 Roof designs shall meet the requirements of the applicable standard.

6.7.2 If the new site location requires a larger design live load than the original site, the adequacy of the existing roof shall be evaluated.

6.8 Seismic Design

Tanks that will be reconstructed in Seismic Zone 2 or greater (see Table E-1 in Appendix E of API Standard 650) shall be checked for seismic stability based on the rules of the applicable standard using the dimensions and thicknesses of the reconstructed tank. Reconstructed tanks shall be built to meet the stability requirements of the applicable standard. Thickened bottom plates under the bottom shell course or anchoring of the tank may be required even if not used on the original tank.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 7—TANK REPAIR AND ALTERATION

7.1 General

7.1.1 The basis for repairs and alterations shall be an API Standard 650 equivalence.

7.1.2 Hydrostatic testing requirements, nondestructive examination requirements, acceptance criteria for the welds, and repairs to shell plate and existing welds are specified in Section 10.

7.1.3 Appendix F summarizes the requirements by method of examination and provides the acceptance standards, inspector qualifications, and procedure requirements. 98 Appendix F is not intended to be used alone to determine the inspection requirements for work covered by this document. The specific requirements as listed in Sections 1 through 10 shall be followed in all cases.

7.2 Removal and Replacement of Shell Plate Material

7.2.1 MINIMUM THICKNESS OF REPLACEMENT SHELL PLATE

The minimum thickness of the replacement shell plate material shall be calculated in accordance with the applicable standard. The thickness of the replacement shell plate shall not be less than the greatest nominal thickness of any plate in the same course adjoining the replacement plate except where the adjoining plate is a thickened insert plate. Any changes from the original design conditions, such as specific gravity, design pressure, liquid level, and shell height, shall be considered.

7.2.2 MINIMUM DIMENSIONS OF REPLACEMENT SHELL PLATE

7.2.2.1 The minimum dimension for a replacement shell plate is 12 inches or 12 times the thickness of the replacement plate, whichever is greater. The replacement plate may be circular, oblong, square with rounded corners, or rectangular with rounded corners except when an entire shell plate is replaced. Refer to Figure 7-1 for typical details of acceptable replacement shell plates.

7.2.2.2 Where one or more entire shell plates or full height segments of shell plates are to be removed and replaced, the minimum spacing requirements specified in Figure 7-1 for vertical weld joints shall be maintained. It is acceptable to remove and replace entire shell plates or full height segments of shell plates by cutting and rewelding along the existing horizontal weld joints. Prior to welding the new vertical joints, the existing horizontal welds shall be cut for a minimum distance of 12 inches beyond the new vertical joints. The vertical joints shall be welded prior to welding the horizontal joints.

7.2.3 WELD JOINT DESIGN

7.2.3.1 Shell replacement plates shall be welded with butt joints with complete penetration and complete fusion, except as permitted for lapped patch shell repairs. 96

7.2.3.2 Weld joint design for replacement shell plates shall be in accordance with 3.1.5.1 through 3.2.5.3 of API Standard 650. Joints in lap-welded shell tanks may be repaired according to the original standard of construction. Lap-welded joint design for lapped patch shell repairs shall meet the requirements of 7.3. 96 Details of welding shall be in accordance with 5.2 of that standard, and Section 9 of this standard.

7.2.3.3 For existing shell plates over $^1/_2$-inch thick, the outer edge of the butt weld attaching the replacement shell plate shall be at least the greater of 8 times the weld thickness or 10 inches from the outer edge of any existing butt-welded shell joints. For existing shell plates $^1/_2$-inch thick and less, the spacing may be reduced to 6 inches from the outer edge of vertical joints or 3 inches from the outer edge of horizontal joints. See Figure 7-1 for minimum dimensions.

For existing shell plates over $^1/_2$-inch thick, the outer edge of the butt weld attaching the replacement shell plate shall be at least the greater of 8 times the weld size or 10 inches from the edge (toe) of the fillet weld attaching the bottom shell course to the bottom except when the replacement shell plate extends to and intersects the bottom-to-shell joint at approximately 90 degrees. For existing shell plates $^1/_2$-inch thick and less, this spacing may be reduced to 6 inches. The edge of any vertical weld joint attaching a replacement plate shall be at least 12 inches from the edge of a weld joint in the annular plate ring or sketch plate. Figure 7-1 has minimum dimensions.

7.2.3.4 To reduce the potential for distortion of an existing tank due to welding a replacement plate into an existing tank shell, fit-up, heat input, and welding sequence must be considered.

7.3 Shell Repairs Using Lap-Welded Patch Plates

7.3.1 Lapped patch shell repairs are an acceptable form of repair for butt-welded, lap-welded, and riveted tank shells, under the conditions outlined in 7.3.2, 7.3.3, and 7.3.4; only when specified by the owner. 96 In addition, the repair details shall comply with the requirements of 7.3.1.1 through 7.3.1.10. These repairs are permanent repairs subject to an ongoing inspection and maintenance program. These requirements may be used to evaluate existing lapped patch shell repairs; however, the plate thickness limits need not apply.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

7.3.1.1 All repair material shall comply with the requirements of the applicable standard of construction and API Standard 653.

7.3.1.2 Lapped patch shell repairs shall not be used on any shell course thickness (original construction) that exceeds $^1/_2$ inch, nor to replace doorsheets or shell plates.

7.3.1.3 Except as permitted in 7.3.3.2 and 7.3.4.3, the repair plate material shall be the smaller of $^1/_2$ inch or the thickness of the shell plate adjacent to the repairs, but not less than $^3/_{16}$ inch.

7.3.1.4 The shape of the repair plate may be circular, oblong, square, or rectangular. All corners, except at the shell-to-bottom joint, shall be rounded to a minimum radius of 2 inches. The nozzle reinforcing plate shapes of API Standard 650 are also acceptable. 96

7.3.1.5 The repair plate may cross any butt-welded vertical or horizontal shell seams that have been ground flush, but must overlap a minimum of 6 inches beyond the shell seam. The weld spacing requirements of Figure 7-1 shall be used as a basis for locating repair plates relative to butt-welded, fillet-welded, and riveted seams and other repair plates.

7.3.1.6 Repair plates may extend to and intersect with the external shell-to-bottom joint if the vertical sides intersect the tank bottom at a 90-degree angle and the shell-to-bottom weld is in conformance with Figure 7-2. Repair plates positioned on the shell interior shall be located such that the toe-to-toe weld clearances are a minimum of 6 inches to the shell-to-bottom weld.

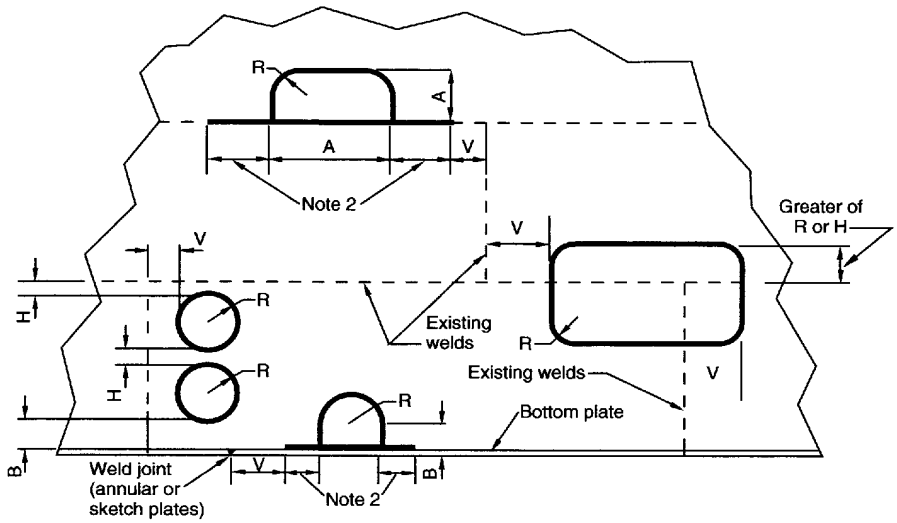
7.3.1.7 The maximum vertical and horizontal dimension of the repair plate is 48 inches and 72 inches, respectively. The minimum repair plate dimension is 6 inches. The repair plate shall be formed to the shell radius.

7.3.1.8 Shell openings and their reinforcements shall not be positioned within a lapped patch shell repair.

7.3.1.9 Prior to application of a lapped patch shell repair, the areas to be welded shall be ultrasonically inspected for plate defects and remaining thickness. 96

7.3.1.10 Repair plates shall not be lapped onto lap-welded shell seams, riveted shell seams, other lapped patch repair plates, distorted areas, or unrepaired cracks or defects.

7.3.2 Lapped patch repair plates may be used for the closure of holes caused by the removal of existing shell openings or the removal of severely corroded or eroded areas. In addition, the following requirements should be satisfied.



Dimension	Minimum weld spacing between edges (toes) of welds for thickness of replacement shell plate, *t*, (inches)	
	$t \leq 0.5$ inch	$t > 0.5$ inch
R	6"	Greater of 6" or 6*t*
B	6"	Greater of 10" or 8*t*
H	3"	Greater of 10" or 8*t*
V	6"	Greater of 10" or 8*t*
A	12"	Greater of 12" or 12*t*

Notes:
1. All weld intersections shall be at approximately 90°.
2. Prior to welding new vertical joints, cut existing horizontal weld for a minimum of 12" beyond the new vertical joints. Weld the horizontal joint last.

Figure 7-1—Acceptable Details for Replacement of Shell Plate Material

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

not meet the current design metal temperature criteria. In addition, the following requirement shall be met:

a. The minimum diameter of the insert plate shall be at least twice the diameter of the penetration or the diameter plus 12 inches, whichever is greater.
b. When reinforcing plates are used, the minimum diameter of the insert plate shall equal the diameter of the reinforcing plate plus 12 inches.
c. If an integral reinforcement design is used, the thickness of the insert plate–at its periphery and for a distance of $2t$ from the edge of the weld (where t is the shell plate thickness)–shall be the same thickness as the shell, or the insert plate shall have a 1:4 taper to match the shell plate thickness.
d. Spacing of welds shall be in accordance with Figure 7-1.
e. The joint between new material of the insert plate and old material of the existing shell plate shall be butt-welded.

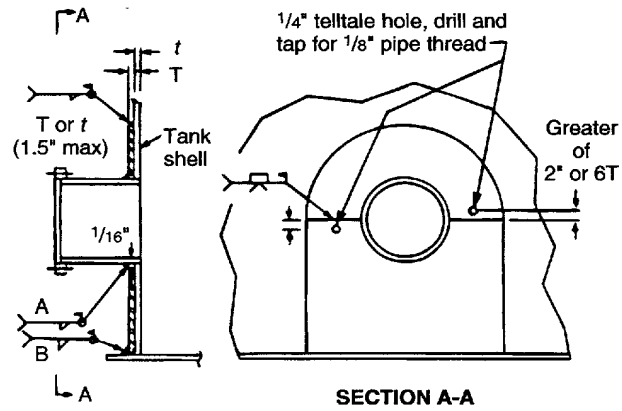
7.9 Alteration of Existing Shell Penetrations

7.9.1 Existing shell penetrations may be modified if the altered details comply with the requirements of API Standard 650, including the requirements for minimum reinforcing area and the requirements for spacing of welds around connections.

7.9.2 When installing a new tank bottom above the existing bottom, it may be necessary to alter existing shell penetrations in the bottom course of a tank shell. If the new bottom is "slotted" through the tank shell several inches above the existing bottom, the spacing between existing welds around penetrations and the new bottom-to-shell weld may not comply with API Standard 650 requirements. Options for altering the penetrations and/or reinforcing plates are given in 7.9.2.1 through 7.9.2.3.

7.9.2.1 The existing reinforcing plate may be trimmed to increase the spacing between the welds provided that the modified detail complies with the requirements of API Standard 650. Care must be exercised during the trimming operation to avoid damaging the shell material beneath the reinforcing plate. The existing weld attaching the portion of the reinforcing plate to be removed shall be completely removed by gouging and grinding.

7.9.2.2 The existing reinforcing plate may be removed and a new reinforcing plate added except that reinforcement plate replacement is not permitted on existing stress relieved assemblies. If it is not known whether the assembly was thermally stressed relieved, then the alteration shall meet the requirements of API Standard 650, paragraph 3.7.4. Care must be exercised when removing the existing reinforcing plate to avoid damaging the shell plate beneath the reinforcing plate. The welds around the perimeter of the reinforcing plate and between the reinforcing plate and neck of the penetration shall be completely removed by gouging and grinding. The new reinforcing plate shall be in accordance with Figure 7-2a. If required to maintain weld spacing, a "tombstone" shaped reinforcing plate may be used. (See Figure 7-3).



SECTION A-A

t and T (maximum)	A	B
9/16"	1/4"	1/4"
3/4"	5/16"	1/4"
15/16"	3/8"	5/16"
1 1/8"	7/16"	5/16"
1 5/16"	1/2"	3/8"
1 9/16"	9/16"	3/8"
1 3/4"	5/8"	3/8"

Note: All details, dimensions, and weld spacing shall be in accordance with the requirements of API Standard 650.

Figure 7-3—Typical Details for Addition of "Tombstone" Shape Reinforcing Plate to Existing Shell Penetration

7.9.2.3 The existing penetration may be moved by cutting the section of the shell containing the fitting and reinforcing plate, and raising the entire assembly to the correct elevation (see Figure 7-4).

7.9.3 Any components of the penetration (neck, flange, and reinforcing plate) that are in serviceable condition after removal may be reused.

7.10 Repair of Tank Bottoms

7.10.1 REPAIRING A PORTION OF TANK BOTTOMS

7.10.1.1 General Repair Requirements

The use of welded-on patch plates for repairing a portion of tank bottoms is permitted within the limitations given in this section and 7.10.1.2. Refer to Figure 7-5 for typical details for welded-on patch plates.

a. The minimum dimension for a welded-on patch plate is 12 inches. The welded-on patch plate may be circular, oblong, or rectangular (rounded corners are recommended).
b. Welded-on patch plates may be placed over existing patches.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

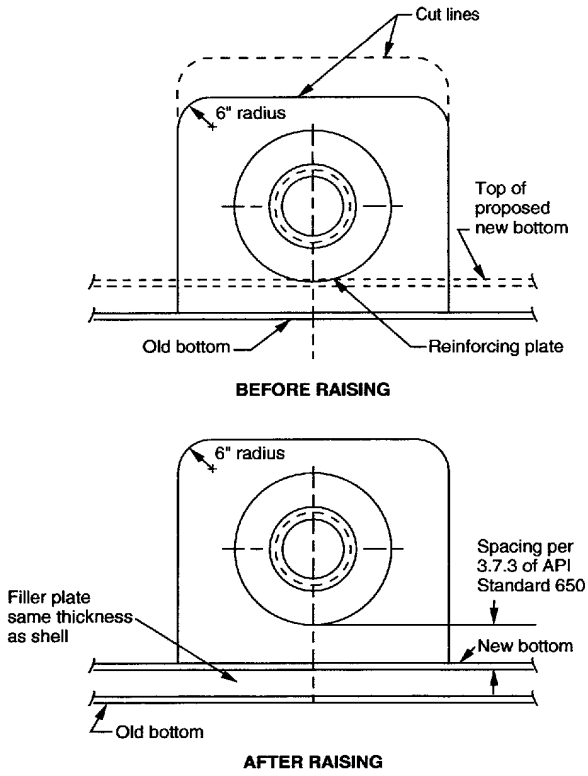


Figure 7-4—Method for Raising Shell Nozzles

c. Welded-on patch plates shall not be placed over areas of the tank bottom that have global dishing, severe local dishing, settlement, or distortion greater than the limits of Appendix B.

Note: If the tank is still undergoing settlement, the addition of welded-on patch plate may not be advisable.

d. Other welded-on plates added to an existing bottom, such as wear and isolation plates, shall be installed in accordance with 7.10.1, and examined in accordance with 10.1.6 and 10.1.7.

e. These repairs are permanent repairs subject to an on-going inspection and maintenance program. 98

7.10.1.2 Repairs Within the Critical Zone

The use of welded-on patch plates is permitted for repairing a portion of tank bottoms within the critical zone (see 1.5 for definition) provided 7.10.1.1 requirements and the following additional requirements are met.

a. Maximum plate thickness for welded-on patch plates within the critical zone is $^1/_4$-inch and must meet the toughness requirements of 2.2.9 of API Standard 650.

b. When a welded-on patch plate is within 6 inches of the shell, the welded-on patch plate shall be "tombstone" shaped. The sides of the "tombstone" shaped welded-on patch plate shall intersect the shell-to-bottom joint at approximately 90 degrees.

c. Perimeter welds on welded-on patch plates within the critical zone shall be two-pass, minimum, and examined per 10.1.7.2 and 10.1.7.3.

d. Installation of a welded-on patch plate by butt-welding to an adjacent existing patch is not permitted in the critical zone.

e. These requirements may be used to evaluate existing welded-on patch plates in the critical zone.

f. Welded-on patch plates over existing patches are not allowed in the critical zone.

g. The bottom plate under the perimeter of a welded-on patch plate shall meet the thickness requirements in 2.4.

Note: The bottom plate thickness at the attachment weld must be at least 0.1 inches thick before welding the welded-on patch plate to the bottom plate. Refer to API Publication 2207 for further information.

7.10.1.2.1 No welding or weld overlays are permitted within the critical zone except for welding of: widely scattered pits (see 2.3.2.2), pinholes, cracks in the bottom plates, the shell-to-bottom weld, welded-on patch plates, or where the bottom plate welded to the shell is being replaced.

7.10.1.2.2 A welded-on patch plate shall not be used if the covered bottom plate minimum remaining thickness at the toe of the internal shell-to-bottom weld will be less than the minimum thickness required by 2.4.7 or 2.4.8 at the next internal inspection. 98

7.10.1.2.3 Welded-on patch plates are not permitted in the critical zone on a tank bottom with an operating temperature exceeding 200°F for carbon steel or 100°F for stainless steel.

7.10.1.2.4 If more extensive repairs are required within the critical zone than those listed in 7.10.1.2, the bottom plate welded to the shell shall be cut out and a new plate shall be installed. Weld spacing requirements shall be in accordance with 3.1.5.4 and 3.1.5.5 of API Standard 650.

7.10.1.3 The use of welded-on patch plates that do not meet the requirements of 7.10.1.1 or 7.10.1.2 is permitted if the repair method has been reviewed and approved by an engineer experienced in storage tank design in accordance with API Standard 650. The review shall consider brittle fracture, stress due to settlement, stress due to shell-bottom discontinuity, metal temperature, fracture mechanics, and the extent and quality of nondestructive examination.

7.10.1.4 The repair of sumps located within the critical zone shall be in accordance with 7.10.1.2.

7.10.2 REPLACEMENT OF ENTIRE TANK BOTTOM 96

7.10.2.1 Requirements governing the installation of a replacement bottom over an existing bottom are given in 7.10.2.1.1 through 7.10.2.1.6. 96

7.10.2.1.1 A suitable noncorrosive material cushion such as sand, gravel, or concrete shall be used between the old bottom and the new bottom. 96

7.10.2.1.2 The shell shall be "slotted" with a uniform cut made parallel to the tank bottom. The cut edges in the slot

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0616746 235 ■

shall be ground to remove all slag and burrs from the cutting operation. The new bottom plate shall be extended outside the shell as required by API Standard 650. All rules for weld spacing shall be followed.

7.10.2.1.3 Voids in the foundation below the old bottom shall be filled with sand, crushed limestone, grout, or concrete.

7.10.2.1.4 Existing shell penetrations shall be raised if the elevation of the new bottom cuts through the reinforcing plate or if the weld space requirements given in 3.7.3 of API Standard 650 are not met.

7.10.2.1.5 For floating roof tanks, the new bottom profile must keep the roof level when it is resting on its support legs in the down position. The support legs can either be jacked up to maintain the original height or be shortened by the same amount as the thickness of the cushion and new bottom plate.

7.10.2.1.6 New bearing plates for floating roof support legs and for fixed roof support columns shall be installed.

7.10.2.2 When removing an existing tank bottom, the tank shell shall be separated from tank bottom either by:

a. Cutting the shell parallel to the tank bottom a minimum of $^1/_2$ inch above the bottom-to-shell weld (cut line B-B as shown in Figure 8-1), or
b. Removing the entire shell-to-bottom attachment weld, including any penetration and heat affected zone by suitable methods such as arc gouging and/or grinding.

If b. is elected, all arc gouged areas of the tank shell shall be magnetic particle examined, and defective areas repaired and re-examined.

7.10.2.3 Installation of a new tank bottom, after removal of the existing tank bottom, shall meet all requirements of API Standard 650. For tanks with shell plate of unknown toughness, which does not meet the exemption criteria of Figure 3-2, new weld joints in the bottom or annular ring shall be spaced at least the greater of 3 inches or $5t$ from existing vertical weld joints in the bottom shell course, where t is the thickness of the bottom shell course, in inches.

7.10.2.4 When planning a bottom replacement, consideration shall be given to removing the old tank bottom or providing other means of preventing potential galvanic corrosion (refer to API RP 651). Also see 2.4.5 regarding bottom leak detection.

7.11 Repair of Fixed Roofs

7.11.1 SUPPORTED CONE ROOFS

7.11.1.1 The minimum thickness of new roof plates shall be $^3/_{16}$ inch plus any corrosion allowance as specified in the repair specifications. In the event roof live loads in excess of 25 pounds per square foot are specified (such as insulation, operating vacuum, high snow loads), the plate thickness shall be based on analysis using the allowable stresses in conformance with 3.10.3 of API Standard 650 (see 7.11.2.2).

7.11.1.2 The roof supports (rafters, girders, columns, and bases) shall be repaired or altered such that under design conditions the resulting stresses do not exceed the stress levels given in 3.10.3 of API Standard 650.

7.11.2 SELF-SUPPORTING ROOFS

7.11.2.1 The nominal thickness of new roof plate shall be $^3/_{16}$ inch or the required plate thickness given in 3.10.5 or 3.10.6 of API Standard 650, plus any specified corrosion allowance, whichever is greater.

7.11.2.2 The details of the roof-to-shell junction shall meet the requirements of 3.10.5 or 3.10.6 or Appendix F of API Standard 650, as applicable, for the intended service. Frangible joint design, or the need for emergency venting devices in accordance with API Standard 2000, shall meet the requirements of 3.10.2.5 of API Standard 650.

7.12 Floating Roofs

7.12.1 EXTERNAL FLOATING ROOFS

Any method of repair is acceptable that will restore the roof to a condition enabling it to perform as required.

7.12.2 INTERNAL FLOATING ROOFS

Repairs to internal floating roofs shall be made in accordance with the original construction drawings, if available. If the original construction drawings are not available, the roof repairs shall be in compliance with the requirements of Appendix H of API Standard 650.

7.12.3 REPAIR OF LEAKS IN PONTOONS

All leaks in pontoons or compartments of double deck floating roofs shall be repaired by rewelding the leaking joints and/or use of patch plates.

7.13 Repair or Replacement of Floating Roof Perimeter Seals

7.13.1 PRIMARY SEALS

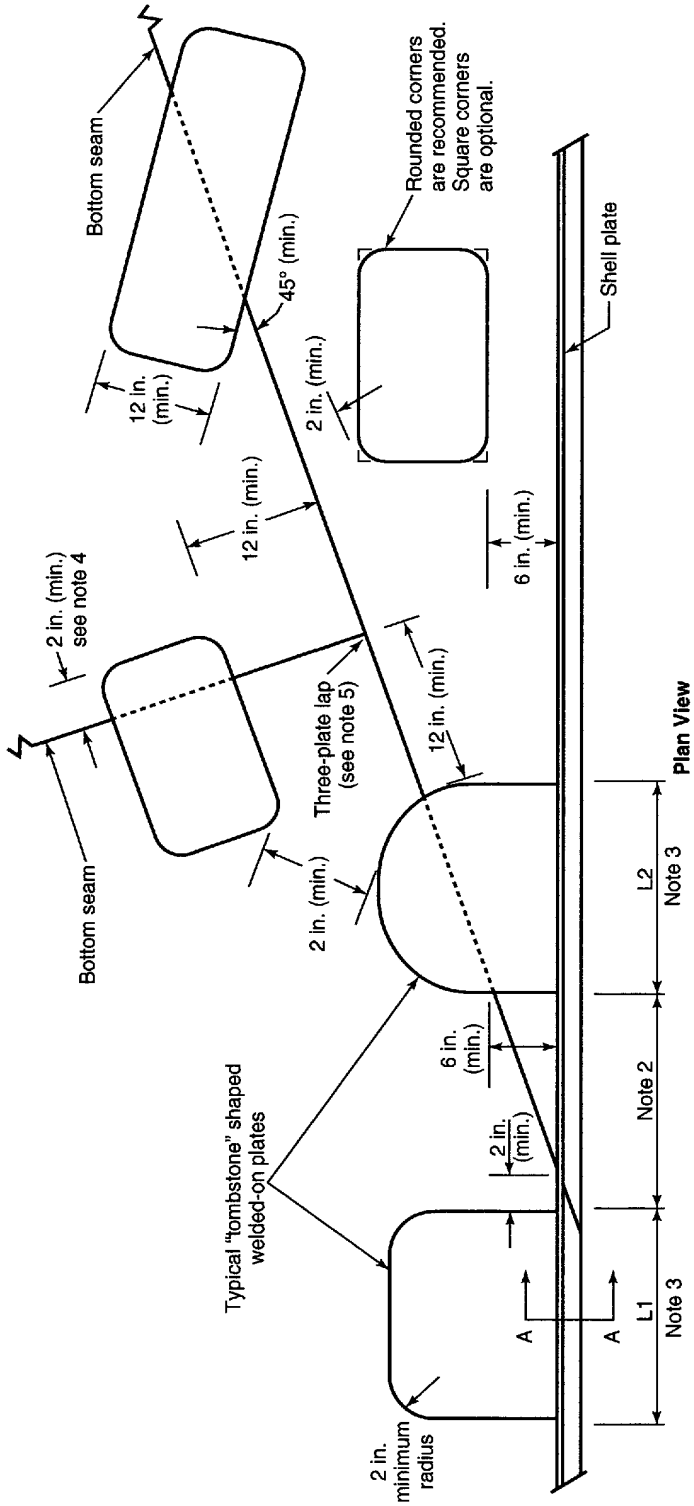
Rim-mounted primary shoe seals and toroidal seal systems can be removed, repaired, or replaced. To minimize evaporation losses and reduce potential hazard to the workers, no more than one-fourth of the roof seal system should be out of an in-service tank at one time. Temporary spacers to keep the roof centered shall be used during the repairs. Primary seal systems mounted partly or fully below the bolting bar or top of the rim usually cannot be reached to allow removal in service. In this case, in-service repairs are limited to replacement of the primary seal fabric.

7.13.2 SECONDARY SEALS

Rim-mounted and shoe-mounted secondary seals may be readily installed, repaired, or replaced while the tank is in service.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0616747 171 ■



Notes:
1. Dimensions given are from toe to toe of fillet welds.
2. Minimum distance between two welded-on patch plates in the critical zone shall be one half of the lesser of L1 or L2.
3. The maximum dimension along the shell for welded-on patch plates in the critical zone is 24 inches.
4. When the edge of a welded-on patch plate is approximately parallel to a bottom seam, the edge shall be held at least 2 inches from weld seam.
5. Patches over three-plate laps shall extend 12 inches minimum beyond the three-plate lap.

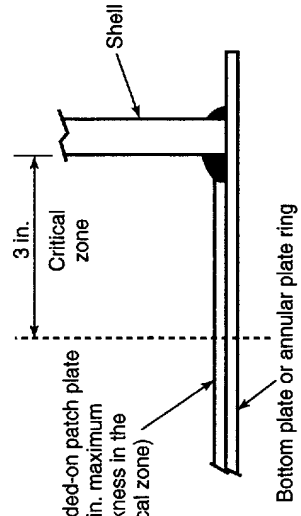


Figure 7-5—Typical Welded-on Patch Plates on Tank Bottom Plates

98

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

96 | **7.13.3 SEAL-TO-SHELL GAP**

Repair and other corrective actions to maintain seal-to-shell gap requirements, include:

a. Adjusting the hanger system on primary shoe seals, and adding foam filler in toroidal seals.
b. Increasing the length of rim mounted secondary seals in the problem area.
c. Replacing all or part of the primary seal system along with possible installation of a rim extension for a secondary seal. This step shall be taken only after checking the annular space variation at several levels from low pump out to high liquid level.

96 | **7.13.4 MECHANICAL DAMAGE**

Damaged parts shall be repaired or replaced. Before taking this action, the cause of the damage shall be identified and corrected. Buckled parts shall be replaced, not straightened. Torn seal fabric shall be replaced.

96 | **7.13.5 DETERIORATION OF SEAL MATERIAL**

Material deterioration results from wear and corrosion on metallic elements, and chemical and weather deterioration of seal fabric. The service life and inspection information shall be used to determine whether a change of material is warranted.

96 | **7.13.6 INSTALLATION OF PRIMARY AND SECONDARY SEALS**

7.13.6.1 The replacement or addition of primary and secondary seals shall be in accordance with the recommendations of the seal manufacturer. In addition, the final installation shall comply with all applicable jurisdictions.

7.13.6.2 If the roof rim thickness is less than 0.10 inch thick, it shall be replaced. The new roof rim shall be $^3/_{16}$-inch thickness, minimum.

7.14 Hot Taps

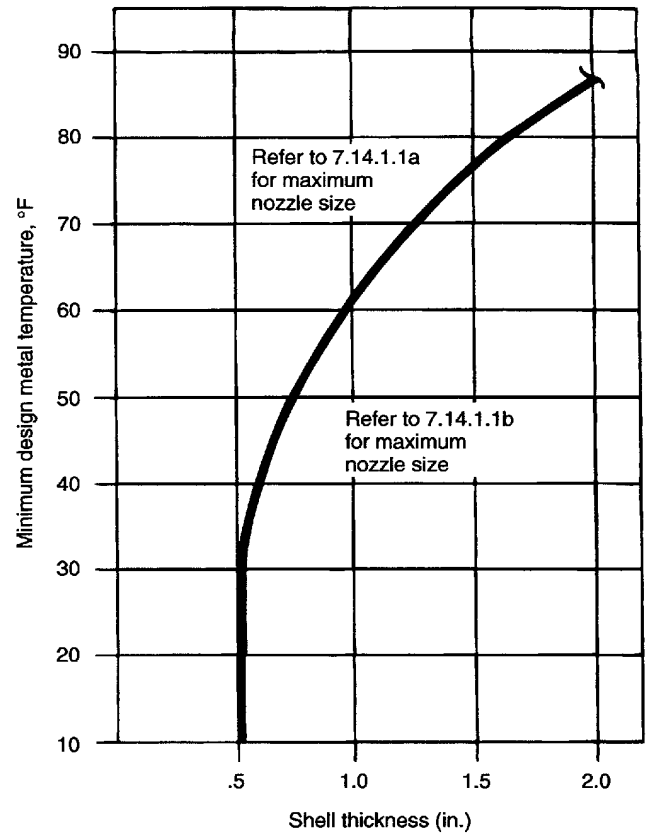
7.14.1 GENERAL

7.14.1.1 The requirements given herein cover the installation of radial hot tap connections on existing in-service tanks constructed with shell material that does not require postweld
96 | heat treatment.

a. For tank shell plates of recognized toughness, shell plates of unknown toughness having a maximum thickness less than or equal to $^1/_2$ inch, or shell plates of unknown toughness having a minimum shell design metal temperature at or above the
98 | curve given in Figure 7-6, the connection size and shell thickness limitations are shown in Table 7-1.

Table 7-1—Hot Tap Connection Sizes and Shell Plate Thicknesses

Connection Size, NPS (inches)	Minimum Shell Plate Thickness (inches)
98 | ≤6	$^3/_{16}$
≤8	$^1/_4$
≤14	$^3/_8$
≤18	$^1/_2$



96

Figure 7-6—Curve for Hot Tapping Tanks With Steels of Unknown Toughness 98

Recognized toughness is defined as meeting or exceeding the toughness requirements of API 650, Sixth Edition, Appendix D or G, API 650, Seventh Edition, or later, or other industry accepted toughness requirements.

b. For tank shell plates of unknown toughness with thickness greater than $^1/_2$ inch and having a minimum shell design metal temperature below the curve given in Figure 7-6 the following limitations apply. 98

1. Nozzles shall be limited to a maximum diameter of 4 inches NPS.
2. The shell plate temperature shall be at or above the minimum shell design metal temperature for the entire hot tapping operation. 96
3. All nozzles shall be reinforced. The reinforcement shall be calculated per API 650, 3.7.2. The minimum thickness of the reinforcing plate shall be equal to the shell plate thickness, and the minimum reinforcing plate diameter shall not be less than the diameter of the shell cutout plus 2 inches.
4. The maximum height of tank liquid above the hot tap location during the hot tapping operation shall be such that the hydrostatic tank shell stress is less than 7,000 pounds per square inch at the elevation of the hot tap.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

7.14.1.2 The minimum height of tank liquid above the hot tap location shall be at least 3 feet during the hot tapping operation.

7.14.1.3 Welding shall be done with low hydrogen electrodes.

7.14.1.4 Hot taps are not permitted on the roof of a tank or within the gas/vapor space of the tank. 96

7.14.1.5 Hot taps shall not be installed on laminated or severely pitted shell plate.

7.14.1.6 Hot taps are not permitted on tanks where the heat of welding may cause environmental cracking (such as caustic cracking or stress corrosion cracking).

7.14.2 HOT TAP PROCEDURES

A hot tap procedure specific to carrying out the work shall be developed and documented. The procedure shall include the practices given in API Publication 2201.

96 7.14.3 PREPARATORY WORK

7.14.3.1 Minimum spacing in any direction (toe-to-toe of welds) between the hot tap and adjacent nozzles shall be equivalent to the square root of RT where R is the tank shell radius, in inches, and T is the shell plate thickness, in inches.

7.14.3.2 Shell plate thickness measurements shall be taken at a minimum of four places along the circumference of the proposed nozzle location.

7.14.4 MATERIAL LIMITATIONS

Hot tap only steels of known acceptable toughness or steels of unknown toughness that have a minimum shell design metal temperature at or above the exemption curve shown in Figure 7-6, unless the additional requirements of 7.14.1.1b are met. 96 98

7.14.5 INSTALLATION PROCEDURE

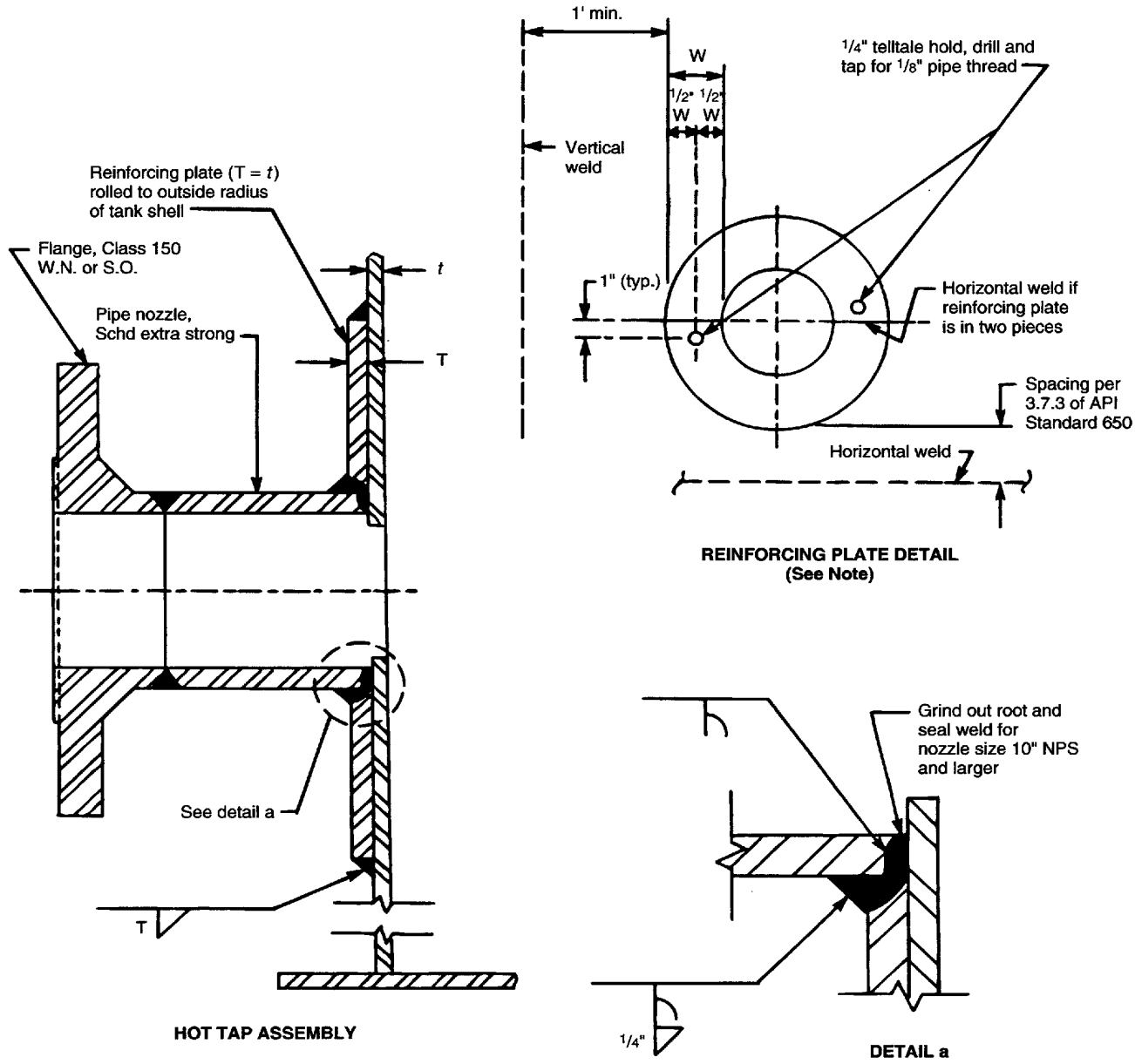
7.14.5.1 Pipe nozzles shall be cut to the contour of the shell and beveled from the outside for a full penetration weld (see Figure 7-7). 98

7.14.5.2 After the pipe is welded, the reinforcing plate shall be installed either in one piece or two pieces with horizontal weld. The reinforcing plate to nozzle shall be installed with a full penetration weld. Care shall be taken to limit the heat input to the welds.

7.14.5.3 After the reinforcing plate has been welded to the shell and nondestructive examination carried out, the pad shall be pneumatically tested by the procedure described in API Standard 650. After the valve has been installed on the flange, a pressure test at least 1.5 times the hydrostatic head shall be performed on the nozzle prior to mounting the hot tap machine—which shall be bolted to the valve.

7.14.5.4 A qualified operator shall operate the hot tap machine and cut the hole in the tank following the hot tap machine manufacturer's procedures.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34



Note: Reinforcing plate diameters as given in Table 3-8 of API Standard 650.

Figure 7-7—Hot Tap for Tanks

98

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 8—DISMANTLING AND RECONSTRUCTION

8.1 General

8.1.1 This section provides procedures for dismantling and reconstructing existing welded tanks that are to be relocated from their original site.

8.1.2 Hydrostatic testing requirements, nondestructive examination requirements, and acceptance criteria for the quality of welds for reconstructed tanks are specified in Section 10.

8.2 Cleaning and Gas Freeing

The tank shall be cleaned and gas-freed prior to commencement of dismantling.

8.3 Dismantling Methods

8.3.1 GENERAL

Roof, shell, and bottom plates may be cut into any size pieces that are readily transportable to the new site for reconstruction.

8.3.2 BOTTOMS

8.3.2.1 Bottom plates that will be reused shall be cut by deseaming of lapwelds; or by cutting alongside of the remaining welds at a minimum of 2 inches away from existing welds, except where cuts cross existing weld seams.

8.3.2.2 If the bottom is to be used, one of the following methods is acceptable:

a. The bottom plates may be cut from the shell along a line A-A and line B-B shown in Figure 8-1, scrapping the welds and the bottom plate directly attached to the shell.

97 b. If the entire bottom is to be reused, the bottom may be cut from the shell on the line C-C leaving the shell with part of the bottom attached.

c. If the tank has an existing butt-welded annular ring, this ring can be left attached to the shell or removed from the shell by cutting out along line B-B or otherwise removing the existing shell to annular ring welds.

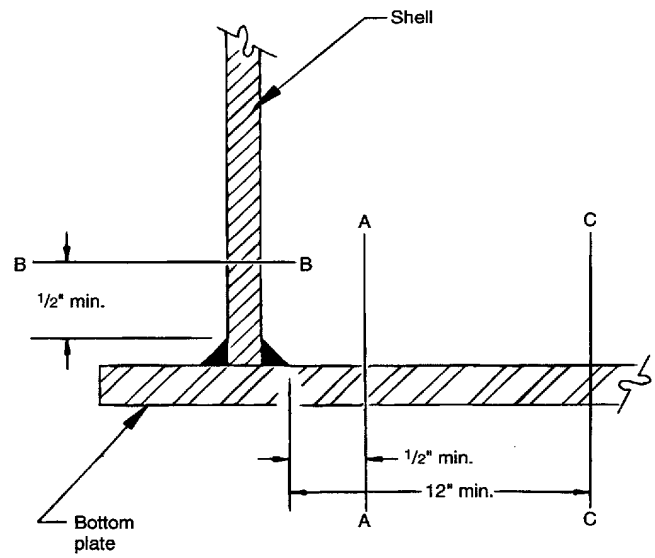
8.3.3 SHELLS

8.3.3.1 Tank shell plates may be dismantled using one of the following methods or a combination thereof:

a. Any shell ring may be dismantled by cutting out existing weld seams and the heat affected zone (HAZ) of the weld.

98 For the purpose of this method, the minimum HAZ to be removed will be on-half of the weld metal width or $^1/_4$ inch, whichever is less, on both sides of the weld seam.

b. Any shell ring $^1/_2$ inch thick or thinner may be dismantled by cutting through the weld without removing the HAZ.



97

Figure 8-1—Tank Shell and Bottom Cut Locations

c. Shell rings may be dismantled by cutting vertical and/or horizontal cuts through the shell a minimum of 6 inches away from existing welds, except where cuts cross existing welds. 98

8.3.3.2 Shell stiffening rings, including wind girders and top angles, may be left attached to the shell plates or may be removed by cutting at the attachment welds. The area where temporary attachments are removed shall be ground flush with the shell plate.

8.3.3.3 The shell shall be cut from the bottom plate along line B-B as shown in Figure 8-1. The existing shell-to-bottom weld connection shall not be reused unless the entire bottom is to be reused intact.

8.3.4 ROOFS

8.3.4.1 Roof plates shall be cut by deseaming of lap welds, or by cutting alongside of the remaining welds at a minimum of 2 inches away from existing welds, except where cuts cross existing welds.

8.3.4.2 Roof supporting structures shall be dismantled by removing bolts (if bolted) or deseaming the structural attachment welds.

8.3.5 PIECE MARKING

8.3.5.1 Shell, bottom, and roof plates shall be marked prior to dismantling for ready identification and placement when the tank is reconstructed. Marking material shall be of a durable type. Drawings showing piece mark locations are also a useful adjunct.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

8.3.5.2 A minimum of two sets of matching center punch marks shall be located on the top and bottom edges of each shell plate to facilitate proper alignment during reconstruction.

8.4 Reconstruction

8.4.1 GENERAL

8.4.1.1 The foundation for the reconstructed tank shall meet the construction tolerances given in 8.5.6.

8.4.1.2 Temporary attachments shall be removed, and the attachment area ground flush with the plate surface.

8.4.2 WELDING

8.4.2.1 Provisions shall be made during the reconstruction of a tank to ensure that weld spacing requirements of Figure 7-1 are maintained. New vertical joints in adjacent shell courses, made in accordance with 8.3.3.1, shall not be aligned but shall be offset from each other a minimum distance of $5t$, where t is the plate thickness of the thicker course at the point of the offset. 98

8.4.2.2 Tanks and their structural attachments shall be welded in accordance with the processes specified in API Standard 650 and the requirements of 8.4.2.3 through 8.4.2.11.

8.4.2.3 No welding of any kind shall be performed when the surfaces of the parts to be welded are wet from rain, snow, or ice; when rain or snow is falling on such surfaces; or during periods of high winds unless the welder and the work are properly shielded. No welding of any kind shall be performed when the temperature of the base metal is less than 0°F. When the temperature of the base metal is between 0°F and 32°F or the thickness is in excess of 1 inch, the base metal within 3 inches of the place where welding is to be started shall be heated to a temperature warm to the hand (approximately 140°F) before welding. (See 8.4.4.3 for preheat requirements for shell plates over $1^1/_2$-inches thick.)

8.4.2.4 Each layer of weld metal of multilayer welding shall be cleaned of slag and other deposits before the next layer is applied.

8.4.2.5 The edges of all welds shall merge with the surface of the plate without a sharp angle. The maximum acceptable undercutting is $^1/_{64}$ inch of the base metal for vertical butt joints. For horizontal butt joints undercutting not exceeding $^1/_{32}$ inch in depth is acceptable.

8.4.2.6 The reinforcement of the new welds on all butt joints on each side of the plate shall not exceed the thicknesses shown in Table 8-1.

Table 8-1—Maximum Thicknesses on New Welds

Plate Thickness (inches)	Maximum Reinforcement Thickness (inches)	
	Vertical Joints	Horizontal Joints
$\leq^1/_2$	$^3/_{32}$	$^1/_8$
$>^1/_2$ through 1	$^1/_8$	$^3/_{16}$
>1	$^3/_{16}$	$^1/_4$

98

8.4.2.7 Tack welds used in the assembly of vertical joints of tank shells shall be removed and shall not remain in the finished joint when the joints are welded manually. When such joints are welded by the submerged-arc process, the tack welds shall be thoroughly cleaned of all welding slag but need not be removed provided that they are sound and are thoroughly fused into the subsequently applied weld beads. Tack welds shall be made using a fillet-weld or butt-weld procedure qualified in accordance with Section IX of the ASME Code. Tack welds to be left in place shall be made by qualified welders.

8.4.2.8 If weldable primer coatings have been applied on surfaces to be welded, they shall be included in welding procedure qualification tests for the brand, formulation, and maximum thickness of primer applied. All other coatings shall be completely removed from the weld area prior to welding.

8.4.2.9 Low-hydrogen electrodes shall be used for manual metal-arc welds, including the attachment of the first shell course to the bottom plates or annular plate ring, as follows:

a. For all welds in shell courses over $^1/_2$-inch thick of API Standard 650 Group I-III materials.
b. For all welds in all shell courses of API Standard 650 Group IV-VI materials.

8.4.2.10 Low-hydrogen electrodes shall be used for welding temporary and new permanent attachments to shells of API Standard 650 Group IV, IVA, V, or VI materials. The welding procedure selected shall not cause underbead cracking; also, the need for preheat of thick plates and the effect of low atmospheric temperature during welding shall be considered.

8.4.2.11 If existing welds are found to be unsatisfactory by the original standard of construction, they shall be repaired in accordance with 7.6. 96

8.4.3 BOTTOMS

8.4.3.1 After the bottom plates are laid out and tacked, they shall be joined by welding the joints in a sequence that results in the least distortion from shrinkage and provides, as nearly as possible, a plane surface.

8.4.3.2 The welding of the shell to the bottom (except for door sheets) shall be completed before the welding of bottom joints.

8.4.3.3 Plates shall be held in close contact at all lap joints during welding.

8.4.4 SHELLS

8.4.4.1 Plates to be joined by butt welding shall be matched accurately and retained in position during welding. Misalignment in completed vertical joints over $^5/_8$-inch thick

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 10—EXAMINATION AND TESTING

10.1 Nondestructive Examinations

10.1.1 GENERAL

10.1.1.1 Nondestructive examination procedures, qualifications, and acceptance criteria shall be prepared for visual, magnetic particle, liquid penetrant, ultrasonic, and radiographic methods in accordance with API Standard 650 and the supplemental requirements given herein.

10.1.1.2 Personnel performing nondestructive examinations shall be qualified in accordance with API Standard 650 and the supplemental requirements given herein.

10.1.2 SHELL PENETRATIONS

10.1.2.1 Ultrasonic examination of shell plate for laminations shall be made in the immediate area affected when:

a. Adding a reinforcing plate to an existing unreinforced penetration.

b. Adding a hot tap connection.

10.1.2.2 Cavities resulting from gouging or grinding operations to remove attachment welds of existing reinforcing plates shall be examined by visual as well as by magnetic particle or liquid penetrant methods.

10.1.2.3 Completed welds attaching nozzle neck to shell, and reinforcing plate to shell and to nozzle neck, shall be examined by the magnetic particle or liquid penetrant methods.

10.1.2.4 Completed welds of stress relieved assemblies shall be examined by visual as well as by magnetic particle or liquid penetrant methods after stress relief, but before hydrostatic test.

10.1.2.5 For penetrations installed using insert plates, the completed butt welds between the insert plate and the shell plate shall be fully radiographed.

10.1.3 REPAIRED WELD FLAWS

10.1.3.1 Cavities resulting from gouging or grinding operations to remove weld defects shall be examined by visual as well as by magnetic particle or liquid penetrant methods.

10.1.3.2 Completed repairs of butt-welds shall be examined over their full length by radiographic or ultrasonic methods.

10.1.3.3 Completed repairs of fillet welds shall be examined over their full length by the appropriate nondestructive examination method listed herein.

10.1.4 TEMPORARY AND PERMANENT ATTACHMENTS TO SHELL PLATES

10.1.4.1 Areas where weld projections (remaining after removal of temporary attachments) have been ground flush with the shell plate shall be examined by the appropriate nondestructive examination method listed herein.

10.1.4.2 Completed welds of new permanent attachments (not including shell-to-bottom welds) and areas where temporary attachments have been removed (API Standard 650 tank shell materials of Groups IV, IVA, V, or VI) shall be examined visually and by the magnetic particle method (or, at the option of the purchaser, by the liquid penetrant method). 96

10.1.5 SHELL PLATE TO SHELL PLATE WELDS

10.1.5.1 New welds attaching existing shell plate to existing or new shell plate shall be examined by visual and by radiographic methods (see 10.2). In addition, for plate thicknesses greater than one inch, the back-gouged surface of the root pass and final pass (each side) shall be examined for its complete length by magnetic particle or liquid penetrant methods. 98

10.1.5.2 New welds joining new shell plate material to new shell plate material (partial or full shell course replacement or addition) need only be examined radiographically in accordance with API 650. 98

10.1.6 SHELL-TO-BOTTOM WELD

10.1.6.1 New welding on the shell-to-bottom joint shall be inspected for its entire length by using a right-angle vacuum box and a solution film, or by applying light diesel oil. Additionally, the first weld pass shall be inspected by applying light diesel oil to the side opposite the first weld pass made. The oil shall be allowed to stand at least four hours (preferably overnight) and then the weld inspected for "wicking" action. The oil shall be removed before the weld is completed. 98

10.1.6.2 The existing weld at the shell-to-bottom joint shall be examined by visual, as well as by magnetic particle or liquid penetrant methods, for the full length under a welded-on patch plate. An additional 6 inches of the shell-to-bottom joint on each side of the welded-on patch plate shall be examined similarly before placement of the repair plate to assure weld integrity and to confirm the absence of weld cracks. 98

10.1.7 BOTTOMS

10.1.7.1 Newly welded bottom joints shall be tested using a vacuum box and a solution film. Every joint shall be tested for its entire length. Alternatively, a suitable tracer gas and compatible detector can be used to test the integrity of welded bottom joints for their entire length provided that an appropriate tracer gas testing procedure has been reviewed and approved by the tank owner/operator. 98

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

10.1.7.2 The weld attaching a welded-on patch plate to the tank bottom shall be visually examined over its full length.

98 **10.1.7.3** The root and final pass of a welded-on patch plate weld in the critical zone (see 1.5.7 for definition) shall be visually examined and examined by either magnetic particle or liquid penetrant method over its full length.

10.2 Radiographs

10.2.1 NUMBER AND LOCATION OF RADIOGRAPHS

The number and location of radiographs shall be in accordance with API Standard 650 and the following additional requirements:

10.2.1.1 For new or repaired shell vertical joints one radiograph shall be taken in every joint.

10.2.1.2 For new or repaired shell horizontal joints one radiograph shall be taken for each 50 feet of horizontal joint, not including those required at intersections of vertical and horizontal joints.

10.2.1.3 For each new or repaired intersection between shell vertical and horizontal joints one radiograph shall be taken.

10.2.1.4 For reconstructed tanks, each butt-welded annular plate joint shall be radiographed in accordance with API Standard 650.

10.2.1.5 For reconstructed tanks, radiographic inspection is required for 25 percent of all junctions of new welds over existing seams.

The purchaser shall, with the consent of the contractor, determine the extent of further inspection and repair that may be 97 required.

Any further inspection or repair of existing welds will be handled by contractual agreement between the purchaser and tank reconstruction contractor.

10.2.1.6 New and replaced shell insert plate and door sheet welds shall be radiographed. For circular replacement plates, a minimum of one radiograph shall be taken regardless of thickness. For square or rectangular replacement plates, at least one radiograph shall be taken in a vertical joint, and at least one in a horizontal joint, and one in each corner. All junctions between repair and existing welds shall be radiographed. If defects are found, 100 percent radiography shall be performed on the repaired weld.

97 **10.2.1.7** The minimum diagnostic length of each radiograph shall be 6 inches.

10.2.2 ACCEPTANCE CRITERIA FOR EXISTING SHELL PLATE TO SHELL PLATE WELDS

If the radiograph of an intersection between a new and old weld detects unacceptable welds by current standards, the existing welds may be evaluated according to the original standard of construction.

10.2.3 MARKING AND IDENTIFICATION OF RADIOGRAPHS

10.2.3.1 Each film shall show an identification of the welder(s) making the weld. A weld map showing location of welds, weld number, radiograph number, welder identification, and grading of each weld is an acceptable alternative to this requirement.

10.2.3.2 Radiographs and radiograph records of all repaired welds shall be marked with the letter "R". 96

10.3 Hydrostatic Testing

10.3.1 WHEN HYDROSTATIC TESTING IS REQUIRED

10.3.1.1 A full hydrostatic test, held for 24 hours, shall be performed on:

a. A reconstructed tank.
b. Any tank that has undergone major repairs or major alterations (see 10.3.1.2) unless exempted by 10.3.2 for the applicable combination of materials, design, and construction features.
c. A tank where an engineering evaluation indicates the need for the hydrostatic test due to an increase in the severity of service. Examples of increased service severity are an increase in operating pressure (such as storing a product with a higher specific gravity), lowering the service temperature (see Figure 3-2), and using tanks that have been damaged. 98

10.3.1.2 The terms major repair and major alteration refer to operations that require cutting, addition, removal and/or replacement of the annular plate ring, the shell-to-bottom weld, or a sizable portion of the shell. Within this context, *major repairs* and *major alterations* would include:

a. The installation of any shell penetration beneath the design liquid level larger than 12 inches NPS, or any bottom penetration located within 12 inches of the shell.
b. The removal and replacement or addition of any shell plate beneath the design liquid level, or any annular plate ring material where the longest dimension of the replacement plate exceeds 12 inches.
c. The complete or partial (more than one-half of the weld thickness) removal and replacement of more than 12 inches of vertical weld joining shell plates, or radial weld joining the annular plate ring.
d. The installation of a new bottom. This does not include new bottoms in tanks where the foundation under the new bottom is not disturbed and either condition 1) or 2) are met:

1) For tanks with annular rings, the annular ring remains intact. 98

2) For tanks without annular rings, the repair does not result in welding on the existing bottom within the critical zone. See 1.5.7 for a definition of the critical zone.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

e. The removal and replacement of any part of the weld attaching the shell to the bottom or to the annular plate ring.
f. Jacking of a tank shell.

10.3.2 WHEN HYDROSTATIC TESTING IS NOT REQUIRED

10.3.2.1 A full hydrostatic test of the tank is not required for major repairs and major alterations when both paragraphs a and b of this section are satisfied, and when either the appropriate parts of 10.3.2.2 through 10.3.2.5 or 10.3.2.6 are met.

a. The repair has been reviewed and approved by an engineer experienced in storage tank design in accordance with API Standard 650. The engineer must concur in writing with taking the hydrostatic testing exemption.
b. The tank owner or operator has authorized the exemption in writing.

10.3.2.2 Shell Repair

10.3.2.2.1 For welds to existing metal, develop welding procedure qualifications based on existing material chemistry, including strength requirements. Welding procedures shall be qualified with existing or similar materials, and shall include impact testing. Impact testing requirements shall follow appropriate portions of API 650, Section 7.2.2 and shall be specified in the repair procedure.

10.3.2.2.2 New materials used for the repair shall meet the current edition of API Standard 650 requirements.

98 **10.3.2.2.3** Existing tank materials in the repair area shall meet at least one of the following requirements:

a. API Standard 650 requirements (Seventh Edition or later).
b. Fall within the "Safe for use" area on Figure 3-2.
c. Stress in the repair area shall not exceed 7,000 psi. This limiting stress shall be calculated as follows:

$$S = \frac{2.6\,H\,D\,G}{t}$$

Where:

S = shell stress in pounds per square foot,

H = tank fill height above the bottom of repair or alteration in feet,

t = shell thickness at area of interest in inches,

D = tank mean diameter in feet,

G = specific gravity of product.

10.3.2.2.4 New vertical and horizontal shell butt-welds shall have complete penetration and fusion.

10.3.2.2.5 The root pass and final pass examination shall be in accordance with 10.1.5. In addition, the finished weld shall be fully radiographed.

Figure 10-1—Deleted 98

10.3.2.2.6 Shell welds for the reinforcing plate-to-nozzle neck and nozzle neck-to-shell joints shall have complete penetration and fusion. The root pass of the nozzle attachment weld shall be back-gouged and examined by magnetic particle or liquid penetrant methods. The completed weld shall be examined by magnetic particle or liquid penetrant methods and by the ultrasonic method. Examination and acceptance criteria for nondestructive examinations shall be in accordance with 10.1.

10.3.2.2.7 See 10.3.2.4 for shell-to-bottom weld restrictions.

10.3.2.2.8 Door sheets shall comply with the requirements of this standard for shell plate installation, except they shall not extend to or intersect the bottom-to-shell joint.

10.3.2.3 Bottom Repair Within the Critical Zone

10.3.2.3.1 Repairs to the annular ring or bottom plates, within the critical zone (see 1.5.7) shall comply with the following:

a. Meet the requirements of 10.3.2.2.1 through 10.3.2.2.3.
b. Be examined visually prior to welding, and examined after the root pass and the final pass by visual as well as magnetic particle or liquid penetrant methods. Annular plate butt-welds shall also be examined by ultrasonic methods after the final pass. Examination and acceptance criteria for nondestructive examinations shall be in accordance with 10.1. 98

10.3.2.4 Shell-To-Bottom Weld Repair

10.3.2.4.1 Repair of the weld attaching the shell to the annular ring or the shell to the bottom plate shall meet one of the following requirements:

a. A portion of the weld (of any length) may be removed and replaced as long as the replaced weld meets the size requirements of API Standard 650, Section 3.1.5.7, and the portion replaced does not represent more than 50 percent of the required weld cross-sectional area.
b. The weld on one side of the shell may be completely removed and replaced for a length not exceeding 12 inches. Shell-to-bottom weld repairs replacing more than 50 percent of the required weld cross-sectional area shall not be closer than 12 inches to each other, including repairs on the opposite side of the shell.

10.3.2.4.2 Repairs shall be examined prior to welding, after the root pass, and after the final pass by visual, as well as magnetic particle or liquid penetrant methods. Examination and acceptance criteria for nondestructive examinations shall be in accordance with 10.1.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

10.3.2.5 Minor Shell Jacking

10.3.2.5.1 Tank shell and critical zone materials shall meet one of the requirements of 10.3.2.2.3

10.3.2.5.1 The engineer shall consider all pertinent variables when exempting a minor shell jacking repair from hydrostatic testing, including but not limited to: the magnitude of jacking required; material; toughness; quality control; inspection before and after repair; material temperature; future foundation stability; and jacking techniques (including controls and measurement). Careful consideration shall be given to potential stresses and damage that may result from jacking. 98

10.3.2.6 Fitness-For-Service Evaluation

The owner/operator may utilize a fitness-for-service or other appropriate evaluation methodology based on established principles and practices to exempt a repair from hydrostatic testing. The procedures and acceptance criteria for conducting an alternative analysis are not included in this standard. This evaluation shall be performed by an engineer experienced in storage tank design and the evaluation methodologies used.

10.4 Leak Tests

New or altered reinforcing plates of shell penetrations shall be given an air leak test in accordance with API Standard 650.

10.5 Measured Settlement During Hydrostatic Testing

10.5.1 INITIAL SURVEY

10.5.1.1 Where settlement is anticipated, a tank receiving a hydrostatic test shall have the foundation checked for settlement.

10.5.1.2 Tank settlement shall initially be surveyed with the tank empty using the number of bottom plate projection elevation measurement points, N, uniformly distributed around the circumference, as indicated by the following formula:

$$N = D/10$$

Where:

N = minimum required number of settlement measurement points, but no less than eight. All fractional values shall be rounded to the next higher whole number. The maximum spacing between settlement measurement points shall be 32 feet. 97

D = tank diameter, in feet.

10.5.1.3 Tank settlement measurements in 10.5.1.2 shall be evaluated for acceptance in accordance with Appendix B.

10.5.2 SURVEY DURING HYDROSTATIC TESTING

Settlement shall be measured during filling and when water reaches 100 percent of test level. Excessive settlement in accordance with Appendix B shall be cause to stop the test for foundation investigation and/or repair.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 11—MARKING AND RECORDKEEPING

11.1 Nameplates

11.1.1 Tanks reconstructed in accordance with this standard shall be identified by a corrosion-resistant metal nameplate similar to that shown in Figure 11-1. Letters and numerals not less than $^{5}/_{32}$ inch high shall be embossed, engraved, or stamped in the plate to indicate information as follows:

a. Reconstructed to API 653.
b. Edition and revision number.
c. Year reconstruction was completed.
d. If known, the original applicable standard and the year of original construction.
e. Nominal diameter.
f. Nominal shell height.
g. Design specific gravity.
h. Maximum permissible operating liquid level.
i. The name of the reconstruction contractor and the assigned serial number or contract number.
j. The owner/operator's tank number.
k. Shell material for each shell course.
l. Maximum operating temperature.
m. Allowable stress used in calculations of each shell course.

11.1.2 The new nameplate shall be attached to the tank shell adjacent to the existing nameplate, if any. An existing nameplate shall be left attached to the tank. Nameplates shall be attached as specified in API Standard 650.

11.2 Recordkeeping

When a tank is evaluated, repaired, altered, or reconstructed in accordance with this standard, the following information, as applicable, shall be made a part of the owner/operator's records for the tank (see 4.8).

11.2.1 Calculations for:

a. Component evaluation for integrity, including brittle fracture considerations (Section 3).
b. Rerating (including liquid level).
c. Repair and alteration considerations.

11.2.2 Construction and repair drawings.

11.2.3 Additional support data including, but not limited to, information pertaining to:

a. Inspections (including thicknesses).
b. Material test reports/certifications.
c. Tests.
d. Radiographs (radiographs shall be retained for at least one year).
e. Brittle fracture considerations.
f. Original tank construction data (date, original standard, etc.).
g. Location and identification (owner/operator's number, serial number).
h. Description of the tank (diameter, height, service).
i. Design conditions (liquid level, specific gravity, allowable stress, unusual design loadings, etc.).
j. Shell material and thickness by course.
k. Tank perimeter elevations.
l. Construction completion record.
m. Basis for hydrostatic test exemption. 98

11.3 Certification

Tanks reconstructed in accordance with this standard shall require documentation of such reconstruction, and certification that the design, reconstruction, inspection, and testing was performed in compliance with this standard. The certification shall contain information as shown in Figure 11-2 for design and/or reconstruction as applicable.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

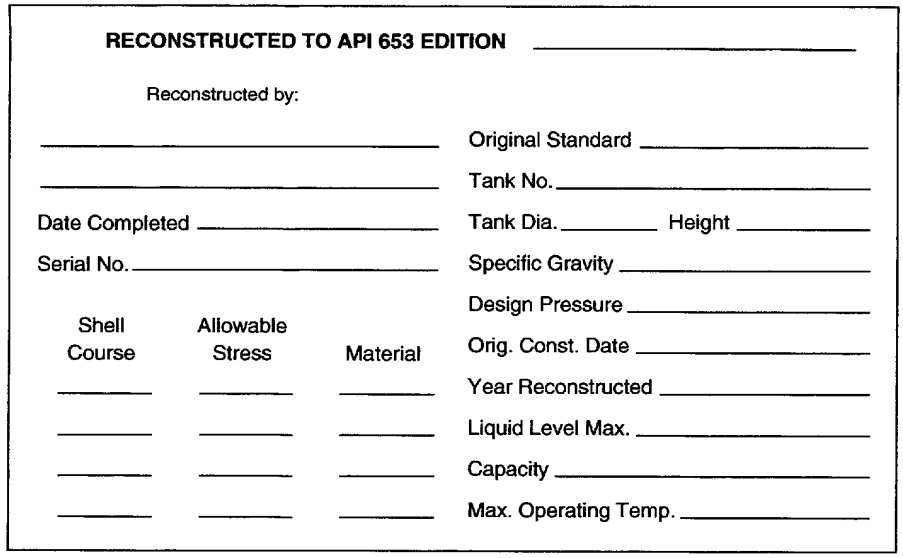


Figure 11-1—Nameplate

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX A—BACKGROUND ON PAST EDITIONS OF API WELDED STORAGE TANK STANDARDS

API published a specification for welded steel storage tanks in 1936 entitled Standard 12C, *API Specification for Welded Oil Storage Tanks.* Fifteen editions of and seven supplements to Standard 12C were published between 1936 and 1961. Standard 12C was replaced by API Standard 650, *Welded Steel Tanks for Oil Storage;* nine editions of and eleven supplements or revision packages to API Standard 650 have been issued. The current edition of API Standard 650 is the Tenth Edition, published in November 1998.

The table below provides a list of editions, supplements, and revisions to Standard 12C and API Standard 650.

Table A-1—Editions of API Standard 650 and Its Precursor, Standard 12C

Standard 12C, *API Specification for Welded Oil Storage Tanks*	
Edition	Date
First	July 1936
Second	October 1937
Supplement 1	April 1938
Supplement 2	September 1938
Supplement 3	April 1939
Third	April 1940
Fourth	March 1941
Fifth	May 1942
Sixth	August 1944
Seventh	August 1946
Supplement 1	September 1947
Eighth	September 1948
Supplement 1	December 1949
Ninth	October 1950
Tenth	September 1951
Eleventh	September 1952
Supplement 1	September 1953
Twelfth	October 1954
Thirteenth	September 1955
Supplement 1	October 1956
Fourteenth	October 1957
Fifteenth	1958

Standard 650, *Welded Steel Tanks for Oil Storage*	
Edition	Date
First	1961
Supplement	1963
Second	1964
Third	1966
Supplement 1	December 1967
Fourth	June 1970
Supplement 1	April 1971
Fifth	July 1973
Supplement 1	October 1973
Supplement 2	April 1974
Supplement 3	March 1975
Sixth	April 1977
Revision 1	May 1978
Revision 2	December 1978
Revision 3	October 1979
Seventh	November 1980
Revision 1	February 1984
Eighth	November 1988
Ninth	July 1993
Addendum 1	December 1994
Addendum 2	December 1995
Addendum 3	December 1996
Addendum 4	December 1997
Tenth	November 1998

98

Ordering information for past editions of API Standard 12C and API Standard 650 is available by calling the API Library at (202) 682-8042.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX B—EVALUATION OF TANK BOTTOM SETTLEMENT

B.1 Introduction

B.1.1 In determining the effects of soil settlement on storage tanks, it is common practice to monitor settlement of the tank bottom. In most cases, such a monitoring program is initiated during the construction and continued during hydrotesting and operations. During operations, settlement measurements should be taken at a planned frequency, based upon an assessment of soil settlement predictions. For existing tanks which do not have initial settlement data, a program of settlement monitoring should be based on prior service history.

B.1.2 If at any time settlement is deemed excessive, the tanks should be emptied and releveled. Releveling of a sizable tank is expensive and rather difficult to achieve. Thus, a decision to relevel a tank is a crucial one, and relies very much on the proper interpretation and evaluation of the monitored settlement data.

B.1.3 Approaches used to correct tank shell and bottom settlement include techniques such as localized repairs of the bottom plates, partial releveling of the tank periphery, and major releveling of the entire tank bottom. Major releveling of the tank, involving total lifting of the tank shell and bottom at one time, can introduce high localized stresses in the structure and impair its integrity. Therefore, when choosing techniques for correcting settlement problems, an alternative to total lifting of the tank shell and bottom should be considered as a first choice. If it is decided to lift the entire tank shell and bottom at one time, it should be done by personnel with demonstrated experience in this technique.

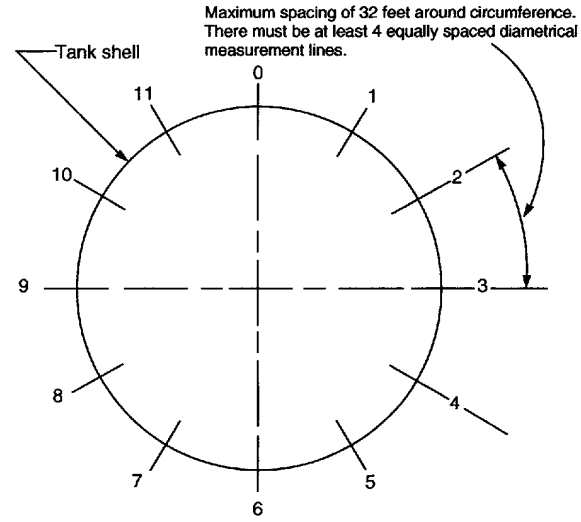
B.2 Types of Settlement

B.2.1 ELEVATION MEASUREMENTS

The principal types of tank settlement consist of settlements which relate to the tank shell and bottom plate. These settlements can be recorded by taking elevation measurements around the tank circumference and across the tank diameter. Figures B-1 and B-2 show recommended locations on a tank shell and bottom plate for settlement measurements. Data obtained from such measurements should be used to evaluate the tank structure. Additional settlement readings may be required to define local depressions.

B.2.2 SHELL SETTLEMENT EVALUATION

Settlement of a tank is considered to be the result of either one or a combination of the following three settlement components.



Notes:
1. There must be at least eight settlement points. The maximum spacing of settlement points is 32 feet around the circumference. 97
2. Points shall be equally spaced around the tank shell. See 10.5.1.2 for method of determining the number of measurement points.

Figure B-1—Measurements of Shell Settlement (External)

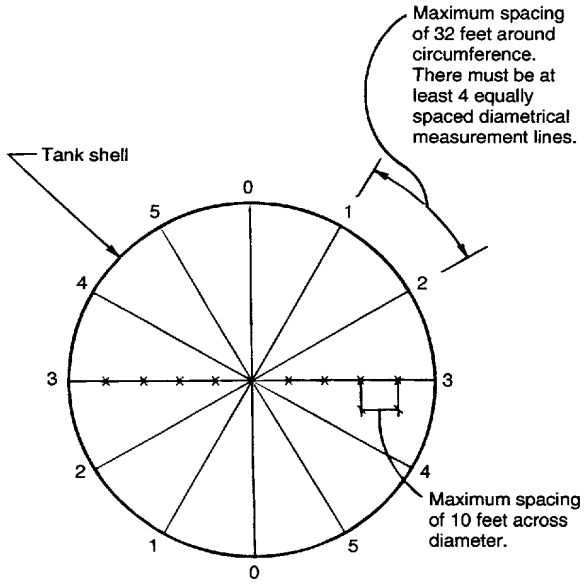
B.2.2.1 Uniform settlement. This component often can be predicted in advance, with sufficient accuracy from soil tests. It may vary in magnitude, depending on the soil characteristics. Uniform settlement of a tank does not induce stresses in the tank structure. However, piping, tank nozzles, and attachments must be given adequate consideration to prevent problems caused by such settlement. 98

B.2.2.2 Rigid body tilting of a tank (planar tilt). This component rotates the tank in a tilted plane. The tilt will cause an increase in the liquid level and, therefore, an increase in the hoop stress in the tank shell. Also, excessive tilting can cause binding of peripheral seals in a floating roof and inhibit roof travel. This type of settlement could affect tank nozzles that have piping attached to them. Figure B-3 shows that the settled location of the tank shell, after rigid body tilt, can be represented by either a cosine or sine wave with respect to its original position in a horizontal plane.

B.2.2.3 Out-of-plane settlement (differential settlement). Due to the fact that a tank is a rather flexible structure, chances are great that the tank shell will settle in a nonplanar configuration, inducing additional stresses in the tank shell. The out-of-plane settlements at the bottom edge lead to a lack of circularity at the top of the tank, and in the case of a floating roof tank, the extent of the induced ovality may impede the

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

97



Note: See 10.5.1.2 for method of determining the number of measurement points.

Figure B-2—Measurements of Bottom Settlement (Internal) Tank Out-of-Service

proper functioning of the floating roof in such a way that releveling is required. Also, such settlements may cause flat spots to develop in the tank shell. This type of settlement could affect tank nozzles that have piping attached to them.

B.2.2.4 While uniform settlement and rigid body tilt of a tank may cause problems as described above, the out-of-plane settlement is the important component to determine and evaluate in order to ensure the structural integrity of the shell and bottom. Based on this principle, a common approach is to determine the magnitudes of the uniform and rigid body tilt component (if any) for each data point on the tank periphery. Once this is carried out, the plane of rigid tilt is then important as a datum from which to measure the magnitudes of the out-of-plane settlements.

A graphical representation illustrating the effect of tank shell settlement components is shown in Figure B-3. The construction of this "settlement plot" has been developed in accordance with the following:

a. The actual edge settlement (in most cases an irregular curve) is plotted using points around the tank circumference as the abscissa.

b. The vertical distance between the abscissa and the lowest point on this curve (point 22) is the minimum settlement, and it is called the uniform settlement component. A line through this point, parallel to the abscissa, provides a new base or datum line for settlement measurements that are called "adjusted settlements."

c. The plane of rigid tilt settlement is represented by the "optimum cosine curve" between the maximum and minimum values of the actual edge settlements. Several methods exist for determining the optimum cosine curve. The least accurate method is by free hand drawing techniques, a kind of trial and error procedure to fit the best cosine curve through the data. A better method is to use the mathematical and graphical capabilities of a computer.

d. The vertical distances between the irregular curve and the cosine curve represent the magnitudes of the out-of-plane settlements (U_i at data point i).

e. The most commonly used and accepted method is to use a computer to solve for constants a, b, and c, to find the optimum cosine curve of the form:

$$Elev_{pred} = (a + b) \times (\cos \theta + c)$$

Where $Elev_{pred}$ is the elevation predicted by the cosine curve at angle theta. A typical starting point for a computer best-fit cosine curve is a least-squares fit where a, b, and c are chosen to minimize the sum of the square of the differences between measured and predicted elevations. The optimum cosine curve is only considered valid (i.e., accurately fits the measured data) if the value R^2 is greater than or equal to 0.9.

$$R^2 = (S_{yy} - SSE)/S_{yy}$$

Where:

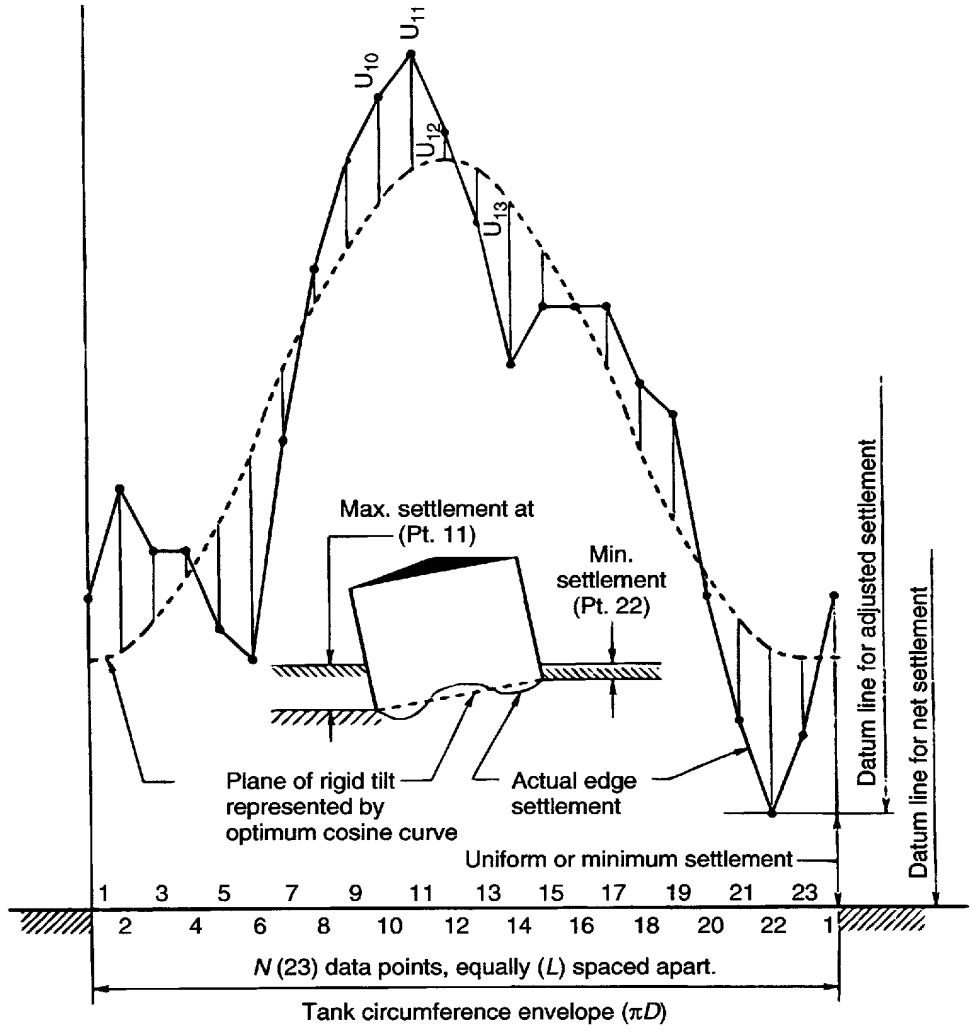
S_{yy} = sum of the squares of the differences between average measured elevation and the measured elevations.

SSE = sum of the square of the differences between the measured and predicted elevations.

98

Obtaining a statistically valid cosine curve may require taking more measurements than minimums shown in Figure B-1. In many cases, the out-of-plane settlement may be concentrated in one or more areas and the least-squares fit underpredicts the local out-of-plane settlement and is not conservative. In these cases, R^2 will typically be less than 0.9. Appropriately selecting a, b, and c will result in very little difference between predicted and measured elevation at all but one or two measurements points, and R^2 will be greater than 0.9. One method of selecting a, b, and c in these cases is to ignore one or two points that do not appear to fit the initially calculated cosine curve, and recalculate the optimum cosine curve. The remaining points will show a good estimate of the true out-of-plane settlement at the worst point. Figure B-4 shows an example where one point at 135 degrees is well off the initial curve, R^2 is 0.87, and S is just less than the maximum allowed. By ignoring the point at 135, and recalculating the optimum cosine curve, R^2 is increased to 0.98 (very good), and the out-of-plane settlement at the point at 135 degrees is just above the maximum allowed.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34



Out-of-plane deflection for point "*i*" is
$S_i = U_i - (1/2\ U_{i-1} + 1/2\ U_{i+1})$, for example
$S_{11} = U_{11} - (1/2 U_{10} + 1/2 U_{12})$

U_i = out-of-plane settlement of point "*i*",
(+) when above cos. curve;
(–) when below cos. curve, for example:
U_{11} = (+)
U_{13} = (–)

Note: See 10.5.1.2 for definition of "N."

Figure B-3—Graphical Representation of Tank Shell Settlement

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

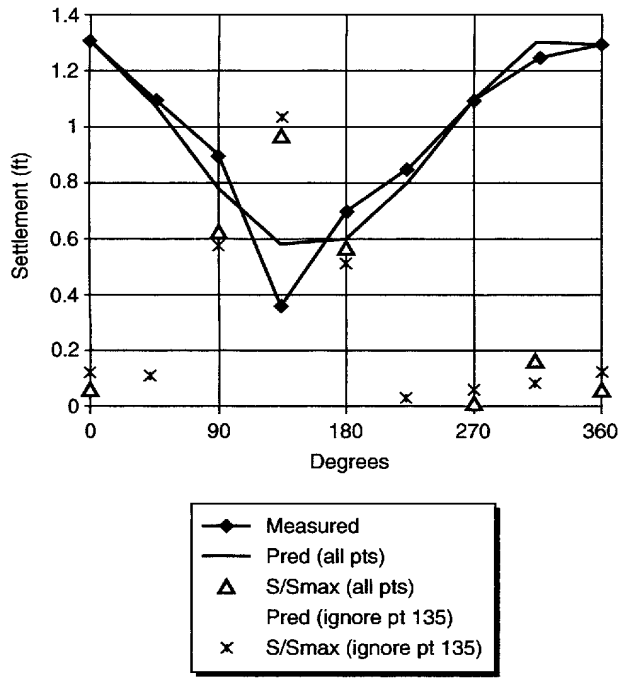


Figure B-4—Out-of-Plane Settlement Measurement Points

f. The vertical distances between the irregular curve and the optimum cosine curve represent the magnitudes of the out-of-plane settlements (U_i at data point i). S_i is the out-of-plane deflection at point i. Refer to Figure B-3.

g. Measurements for out-of-plane settlement must be taken carefully. In cases of distortion or corrosion of the tank bottom extending beyond the shell, measurements taken near lap welds in the tank bottom can result in significant errors in measured elevation. Repaired or replaced bottom plates, or new slotted-in bottoms may not have been installed parallel to the bottom shell course. In some cases, more consistent and accurate results may be obtained by surveying the elevation of the weld between the first and second courses.

Note: When using the optimum cosine curve described in B.2.3.9 d, taking measurements around the shell will result in a more accurate cosine curve. However, using all of these measurement points in the equation shown in B.3.2 will result in very small allowable settlements, S_{max}, since the arc length L between measurement points is small. It is acceptable to use as many measurement points as needed to develop the optimum cosine curve, but only use a subset of these points spaced approximately 30 feet (eight minimum) when calculating S and S_{max}. The points used must include the points furthest from the optimum cosine curve. For example, if eight points are required, but 16 measurements are taken, and the arc length between measurements is only 15 feet, calculate the optimum cosine curve using all 16 points, but use only eight points to calculate S. The equations in Figure B-3 would be revised to read:

$$S_i = U_i - ({}^1/_2\, U_{i-2} + {}^1/_2\, U_{i+2})$$

$$S_{11} = U_{11} - ({}^1/_2\, U_9 + {}^1/_2\, U_{13})$$

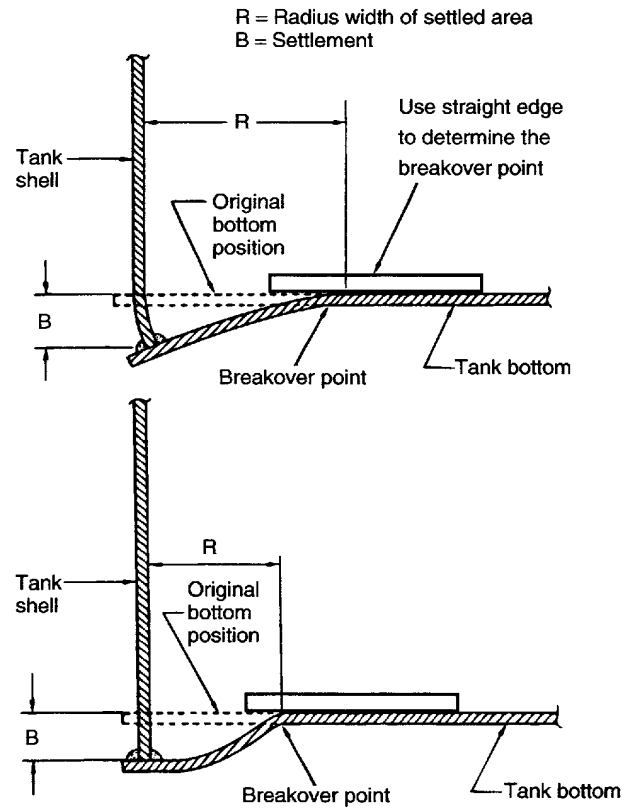


Figure B-5—Edge Settlement

h. If out-of-measured out-of-plane settlement exceeds the limits described in B.3.2 using the optimum cosine curve method, a more rigorous evaluation may be performed in lieu of repairs. This evaluation must be done by an engineer experienced in tank settlement analysis.[1]

B.2.3 EDGE SETTLEMENT

B.2.3.1 Edge settlement occurs when the tank shell settles sharply around the periphery, resulting in deformation of the bottom plate near the shell-to-bottom corner junction. Figure B-5 illustrates this settlement.

B.2.3.2 The formula given in B.3.3 can be used to evaluate this criteria. Alternatively, a rigorous stress analysis can be carried out for the deformed profile. Measure the edge settlement carefully.

a. Measurements taken when the bottom is not in contact with the soil or foundation under the tank can overestimate or underestimate edge settlement significantly. If the measured

[1] One such method is to take closely spaced measurements (approximately 5–10 feet), fit the measured data with piecewise cubic splines to accurately represent the geometry of the shell. This curve is used to develop a three-dimensional finite element model of the tank shell. The primary and secondary stresses are then evaluated using methods in ASME Section VIII, Div. 2, and the allowable stress defined in API 653.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

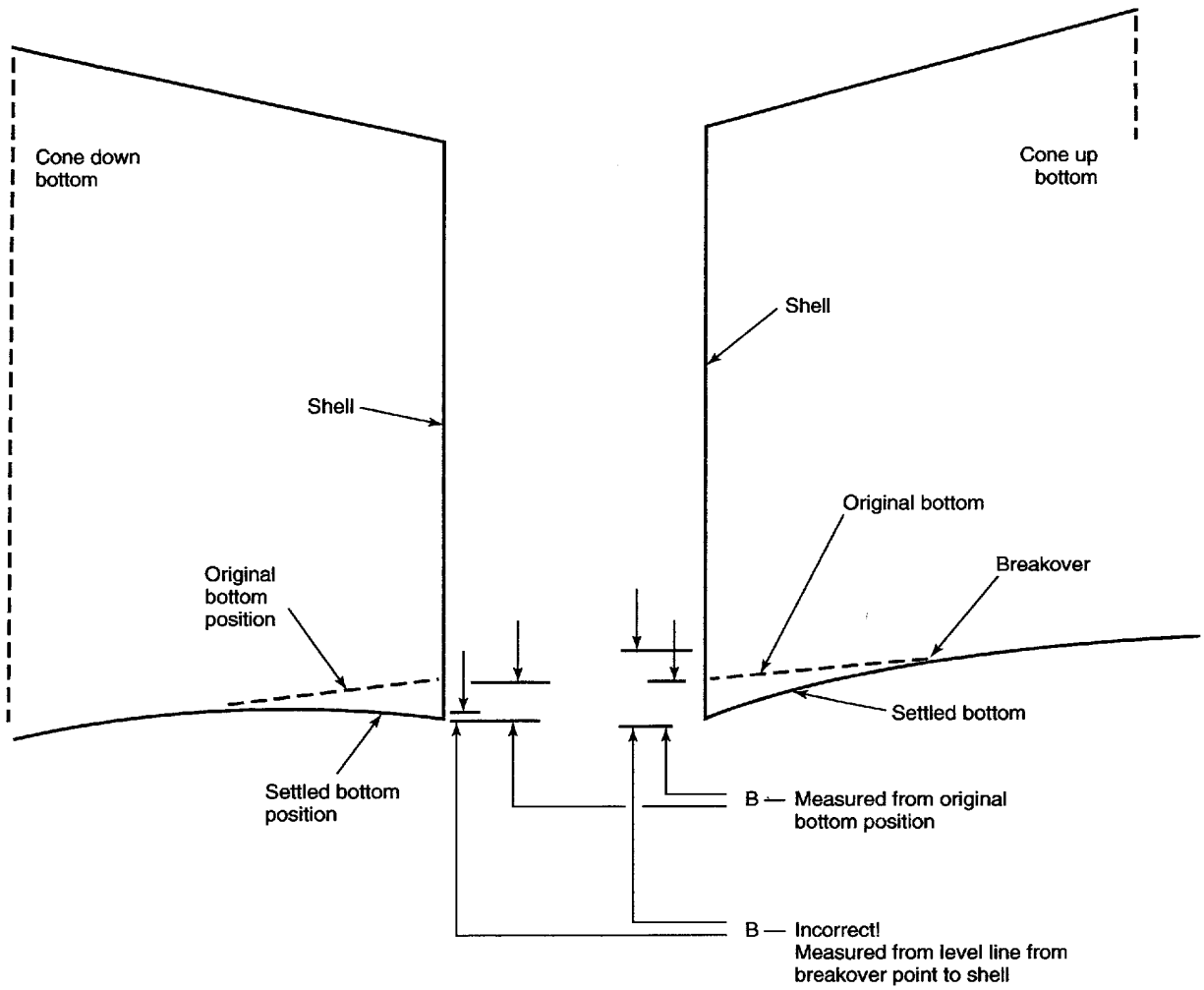


Figure B-6—Correction for Measured Edge Settlement

settlement is near the maximum allowable settlement, consider repeating the measurement with the bottom forced down to the soil, e.g. standing on it, or take an additional set of measurements in the same area, where the bottom is in firm contact with the soil.

b. Locating the "breakover point" where the settled area begins requires some judgment. Placing a straight edge on the unsettled floor as shown in Figure B-5, and observing where the floor separates from the straight edge will help define the breakover point.

c. If the tank floor is cone up or cone down, the settlement B, should be measured from a projection of the unsettled floor, not from level. See Figure B-6.

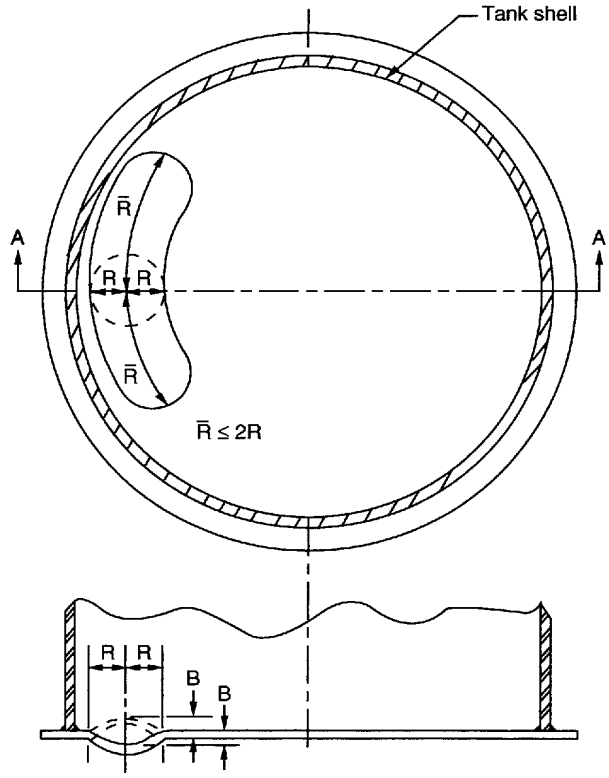
B.2.3.3 The measured edge settlement B is defined as shown in Figure B-5. B_{ew} is defined as the allowable edge settlement in an area where there is a floor lap weld in the settled area that is essentially parallel (± 20 degrees) to the shell. B_e is defined as the allowable settlement in an area with no floor welds, or only butt-welds in the floor, or lap welds in the floor that are essentially perpendicular (± 20 degrees) to the shell. Refer to Figure B-4.

B.2.3.4 Section B.3.4 provides methods for evaluation of measured edge settlement B against allowable edge settlement B_{ew} and B_e. Since B_{ew} is more conservative the B_e, the simplest approach is to initially evaluate measured settlement B against B_{ew} for all settled areas. If all areas meet this criteria, the settlement is OK and no further evaluation is necessary. If necessary, different settled areas can be evaluated separately against B_{ew} and B_e. For areas containing lap welds at an arbitrary angle to the shell, interpolation to find an allowable settlement between B_{ew} and B_e based on the angle of the weld to the shell is allowed.

B.2.4 BOTTOM SETTLEMENT NEAR TANK SHELL

B.2.4.1 Figure B-7 illustrates this type of settlement.

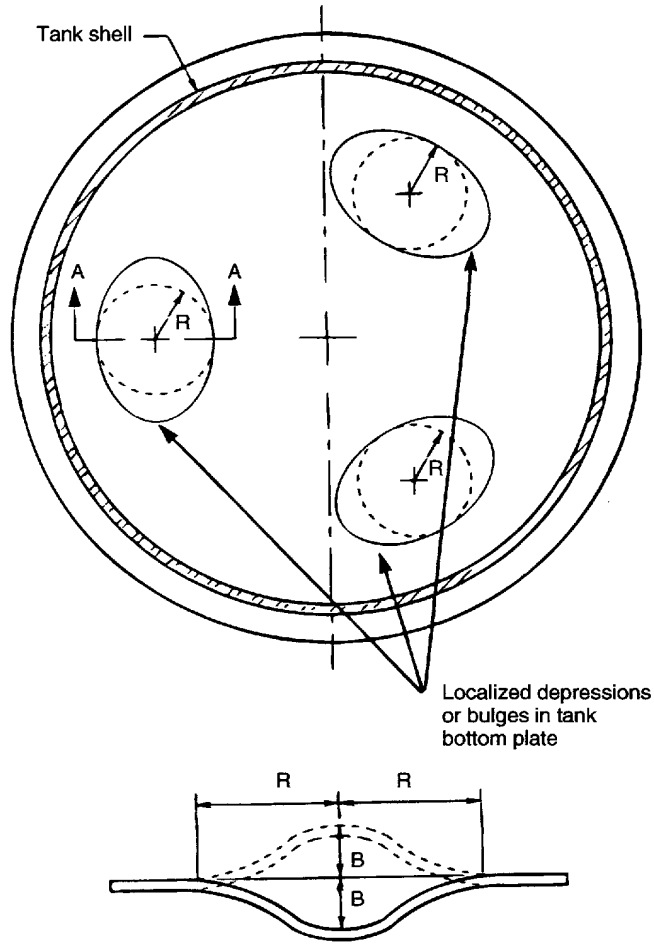
COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34



SECTION A-A

R = Radius of inscribed circle in bulged or depressed area
B = Settlement or depression depth or height of bulge

98 | Figure B-7—Bottom Settlement Near Shell



R = Radius of inscribed circle in bulged or depressed area

B = Settlement depth or height of bulge

SECTION A-A

Figure B-8—Localized Bottom Depressions or Bulges Remote From Shell | 98

B.2.4.2 The formula given in B.3.3 can be used to evaluate this criteria. Alternatively, a rigorous stress analysis can be carried out for the deformed profile.

B.2.5 LOCALIZED BOTTOM SETTLEMENT REMOTE FROM TANK SHELL

B.2.5.1 These are depressions (or bulges) that occur in a random manner, remote from the shell (see Figure B-8). 98 |

B.2.5.2 Acceptability of these localized settlements is dependent upon localized stresses in the bottom plate, design and quality of the lap welds (single pass or multipass), and voids below the bottom plate. The formula given in B.3.3 can be used to evaluate this criteria. These limits are applicable to tank bottoms which have single pass lap-welded joints.

B.3 Determination of Acceptable Settlement

B.3.1 GENERAL

For existing tanks with history of successful service, it may be possible to accept greater settlement and distortion of the foundation from a true plane than new tank construction standards allow. Each tank must be evaluated based on service conditions, materials of construction, soil characteristics, tank foundation design, and tank service history. The methods discussed in following sections are not mandatory and are an approximation of the maximum permissible settlement. However, experience has shown that if settlements exceed the following requirements, further assessment or repair is required.

B.3.2 SHELL SETTLEMENT

From the measurements described in B.2, determine the maximum out-of-plane deflection. Use the following formula to calculate the maximum permissible out-of-plane deflection (see note below.)

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

$$|S| \le \frac{(L^2 \times Y \times 11)}{2[(E \times H)]}$$

Where:

S = deflection, in feet (out of plane distortion),
L = arc length between measurement points, in feet,
Y = yield strength, in pounds per square inch,
E = Young's modulus, in pounds per square inch,
H = tank height, in feet.

Note: This formula is based on "Criteria for Settlement of Tanks," W. Allen Marr, M. ASCE, Jose A. Ramos, and T. William Lambe, F. ASCE, *Journal of Geotechnical Engineering Division, Proceedings of the American Society of Civil Engineers*, Vol. 108, August 1982.

B.3.3 INTERNAL BOTTOM SETTLEMENTS OR BULGES

Measure the bulge or depression. The permissible bulge or depression is given by the following formula (see note below).

98 $$B_B = 0.37R$$

Where:

98 B_B = maximum height of bulge or depth of local depression, inches,
R = radius of inscribed circle in bulged area or local depression, in feet.

98 Figure B-9 is a graphical representation of this formula.

Note: This formula is based on "Criteria for Settlement of Tanks," W. Allen Marr, M. ASCE, Jose A. Ramos, and T. William Lambe, F. ASCE, *Journal of Geotechnical Engineering Division, Proceedings of the American Society of Civil Engineers*, Vol. 108, August 1982.

B.3.4 Edge Settlement

B.3.4.1

a. Maximum allowable settlement B_{ew} is shown in Figure B-10 for settled areas that include floor lap welds essentially parallel to the shell (± 20 degrees). In settled areas where the measured settlement B exceeds 75 percent of allowed settlement B_{ew}, all shell-to-bottom welds and floor welds should
98 be inspected visually and with magnetic particle examination or liquid penetrant examination. All indications should be repaired, or evaluated for risk of brittle fracture, and/or fatigue failure prior to returning the tank to service.
b. For settled areas where measured settlement B exceeds 75 percent of B_{ew}, any welds within 12 inches of either side of the breakover area (see Figure B-5) should be examined visually. Any suspect areas should be examined with either magnetic particle examination or liquid penetrant examination. All indications should be repaired or evaluated for risk of fatigue prior to returning the tank to service.

B.3.4.2 Maximum allowable settlement B_e is shown in Figure B-11 for areas of edge settlement with no welds, butt welds, or lap welds in the floor that are essentially radial to the shell (± 20 degrees). In settled areas where the measured settlement exceeds 75 percent of the allowed settlement, all shell-to-bottom welds and floor welds should be inspected visually and with magnetic particle examination or liquid penetrant examination. All indications should be repaired or evaluated for risk of brittle fracture and/or fatigue prior to returning the tank to service.

B.3.4.3 Maximum allowable settlement for areas of edge settlement with a lap weld at an arbitrary angle to the shell may be interpolated from B_e and B_{ew} from Figures B-8 and B-9, and the following formula:

$$B_\alpha = B_e - (B_e - B_{ew}) \times \sin \alpha$$

Where α is the angle of the weld to a tank centerline and B_α is the allowable settlement for an area with a weld at that angle (see Figure B-12).

B.3.4.4 In general, settlement occurs slowly, and for most existing tanks, the majority of settlement is presumed to have occurred in the first few years of service. Significant additional settlement will not be expected after the initial inspections. Therefore, typical practice is to compare the measured edge settlement with the maximum allowable edge settlement B_{ew} and B_e, and not include allowance for additional settlement during subsequent operation. Note that erosion of the pad adjacent to the tank may cause local settlement. In this case the settlement will continue unless the pad is repaired and future erosion prevented. For cases where significant additional settlement is expected, an engineer experienced in tank settlement evaluation should evaluate the settlement expected at the next inspection with the limits in B.3.4. This is analogous to a corrosion allowance for components expected to corrode. 98

B.3.4.5 The edge settlement limits described in B.2.3.1 through B.2.3.5 were developed for typical $^1/_4$-inch thick tank bottoms with minimal corrosion.

a. Edge settlement limits can be applied with reasonable accuracy to $^5/_{16}$ and $^3/_8$-inch thick tank bottoms.
b. Edge settlement limits can be applied with reasonable accuracy to bottoms with general corrosion, as long as the areas near all welds are thicker than $^3/_{16}$ inch.
c. Edge settlement limits can be applied with reasonable accuracy to bottoms with local corrosion, if all locally thin areas in the settled area (closer than "R" to the shell) thinner than $^3/_{16}$ inch are smaller than 12 inches in diameter and the thin area does not include a weld.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

d. Settlement is presumed to be slow, and a small amount of additional settlement is expected to occur prior to the next inspection.[2]

B.3.4.6 Edge settlement increases secondary stress at the bottom-to-shell weld. If weld repairs are made to the bottom-to-shell weld in an area where settlement exceeds $^1/_2$ of B_e, these additional stresses should be evaluated by an engineer experienced in tank settlement evaluation before waiving a hydrotest per 10.3.

B.4 Repairs

B.4.1 If it is determined that settlements have occurred which are beyond the permissible limits established in the previous sections, then consideration should be given to making repairs or a rigorous stress analysis should be performed to evaluate the deformed profile. Various repair techniques have been discussed above. The judgment on repairs should be tempered with knowledge of tank service history, previous repairs, previous inspections, tank foundation conditions, soil characteristics, the material of construction, and estimates of future settlement. See 7.10 for suggested repair details.

B.4.2 For tanks with edge settlement exceeding the limits and assumptions given in B.2.3.1 through B.2.3.7, the tank should be repaired. Any plate exceeding acceptable strains (typically 2–3 percent) should be replaced. Releveling the tank will not remove the plastic strain, so leveling the tank without replacing the strain may not be a sufficient repair. Welds in the area of the high strains should be removed and replaced, or be subjected to a fitness-for-service evaluation by an engineer experienced in tank settlement evaluation. The condition leading to the unacceptable settlement should be corrected. Depending on the severity and location of the settlement, required repairs may include regrading the soil under the tank bottom, and/or repairing the foundation. Jacking and releveling the shell may be required to prevent additional settlement damage. Jacking and leveling are usually done in conjunction with, not instead of, replacing damaged plate and welds. In lieu of repairs, a detailed analysis of the settled area may be performed by an engineer experienced in tank design and settlement evaluation. The analysis should consider primary and secondary stress and the risk of brittle fracture.

[2] The limits in this section are based on methods described in references included in 1) "Criteria for Settlement of Tanks", W. A. Marr, J. A. Ramos, and T. W. Williams; *Journal of Geotechnical Engineering Division*, Proceedings of the American Society of Civil Engineers, Vol 108, August 1982; 2) "Non-Linear Finite Element Analysis of Edge Settlement in Riveted Aboveground Storage Tanks" J. Andreani, *Fitness for Service and Decisions for Petroleum and Chemical Equipment*, ASME, 1995; 3) "An Evaluation of procedures for Determining the Fitness for Service of Settled Aboveground Storage Tanks", J. L. Andreani, D. A. Osage, P. D. Parikh, and J. A. Horwege, ASME PVP Conference 6/95.

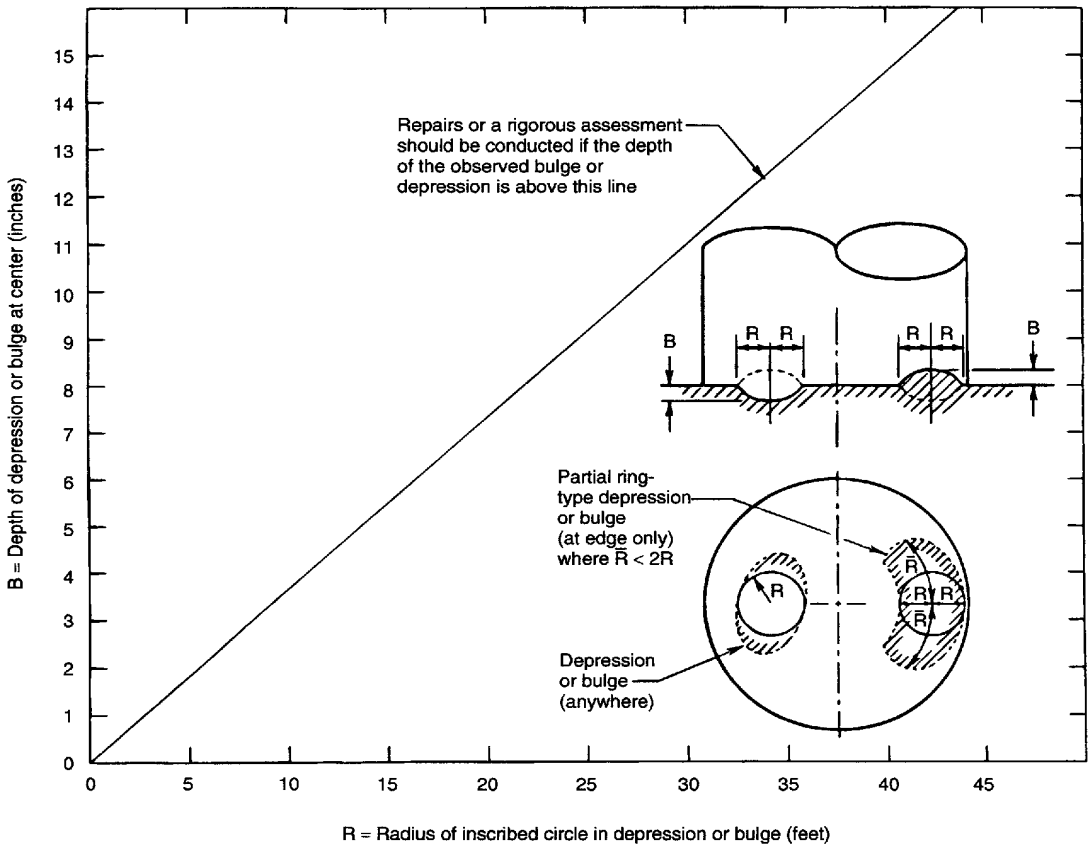
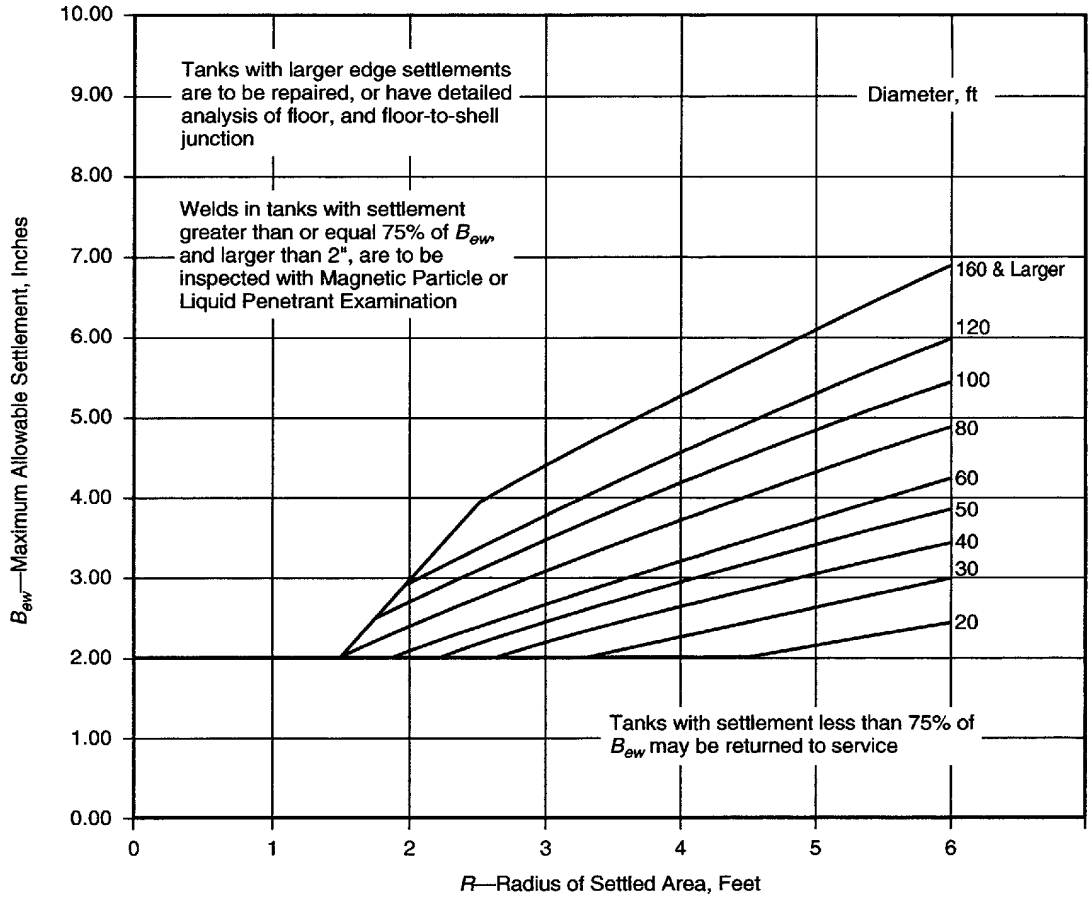


Figure B-9—Localized Tank Bottom Settlement Limits for Single Pass Welds

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34



98

Figure B-10—Maximum Allowable Edge Settlement for Areas with Bottom Lap Welds Approximately Parallel to the Shell

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

98

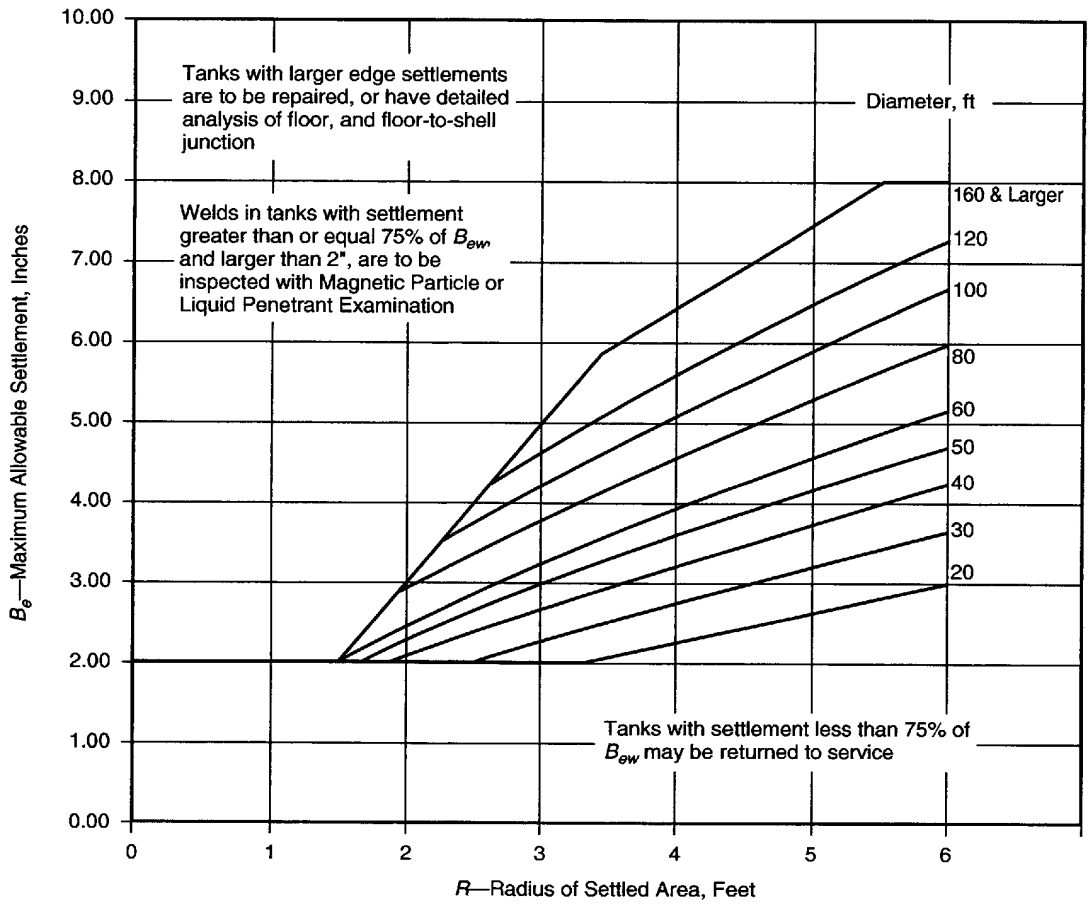
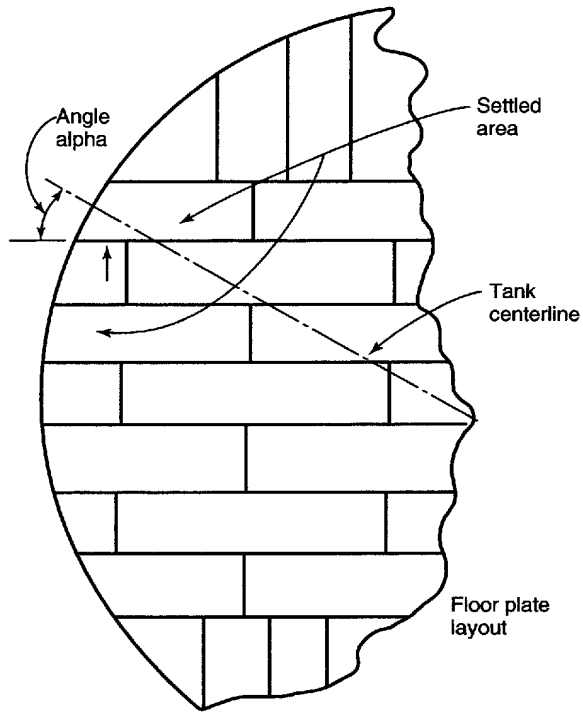


Figure B-11—Maximum Allowable Edge Settlement for Areas with Bottom Lap Welds Approximately Perpendicular to the Shell

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34



98

Figure B-12—Edge Settlement with a Lap Weld at an Arbitrary Angle to the Shell

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX D—AUTHORIZED INSPECTOR CERTIFICATION

D.1 Examination/Grading/Validation

D.1.1 EXAMINATION

A written examination to certify inspector candidates within the scope of this standard shall be administered by a third party designated by the American Petroleum Institute. The examination shall be based on the content of the latest edition of this standard. API RP 575, ASME Section V and IX, and the applicable section of API Standard 650, API RP 651, API RP 652, API IRE Chapter II[1], and API RP 576.

D.1.2 GRADING

The third party testing agency shall administer and grade the examinations using an equating system that statistically adjusts the passing score of each individual examination administration to compensate for any potential differences in the difficulty levels of the exams. Applicants may appeal any examination grievance to the American Petroleum Institute within ninety days of receipt of their grades.

D.1.3 VALIDATION

The examination shall be validated as to its suitability for determining the applicant's level of knowledge on how to inspect aboveground storage tanks.

[1] API IRE Chapter II is under revision and will be reissued as API Recommended Practice 571, *Recognition of Conditions Causing Deterioration or Failure.*

D.2 Examination Methods/Types

The examination shall be written.

D.3 Certification

An API certificate will be issued upon compliance with the provisions of 4.10 and D.1. 98

D.4 Certification Agency

The American Petroleum Institute shall be the certifying agency.

D.5 Recertification

D.5.1 All Authorized Inspectors will be required to provide notarized documentation of employment as an API 653 Inspector whose regular duties routinely have included atmospheric storage tank inspection during the previous three-year certification period; otherwise, re-examination is required.

D.5.2 Recertification by written examination is required for Authorized Inspectors who qualified under D.3.2.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX E—TECHNICAL INQUIRIES

E.1 Introduction

API will consider written requests for interpretations of Standard 653. API staff will make such interpretations in writing after consulting, if necessary, with the appropriate committee officers and committee members. The API committee responsible for maintaining Standard 653 meets regularly to consider written requests for interpretations and revisions and to develop new criteria dictated by technological development. The committee's activities in this regard are limited strictly to interpretations of the standard and to the consideration of revisions to the present standard on the basis of new data or technology. As a matter of policy, API does not approve, certify, rate, or endorse any item, construction, proprietary device, or activity, and accordingly, inquiries that require such consideration will be returned. Moreover, API does not act as a consultant on specific engineering problems or on the general understanding or application of the standard. If it is the opinion of the committee, based on the inquiry information submitted, that the inquirer should seek other assistance, the inquiry will be returned with the recommendation that such assistance be obtained. All inquiries that do not provide the information needed for the committee's full understanding will be returned.

E.2 Inquiry Format

E.2.1 Inquiries shall be limited strictly to requests for interpretation of the standard or to the consideration of revisions to the standard on the basis of new data or technology. Inquiries shall be submitted in the format described in E.2.2 through E.2.5.

E.2.2 The scope of an inquiry shall be limited to a single subject or a group of closely related subjects. An inquiry concerning two or more unrelated subjects will be returned.

E.2.3 An inquiry shall start with a background section that states the purpose of the inquiry, which would be either to obtain an interpretation of the standard or to propose a revision to the standard. The background section shall concisely provide the information needed for the committee's understanding of the inquiry (with sketches as necessary) and shall cite the applicable edition, revision, paragraphs, figures, and tables.

E.2.4 After the background section, an inquiry's main section shall state the inquiry as a condensed, precise question, omitting superfluous background information and, where appropriate, posing the question so that the reply could take the form of "yes" or "no" (perhaps with provisos). This inquiry statement should be technically and editorially correct. The inquirer shall state what he believes the standard requires. If he believes a revision to the standard is needed, he shall provide recommended wording.

E.2.5 The inquirer shall include his name and mailing address. The inquiry should be typed; however, legible handwritten inquiries will be considered. Inquiries should be submitted to the general manager, Downstream Segment, American Petroleum Institute, 1220 L Street, N.W., Washington, D.C. 20005. 98

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX F—NDE REQUIREMENTS SUMMARY

F.1 Introduction

This appendix is a summary of all nondestructive examination requirements for repairs and reconstruction of tanks. It is provided only as a guide to assure that the proper inspections are performed and that the acceptance standards, inspector qualifications, and procedure requirements are followed.

F.2 Visual Examination

F.2.1 Visual examinations are required for:

a. Cavities from removal of repads (see API 653, Section 10.1.2.2).
b. Completed welds of stress-relieved assemblies after stress relief but before hydrostatic testing (see API 653, Section 10.1.2.4).
c. All fillet welds and completed repairs of fillet welds (see API 650, Section 5.3.2.2 and API 653, Section 10.1.3.3).
d. Completed welds of new permanent attachments and areas of removed temporary attachments (see API 653, Section 10.1.4.2)
e. New shell plate-to-shell plate welds (see API 653, Section 10.1.5)
f. Tack welds left in place (see API 650, Section 5.2.1.8).

F.2.2 The examination acceptance standard is API 650, Section 6.5. 98

F.2.3 There are no requirements for inspector qualifications.

F.2.4 There are no procedure requirements.

F.3 Magnetic Particle and Liquid Penetrant Examination

F.3.1 Magnetic particle or liquid penetrant examinations are required for:

a. Cavities from removing existing reinforcing pad welds (see API 653, Section 10.1.2.2).

b. New welds of nozzle neck-to-shell, nozzle neck-to-repad, and repad-to-shell (see API 653, Section 10.1.2.3).

c. Completed welds of stress-relieved assemblies after stress relief, before hydrostatic testing (see API 653, Section 10.1.2.4).

d. Cavities from removal of weld defects (see API 653, Section 10.1.3.1).

e. Completed welds of new permanent attachments and areas of removed temporary attachments on API 650 material groups IV, IVA, V, or VI (see API 653, Section 10.1.4.2).

f. The back-gouged surface of the root pass and the final surface of new shell plate welds where the shell is thicker than one inch (see API 653, Section 10.1.5).

g. The back-gouged surface of root pass of full penetration nozzle neck-to-shell and repad welds as required by API 653, Section 10.3.2.2.6, specific hydrotest exemption requirement.
h. The back-gouged surface of the root pass and final surface of new vertical and horizontal shell joints as required by API 653, Section 10.3.2.2.5, specific hydrotest exemption requirement.

F.3.2 Magnetic particle examination alone is required for weld removal areas of the bottom-to-shell welds when removing a bottom (see API 653, Section 7.10.2.2 b).

F.3.3 The magnetic particle inspection acceptance standard is ASME Section V, Article 7. The acceptance standards for the removal and repair of defects shall be in accordance with ASME Section VIII, Appendix 6, Paragraph 6-3, 6-4, and 6-5.

F.3.4 The liquid penetrant inspection acceptance standards is ASME Section V, Article 6. The acceptance standards for the removal and repair of defects shall be in accordance with ASME Section VIII, Appendix 8, Paragraphs 8-3, 8-4, and 8-5.

F.3.5 The requirements for inspector qualifications shall follow API 650, Section 6.2.3, requiring an inspector with adequate vision and competent in the examination, interpretation, and evaluation of results.

F.3.6 Procedure requirements shall follow ASME Section V. 98

F.4 Ultrasonic Examination

F.4.1 Ultrasonic examinations are required for:

a. Shell areas over which lap patch plates are to be welded (see API 653, Section 7.3.1.9).
b. Shell areas over which new reinforcing or hot tap nozzles are to be welded (see API 653, Section 10.1.2.1).
c. Completed repairs of butt-welds unless radiographed (see API 653, Section 10.1.3.2).
d. The full penetration nozzle neck-to-shell and repad welds as required by API 653, Section 10.3.2.2.6, specific hydrotest exemption requirement.

F.4.2 The ultrasonic inspection acceptance standard shall be agreed upon by the purchaser and manufacturer (see API 650, Section 6.3.4).

F.4.3 The requirements for inspector qualifications shall follow ASNT Level II or III, according to ASNT SNT-TC-1A. Level I personnel may be used if they are given written acceptance/rejection criteria prepared by Level II or Level III personnel, and are under the direct supervision of Level II or Level III personnel.

F.4.4 Procedure requirements shall follow ASME Section V.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

F.5 Vacuum Box Testing

F.5.1 Vacuum box testing is required for:

a. New shell-to-bottom welds, unless diesel tested (see API 653, Section 10.1.6).
b. New bottom welds, unless tracer gas tested (see API 653, Section 10.1.7).
c. New roof plate welds for tanks designed to be gas tight (see API 650, Section 5.3.7).

F.5.2 There are no requirements for inspector qualifications.

F.5.3 Procedure requirements shall follow API 650, Section 5.3.3 (3 psig minimum).

F.6 Tracer Gas Testing

F.6.1 Tracer gas testing is required for new bottom welds, unless vacuum box tested (see API 653, Section 10.1.7).

F.6.2 There are no requirements for inspector qualifications.

98 **F.6.3** There are no procedure requirements.

F.7 Diesel Oil Test

F.7.1 Diesel oil testing is required for:

a. The first pass of new shell-to-bottom welds and the final new shell-to-bottom weld, unless the final weld is vacuum box tested (see API 653, Section 10.1.7).
b. Floating roof deck seams and other joints required to be liquid tight or vapor tight (see API 650, Section H.7.2 and C.4.2).

F.7.2 There are no requirements for inspector qualifications.

F.7.3 There are no procedure requirements.

F.8 Air Leak Testing

F.8.1 Air leak testing is required for repad-to-shell, repad-to-nozzle, and nozzle-to-shell welds of new or altered shell nozzles (see API 653, Section 10.4).

F.8.2 There are no requirements for inspector qualifications.

F.8.3 Procedure requirements shall follow API 650, Section 5.3.5.

F.9 Radiographic Examination

F.9.1 Radiographic examinations are required for:

a. Insert plates used for installing penetrations shall be fully radiographed (see API 653, Section 10.1.2.5).

b. Repairs to butt-welds, unless ultrasonically examined (see API 653, Section 10.1.3.2).

c. Horizontal welds of shell-to-shell plate welds—one per 50 feet, including junctions (see API 653, Section 10.2.1.2).

d. Each junction between new or repaired intersections of shell vertical and horizontal seams—one radiograph required (see API 653, Section 10.2.1.3).

e. Each butt-welded annular plate joint of reconstructed tanks (see API 653, Section 10.2.1.4).

f. New annular plate joints (see API 650, Section 6.1.1). 98

g. New and replaced shell insert plate and door sheet welds (see API 653, Section 10.2.1.5).

h. Vertical and horizontal shell joints as required by API 653, Section 10.3.2.2.5, specific hydrotest exemption requirement.

F.9.2 The inspection acceptance standard is ASME Section VIII, Paragraph UW-51(b).

F.9.3 The requirements for inspector qualifications shall follow ASNT Level II or III, according to ASNT SNT-TC-1A. Level I personnel may be used if they are given written acceptance/rejection criteria prepared by Level II or Level III personnel. These procedures must contain the applicable requirements of ASME Section V, Article 2. All Level I personnel must be under the direct supervision of Level II or Level III personnel.

F.9.4 Procedure requirements shall follow ASME Section V, Article 2.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

The American Petroleum Institute provides additional resources and programs to industry which are based on API Standards. For more information, contact:

• Training and Seminars	Ph: 202-682-8490 Fax: 202-682-8222
• Inspector Certification Programs	Ph: 202-682-8161 Fax: 202-962-4739
• American Petroleum Institute Quality Registrar	Ph: 202-962-4791 Fax: 202-682-8070
• Monogram Licensing Program	Ph: 202-962-4791 Fax: 202-682-8070
• Engine Oil Licensing and Certification System	Ph: 202-682-8233 Fax: 202-962-4739
• Petroleum Test Laboratory Accreditation Program	Ph: 202-682-8064 Fax: 202-962-4739

In addition, petroleum industry technical, patent, and business information is available online through API EnCompass™. Call 212-366-4040 or fax 212-366-4298 to discover more.

To obtain a free copy of the API Publications, Programs, and Services Catalog, call 202-682-8375 or fax your request to 202-962-4776. Or see the online interactive version of the catalog on our World Wide Web site – http://www.api.org.

American Petroleum Institute

Helping You Get The Job Done Right.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Additional copies available from API Publications and Distribution:
(202) 682-8375

Information about API Publications, Programs and Services
is available on the World Wide Web at: http://www.api.org

American
Petroleum
Institute

1220 L Street, Northwest
Washington, D.C. 20005-4070
202-682-8000

Order No. C653A3

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005-4070
202-682-8000

Date: December 1997
To: Purchasers of API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction,* Second Edition
Re: Addendum 2

This package contains Addendum 2 of API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction,* Second Edition. The addendum consists of the pages that have changed since the December 1996 printing of Addendum 1.

To update your copy of API Standard 653, replace the following pages as indicated:

Part of Book Changed	Old Pages to be Replaced	New Pages
Cover	Front and back covers	Front and back covers
Front matter	Title page to Addendum 1 v-xi	Title Page to Addendum 2 v-xi (+ blank)
Section 1	1-1–1-2	1-1–1-2
Section 2	2-1–2-4	2-1–2-4
Section 7	7-3–7-6	7-3–7-6
Section 8	8-1–8-3	8-1–8-3 (+ blank)
Section 10	10-1–10-3	10-1–10-3 (+ blank)
Appendix B	B-1–B-2	B-1–B-2

The parts of the text, tables, and figures that contain changes are indicated by a vertical bar and a small "97" in the margin.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607707 584 ■

Tank Inspection, Repair, Alteration, and Reconstruction

API STANDARD 653
SECOND EDITION, DECEMBER 1995
ADDENDUM 1, DECEMBER 1996
ADDENDUM 2, DECEMBER 1997



American Petroleum Institute

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34
Information Handling Services, 2000

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607708 410 ■

Tank Inspection, Repair, Alteration, and Reconstruction

Manufacturing, Distribution and Marketing Department

API STANDARD 653
SECOND EDITION, DECEMBER 1995
ADDENDUM 1, DECEMBER 1996
ADDENDUM 2, DECEMBER 1997



American
Petroleum
Institute

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34
Information Handling Services, 2000

SPECIAL NOTES

1. API PUBLICATIONS NECESSARILY ADDRESS PROBLEMS OF A GENERAL NATURE. WITH RESPECT TO PARTICULAR CIRCUMSTANCES, LOCAL, STATE, AND FEDERAL LAWS AND REGULATIONS SHOULD BE REVIEWED.

2. API IS NOT UNDERTAKING TO MEET THE DUTIES OF EMPLOYERS, MANUFACTURERS, OR SUPPLIERS TO WARN AND PROPERLY TRAIN AND EQUIP THEIR EMPLOYEES, AND OTHERS EXPOSED, CONCERNING HEALTH AND SAFETY RISKS AND PRECAUTIONS, NOR UNDERTAKING THEIR OBLIGATIONS UNDER LOCAL, STATE, OR FEDERAL LAWS.

3. INFORMATION CONCERNING SAFETY AND HEALTH RISKS AND PROPER PRECAUTIONS WITH RESPECT TO PARTICULAR MATERIALS AND CONDITIONS SHOULD BE OBTAINED FROM THE EMPLOYER, THE MANUFACTURER OR SUPPLIER OF THAT MATERIAL, OR THE MATERIAL SAFETY DATA SHEET.

4. NOTHING CONTAINED IN ANY API PUBLICATION IS TO BE CONSTRUED AS GRANTING ANY RIGHT, BY IMPLICATION OR OTHERWISE, FOR THE MANUFACTURE, SALE, OR USE OF ANY METHOD, APPARATUS, OR PRODUCT COVERED BY LETTERS PATENT. NEITHER SHOULD ANYTHING CONTAINED IN THE PUBLICATION BE CONSTRUED AS INSURING ANYONE AGAINST LIABILITY FOR INFRINGEMENT OF LETTERS PATENT.

5. GENERALLY, API STANDARDS ARE REVIEWED AND REVISED, REAFFIRMED, OR WITHDRAWN AT LEAST EVERY FIVE YEARS. SOMETIMES A ONE-TIME EXTENSION OF UP TO TWO YEARS WILL BE ADDED TO THIS REVIEW CYCLE. THIS PUBLICATION WILL NO LONGER BE IN EFFECT FIVE YEARS AFTER ITS PUBLICATION DATE AS AN OPERATIVE API STANDARD OR, WHERE AN EXTENSION HAS BEEN GRANTED, UPON REPUBLICATION. STATUS OF THE PUBLICATION CAN BE ASCERTAINED FROM THE API AUTHORING DEPARTMENT [TELEPHONE (202) 682-8000]. A CATALOG OF API PUBLICATIONS AND MATERIALS IS PUBLISHED ANNUALLY AND UPDATED QUARTERLY BY API, 1220 L STREET, N.W., WASHINGTON, D.C. 20005.

All rights reserved. No part of this work may be reproduced, stored in a retrieval system, or transmitted by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior written permission from the publisher. Contact API Publications Manager, 1220 L Street, N.W., Washington, D.C. 20005.

Copyright © 1995, 1996, 1997 American Petroleum Institute

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

CONTENTS

Page

SECTION 1—INTRODUCTION
1.1 General 1-1
1.2 Compliance With This Standard 1-1
1.3 Jurisdiction 1-1
1.4 Safe Working Practices 1-1
1.5 Definitions 1-1
97 | 1.6 Referenced Publications 1-2

SECTION 2—SUITABILITY FOR SERVICE
2.1 General 2-1
2.2 Tank Roof Evaluation 2-1
2.2.1 General 2-1
97 | 2.2.2 Fixed Roofs 2-1
2.2.3 Floating Roofs 2-1
2.2.4 Change of Service 2-1
2.3 Tank Shell Evaluation 2-2
2.3.1 General 2-2
2.3.2 Actual Thickness Determination 2-2
97 | 2.3.3 Minimum Thickness Calculation for Welded Tank Shell 2-3
2.3.4 Minimum Thickness Calculation for Riveted Tank Shell 2-4
2.3.5 Distortions 2-4
2.3.6 Flaws 2-5
2.3.7 Wind Girders and Shell Stiffeners 2-5
2.3.8 Shell Welds 2-5
2.3.9 Shell Penetrations 2-5
2.4 Tank Bottom Evaluation 2-5
2.4.1 General 2-5
2.4.2 Causes of Bottom Failure 2-5
2.4.3 Cathodic Protection of Tank Bottoms 2-5
2.4.4 Internal Lining Protection of Tank Bottoms 2-6
2.4.5 Bottom Leak Detection 2-6
2.4.6 Bottom Plate Thickness Measurements 2-6
2.4.7 Minimum Thickness for Tank Bottom Plate 2-6
2.4.8 Minimum Thickness for Annular Plate Ring 2-7
2.5 Tank Foundation Evaluation 2-7
2.5.1 General 2-7
2.5.2 Foundation Repair or Replacement 2-7
2.5.3 Anchor Bolts 2-7

SECTION 3—BRITTLE FRACTURE CONSIDERATIONS
3.1 General 3-1
3.2 Basic Considerations 3-1
3.3 Notes for Figure 3 3-1

SECTION 4—INSPECTION
4.1 General 4-1
4.2 Inspection Frequency Considerations 4-1
96 | 4.3 Inspections from the Outside of the Tank 4-1

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

4.3.1 Routine In-Service Inspections 4-1
4.3.2 External Inspection 4-1
96 4.3.3 Untrasonic Thickness Inspection 4-1
4.3.4 Cathodic Protection Surveys 4-2
4.4 Internal Inspection 4-2
4.4.1 General 4-2
4.4.2 Inspection Intervals 4-2
4.4.3 Alternative Internal Inspection Interval 4-2
4.5 Alternative to Internal Inspection to Determine Bottom Thickness 4-2
4.6 Preparatory Work for Internal Inspection 4-2
4.7 Inspection Checklists 4-2
4.8 Records 4-2
4.8.1 General 4-2
4.8.2 Construction Records 4-3
4.8.3 Inspection History 4-3
4.8.4 Repair/Alteration History 4-3
4.9 Reports 4-3
4.10 Inspector Qualifications 4-3

SECTION 5—MATERIALS
5.1 General 5-1
5.2 New Materials 5-1
5.3 Original Materials for Reconstructed Tanks 5-1
5.3.1 Shell and Bottom Plates Welded to the Shell 5-1
5.3.2 Structural 5-1
5.3.3 Flanges and Fasteners 5-1
5.3.4 Roof, Bottom, and Plate Wind Girders 5-1
5.4 Welding Consumables 5-1

SECTION 6—DESIGN CONSIDERATIONS FOR RECONSTRUCTED TANKS
6.1 General 6-1
6.2 New Weld Joints 6-1
6.3 Existing Weld Joints 6-1
6.4 Shell Design 6-1
6.5 Shell Penetrations 6-1
6.6 Wind Girders and Shell Stability 6-1
6.7 Roofs 6-1
6.8 Seismic Design 6-1

SECTION 7—TANK REPAIR AND ALTERATION
7.1 General 7-1
7.2 Removal and Replacement of Shell Plate Material 7-1
7.2.1 Minimum Thickness of Replacement Shell Plate 7-1
7.2.2 Minimum Dimensions of Replacement Shell Plate 7-1
7.2.3 Weld Joint Design 7-1
7.3 Shell Repairs Using Lap Welded Patch Plates 7-1
7.4 Repair of Defects in Shell Plate Material 7-4
7.5 Alteration of Tank Shells to Change Shell Height 7-4
96 7.6 Repair of Defective Welds 7-4
7.7 Repair of Shell Penetrations 7-4
97 7.8 Addition or Replacement of Shell Penetrations 7-4

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

7.9 Alteration of Existing Shell Penetrations 7-5
97 7.10 Repair of Tank Bottoms 7-5
7.10.1 Repairing a Portion of Tank Bottoms 7-5
7.10.2 Replacement of Entire Tank Bottom 7-6
7.11 Repair of Fixed Roofs 7-6
7.11.1 Supported Cone Roofs 7-6
7.11.2 Self-Supporting Roofs 7-7
7.12 Floating Roofs 7-7
7.12.1 External Floating Roofs 7-7
7.12.2 Internal Floating Roofs 7-7
7.12.3 Repair of Leaks in Pontoons 7-7
96 7.13 Repair or Replacement of Floating Roof Perimeter Seals 7-7
7.13.1 Primary Seals 7-7
7.13.2 Secondary Seals 7-7
7.13.3 Seal-to-Shell Gap 7-7
7.13.4 Mechanical Damage 7-7
7.13.5 Deterioration of Seal Material 7-8
7.13.6 Installation of Primary and Secondary Seals 7-8
7.14 Hot Taps 7-8
7.14.1 General 7-8
7.14.2 Hot Tap Procedures 7-8
7.14.3 Preparatory Work 7-8
7.14.4 Material Limitations 7-8
7.14.5 Installation Procedure 7-8

SECTION 8—DISMANTLING AND RECONSTRUCTION
8.1 General 8-1
8.2 Cleaning and Gas Freeing 8-1
8.3 Dismantling Methods 8-1
8.3.1 General 8-1
8.3.2 Bottoms 8-1
97 8.3.3 Shells 8-1
8.3.4 Roofs 8-1
8.3.5 Piece Marking 8-1
8.4 Reconstruction 8-2
8.4.1 General 8-2
97 8.4.2 Welding 8-2
8.4.3 Bottoms 8-2
8.4.4 Shells 8-2
8.4.5 Roofs 8-3
8.5 Dimensional Tolerances 8-3
8.5.1 General 8-3
97 8.5.2 Plumbness 8-3
8.5.3 Roundness 8-3
8.5.4 Peaking 8-3
8.5.5 Banding 8-3
8.5.6 Foundations 8-3

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

SECTION 9—WELDING
9.1 Welding Qualifications 9-1
9.2 Identification and Records 9-1

SECTION 10—EXAMINATION AND TESTING
10.1 Nondestructive Examinations 10-1
10.1.1 General 10-1
10.1.2 Shell Penetrations 10-1
10.1.3 Repaired Weld Flaws 10-1
10.1.4 Temporary and Permanent Attachments to Shell Plates 10-1
10.1.5 Shell Plate to Shell Plate Welds 10-1
10.1.6 Shell-to-Bottom Weld 10-1
10.1.7 Bottoms 10-1
10.2 Radiographs 10-1
97 | 10.2.1 Number and Location of Radiographs 10-1
10.2.2 Acceptance Criteria for Existing Shell Plate to Shell Plate Welds 10-2
10.2.3 Marking and Identification of Radiographs 10-2
10.3 Hydrostatic Testing 10-2
10.3.1 When Hydrostatic Testing Required 10-2
97 | 10.3.2 When Hydrostatic Testing Not Required 10-2
10.4 Leak Tests 10-3
10.5 Measured Settlement During Hydrostatic Testing 10-3
97 | 10.5.1 Initial Survey 10-3
10.5.2 Survey During Hydrostatic Testing 10-3

SECTION 11—MARKING AND RECORDKEEPING
11.1 Nameplates 11-1
11.2 Recordkeeping 11-1
11.3 Certification 11-1

APPENDIX A—BACKGROUND ON PAST EDITIONS OF API WELDED STORAGE TANK STANDARDS A-1
APPENDIX B—EVALUATION OF TANK BOTTOM SETTLEMENT B-1
APPENDIX C—CHECKLISTS FOR TANK INSPECTION C-1
APPENDIX D—AUTHORIZED INSPECTOR CERTIFICATION D-1
APPENDIX E—TECHNICAL INQUIRIES E-1

Figures
2-1—Inspection of Corrosion Areas 2-2
2-2—Pit Measurement 2-3
3-1—Brittle Fracture Considerations 3-2
3-2—Exemption Curve for Tanks Constructed From Carbon Steel of Unknown Toughness 3-3
7-1—Acceptable Details for Replacement of Shell Plate Material 7-2
7-2—Lapped Patch Repair Plates at the External Shell-to-Bottom Joint 7-3
96 | 7-2a—Typical Details for Addition of Reinforcing Plate to Existing Shell Penetration 7-4
7-3—Typical Details for Addition of "Tombstone" Shape Reinforcing Plate to Existing Shell Penetration 7-5
7-4—Method for Raising Shell Nozzles 7-6
7-5—Exemption Curve for Hot Tapping Tanks With Steels of Unknown Toughness 7-8

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

Figures

7-6—Hot Tap for Tanks 7-9
97 | 8-1—Tank Shell and Bottom Cut Locations 8-1
10-1—Exemption Curve for Hydrostatic Testing of Tanks 10-3
11-1—Nameplate 11-2
11-2—Certification Forms 11-3
B-1—Measurements of Shell Settlement (External) B-1
97 | B-2—Measurements of Bottom Settlement (Internal) Tank Out-of-Service B-2
B-3—Graphical Representation of Tank Shell Settlement B-3
B-4—Edge Settlement B-4
B-5—Bottom Settlement Near Shell B-4
B-6—Localized Bottom Depressions or Bulges Remote From Shell B-5
B-7—Localized Tank Bottom Settlement Limits for Single Pass Welds B-6

Tables

2-1—Joint Efficiencies for Welded Joints 2-4
2-2—Joint Efficiencies for Riveted Joints 2-4
2-3—Annular Bottom Plate Thicknesses (Inches)
(Product Specific Gravity <1.0) 2-7
4-1—Bottom Plate Minimum Thickness 4-3
96 | 7-1—Hot Tap Connection Sizes and Shell Plate Thicknesses 7-8
8-1—Maximum Thicknesses on New Welds 8-2
8-2—Radii Tolerances 8-3
10-1—Conditions for Exemption From Tank Hydrostatic Test for
Major Repairs and Alterations 10-3
A-1—Editions of API Standard 650 and Its Precursor, Standard 12C A-1

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Tank Inspection, Repair, Alteration, and Reconstruction

SECTION 1—INTRODUCTION

1.1 General

1.1.1 This standard covers carbon and low alloy steel tanks built to API Standard 650 and its predecessor 12C. This standard provides minimum requirements for maintaining the integrity of welded or riveted, non-refrigerated, atmospheric pressure, aboveground storage tanks after they have been placed in service. It covers the maintenance inspection, repair, alteration, relocation, and reconstruction of such tanks.

1.1.2 The scope is limited to the tank foundation, bottom, shell, structure, roof, attached appurtenances, and nozzles to the face of the first flange, first threaded joint, or first welding-end connection. Many of the design, welding, examination, and material requirements of API Standard 650 can be applied in the maintenance inspection, rating, repair, and alteration of in-service tanks. In the case of apparent conflicts between the requirements of this standard and API Standard 650 or its predecessor API 12C, this standard shall govern for tanks that have been placed in service.

1.1.3 This standard employs the principles of API Standard 650; however, storage tank owner/operators, based on consideration of specific construction and operating details, may apply this standard to any steel tank constructed in accordance with a tank specification.

1.1.4 This standard is intended for use by organizations that maintain or have access to engineering and inspection personnel technically trained and experienced in tank design, fabrication, repair, construction, and inspection.

1.1.5 This standard does not contain rules or guidelines to cover all the varied conditions which may occur in an existing tank. When design and construction details are not given, and are not available in the standard to which the tank was originally constructed, then details that will provide a level of integrity equal to the level provided by the current edition of API Standard 650 must be used.

1.2 Compliance With This Standard

The owner/operator has ultimate responsibility for complying with the provisions of this standard. Should a party other than the owner/operator be assigned certain tasks, such as relocating and reconstructing a tank, the limits of responsibility for each party shall be defined by the owner/operator prior to commencing work.

1.3 Jurisdiction

If any provision of this standards presents a direct or implied conflict with any statutory regulation, the regulation shall govern. However, if the requirements of this standard are more stringent than the requirements of the regulation, then the requirements of this standard shall govern.

1.4 Safe Working Practices

An assessment shall be made of the potential hazards to which personnel may be exposed when conducting internal tank inspections, making repairs, or dismantling tanks. Procedures shall be developed according to the guidelines given in API Publications 2015, 2015A, 2015B, 2207, and 2217 that will include safeguard for personnel health and safety, prevention of accidental fires and explosions, and the prevention of property damage.

Special procedures may need to be developed for certain activities described in this standard that are not fully covered by the referenced API publications; for example, safety precautions for personnel accessing floating roof tanks that are in service, or gas freeing the bottom side of a tank. Finally, procedures must comply with any federal or state safety regulations pertaining to "confined spaces" or any other relevant provisions.

1.5 Definitions

1.5.1 *Alteration* means any work on a tank involving cutting, burning, welding, or heating operations that changes the physical dimensions and/or configuration of a tank. Typical examples of alterations include:

a. The addition of manways and nozzles greater than 12-inch nominal pipe size (NPS).
b. An increase or decrease in tank shell height.

1.5.2 *Applicable standard* refers to the original standard of construction, such as API standards or specifications or Underwriter Laboratories (UL) standards, unless the original standard of construction has been superseded or withdrawn from publication; in this event, applicable standard means the current edition of the appropriate standard. See Appendix A for background on editions of API welded storage tank standards.

1.5.3 *Atmospheric pressure* is used to describe tanks designed to withstand an internal pressure up to but not exceeding 2 $^{1}/_{2}$ pounds per square inch gauge.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607716 597 ■

1.5.4 An *Authorized Inspection Agency* is one of the following organizations or individuals.

a. The inspection organization of the jurisdiction in which the aboveground storage tank is operated.
b. The inspection organization of an insurance company which is licensed or registered to and does write aboveground storage tank insurance.
c. An owner or operator of one or more aboveground storage tank(s) who maintains an inspection organization for activities relating only to his equipment, and not for aboveground storage tanks intended for sale or resale.
d. An independent organization or individual under contract to and under the direction of an owner or operator and recognized or otherwise not prohibited by the jurisdiction in which the aboveground storage tank is operated. The owner or operator's inspection program shall provide the controls necessary for use by Authorized Inspectors contracted to inspect aboveground storage tanks.

1.5.5 *Change in service* means a change from previous operating conditions involving different properties of the stored product such as specific gravity or corrosivity and/or different service conditions of temperature and/or pressure.

1.5.6 *Hot tap* identifies a procedure for installing a nozzle in the shell of a tank that is in service.

1.5.7 An *Inspector* as used in this document is an Authorized Inspector and is an employee of an Authorized Inspection Agency who is qualified and certified to perform tank inspections under this standard.

1.5.8 *Owner/operator* refers to the legal entity having both control of and/or responsibility for the operation and maintenance of an existing storage tank.

1.5.9 *Reconstruction* means the work necessary to reassemble a tank that has been dismantled and relocated to a new site.

1.5.10 *Reconstruction organization* refers to that organization having assigned responsibility by the owner/operator to design and/or reconstruct a tank.

1.5.11 *Repair* means any work necessary to maintain or restore a tank to a condition suitable for safe operation. Typical examples of repairs includes:

a. Removal and replacement of material (such as roof, shell, or bottom material, including weld metal) to maintain tank integrity.
b. Re-leveling and/or jacking of a tank shell, bottom, or roof.
c. Addition of reinforcing plates to existing shell penetrations.
d. Repair of flaws, such as tears or gouges, by grinding and/or gouging followed by welding.

1.6 Referenced Publications

The following standards, codes, publications, and specifications are cited in this standard. The latest edition or revision shall be used unless otherwise noted.

API

Std 620	*Design and Construction of Large, Welded Low-Pressure Storage Tanks*
Std 650	*Welded Steel Tanks for Oil Storage*
RP 651	*Cathodic Protection of Aboveground Petroleum Storage Tanks*
RP 652	*Lining of Aboveground Petroleum Storage Tank Bottoms*
Std 2000	*Venting Atmospheric and Low-Pressure Storage Tanks (Non-refrigerated and Refrigerated)*
RP 2003	*Protection Against Ignitions Arising Out of Static, Lightning, and Stray Currents*
Publ 2015	*Cleaning Petroleum Storage Tanks*
Publ 2015A	*Guide for Controlling the Lead Hazard Associated with Tank Entry and Cleaning*
Publ 2015B	*Cleaning Open-Top and Covered Floating-Roof Tanks*
Publ 2201	*Procedure for Welding or Hot Tapping on Equipment Containing Flammables*
Publ 2207	*Preparing Tank Bottoms for Hot Work*
Publ 2217A	*Guidelines for Work in Inert Confined Spaces in the Petroleum Industry*

97

ASME[1]

Boiler and Pressure Vessel Code, Section V, "Nondestructive Examination"; Section VIII, "Pressure Vessels" Alternative Rules, Division 2; Section IX, "Welding and Brazing Qualifications"

ASTM[2]

A6	*General Requirements for Rolled Steel Plates, Shapes, Sheet Piling, and Bars for Structural Use*
A20	*General Requirements for Steel Plates for Pressure Vessels*
A36	*Structural Steel*
A370	*Standard Test Methods and Definitions for Mechanical Testing of Steel Products*

[1] American Society for Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[2] American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

97

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 2—SUITABILITY FOR SERVICE

2.1 General

2.1.1 When the results of a tank inspection show that a change has occurred from the original physical condition of that tank, an evaluation shall be made to determine its suitability for continued use.

2.1.2 This section provides an evaluation of the suitability of an existing tank for continued service, or for a change of service, or when making decisions involving repairs, alterations, dismantling, relocating, or reconstructing an existing tank.

2.1.3 The following list of factors for consideration is not all-inclusive for all situations, nor is it intended to be a substitute for the engineering analysis and judgements required for each situation:

a. Internal corrosion due to the product stored or water bottoms.
b. External corrosion due to environmental exposure.
c. Stress levels and allowable stress levels.
d. Properties of the stored product such as specific gravity, temperature, and corrosivity.
e. Metal design temperatures at the service location of the tank.
f. External roof live load, wind, and seismic loadings.
g. Tank foundation, soil, and settlement conditions.
h. Chemical analysis and mechanical properties of the materials of construction.
i. Distortions of the existing tank.
j. Operating conditions such as filling/emptying rates and frequency.

2.2 Tank Roof Evaluation

2.2.1 GENERAL

2.2.1.1 The structural integrity of the roof and roof support system shall be verified.

2.2.1.2 Roof plates corroded to an average thickness of less than 0.09 inch in any 100 square inch area or roof plates with any holes through the roof plate shall be repaired or replaced.

2.2.2 FIXED ROOFS

Roof support members (rafters, girders, columns, and bases) shall be inspected for soundness by a method acceptable to the responsible inspector; distorted (such as out-of-plumb columns), corroded, and damaged members shall be evaluated and repaired or replaced if necessary. Particular attention must be given to the possibility of severe internal corrosion of pipe columns (corrosion may not be evidenced by external visual inspection).

97

2.2.3 FLOATING ROOFS

2.2.3.1 Areas of roof plates and pontoons exhibiting cracks or punctures shall be repaired or the affected sections replaced. Holes through roof plates shall be repaired or replaced.

2.2.3.2 Areas that are pitted shall be evaluated to determine the likelihood of through-pitting occurring before the next scheduled internal inspection. If so, the affected areas shall be repaired or replaced.

2.2.3.3 Roof support systems, perimeter seal systems, appurtenances such as a roof rolling ladder, anti-rotation devices, water drain systems, and venting systems shall be evaluated for needed repairs or replacements.

2.2.3.4 Guidance for the evaluation of existing floating roofs shall be based on the criteria of API Standard 650, Appendix C for external floating roofs, and Appendix H for internal floating roofs. However, upgrading to meet this standard is not mandatory.

2.2.4 CHANGE OF SERVICE

2.2.4.1 Internal Pressure

All requirements of the applicable standard (for example, API 650, Appendix F) shall be considered in the evaluation and subsequent modifications to the tank roof and roof-to-shell junction.

2.2.4.2 External Pressure

As applicable, the roof support structure (if any), and the roof-to-shell junction shall be evaluated for the effects of a design partial vacuum. The criteria outlined in API Standard 620 shall be used.

2.2.4.3 Operation at Elevated Temperature

All requirements of API Standard 650, Appendix M, shall be considered before changing the service of a tank to operation at temperatures above 200°F.

2.2.4.4 Operation at Lower Temperature Than Original Design

If the operating temperature is changed to a lower temperature than the original design, the requirements of the

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607718 36T ■

applicable standard or API Standard 650 for the lower temperature shall be met.

2.2.4.5 Normal and Emergency Venting

Effects of change of service on normal and emergency venting shall be considered.

2.3 Tank Shell Evaluation

2.3.1 GENERAL

2.3.1.1 Flaws, deterioration, or other conditions (for example, change of service, relocation, corrosion greater than the original corrosion allowance) that might adversely affect the performance or structural integrity of the shell of an existing tank must be evaluated and a determination made regarding suitability for intended service.

2.3.1.2 The evaluation of the existing tank shell shall be conducted by personnel experienced in tank design and shall include an analysis of the shell for the intended design conditions, based on existing shell plate thickness and material. The analysis shall take into consideration all anticipated loading conditions and combinations, including pressure due to fluid static head, internal and external pressure, wind loads, seismic loads, roof live loads, nozzle loads, settlement, and attachment loads.

2.3.1.3 Shell corrosion occurs in many forms and varying degrees of severity and may result in a generally uniform loss of metal over a large surface area or in localized areas. Pitting may also occur. Each case must be treated as a unique situation and a thorough inspection conducted to determine the nature and extent of corrosion prior to developing a repair procedure. Pitting does not normally represent a significant threat to the overall structural integrity of a shell unless present in a severe form with pits in close proximity to one another. Criteria for evaluating both general corrosion and pitting are defined below.

2.3.1.4 Methods for determining the minimum shell thickness suitable for continued operation are given in 2.3.2, 2.3.3, and 2.3.4. (See Section 4 for frequency of inspection.)

2.3.1.5 If the requirements of 2.3.3 (welded) or 2.3.4 (riveted) cannot be satisfied, the corroded or damaged areas shall be repaired, or the allowable liquid level of the tank reduced, or the tank retired. The allowable liquid level for the continued use of a tank may be established by using the formulas for a minimum acceptable thickness (see 2.3.3.1 and 2.3.4.1) and solving for height H. The actual thickness, as determined by inspection, minus the corrosion allowance shall be used to establish the liquid level limit. The maximum design liquid level shall not be exceeded.

2.3.2 ACTUAL THICKNESS DETERMINATION

2.3.2.1 For determining the controlling thicknesses in each shell course when there are corroded areas of considerable size, measured thicknesses shall be averaged in accordance with the following procedure (see Figure 2-1):

a. For each area, the inspector shall determine the minimum thickness, t_2, at any point in the corroded area, excluding widely scattered pits (see 2.3.2.2).

b. Then, calculate the critical length, L:

$L = 3.7\sqrt{Dt_2}$, but not more than 40 inches.

Where:

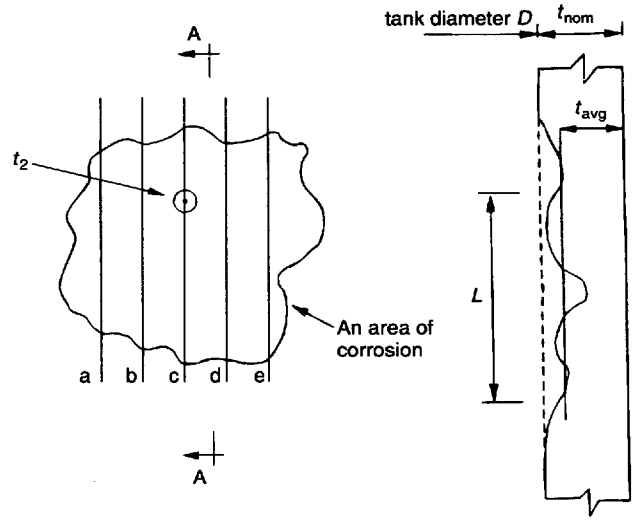
L = The maximum vertical length, in inches, over which hoop stresses are assumed to "average out" around local discontinuities.

Note: The actual vertical length of the corroded area may *exceed L.*

D = Tank diameter, in feet.

t_2 = The least thickness, in inches, in an area of corrosion, exclusive of pits.

c. Then, the inspector shall visually or otherwise decide which vertical plane(s) in the area is likely to be the most affected by corrosion. Profile measurements shall be taken along each vertical plane for a distance, L. In the plane(s),



Legend:
a–e are inspection planes selected by inspector.

t_2 = least min. thickness in entire area, exclusive of pits.

SECTION A-A
Profile along plane "c", the plane having the lowest average thickness, t_1.

Procedure:
1. Determine t_2.
2. Calculate $L = 3.7\sqrt{Dt_2}$, but not more than 40".
3. Locate "L" to get minimum t_{avg}, which is t_1.

Figure 2-1—Inspection of Corrosion Areas

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607719 2T6 ■

determine the lowest average thickness, t_1, averaged over a length of L, using at least five equally spaced measurements over length L.
d. Refer to 2.3.3.1 for minimum permitted values for t_1 and t_2. The additional loads in 2.3.3.4 shall also be considered.

2.3.2.2 Widely scattered pits may be ignored provided that:

a. No pit depth results in the remaining shell thickness being less than one-half the minimum acceptable tank shell thickness exclusive of the corrosion allowance; and
b. The sum of their dimensions along any vertical line does not exceed 2 inches in an 8-inch length (see Figure 2-2).

2.3.3 MINIMUM THICKNESS CALCULATION FOR WELDED TANK SHELL

2.3.3.1 The minimum acceptable shell plate thickness for tanks with diameters equal to or less than 200 feet shall be calculated as follows:

$$t_{min} = \frac{2.6\,DHG}{SE}$$

Where:

t_{min} = The minimum acceptable thickness, in inches for each course as calculated from the above formula; however, t_{min} shall not be less than 0.1 inch for any tank course.
D = Nominal diameter of tank, in feet.
H, H_t = Height, in feet, from the bottom of the length L (see 2.3.2.1) for the most severely corroded area in each shell course to the maximum liquid level. 97
G = Highest specific gravity of the contents.
S = Maximum allowable stress in pounds per square inch; use the smaller of $0.80Y$ or $0.429T$ for bottom and second course; use the smaller of $0.88Y$ or $0.472T$ for all other courses. Note: For reconstructed tanks, S shall be in accordance with the current applicable standard.
Y = Specified minimum yield strength of the plate; use 30,000 pounds per square inch if not known.
T = The smaller of the specified minimum tensile strength of the plate or 80,000 pounds per square inch; use 55,000 pounds per square inch if not known.
E = Original joint efficiency for the tank. Use Table 2-1 if original E is unknown. $E = 1.0$ when evaluating the retirement thickness in a corroded plate, when away from welds or joints by at least the greater of one inch or twice the plate thickness.

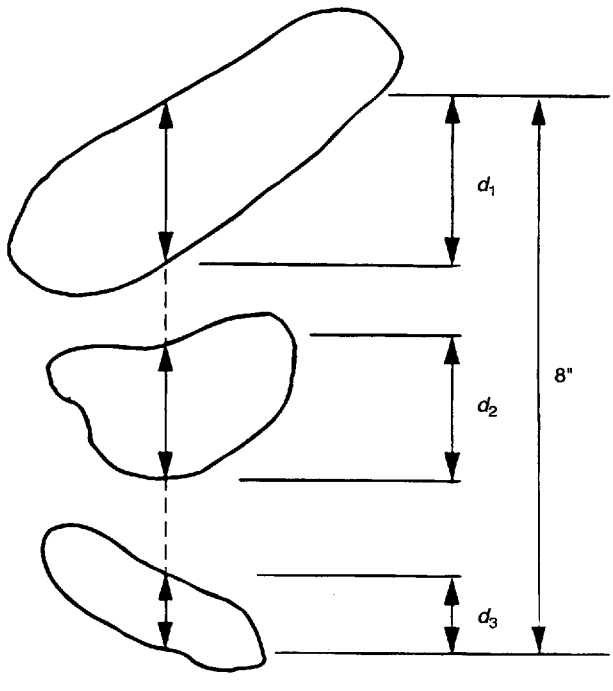


Figure 2-2—Pit Measurement

If the tank will be hydrostatically tested, after determining t_{min}, the hydrostatic test height, H_t, shall be:

$$H_t = \frac{S_t E t_{min}}{2.6D}$$

Where:

S_t = Maximum allowable hydrostatic test stress in pounds per square inch; use the smaller of $0.88Y$ or $0.472T$ for bottom and second courses; use the smaller of $0.9Y$ or $0.519T$ for all other courses. 97

Notes:
1. Depending on the specific gravity of the content used to determine t_{min}, H_t may be less than H. Testing the tank to H may yield the corroded area.
2. If H_t is less than H, owner/operator shall determine the consequence and acceptability of operating the tank to H, its maximum design liquid level. Repairs to shell sections above H_t shall comply with the requirements of 10.3.2.
3. For reconstructed tanks, S_t shall be per the current applicable standard.

The criteria for continued operation is:

a. The value t_1 (see 2.3.2.1.c) shall be greater than or equal to t_{min} (see above), subject to verification of all other loadings listed below; and
b. The value t_2 (see 2.3.2.1.b) shall be greater than or equal to 60 percent of t_{min}.

Any corrosion allowance for service until the time of the next tank inspection shall be added to the two required thicknesses (t_{min} and $0.60\,t_{min}$) in a. and b. above.

2.3.3.2 Alternatively, the minimum acceptable shell plate thickness for tanks with diameters equal to or less than 200 feet may be calculated in accordance with the variable design point method in API Standard 650, 3.6.4, substituting "S × E" for "S"; E and S may be defined as in 2.3.3.1.

2.3.3.3 The variable design point method shall be used for tanks greater than 200 feet in diameter, with all variables defined as in 2.3.3.1.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607720 T18 ■

Table 2-1—Joint Efficiencies for Welded Joints

Standard	Edition & Year	Type of Joint	Joint Efficiency E	Applicability or Limits
API 650	7th & Later	Butt	1.00	Basic Standard
	(1980–Present)	Butt	0.85	Appendix A–Spot RT
		Butt	0.70	Appendix A–No RT
	1st–6th	Butt	0.85	Basic Standard
	(1961–1978)	Butt	1.00	Appendices D&G
API 12C	14th & 15th (1957–1958)	Butt	0.85	
	3rd–13th (1940–1956)	Lap[a]	0.75	3/8 inch max. *t*
		Butt[c]	0.85	
	1st & 2nd (1936–1939)	Lap[a]	0.70	7/16 inch max. *t*
		Lap[b]	0.50 + k/5	1/4 inch max. *t*
		Butt[c]	0.85	
Unknown		Lap[a]	0.70	7/16 inch max. *t*
		Lap[b]	0.50 + k/5	1/4 inch max. *t*
		Butt[c]	0.70	
		Lap[d]	0.35	

[a]Full double lap welded.
[b]Full fillet weld with at least 25 percent intermittent full fillet opposite side; k = percent of intermittent weld expressed in decimal form.
[c]Single butt-welded joints with a back-up bar were permitted from the years of 1936 to 1940 and 1948 to 1954.
[d]Single lap welded only.

2.3.3.4 The thickness determinations of 2.3.3.1, 2.3.3.2, and 2.3.3.3 consider liquid loading only. All other loads shall also be evaluated according to the original standard of construction; and engineering judgement shall be used to evaluate different conditions or new information. As applicable, the following loadings shall be taken into account:

a. Wind-induced buckling.
b. Seismic loads.
c. Operation at temperatures over 200°F.
d. Vacuum-induced external pressure.
e. External loads caused by piping, tank-mounted equipment, hold down lugs, etc.
f. Wind-induced overturning.
g. Loads due to settlement.

2.3.3.5 As an alternative to the procedures described above, any thinning of the tank shell below minimum required wall thickness due to corrosion or other wastage may be evaluated to determine the adequacy for continued service by employing the "design by analysis methods," defined in Section VIII, Division 2, Appendix 4 of the ASME Code. When using these criteria, the stress value used in the original tank design shall be substituted for the S_m value of Division 2, if the design stress is less than or equal to the lesser of $^2/_3Y$ (Specified Minimum Yield Strength) or $^1/_3T$ (Specified Minimum Tensile Strength). If the original design stress is greater than $^2/_3Y$ or $^1/_3T$, then the lesser of $^2/_3Y$ or $^1/_3T$ shall be substituted for S_m.

2.3.4 MINIMUM THICKNESS CALCULATION FOR RIVETED TANK SHELL

2.3.4.1 The minimum acceptable thickness for riveted tank shells shall be calculated using the formula of 2.3.3.1 except that the following allowable stress criteria and joint efficiencies shall be used:

S = 21,000 pounds per square inch

E = 1.0 for shell plate 6 inches or more away from rivets. See Table 2-2 for joint efficiencies for locations within 6 inches or rivets.

2.3.4.2 The rivet joint efficiencies given in Table 2-2 are conservative minimums for riveted tank construction details and are included to simplify riveted tank evaluations. However, in some cases it may be advantageous to calculate the actual rivet joint efficiencies using computational methods applicable to lap and butt type riveted joints. When this alternative of calculated joint efficiencies is used, the following maximum allowable stresses shall apply:

a. For the maximum tensile stress in net section of plate, use the lesser of $0.80Y$ or $0.429T$; use 21,000 pounds per square inch if T or Y is unknown. | 97
b. For the maximum shear in net section of rivet, use 16,000 pounds per square inch.
c. For the maximum bearing stress on plates or rivets, use 32,000 pounds per square inch for rivets in single shear, and 35,000 pounds per square inch for rivets in double shear.

2.3.4.3 For tanks with riveted joints, consideration shall be given to whether, and to what extent, corrosion affects such joints. If calculations show that excess thickness exists, this excess may be taken as corrosion allowance.

2.3.4.4 Non-liquid loads (see 2.3.3.4) shall also be considered in the analysis of riveted tanks.

2.3.5 DISTORTIONS

2.3.5.1 Shell distortions include out-of-roundness, buckled areas, flat spots, and peaking and banding at welded joints.

Table 2-2—Joint Efficiencies for Riveted Joints

Type of Joint	Number of Rivet Rows	Joint Efficiency E
Lap	1	0.45
Lap	2	0.60
Lap	3	0.70
Lap	4	0.75
Butt[a]	2[b]	0.75
Butt	3[b]	0.85
Butt	4[b]	0.90
Butt	5[b]	0.91
Butt	6[b]	0.92

[a]All butt joints listed have butt straps both inside and outside.
[b]Number of row on each side of joint center line.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607721 954 ■

7.3.2.1 The welding shall be continuous on the outer perimeter of the repair plate and the inner perimeter of the hole in the shell plate. The minimum hole diameter is 2 inches. Shell openings due to plate removal shall have a minimum corner radius of 2 inches.

7.3.2.2 Nozzle necks and reinforcing plates shall be entirely removed prior to installation of a repair plate.

7.3.2.3 The repair plate thickness selection shall be based upon a design that conforms to the applicable standard of construction and API Standard 653, using a joint efficiency not exceeding 0.70. The welds of the repair plate shall be full fillet welds. The overlap of repair plate onto the shell shall not exceed 8 times the shell thickness. The minimum overlap is 1 inch.

7.3.2.4 The repair plate thickness shall not exceed the nominal thickness of the shell plate adjacent to the repair.

7.3.3 Lapped patch repair plates may be used to reinforce areas of severely deteriorated shell plates that are not able to resist the service loads to which the tank is to be subjected. Lapped patch repair plates may also be used for shell plates that are below the retirement thickness, providing the following additional requirements are satisfied.

96 **7.3.3.1** The selection of the repair plate thickness shall be based upon a design that conforms to the applicable standard of construction and API Standard 653, using a joint efficiency not exceeding 0.35. The perimeter weld shall be a full fillet weld.

7.3.3.2 The repair plate thickness shall not exceed the shell plate thickness at the perimeter of the repair plate by more than one-third, but no more than $^1/_8$ inch. The repair plate thickness shall not exceed $^1/_2$ inch.

7.3.3.3 The remaining strength of the deteriorated areas under the repair plate shall not be considered as effective in carrying the calculated service or hydrotest loads.

7.3.4 Lapped patch repair plates may be used to repair small shell leaks, or minimize the potential from leaks from severely isolated or widely scattered pitting if the following requirements are satisfied.

7.3.4.1 The existing shell thickness, excluding the holes and pitting, meets the minimum acceptable shell thickness as determined by 2.3.2 and 2.3.3.

7.3.4.2 The repair plate is designed to withstand the hydrostatic pressure load between the repair plate and the shell assuming a hole exists in the shell using a joint efficiency of 0.35.

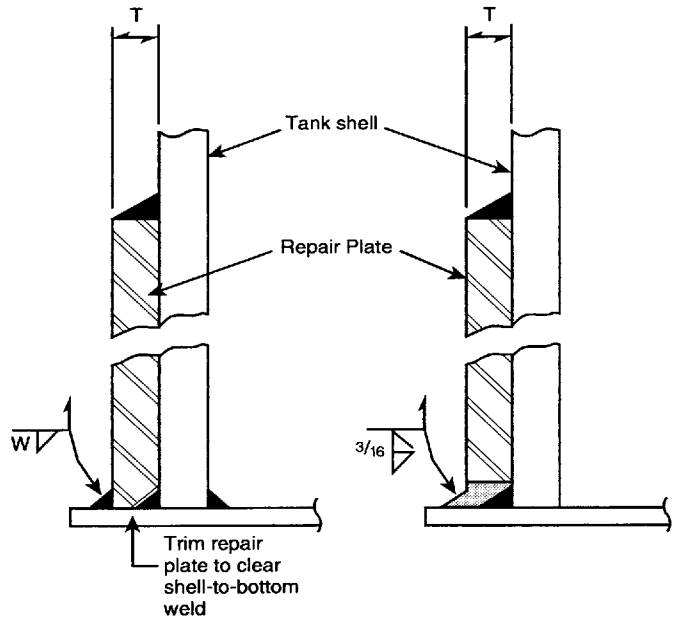


Figure 7-2—Lapped Patch Repair Plates at the External Shell-to-Bottom Joint

7.3.4.3 The repair plate thickness shall not exceed the shell plate thickness at the perimeter of the repair plate by more than one-third, but no more than $^1/_8$ inch. The repair plate thickness shall be no thinner than $^3/_{16}$ inch nor thicker than $^1/_2$ inch. A full fillet perimeter weld is required.

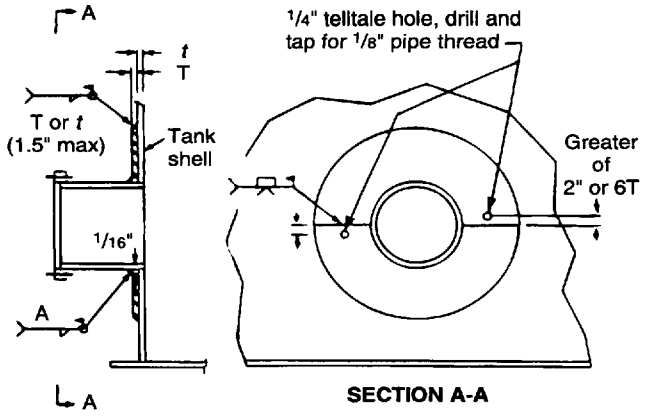
7.3.4.4 This repair method shall not be used if exposure of the fillet welds to the product will produce crevice corrosion or if a corrosion cell between the shell plate and repair plate is likely to occur.

96 **7.3.4.5** This repair method shall not be used to repair shell leaks if the presence of product between the shell plate and repair plate will prevent gas freeing from the tank to perform hot work.

7.3.4.6 The existing shell plate under the repair plate shall be evaluated at each future inspection to ensure it satisfies the requirements of 7.3.4.1. If the existing shell plate thickness does not satisfy 7.3.4.1 or the repair plate does not satisfy 7.3.3, the area is to be repaired in accordance with 7.2 or 7.3.2.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607722 890 ■



t and T (maximum)	A
9/16"	1/4"
3/4"	5/16"
15/16"	3/8"
1 1/8"	7/16"
1 5/16"	1/2"
1 9/16"	9/16"
1 3/4"	5/8"

Note: All details, dimensions, and weld spacing shall be in accordance with the requirements of API Standard 650.

Figure 7-2a—Typical Details for Addition of Reinforcing Plate to Existing Shell Penetration

7.4 Repair of Defects in Shell Plate Material

The need for repairing flaws such as cracks, gouges or tears (such as those often remaining after the removal of temporary attachments) discovered during an inspection of the tank shell shall be determined on an individual case basis. In areas of the shell where the plate thickness exceeds that required by design conditions, it is permissible to grind scars to a smooth contour so long as the remaining thickness is adequate for the design conditions. Where grinding to a smoothly contoured surface will result in unacceptable metal thickness, weld metal must be added to repair the scar using a qualified weld procedure.

7.5 Alteration of Tank Shells to Change Shell Height

Tank shells may be altered by adding new plate material to increase the height of the tank shell. The modified shell height shall be in accordance with the requirements of the applicable standard and shall take into consideration all anticipated loadings such as wind and seismic.

7.6 Repair of Defective Welds

Types of weld flaws and nonconformances that need repair are described in 7.6.1 through 7.6.4. | 96

7.6.1 Cracks, lack of fusion, and rejectable slag and porosity that need repair shall be removed completely by gouging and/or grinding and the resulting cavity properly prepared for welding.

7.6.2 Generally, it is not necessary to remove existing weld reinforcement in excess of that allowed by API Standard 650 when discovered on an existing tank with a satisfactory service history. However, if operating conditions are such that the excessive weld reinforcement may be deleterious (such as for a floating roof with flexible seals), consideration shall be given to repairing the welds by grinding.

7.6.3 Existing weld undercut deemed unacceptable based upon suitability for service considerations shall be repaired by additional weld metal, or grinding, as appropriate.

7.6.4 Welded joints that have experienced loss of metal due to corrosion may be repaired by welding.

7.6.5 Arc strikes discovered in or adjacent to welded joints shall be repaired by grinding and/or welding. Arc strikes repaired by welding shall be ground flush with the plate.

7.7 Repair of Shell Penetrations

7.7.1 Repairs to existing shell penetrations shall be in compliance with API Standard 650.

7.7.2 Reinforcing plates may be added to existing unreinforced nozzles when deemed appropriate. The reinforcing plate shall meet all dimensional and weld spacing requirements of API Standard 650. See Figures 7-2a and 7-3 for acceptable details. | 96

7.7.3 As a alternative, the reinforcing plates may be added to the inside of the tank provided that sufficient nozzle projection exists.

7.8 Addition or Replacement of Shell Penetrations

7.8.1 New shell penetrations (addition or replacement) shall be in accordance with requirements of API Standard 650 and 7.8.2 of this standard. | 97

7.8.2 Penetrations larger than 2 inches NPS shall be installed with the use of an insert plate if the shell plate thickness is greater than 1/2 inch and the shell plate material does

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607723 727 ■

not meet the current design metal temperature criteria. In addition, the following requirement shall be met:

a. The minimum diameter of the insert plate shall be at least twice the diameter of the penetration or the diameter plus 12 inches, whichever is greater.
b. When reinforcing plates are used, the minimum diameter of the insert plate shall equal the diameter of the reinforcing plate plus 12 inches.
c. If an integral reinforcement design is used, the thickness of the insert plate–at its periphery and for a distance of $2t$ from the edge of the weld (where t is the shell plate thickness)–shall be the same thickness as the shell, or the insert plate shall have a 1:4 taper to match the shell plate thickness.
d. Spacing of welds shall be in accordance with Figure 7-1.
e. The joint between new material of the insert plate and old material of the existing shell plate shall be butt-welded.

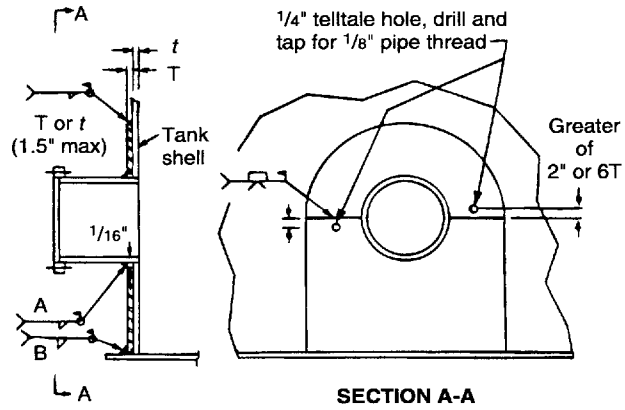
7.9 Alteration of Existing Shell Penetrations

7.9.1 Existing shell penetrations may be modified if the altered details comply with the requirements of API Standard 650, including the requirements for minimum reinforcing area and the requirements for spacing of welds around connections.

7.9.2 When installing a new tank bottom above the existing bottom, it may be necessary to alter existing shell penetrations in the bottom course of a tank shell. If the new bottom is "slotted" through the tank shell several inches above the existing bottom, the spacing between existing welds around penetrations and the new bottom-to-shell weld may not comply with API Standard 650 requirements. Options for altering the penetrations and/or reinforcing
96 | plates are given in 7.9.2.1 through 7.9.2.3.

7.9.2.1 The existing reinforcing plate may be trimmed to increase the spacing between the welds provided that the modified detail complies with the requirements of API Standard 650. Care must be exercised during the trimming operation to avoid damaging the shell material beneath the reinforcing plate. The existing weld attaching the portion of the reinforcing plate to be removed shall be completely removed by gouging and grinding.

7.9.2.2 The existing reinforcing plate may be removed and a new reinforcing plate added except that reinforcement plate replacement is not permitted on existing stress relieved assem-
97 | blies. If it is not known whether the assembly was thermally stressed relieved, then the alteration shall meet the requirements of API Standard 650, paragraph 3.7.4. Care must be exercised when removing the existing reinforcing plate to avoid damaging the shell plate beneath the reinforcing plate. The welds around the perimeter of the reinforcing plate and between the reinforcing plate and neck of the penetration shall be completely removed by gouging and grinding. The new rein-
96 | forcing plate shall be in accordance with Figure 7-2a. If required to maintain weld spacing, a "tombstone" shaped reinforcing plate may be used. (See Figure 7-3).



t and T (maximum)	A	B
9/16"	1/4"	1/4"
3/4"	5/16"	1/4"
15/16"	3/8"	5/16"
1 1/8"	7/16"	5/16"
1 5/16"	1/2"	3/8"
1 9/16"	9/16"	3/8"
1 3/4"	5/8"	3/8"

Note: All details, dimensions, and weld spacing shall be in accordance with the requirements of API Standard 650.

Figure 7-3—Typical Details for Addition of "Tombstone" Shape Reinforcing Plate to Existing Shell Penetration

7.9.2.3 The existing penetration may be moved by cutting the section of the shell containing the fitting and reinforcing plate, and raising the entire assembly to the correct elevation (see Figure 7-4).

7.9.3 Any components of the penetration (neck, flange, and reinforcing plate) that are in serviceable condition after removal may be reused.

7.10 Repair of Tank Bottoms

7.10.1 REPAIRING A PORTION OF TANK BOTTOMS

7.10.1.1 The critical zone for repairs to tank bottoms is within the annular plate ring, within 12 inches of the shell, or within 12 inches of the inside edge of the annular plate ring. No welding, welded-on patch plates or weld overlays are permitted within the critical zone except for welding of: widely scattered pits (see 2.3.2.2), cracks in the bottom plates, the shell-to-bottom weld, or where the bottom or annular plate is being replaced.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607724 663 ■

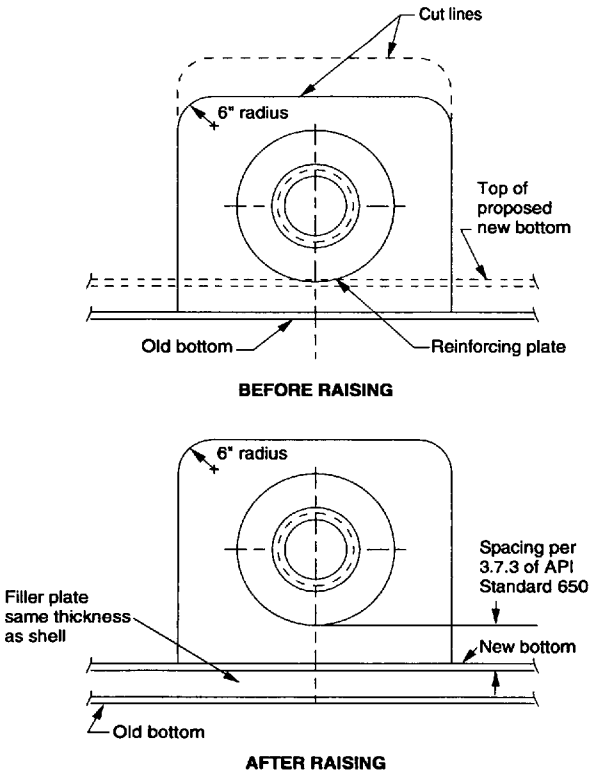


Figure 7-4—Method for Raising Shell Nozzles

7.10.1.2 If more extensive repairs are required within the critical zone than those listed in 7.10.1.1, the bottom plate under the bottom shell course shall be cut out and a new plate shall be installed. Weld spacing requirements shall be in accordance with 3.1.5.4 and 3.1.5.5 of API Standard 650 and 7.10.2.3 of API Standard 653.

7.10.1.3 The repair of sumps located within the critical zone shall be in accordance with 7.10.1.2.

7.10.2 REPLACEMENT OF ENTIRE TANK BOTTOM

7.10.2.1 Requirements governing the installation of a replacement bottom over an existing bottom are given in 7.10.2.1.1 through 7.10.2.1.6.

7.10.2.1.1 A suitable non-corrosive material cushion such as sand, gravel, or concrete shall be used between the old bottom and the new bottom.

7.10.2.1.2 The shell shall be "slotted" with a uniform cut made parallel to the tank bottom. The cut edges in the slot shall be ground to remove all slag and burrs from the cutting operation. The new bottom plate shall be extended outside the shell as required by API Standard 650. All rules for weld spacing shall be followed.

7.10.2.1.3 Voids in the foundation below the old bottom shall be filled with sand, crushed limestone, grout, or concrete.

7.10.2.1.4 Existing shell penetrations shall be raised if the elevation of the new bottom cuts through the reinforcing plate or if the weld space requirements given in 3.7.3 of API Standard 650 are not met.

7.10.2.1.5 For floating roof tanks, the new bottom profile must keep the roof level when it is resting on its support legs in the down position. The support legs can either be jacked up to maintain the original height or be shortened by the same amount as the thickness of the cushion and new bottom plate.

7.10.2.1.6 New bearing plates for floating roof support legs and for fixed roof support columns shall be installed.

7.10.2.2 When removing an existing tank bottom, the tank shell shall be separated from tank bottom either by:

a. Cutting the shell parallel to the tank bottom a minimum of $^1/_2$ inch above the bottom to shell weld (cut line B-B as shown in Figure 8-1), or
b. Removing the entire shell to bottom attachment weld, including any penetration and heat affected zone by suitable methods such as arc gouging and/or grinding.

If b. is elected, all arc gouged areas of the tank shell shall be magnetic particle examined, and defective areas repaired and re-examined.

7.10.2.3 Installation of a new tank bottom, after removal of the existing tank bottom, shall meet all requirements of API Standard 650. For tanks with shell plate of unknown toughness, which does not meet the exemption criteria of Figure 10-1, new weld joints in the bottom or annular ring shall be spaced at least the greater of 3 inches or $5t$ from existing vertical weld joints in the bottom shell course, where t is the thickness or bottom shell course, in inches.

7.10.2.4 When planning a bottom replacement, consideration shall be given to removing the old tank bottom or providing other means of preventing potential galvanic corrosion (refer to API RP 651). Also see 2.4.5 regarding bottom leak detection.

7.11 Repair of Fixed Roofs

7.11.1 SUPPORTED CONE ROOFS

7.11.1.1 The minimum thickness of new roof plates shall be $^3/_{16}$ inch plus any corrosion allowance as specified in the repair specifications. In the event roof live loads in excess of 25 pounds per square foot are specified (such as insulation, operating vacuum, high snow loads), the plate thickness shall be based on analysis using the allowable stresses in conformance with 3.10.3 of API Standard 650 (see 7.11.2.2).

7.11.1.2 The roof supports (rafters, girders, columns, and bases) shall be repaired or altered such that under design conditions the resulting stresses do not exceed the stress levels given in 3.10.3 of API Standard 650.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 8—DISMANTLING AND RECONSTRUCTION

8.1 General

8.1.1 This section provides procedures for dismantling and reconstructing existing welded tanks that are to be relocated from their original site.

8.1.2 Hydrostatic testing requirements, nondestructive examination requirements, and acceptance criteria for the quality of welds for reconstructed tanks are specified in Section 10.

8.2 Cleaning and Gas Freeing

The tank shall be cleaned and gas-freed prior to commencement of dismantling.

8.3 Dismantling Methods

8.3.1 GENERAL

Roof, shell, and bottom plates may be cut into any size pieces that are readily transportable to the new site for reconstruction.

8.3.2 BOTTOMS

8.3.2.1 Bottom plates that will be reused shall be cut by deseaming of lapwelds; or by cutting alongside of the remaining welds at a minimum of 2 inches away from existing welds, except where cuts cross existing weld seams.

8.3.2.2 If the bottom is to be used, one of the following methods is acceptable:

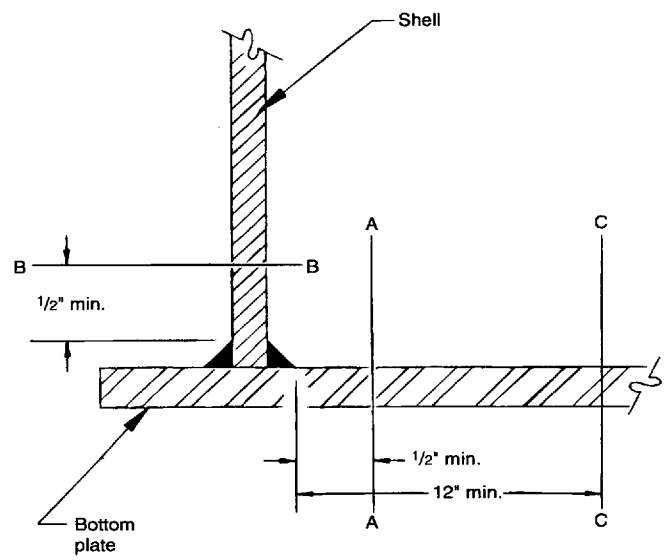
a. The bottom plates may be cut from the shell along a line A-A and line B-B shown in Figure 8-1, scrapping the welds and the bottom plate directly attached to the shell.
b. If the entire bottom is to be reused, the bottom may be cut from the shell on the line C-C leaving the shell with part of the bottom attached.
c. If the tank has an existing butt welded annular ring, this ring can be left attached to the shell or removed from the shell by cutting out along line B-B or otherwise removing the existing shell to annular ring welds.

97

8.3.3 SHELLS

8.3.3.1 Shell plates shall be cut in accordance with one of the following methods or a combination thereof:

a. Cuts made to remove existing welds and heat affected zones, except shell plates 1/2 inch or thinner can be cut through the welds leaving part of the weld and part or all of the heat affected zones.
b. Cuts made a minimum of 6 inches away from existing welds except where cuts cross existing welds.
c. Combinations of a and b.



97

Figure 8-1—Tank Shell and Bottom Cut Locations

8.3.3.2 Shell stiffening rings, including wind girders and top angles, may be left attached to the shell plates or may be removed by cutting at the attachment welds. The area where temporary attachments are removed shall be ground flush with the shell plate.

8.3.3.3 The shell shall be cut from the bottom plate along line B-B as shown in Figure 8-1. The existing shell-to-bottom weld connection shall not be reused unless the entire bottom is to be reused intact.

8.3.4 ROOFS

8.3.4.1 Roof plates shall be cut by deseaming of lap welds, or by cutting alongside of the remaining welds at a minimum of 2 inches away from existing welds, except where cuts cross existing welds.

8.3.4.2 Roof supporting structures shall be dismantled by removing bolts (if bolted) or deseaming the structural attachment welds.

8.3.5 PIECE MARKING

8.3.5.1 Shell, bottom, and roof plates shall be marked prior to dismantling for ready identification and placement when the tank is reconstructed. Marking material shall be of a durable type. Drawings showing piece mark locations are also a useful adjunct.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0608715 548 ■

8.3.5.2 A minimum of two sets of matching center punch marks shall be located on the top and bottom edges of each shell plate to facilitate proper alignment during reconstruction.

8.4 Reconstruction

8.4.1 GENERAL

8.4.1.1 The foundation for the reconstructed tank shall meet the construction tolerances given in 8.5.6.

8.4.1.2 Temporary attachments shall be removed, and the attachment area ground flush with the plate surface.

8.4.2 WELDING

8.4.2.1 Provisions shall be made during the reconstruction of a tank to ensure that weld spacing requirements of Figure 7-1 are maintained. 97

8.4.2.2 Tanks and their structural attachments shall be welded in accordance with the processes specified in API Standard 650 and the requirements of 8.4.2.3 through 8.4.2.11.

8.4.2.3 No welding of any kind shall be performed when the surfaces of the parts to be welded are wet from rain, snow, or ice; when rain or snow is falling on such surfaces; or during periods of high winds unless the welder and the work are properly shielded. No welding of any kind shall be performed when the temperature of the base metal is less than 0°F. When the temperature of the base metal is between 0°F and 32°F or the thickness is in excess of 1 inch, the base metal within 3 inches of the place where welding is to be started shall be heated to a temperature warm to the hand (approximately 140°F) before welding. (See 8.4.4.3 for preheat requirements for shell plates over $1^1/_2$ inches thick.)

8.4.2.4 Each layer of weld metal of multilayer welding shall be cleaned of slag and other deposits before the next layer is applied.

8.4.2.5 The edges of all welds shall merge with the surface of the plate without a sharp angle. The maximum acceptable undercutting is $^1/_{64}$ inch of the base metal for vertical butt joints. For horizontal butt joints undercutting not exceeding $^1/_{32}$ inch in depth is acceptable.

8.4.2.6 The reinforcement of the new welds on all butt joints on each side of the plate shall not exceed the thicknesses shown in Table 8-1.

Table 8-1—Maximum Thicknesses on New Welds

Plate Thickness (inches)	Maximum Reinforcement Thickness (inches)	
	Vertical Joints	Horizontal Joints
≤1/2	3/22	1/8
>1/2 through 1	1/8	3/16
>1	3/16	1/4

8.4.2.7 Tack welds used in the assembly of vertical joints of tank shells shall be removed and shall not remain in the finished joint when the joints are welded manually. When such joints are welded by the submerged-arc process, the tack welds shall be thoroughly cleaned of all welding slag but need not be removed provided that they are sound and are thoroughly fused into the subsequently applied weld beads. Tack welds shall be made using a fillet-weld or butt-weld procedure qualified in accordance with Section IX of the ASME Code. Tack welds to be left in place shall be made by qualified welders.

8.4.2.8 If weldable primer coatings have been applied on surfaces to be welded, they shall be included in welding procedure qualification tests for the brand, formulation, and maximum thickness of primer applied. All other coatings shall be completely removed from the weld area prior to welding.

8.4.2.9 Low-hydrogen electrodes shall be used for manual metal-arc welds, including the attachment of the first shell course to the bottom plates or annular plate ring, as follows:

a. For all welds in shell courses over 0.5 inch thick of API Standard 650 Group I-III materials.
b. For all welds in all shell courses of API Standard 650 Group IV-VI materials.

8.4.2.10 Low-hydrogen electrodes shall be used for welding temporary and new permanent attachments to shells of API Standard 650 Group IV, IVA, V, or VI materials. The welding procedure selected shall not cause underbead cracking; also, the need for preheat of thick plates and the effect of low atmospheric temperature during welding shall be considered.

8.4.2.11 If existing welds are found to be unsatisfactory by the original standard of construction, they shall be repaired in accordance with 7.6. 96

8.4.3 BOTTOMS

8.4.3.1 After the bottom plates are laid out and tacked, they shall be joined by welding the joints in a sequence that results in the least distortion from shrinkage and provides as nearly as possible a plane surface.

8.4.3.2 The welding of the shell to the bottom (except for door sheets) shall be completed before the welding of bottom joints.

8.4.3.3 Plates shall be held in close contact at all lap joints during welding.

8.4.4 SHELLS

8.4.4.1 Plates to be joined by butt welding shall be matched accurately and retained in position during welding. Misalignment in completed vertical joints over 5/8 inch thick

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607726 436 ■

shall not exceed 10 percent of the plate thickness, with a maximum of 1/8 inch. Misalignment in joint 5/8 inch thick or less shall not exceed 1/16 inch. Vertical joints shall be completed before the lower horizontal weld is made.

8.4.4.2 In completed horizontal butt joints, the upper plate shall not project beyond the face of the lower plate at any point by more than 20 percent of the thickness of the upper plate, with a maximum projection of 1/8 inch, except that a projection of 1/16 inch is acceptable for upper plates less than 5/16 inch thick.

8.4.4.3 For horizontal and vertical joints in tank shell courses constructed of material over 1 1/2 inches thick (based on the thickness of the thicker plate at the joint), multipass weld procedures are required, with no pass more than 3/4 inch thick permitted. A minimum preheat of 200°F is required of these welds.

8.4.5 ROOFS

This standard does not include special stipulations for reconstruction thereof, except that the structural framing (such as rafters and girders) must be reasonably true to line and surface. Other requirements shall be in accordance with the applicable standard.

8.5 Dimensional Tolerances

8.5.1 GENERAL

8.5.1.1 The tolerances given in this section have been established to produce a reconstructed tank of acceptable appearance and structural integrity and to permit proper functioning of floating roofs and seals.

8.5.1.2 Measurements to verify these tolerances shall be taken before the hydrostatic test of the reconstructed tank.

8.5.2 PLUMBNESS

8.5.2.1 The maximum out-of-plumbness of the top of the shell relative to the bottom of the shell shall not exceed $^1/_{100}$ of the total tank height, with a maximum of 5 inches. The $^1/_{100}$ criteria, with a maximum of 5 inches, shall also apply to fixed roof columns. For tanks with internal floating roofs, apply the criteria of this section or Appendix H of API Standard 650, whichever is more stringent. 97

8.5.2.2 The out-of-plumbness in one shell course shall not exceed the values specified for mill tolerances in ASTM A6 or A20, whichever is applicable.

8.5.3 ROUNDNESS

Radii measured at 1 foot above the shell-to-bottom weld shall not exceed the tolerances shown in Table 8-2.

Table 8-2—Radii Tolerances

Tank Diameter (feet)	Radius Tolerances (inches)
< 40	± 1/2
40 to < 150	± 3/4
150 to < 250	± 1
≥ 250	± 1 1/4

Radius tolerances measured higher than one foot above the shell-to-bottom weld shall not exceed three times the tolerances given above.

8.5.4 PEAKING

With a horizontal sweep board 36 inches long, peaking shall not exceed 1/2 inch. The sweep board shall be made to the true outside radius of the tank.

8.5.5 BANDING

With a vertical sweep board 36 inches long, banding shall not exceed one inch.

8.5.6 FOUNDATIONS

8.5.6.1 To achieve the tolerances specified in 8.5.1 through 8.5.5, it is essential that foundations true to a plane be provided for the tank reconstruction. The foundation shall have adequate bearing capacity to maintain the trueness of the foundation.

8.5.6.2 Where foundations true to a horizontal plane are specified, tolerances shall be as follows:

a. Where concrete ringwalls are provided under the shell, the top of the ringwall shall be level within ±1/8 inch in any 30 feet of the circumference and within ±1/4 inch in the total circumference measured from the average elevation.
b. Where concrete ringwalls are not provided, the foundation under the shell shall be level within ±1/8 inch in any 10 feet of circumference and within ±1/2 inch in the total circumference measured from the average elevation.

8.5.6.3 For foundations specified to be sloped from a horizontal plan, elevation differences about the circumference shall be calculated from the specified high point. Actual elevation differences about the circumference shall be determined from the actual elevation of the specified high point. The actual elevation differences shall not deviate from the calculated differences by more than the following tolerances:

a. Where concrete ringwalls are provided ±1/8 inch in any 30 feet of circumference and ±1/4 inch in the total circumference.
b. Where concrete ringwalls are not provided, ±1/8 inch in any 10 feet of circumference and ±1/2 inch in the total circumference.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 10—EXAMINATION AND TESTING

10.1 Nondestructive Examinations

10.1.1 GENERAL

10.1.1.1 Nondestructive examination procedures, qualifications, and acceptance criteria shall be prepared for visual, magnetic particle, liquid penetrant, ultrasonic, and radiographic methods in accordance with API Standard 650 and the supplemental requirements given herein.

10.1.1.2 Personnel performing nondestructive examinations shall be qualified in accordance with API Standard 650 and the supplemental requirements given herein.

10.1.2 SHELL PENETRATIONS

10.1.2.1 Ultrasonic examination of shell plate for laminations shall be made in the immediate area affected when:

a. Adding a reinforcing plate to an existing unreinforced penetration.

b. Adding a hot tap connection.

10.1.2.2 Cavities resulting from gouging or grinding operations to remove attachment welds of existing reinforcing plates shall be examined by visual as well as by magnetic particle or liquid penetrant methods.

10.1.2.3 Completed welds attaching nozzle neck to shell, and reinforcing plate to shell and to nozzle neck, shall be examined by the magnetic particle or liquid penetrant methods.

10.1.2.4 Completed welds of stress relieved assemblies shall be examined by visual as well as by magnetic particle or liquid penetrant methods after stress relief, but before hydrostatic test.

10.1.2.5 For penetrations installed using insert plates, the completed butt welds between the insert plate and the shell plate shall be fully radiographed.

10.1.3 REPAIRED WELD FLAWS

10.1.3.1 Cavities resulting from gouging or grinding operations to remove weld defects shall be examined by visual as well as by magnetic particle or liquid penetrant methods.

10.1.3.2 Completed repairs of butt welds shall be examined over their full length by radiographic or ultrasonic methods.

10.1.3.3 Completed repairs of fillet welds shall be examined over their full length by the appropriate nondestructive examination method listed herein.

10.1.4 TEMPORARY AND PERMANENT ATTACHMENTS TO SHELL PLATES

10.1.4.1 Areas where weld projections (remaining after removal of temporary attachments) have been ground flush with the shell plate shall be examined by the appropriate nondestructive examination method listed herein.

10.1.4.2 Completed welds of new permanent attachments (not including shell-to-bottom welds) and areas where temporary attachments have been removed (API Standard 650 tank shell materials of Groups IV, IVA, V, or VI) shall be examined visually and by the magnetic particle method (or, at the option of the purchaser, by the liquid penetrant method). 96

10.1.5 SHELL PLATE TO SHELL PLATE WELDS

New welds attaching shell plate to shell plate shall be examined by visual and by radiographic methods (see 10.2). In addition, for plate thicknesses greater than one inch, the back-gouged surface of the root pass and final pass (each side) shall be examined for its complete length by magnetic particle or liquid penetrant methods.

10.1.6 SHELL-TO-BOTTOM WELD

New welding on the shell-to-bottom joint shall be inspected for its entire length by a right-angle vacuum box and a solution film, or by applying light diesel oil. Additionally, the first weld pass shall be inspected by applying light diesel oil to the side opposite the first weld pass made. The oil shall be allowed to stand at least four hours (preferably overnight) and then the weld inspected for "wicking" action. The oil shall be removed before the weld is completed.

10.1.7 BOTTOMS

Newly welded bottom joints shall be tested using a vacuum box and a solution film, or a tracer gas and detector. Every joint shall be tested for its entire length.

10.2 Radiographs

10.2.1 NUMBER AND LOCATION OF RADIOGRAPHS

The number and location of radiographs shall be in accordance with API Standard 650 and the following additional requirements:

10.2.1.1 For new or repaired shell vertical joints one radiograph shall be taken in every joint.

10.2.1.2 For new or repaired shell horizontal joints one radiograph shall be taken for each 50 feet of horizontal joint, not including those required at intersections of vertical and horizontal joints.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607728 209 ■

10.2.1.3 For each new or repaired intersection between shell vertical and horizontal joints one radiograph shall be taken.

10.2.1.4 For reconstructed tanks, each butt-welded annular plate joint shall be radiographed in accordance with API Standard 650.

10.2.1.5 For reconstructed tanks, radiographic inspection is required for 25 percent of all junctions of new welds over existing seams.

The purchaser shall, with the consent of the contractor, determine the extent of further inspection and repair that may be required. 97

Any further inspection or repair of existing welds will be handled by contractual agreement between the purchaser and tank reconstruction contractor.

10.2.1.6 New and replaced shell insert plate and door sheet welds shall be radiographed. For circular replacement plates, a minimum of one radiograph shall be taken regardless of thickness. For square or rectangular replacement plates, at least one radiograph shall be taken in a vertical joint, and at least one in a horizontal joint, and one in each corner. All junctions between repair and existing welds shall be radiographed. If defects are found, 100 percent radiography shall be performed on the repaired weld.

97 **10.2.1.7** The minimum diagnostic length of each radiograph shall be 6 inches.

10.2.2 ACCEPTANCE CRITERIA FOR EXISTING SHELL PLATE TO SHELL PLATE WELDS

If the radiograph of an intersection between a new and old weld detects unacceptable welds by current standards, the existing welds may be evaluated according to the original standard of construction.

10.2.3 MARKING AND IDENTIFICATION OF RADIOGRAPHS

10.2.3.1 Each film shall show an identification of the welder(s) making the weld. A weld map showing location of welds, weld number, radiograph number, welder identification, and grading of each weld is an acceptable alternative to this requirement.

96 **10.2.3.2** Radiographs and radiograph records of all repaired welds shall be marked with the letter "R".

10.3 Hydrostatic Testing

10.3.1 WHEN HYDROSTATIC TESTING REQUIRED

10.3.1.1 A full hydrostatic test, held for 24 hours, shall be performed on:

a. A reconstructed tank.
b. Any tank that has undergone major repairs or major alterations (see 10.3.1.2) unless exempted by 10.3.2 for the applicable combination of materials, design, and construction features.

10.3.1.2 The terms major repair and major alteration refer to operations that require cutting, addition, removal and/or replacement of the annular plate ring, the shell-to-bottom weld, or a sizable portion of the shell. Within this context, *major repairs* and *major alterations* would include:

a. The installation of any shell penetration beneath the design liquid level larger than 12 inches NPS, or any bottom penetration located within 12 inches of the shell.
b. The removal and replacement or addition of any shell plate beneath the design liquid level, or any annular plate ring material where the longest dimension of the replacement plate exceeds 12 inches.
c. The complete or partial (more than one-half of the weld thickness) removal and replacement of more than 12 inches of vertical weld joining shell plates, or radial weld joining the annular plate ring.
d. The installation of a new bottom.
e. The removal and replacement of any part of the weld attaching the shell to the bottom or to the annular plate ring.
f. Partial or complete jacking of a tank shell.

10.3.2 WHEN HYDROSTATIC TESTING NOT REQUIRED

10.3.2.1 A full hydrostatic test of the tank is not required for major repairs and major alterations when any of the basic conditions listed in Column A of Table 10-1, in combination with the specific requirements of Column B, are satisfied. (See Figure 10-1.)

10.3.2.2 Specific requirements that must be satisfied in establishing an exemption basis from hydrostatic testing are identified for use in Column B of Table 10-1.

a. The repair method has been reviewed and approved by an engineer experienced in storage tank design in accordance with API Standard 650.
b. Materials used for the repair shall meet API Standard 650 requirements.
c. Vertical and horizontal shell joint welds shall have complete penetration and complete fusion. The root pass and final pass examination shall be in accordance with 10.1.5. In addition, the finished weld shall be fully radiographed.
d. Shell penetrations shall be installed with complete penetration and complete fusion weld for the reinforcing plate to nozzle neck and nozzle neck to shell joints. The root pass of the nozzle attachment weld shall be back-gouged and examined by magnetic particle or liquid penetrant methods; the completed weld shall be examined by the ultrasonic method. Examination and acceptance criteria for nondestructive examinations shall be in accordance with 10.1.1.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

10.3.2.3 A full hydrostatic test is not required after replacement of a door sheet provided that both of the following conditions are met:

97| a. Door sheet is in accordance with the requirements of 7.2, except that it shall not extend to or intersect the bottom to the shell joint.
b. The tank is otherwise exempt from hydrostatic testing as permitted by 10.3.2.1.

10.4 Leak Tests

New or altered reinforcing plates of shell penetrations shall be given an air leak test in accordance with API Standard 650.

10.5 Measured Settlement During Hydrostatic Testing

10.5.1 INITIAL SURVEY

10.5.1.1 Where settlement is anticipated, a tank receiving a hydrostatic test shall have the foundation checked for settlement.

10.5.1.2 Tank settlement shall initially be surveyed with the tank empty using the number of bottom plate projection elevation measurement points, N, uniformly distributed around the circumference, as indicated by the following formula:

$$N = D/10$$

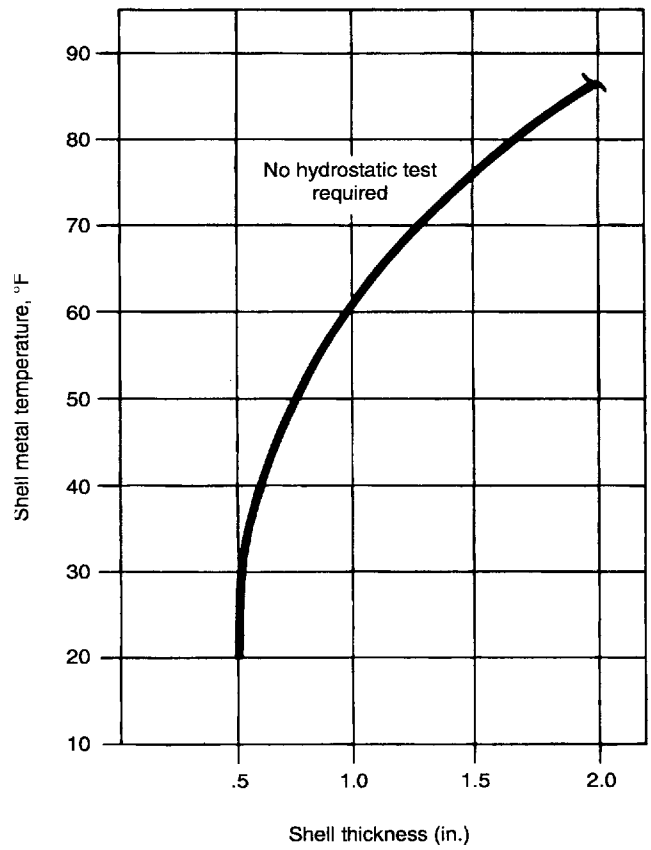
Where:

N = minimum required number of settlement measurement points, but no less than eight. All fractional values shall be rounded to the next higher whole number. The maximum spacing between
97| settlement measurement points shall be 32 feet.
D = tank diameter, in feet.

10.5.1.3 Tank settlement measurements in 10.5.1.2 shall be evaluated for acceptance in accordance with Appendix B.

10.5.2 SURVEY DURING HYDROSTATIC TESTING

Settlement shall be measured during filling and when water reaches 100 percent of test level. Excessive settlement in accordance with Appendix B shall be cause to stop the test for foundation investigation and/or repair.



Note: For unheated tanks, the shell metal temperature shall equal the design metal temperature as defined in 2.2.9.3 of API Standard 650. The original nominal thickness for the thickest tank shell plate shall be used for the assessment.

Figure 10-1—Exemption Curve for Hydrostatic Testing of Tanks

Table 10-1—Conditions for Exemption From Tank Hydrostatic Test for Major Repairs and Alterations

A Basic Conditions	B Specific Requirements (see 10.3.2.2 for subitem identification)
1. Tank material of unknown toughness, but satisfies the exemption criteria of Figure 10-1.	a, b, c, d
2. Tank material meets the toughness requirements of API Standard 650, seventh edition or later.	a, b, c, d
3. Tank repairs limited to bottom plates or annular plate ring (excluding shell-to-bottom weld).	a, b

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX B—EVALUATION OF TANK BOTTOM SETTLEMENT

B.1 Introduction

B.1.1 In determining the effects of soil settlement on storage tanks, it is common practice to monitor settlement of the tank bottom. In most cases, such a monitoring program is initiated during the construction and continued during hydrotesting and operations. During operations, settlement measurements should be taken at a planned frequency, based upon an assessment of soil settlement predictions. For existing tanks which do not have initial settlement data, a program of settlement monitoring should be based on prior service history.

B.1.2 If at any time settlement is deemed excessive, the tanks should be emptied and releveled. Releveling of a sizable tank is expensive and rather difficult to achieve. Thus, a decision to relevel a tank is a crucial one, and relies very much on the proper interpretation and evaluation of the monitored settlement data.

B.1.3 Approaches used to correct tank shell and bottom settlement include techniques such as localized repairs of the bottom plates, partial releveling of the tank periphery, and major releveling of the entire tank bottom. Major releveling of the tank, involving total lifting of the tank shell and bottom at one time, can introduce high localized stresses in the structure and impair its integrity. Therefore, when choosing techniques for correcting settlement problems, an alternative to total lifting of the tank shell and bottom should be considered as a first choice. If it is decided to lift the entire tank shell and bottom at one time, it should be done by personnel with demonstrated experience in this technique.

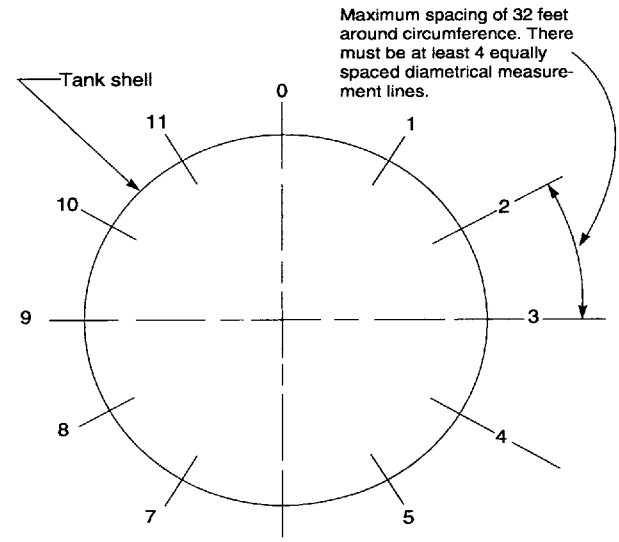
B.2 Types of Settlement

B.2.1 ELEVATION MEASUREMENTS

The principal types of tank settlement consist of settlements which relate to the tank shell and bottom plate. These settlements can be recorded by taking elevation measurements around the tank circumference and across the tank diameter. Figures B-1 and B-2 show recommended locations on a tank shell and bottom plate for settlement measurements. Data obtained from such measurements should be used to evaluate the tank structure. Additional settlement readings may be required to define local depressions.

B.2.2 SHELL SETTLEMENT EVALUATION

Settlement of a tank is considered to be the result of either one or a combination of the following three settlement components.



Notes:
1. There must be at least eight settlement points. The maximum spacing of settlement points is 32 feet around the circumference. 97
2. Points shall be equally spaced around the tank shell. See 10.5.1.2 for method of determining the number of measurement points.

Figure B-1—Measurements of Shell Settlement (External)

B.2.2.1 Uniform settlement. This component often can be predicted in advance, with sufficient accuracy from soil tests. It may vary in magnitude, depending on the soil characteristics. Uniform settlement of a tank does not induce stresses in the tank structure. However, piping and attachments must be given adequate consideration to prevent problems caused by such settlement.

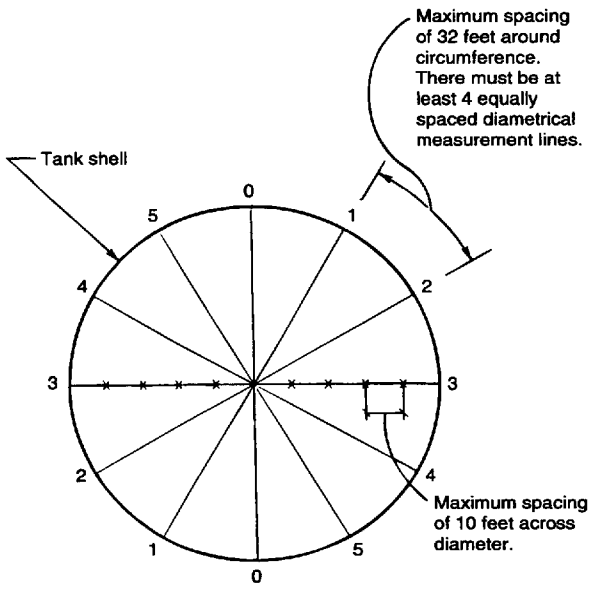
B.2.2.2 Rigid body tilting of a tank (planar tilt). This component rotates the tank in a tilted plane. The tilt will cause an increase in the liquid level and, therefore, an increase in the hoop stress in the tank shell. Also, excessive tilting can cause binding of peripheral seals in a floating roof and inhibit roof travel. This type of settlement could affect tank nozzles that have piping attached to them. Figure B-3 shows that the settled location of the tank shell, after rigid body tilt, can be represented by either a cosine or sine wave with respect to its original position in a horizontal plane.

B.2.2.3 Out-of-plane settlement (differential settlement). Due to the fact that a tank is a rather flexible structure, chances are great that the tank shell will settle in a non-planar con-

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607731 8T3 ■

97



Note: See 10.5.1.2 for method of determining the number of measurement points.

Figure B-2—Measurements of Bottom Settlement (Internal) Tank Out-of-Service

figuration, inducing additional stresses in the tank shell. The out-of-plane settlements at the bottom edge lead to a lack of circularity at the top of the tank, and in the case of a floating roof tank, the extent of the induced ovality may impede the proper functioning of the floating roof in such a way that releveling is required. Also, such settlements may cause flat spots to develop in the tank shell. This type of settlement could affect tank nozzles that have piping attached to them.

B.2.2.4 While uniform settlement and rigid body tilt of a tank may cause problems as described above, the out-of-plane settlement is the important component to determine and evaluate in order to ensure the structural integrity of the shell and bottom. Based on this principle, a common approach is to determine the magnitudes of the uniform and rigid body tilt component (if any) for each data point on the tank periphery. Once this is carried out, the plane of rigid tilt is then important as a datum from which to measure the magnitudes of the out-of-plane settlements.

A graphical representation illustrating the effect of tank shell settlement components is shown in Figure B-3. The construction of this "settlement plot" has been developed in accordance with the following:

a. The actual edge settlement (in most cases an irregular curve) is plotted using points around the tank circumference as the abscissa.

b. The vertical distance between the abscissa and the lowest point on this curve (point 22) is the minimum settlement, and it is called the uniform settlement component. A line through this point, parallel to the abscissa, provides a new base or datum line for settlement measurements which are called "adjusted settlements."

c. The plane of rigid tilt settlement is represented by the "optimum cosine curve" between the maximum and minimum values of the actual edge settlements. Several methods exist for determining the optimum cosine curve. The least accurate method is by free hand drawing techniques, a kind of trial and error procedure to fit the best cosine curve through the data. A better method is to use the mathematical and graphical capabilities of a computer.

d. The vertical distances between the irregular curve and the cosine curve represent the magnitudes of the out-of-plane settlements (U_i at data point i).

B.2.2.5 A method of determining the acceptable settlement is given in B.3.2. As an alternative, a rigorous stress analysis could be carried out.

B.2.3 EDGE SETTLEMENT

B.2.3.1 Edge settlement occurs when the tank shell settles sharply around the periphery, resulting in deformation of the bottom plate near the shell-to-bottom corner junction. Figure B-4 illustrates this settlement.

B.2.3.2 The formula given in B.3.3 can be used to evaluate this criteria. Alternatively, a rigorous stress analysis can be carried out for the deformed profile.

B.2.4 BOTTOM SETTLEMENT NEAR TANK SHELL

B.2.4.1 Figure B-5 illustrates this type of settlement.

B.2.4.2 The formula given in B.3.3 can be used to evaluate this criteria. Alternatively, a rigorous stress analysis can be carried out for the deformed profile.

B.2.5 LOCALIZED BOTTOM SETTLEMENT REMOTE FROM TANK SHELL

B.2.5.1 These are depressions (or bulges) that occur in a random manner, remote from the shell (see Figure B-6).

B.2.5.2 Acceptability of these localized settlements is dependent upon localized stresses in the bottom plate, design and quality of the lap welds (single pass or multipass), and voids below the bottom plate. The formula given in B.3.3 can be used to evaluate this criteria. These limits are applicable to tank bottoms which have single pass lap welded joints.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Additional copies available from API Publications and Distribution:
(202) 682-8375

Information about API Publications, Programs and Services is
available on the World Wide Web at: http://www.api.org

American
Petroleum
Institute

1220 L Street, Northwest
Washington, D.C. 20005-4070
202-682-8000

Order No.: C653A2

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34

Information Handling Services, 2000

Reproduced By GLOBAL ENGINEERING DOCUMENTS With The Permission of API Under Royalty Agreement

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005-4070
202-682-8000

Date: December 1996

To: Purchasers of API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*, Second Edition

Re: Addendum 1

This package contains Addendum 1 of API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction,* Second Edition. This addendum also incorporates the corrected pages that were part of a 1996 errata after the December 1995 printing of the Second Edition.

To update your copy of API Standard 653, replace the following pages as indicated:

Part of Book Changed	Old Pages to be Replaced	New Pages
Cover	Front cover	Front cover to Addendum 1
Front matter	Title page v thru ix, and blank	Title Page to Addendum 1 v thru ix, and blank
Section 1	1-1 and 1-2	1-1 and 1-2
Section 2	2-1 thru 2-7, and blank	2-1 thru 2-8
Section 4	4-1 thru 4-3, and blank	4-1 thru 4-3, and blank
Section 5	5-1, and blank	5-1, and blank
Section 6	6-1, and blank	6-1, and blank
Section 7	7-1 thru 7-8	7-1 thru 7-9, and blank
Section 8	8-1 and 8-2	8-1 and 8-2
Section 10	10-1 and 10-2	10-1 thru 10-3, and blank
Section 11	11-1 thru 11-3	11-1 thru 11-3, and blank
Appendix B	B-1 thru B-6	B-1 thru B-6

The parts of the text, tables, and figures that contain changes are indicated by a vertical bar and a small "96" in the margin.

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34
Information Handling Services, 2000

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607734 502 ■

Tank Inspection, Repair, Alteration, and Reconstruction

API STANDARD 653
SECOND EDITION, DECEMBER 1995
ADDENDUM 1, DECEMBER 1996



American
Petroleum
Institute

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34
Information Handling Services, 2000

Tank Inspection, Repair, Alteration, and Reconstruction

Manufacturing, Distribution and Marketing Department

API STANDARD 653
SECOND EDITION, DECEMBER 1995
ADDENDUM 1, DECEMBER 1996



American Petroleum Institute

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34
Information Handling Services, 2000

SPECIAL NOTES

1. API PUBLICATIONS NECESSARILY ADDRESS PROBLEMS OF A GENERAL NATURE. WITH RESPECT TO PARTICULAR CIRCUMSTANCES, LOCAL, STATE, AND FEDERAL LAWS AND REGULATIONS SHOULD BE REVIEWED.

2. API IS NOT UNDERTAKING TO MEET THE DUTIES OF EMPLOYERS, MANUFACTURERS, OR SUPPLIERS TO WARN AND PROPERLY TRAIN AND EQUIP THEIR EMPLOYEES, AND OTHERS EXPOSED, CONCERNING HEALTH AND SAFETY RISKS AND PRECAUTIONS, NOR UNDERTAKING THEIR OBLIGATIONS UNDER LOCAL, STATE, OR FEDERAL LAWS.

3. INFORMATION CONCERNING SAFETY AND HEALTH RISKS AND PROPER PRECAUTIONS WITH RESPECT TO PARTICULAR MATERIALS AND CONDITIONS SHOULD BE OBTAINED FROM THE EMPLOYER, THE MANUFACTURER OR SUPPLIER OF THAT MATERIAL, OR THE MATERIAL SAFETY DATA SHEET.

4. NOTHING CONTAINED IN ANY API PUBLICATION IS TO BE CONSTRUED AS GRANTING ANY RIGHT, BY IMPLICATION OR OTHERWISE, FOR THE MANUFACTURE, SALE, OR USE OF ANY METHOD, APPARATUS, OR PRODUCT COVERED BY LETTERS PATENT. NEITHER SHOULD ANYTHING CONTAINED IN THE PUBLICATION BE CONSTRUED AS INSURING ANYONE AGAINST LIABILITY FOR INFRINGEMENT OF LETTERS PATENT.

5. GENERALLY, API STANDARDS ARE REVIEWED AND REVISED, REAFFIRMED, OR WITHDRAWN AT LEAST EVERY FIVE YEARS. SOMETIMES A ONE-TIME EXTENSION OF UP TO TWO YEARS WILL BE ADDED TO THIS REVIEW CYCLE. THIS PUBLICATION WILL NO LONGER BE IN EFFECT FIVE YEARS AFTER ITS PUBLICATION DATE AS AN OPERATIVE API STANDARD OR, WHERE AN EXTENSION HAS BEEN GRANTED, UPON REPUBLICATION. STATUS OF THE PUBLICATION CAN BE ASCERTAINED FROM THE API AUTHORING DEPARTMENT [TELEPHONE (202) 682-8000]. A CATALOG OF API PUBLICATIONS AND MATERIALS IS PUBLISHED ANNUALLY AND UPDATED QUARTERLY BY API, 1220 L STREET, N.W., WASHINGTON, D.C. 20005.

All rights reserved. No part of this work may be reproduced, stored in a retrieval system, or transmitted by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior written permission from the publisher. Contact API Publications Manager, 1220 L Street, N.W., Washington, D.C. 20005.

Copyright © 1995 American Petroleum Institute

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

CONTENTS

Page

SECTION 1—INTRODUCTION
1.1 General 1-1
1.2 Compliance With This Standard 1-1
1.3 Jurisdiction 1-1
1.4 Safe Working Practices 1-1
1.5 Definitions 1-1
1.6 Referenced Publications 1-2

SECTION 2—SUITABILITY FOR SERVICE
2.1 General 2-1
2.2 Tank Roof Evaluation 2-1
2.2.1 General 2-1
2.2.2 Fixed Roofs 2-1
2.2.3 Floating Roofs 2-1
2.2.4 Change of Service 2-1
2.3 Tank Shell Evaluation 2-2
2.3.1 General 2-2
2.3.2 Actual Thickness Determination 2-2
2.3.3 Minimum Thickness Calculation for Welded Tank Shell 2-3
2.3.4 Minimum Thickness Calculation for Riveted Tank Shell 2-4
2.3.5 Distortions 2-4
2.3.6 Flaws 2-5
2.3.7 Wind Girders and Shell Stiffeners 2-5
2.3.8 Shell Welds 2-5
2.3.9 Shell Penetrations 2-5
2.4 Tank Bottom Evaluation 2-5
2.4.1 General 2-5
2.4.2 Causes of Bottom Failure 2-5
2.4.3 Cathodic Protection of Tank Bottoms 2-5
2.4.4 Internal Lining Protection of Tank Bottoms 2-6
2.4.5 Bottom Leak Detection 2-6
2.4.6 Bottom Plate Thickness Measurements 2-6
2.4.7 Minimum Thickness for Tank Bottom Plate 2-6
2.4.8 Minimum Thickness for Annular Plate Ring 2-7
2.5 Tank Foundation Evaluation 2-7
2.5.1 General 2-7
2.5.2 Foundation Repair or Replacement 2-7
2.5.3 Anchor Bolts 2-7

SECTION 3—BRITTLE FRACTURE CONSIDERATIONS
3.1 General 3-1
3.2 Basic Considerations 3-1
3.3 Notes for Figure 3 3-1

SECTION 4—INSPECTION
4.1 General 4-1
4.2 Inspection Frequency Considerations 4-1
96 | 4.3 Inspections from the Outside of the Tank 4-1

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

		Page
	4.3.1 Routine In-Service Inspections	4-1
	4.3.2 External Inspection	4-1
96	4.3.3 Untrasonic Thickness Inspection	4-1
	4.3.4 Cathodic Protection Surveys	4-2
	4.4 Internal Inspection	4-2
	4.4.1 General	4-2
	4.4.2 Inspection Intervals	4-2
	4.4.3 Alternative Internal Inspection Interval	4-2
	4.5 Alternative to Internal Inspection to Determine Bottom Thickness	4-2
	4.6 Preparatory Work for Internal Inspection	4-2
	4.7 Inspection Checklists	4-2
	4.8 Records	4-2
	4.8.1 General	4-2
	4.8.2 Construction Records	4-3
	4.8.3 Inspection History	4-3
	4.8.4 Repair/Alteration History	4-3
	4.9 Reports	4-3
	4.10 Inspector Qualifications	4-3

SECTION 5—MATERIALS

	5.1 General	5-1
	5.2 New Materials	5-1
	5.3 Original Materials for Reconstructed Tanks	5-1
	5.3.1 Shell and Bottom Plates Welded to the Shell	5-1
	5.3.2 Structural	5-1
	5.3.3 Flanges and Fasteners	5-1
	5.3.4 Roof, Bottom, and Plate Wind Girders	5-1
	5.4 Welding Consumables	5-1

SECTION 6—DESIGN CONSIDERATIONS FOR RECONSTRUCTED TANKS

	6.1 General	6-1
	6.2 New Weld Joints	6-1
	6.3 Existing Weld Joints	6-1
	6.4 Shell Design	6-1
	6.5 Shell Penetrations	6-1
	6.6 Wind Girders and Shell Stability	6-1
	6.7 Roofs	6-1
	6.8 Seismic Design	6-1

SECTION 7—TANK REPAIR AND ALTERATION

	7.1 General	7-1
	7.2 Removal and Replacement of Shell Plate Material	7-1
	7.2.1 Minimum Thickness of Replacement Shell Plate	7-1
	7.2.2 Minimum Dimensions of Replacement Shell Plate	7-1
	7.2.3 Weld Joint Design	7-1
	7.3 Shell Repairs Using Lap Welded Patch Plates	7-1
	7.4 Repair of Defects in Shell Plate Material	7-4
	7.5 Alteration of Tank Shells to Change Shell Height	7-4
96	7.6 Repair of Defective Welds	7-4
	7.7 Repair of Shell Penetrations	7-4
	7.8 Addition or Replacement of Shell Penetrations	7-4

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

7.9 Alteration of Existing Shell Penetrations 7-5
7.10 Repair of Tank Bottoms 7-5
7.10.1 Repairing a Portion of Tank Bottoms 7-5
7.10.2 Replacement of Entire Tank Bottom 7-6
7.11 Repair of Fixed Roofs 7-6
7.11.1 Supported Cone Roofs 7-6
7.11.2 Self-Supporting Roofs 7-7
7.12 Floating Roofs 7-7
7.12.1 External Floating Roofs 7-7
7.12.2 Internal Floating Roofs 7-7
7.12.3 Repair of Leaks in Pontoons 7-7
96 7.13 Repair or Replacement of Floating Roof Perimeter Seals 7-7
7.13.1 Primary Seals 7-7
7.13.2 Secondary Seals 7-7
7.13.3 Seal-to-Shell Gap 7-7
7.13.4 Mechanical Damage 7-7
7.13.5 Deterioration of Seal Material 7-8
7.13.6 Installation of Primary and Secondary Seals 7-8
7.14 Hot Taps 7-8
7.14.1 General 7-8
7.14.2 Hot Tap Procedures 7-8
7.14.3 Preparatory Work 7-8
7.14.4 Material Limitations 7-8
7.14.5 Installation Procedure 7-8

SECTION 8—DISMANTLING AND RECONSTRUCTION
8.1 General 8-1
8.2 Cleaning and Gas Freeing 8-1
8.3 Dismantling Methods 8-1
8.3.1 General 8-1
8.3.2 Bottoms 8-1
8.3.3 Shells 8-1
8.3.4 Roofs 8-1
8.3.5 Piece Marking 8-1
8.4 Reconstruction 8-1
8.4.1 General 8-1
8.4.2 Welding 8-2
8.4.3 Bottoms 8-2
8.4.4 Shells 8-3
8.4.5 Roofs 8-3
8.5 Dimensional Tolerances 8-3
8.5.1 General 8-3
8.5.2 Plumbness 8-3
8.5.3 Roundness 8-3
8.5.4 Peaking 8-3
8.5.5 Banding 8-3
8.5.6 Foundations 8-3

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

SECTION 9—WELDING
9.1 Welding Qualifications 9-1
9.2 Identification and Records 9-1

SECTION 10—EXAMINATION AND TESTING
10.1 Nondestructive Examinations 10-1
10.1.1 General 10-1
10.1.2 Shell Penetrations 10-1
10.1.3 Repaired Weld Flaws 10-1
10.1.4 Temporary and Permanent Attachments to Shell Plates 10-1
10.1.5 Shell Plate to Shell Plate Welds 10-1
10.1.6 Shell-to-Bottom Weld 10-1
10.1.7 Bottoms 10-1
10.2 Radiographs 10-1
10.2.1 Number and Location of Radiographs 10-1
10.2.2 Acceptance Criteria for Existing Shell Plate to Shell Plate Welds 10-2
10.2.3 Marking and Identification of Radiographs 10-2
10.3 Hydrostatic Testing 10-2
10.3.1 When Hydrostatic Testing Required 10-2
10.3.2 When Hydrostatic Testing Not Required 10-2
10.4 Leak Tests 10-3
10.5 Measured Settlement During Hydrostatic Testing 10-3
10.5.1 Initial Survey 10-3
10.5.2 Survey During Hydrostatic Testing 10-3

SECTION 11—MARKING AND RECORDKEEPING
11.1 Nameplates 11-1
11.2 Recordkeeping 11-1
11.3 Certification 11-1

APPENDIX A—BACKGROUND ON PAST EDITIONS OF API WELDED STORAGE TANK STANDARDS A-1
APPENDIX B—EVALUATION OF TANK BOTTOM SETTLEMENT B-1
APPENDIX C—CHECKLISTS FOR TANK INSPECTION C-1
APPENDIX D—AUTHORIZED INSPECTOR CERTIFICATION D-1
APPENDIX E—TECHNICAL INQUIRIES E-1

Figures
2-1—Inspection of Corrosion Areas 2-2
2-2—Pit Measurement 2-3
3-1—Brittle Fracture Considerations 3-2
3-2—Exemption Curve for Tanks Constructed From Carbon Steel of Unknown Toughness 3-3
7-1—Acceptable Details for Replacement of Shell Plate Material 7-2
7-2—Lapped Patch Repair Plates at the External Shell-to-Bottom Joint 7-3
7-2a—Typical Details for Addition of Reinforcing Plate to Existing Shell Penetration 7-4
7-3—Typical Details for Addition of "Tombstone" Shape Reinforcing Plate to Existing Shell Penetration 7-5
7-4—Method for Raising Shell Nozzles 7-6
7-5—Exemption Curve for Hot Tapping Tanks With Steels of Unknown Toughness 7-8

96

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

Figures
7-6—Hot Tap for Tanks 7-9
8-1—Tank Shell and Bottom Cut Locations 8-1
10-1—Exemption Curve for Hydrostatic Testing of Tanks 10-3
11-1—Nameplate 11-2
11-2—Certification Forms 11-3
B-1—Measurements of Shell Settlement (External) B-1
B-2—Measurements of Bottom Settlement (Internal) Tank Out-of-Service B-2
B-3—Graphical Representation of Tank Shell Settlement B-3
B-4—Edge Settlement B-4
B-5—Bottom Settlement Near Shell B-4
B-6—Localized Bottom Depressions or Bulges Remote From Shell B-5
B-7—Localized Tank Bottom Settlement Limits for Single Pass Welds B-6

Tables
2-1—Joint Efficiencies for Welded Joints 2-4
2-2—Joint Efficiencies for Riveted Joints 2-4
2-3—Annular Bottom Plate Thicknesses (Inches)
(Product Specific Gravity <1.0) 2-7
4-1—Bottom Plate Minimum Thickness 4-3
96 | 7-1—Hot Tap Connection Sizes and Shell Plate Thicknesses 7-8
8-1—Maximum Thicknesses on New Welds 8-2
8-2—Radii Tolerances 8-3
10-1—Conditions for Exemption From Tank Hydrostatic Test for
Major Repairs and Alterations 10-3
A-1—Editions of API Standard 650 and Its Precursor, Standard 12C A-1

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Tank Inspection, Repair, Alteration, and Reconstruction

SECTION 1—INTRODUCTION

1.1 General

1.1.1 This standard covers carbon and low alloy steel tanks built to API Standard 650 and its predecessor 12C. This standard provides minimum requirements for maintaining the integrity of welded or riveted, non-refrigerated, atmospheric pressure, aboveground storage tanks after they have been placed in service. It covers the maintenance inspection, repair, alteration, relocation, and reconstruction of such tanks.

1.1.2 The scope is limited to the tank foundation, bottom, shell, structure, roof, attached appurtenances, and nozzles to the face of the first flange, first threaded joint, or first welding-end connection. Many of the design, welding, examination, and material requirements of API Standard 650 can be applied in the maintenance inspection, rating, repair, and alteration of in-service tanks. In the case of apparent conflicts between the requirements of this standard and API Standard 650 or its predecessor API 12C, this standard shall govern for tanks that have been placed in service.

1.1.3 This standard employs the principles of API Standard 650; however, storage tank owner/operators, based on consideration of specific construction and operating details, may apply this standard to any steel tank constructed in accordance with a tank specification.

1.1.4 This standard is intended for use by organizations that maintain or have access to engineering and inspection personnel technically trained and experienced in tank design, fabrication, repair, construction, and inspection.

1.1.5 This standard does not contain rules or guidelines to cover all the varied conditions which may occur in an existing tank. When design and construction details are not given, and are not available in the standard to which the tank was originally constructed, then details that will provide a level of integrity equal to the level provided by the current edition of API Standard 650 must be used.

1.2 Compliance With This Standard

The owner/operator has ultimate responsibility for complying with the provisions of this standard. Should a party other than the owner/operator be assigned certain tasks, such as relocating and reconstructing a tank, the limits of responsibility for each party shall be defined by the owner/operator prior to commencing work.

1.3 Jurisdiction

If any provision of this standards presents a direct or implied conflict with any statutory regulation, the regulation shall govern. However, if the requirements of this standard are more stringent than the requirements of the regulation, then the requirements of this standard shall govern.

1.4 Safe Working Practices

An assessment shall be made of the potential hazards to which personnel may be exposed when conducting internal tank inspections, making repairs, or dismantling tanks. Procedures shall be developed according to the guidelines given in API Publications 2015, 2015A, 2015B, 2207, and 2217 that will include safeguard for personnel health and safety, prevention of accidental fires and explosions, and the prevention of property damage.

Special procedures may need to be developed for certain activities described in this standard that are not fully covered by the referenced API publications; for example, safety precautions for personnel accessing floating roof tanks that are in service, or gas freeing the bottom side of a tank. Finally, procedures must comply with any federal or state safety regulations pertaining to "confined spaces" or any other relevant provisions.

1.5 Definitions

1.5.1 *Alteration* means any work on a tank involving cutting, burning, welding, or heating operations that changes the physical dimensions and/or configuration of a tank. Typical examples of alterations include:

a. The addition of manways and nozzles greater than 12-inch nominal pipe size (NPS).
b. An increase or decrease in tank shell height.

1.5.2 *Applicable standard* refers to the original standard of construction, such as API standards or specifications or Underwriter Laboratories (UL) standards, unless the original standard of construction has been superseded or withdrawn from publication; in this event, applicable standard means the current edition of the appropriate standard. See Appendix A for background on editions of API welded storage tank standards.

1.5.3 *Atmospheric pressure* is used to describe tanks designed to withstand an internal pressure up to but not exceeding 2 $^1/_2$ pounds per square inch gauge.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607743 515 ■

1.5.4 An *Authorized Inspection Agency* is one of the following organizations or individuals.

a. The inspection organization of the jurisdiction in which the aboveground storage tank is operated.
b. The inspection organization of an insurance company which is licensed or registered to and does write aboveground storage tank insurance.
c. An owner or operator of one or more aboveground storage tank(s) who maintains an inspection organization for activities relating only to his equipment, and not for aboveground storage tanks intended for sale or resale.
d. An independent organization or individual under contract to and under the direction of an owner or operator and recognized or otherwise not prohibited by the jurisdiction in which the aboveground storage tank is operated. The owner or operator's inspection program shall provide the controls necessary for use by Authorized Inspectors contracted to inspect aboveground storage tanks.

1.5.5 *Change in service* means a change from previous operating conditions involving different properties of the stored product such as specific gravity or corrosivity and/or different service conditions of temperature and/or pressure.

1.5.6 *Hot tap* identifies a procedure for installing a nozzle in the shell of a tank that is in service.

1.5.7 An *Inspector* as used in this document is an Authorized Inspector and is an employee of an Authorized Inspection Agency who is qualified and certified to perform tank inspections under this standard.

1.5.8 *Owner/operator* refers to the legal entity having both control of and/or responsibility for the operation and maintenance of an existing storage tank.

1.5.9 *Reconstruction* means the work necessary to reassemble a tank that has been dismantled and relocated to a new site.

1.5.10 *Reconstruction organization* refers to that organization having assigned responsibility by the owner/operator to design and/or reconstruct a tank.

1.5.11 *Repair* means any work necessary to maintain or restore a tank to a condition suitable for safe operation. Typical examples of repairs includes:

a. Removal and replacement of material (such as roof, shell, or bottom material, including weld metal) to maintain tank integrity.
b. Re-leveling and/or jacking of a tank shell, bottom, or roof.
c. Addition of reinforcing plates to existing shell penetrations.
d. Repair of flaws, such as tears or gouges, by grinding and/or gouging followed by welding.

1.6 Referenced Publications

The following standards, codes, publications, and specifications are cited in this standard. The latest edition or revision shall be used unless otherwise noted.

API

Std 620	*Design and Construction of Large, Welded Low-Pressure Storage Tanks*
Std 650	*Welded Steel Tanks for Oil Storage*
RP 651	*Cathodic Protection of Aboveground Petroleum Storage Tanks*
RP 652	*Lining of Aboveground Petroleum Storage Tank Bottoms*
Std 2000	*Venting Atmospheric and Low-Pressure Storage Tanks (Non-refrigerated and Refrigerated)*
RP 2003	*Protection Against Ignitions Arising Out of Static, Lightning, and Stray Currents*
Publ 2015	*Cleaning Petroleum Storage Tanks*
Publ 2015A	*Guide for Controlling the Lead Hazard Associated with Tank Entry and Cleaning*
Publ 2015B	*Cleaning Open-Top and Covered Floating-Roof Tanks*
Publ 2201	*Procedure for Welding or Hot Tapping on Equipment Containing Flammables*
Publ 2207	*Preparing Tank Bottoms for Hot Work*
Publ 2217	*Guidelines for Confined Space Work in the Petroleum Industry*

ASME[1]

Boiler and Pressure Vessel Code, Section V, "Nondestructive Examination"; Section VIII, "Pressure Vessels" Alternative Rules, Division 2; Section IX, "Welding and Brazing Qualifications"

ASTM[2]

A6	*General Requirements for Rolled Steel Plates, Shapes, Sheet Piling, and Bars for Structural Use*
A20	*General Requirements for Steel Plates for Pressure Vessels*
A36	*Structural Steel*
A370	*Standard Test Methods and Definitions for Mechanical Testing of Steel Products*

[1] American Society for Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[2] American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 2—SUITABILITY FOR SERVICE

2.1 General

2.1.1 When the results of a tank inspection show that a change has occurred from the original physical condition of that tank, an evaluation shall be made to determine its suitability for continued use.

2.1.2 This section provides an evaluation of the suitability of an existing tank for continued service, or for a change of service, or when making decisions involving repairs, alterations, dismantling, relocating, or reconstructing an existing tank.

2.1.3 The following list of factors for consideration is not all-inclusive for all situations, nor is it intended to be a substitute for the engineering analysis and judgements required for each situation:

a. Internal corrosion due to the product stored or water bottoms.
b. External corrosion due to environmental exposure.
c. Stress levels and allowable stress levels.
d. Properties of the stored product such as specific gravity, temperature, and corrosivity.
e. Metal design temperatures at the service location of the tank.
f. External roof live load, wind, and seismic loadings.
g. Tank foundation, soil, and settlement conditions.
h. Chemical analysis and mechanical properties of the materials of construction.
i. Distortions of the existing tank.
j. Operating conditions such as filling/emptying rates and frequency.

2.2 Tank Roof Evaluation

2.2.1 GENERAL

2.2.1.1 The structural integrity of the roof and roof support system shall be verified.

2.2.1.2 Roof plates corroded to an average thickness of less than 0.09 inch in any 100 square inch area or roof plates with any holes through the roof plate shall be repaired or replaced.

2.2.2 FIXED ROOFS

Roof support members (rafters, girders, columns, and bases) shall be inspected for soundness by a method acceptable to the responsible inspector; corroded and damaged members shall be evaluated and repaired or replaced if necessary. Particular attention must be given to the possibility of severe internal corrosion of pipe columns (corrosion may not be evidenced by external visual inspection).

2.2.3 FLOATING ROOFS

2.2.3.1 Areas of roof plates and pontoons exhibiting cracks or punctures shall be repaired or the affected sections replaced. Holes through roof plates shall be repaired or replaced.

2.2.3.2 Areas that are pitted shall be evaluated to determine the likelihood of through-pitting occurring before the next scheduled internal inspection. If so, the affected areas shall be repaired or replaced.

2.2.3.3 Roof support systems, perimeter seal systems, appurtenances such as a roof rolling ladder, anti-rotation devices, water drain systems, and venting systems shall be evaluated for needed repairs or replacements.

2.2.3.4 Guidance for the evaluation of existing floating roofs shall be based on the criteria of API Standard 650, Appendix C for external floating roofs, and Appendix H for internal floating roofs. However, upgrading to meet this standard is not mandatory.

2.2.4 CHANGE OF SERVICE

2.2.4.1 Internal Pressure

All requirements of the applicable standard (for example, API 650, Appendix F) shall be considered in the evaluation and subsequent modifications to the tank roof and roof-to-shell junction.

2.2.4.2 External Pressure

As applicable, the roof support structure (if any), and the roof-to-shell junction shall be evaluated for the effects of a design partial vacuum. The criteria outlined in API Standard 620 shall be used.

2.2.4.3 Operation at Elevated Temperature

All requirements of API Standard 650, Appendix M, shall be considered before changing the service of a tank to operation at temperatures above 200°F.

2.2.4.4 Operation at Lower Temperature Than Original Design

If the operating temperature is changed to a lower temperature than the original design, the requirements of the

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607745 398 ■

applicable standard or API Standard 650 for the lower temperature shall be met.

2.2.4.5 Normal and Emergency Venting

Effects of change of service on normal and emergency venting shall be considered.

2.3 Tank Shell Evaluation

2.3.1 GENERAL

2.3.1.1 Flaws, deterioration, or other conditions (for example, change of service, relocation, corrosion greater than the original corrosion allowance) that might adversely affect the performance or structural integrity of the shell of an existing tank must be evaluated and a determination made regarding suitability for intended service.

2.3.1.2 The evaluation of the existing tank shell shall be conducted by personnel experienced in tank design and shall include an analysis of the shell for the intended design conditions, based on existing shell plate thickness and material. The analysis shall take into consideration all anticipated loading conditions and combinations, including pressure due to fluid static head, internal and external pressure, wind loads, seismic loads, roof live loads, nozzle loads, settlement, and attachment loads.

2.3.1.3 Shell corrosion occurs in many forms and varying degrees of severity and may result in a generally uniform loss of metal over a large surface area or in localized areas. Pitting may also occur. Each case must be treated as a unique situation and a thorough inspection conducted to determine the nature and extent of corrosion prior to developing a repair procedure. Pitting does not normally represent a significant threat to the overall structural integrity of a shell unless present in a severe form with pits in close proximity to one another. Criteria for evaluating both general corrosion and pitting are defined below.

2.3.1.4 Methods for determining the minimum shell thickness suitable for continued operation are given in 2.3.2, 2.3.3, and 2.3.4. (See Section 4 for frequency of inspection.)

2.3.1.5 If the requirements of 2.3.3 (welded) or 2.3.4 (riveted) cannot be satisfied, the corroded or damaged areas shall be repaired, or the allowable liquid level of the tank reduced, or the tank retired. The allowable liquid level for the continued use of a tank may be established by using the formulas for a minimum acceptable thickness (see 2.3.3.1 and 2.3.4.1) and solving for height H. The actual thickness, as determined by inspection, minus the corrosion allowance shall be used to establish the liquid level limit. The maximum design liquid level shall not be exceeded.

2.3.2 ACTUAL THICKNESS DETERMINATION

2.3.2.1 For determining the controlling thicknesses in each shell course when there are corroded areas of considerable size, measured thicknesses shall be averaged in accordance with the following procedure (see Figure 2-1):

a. For each area, the inspector shall determine the minimum thickness, t_2, at any point in the corroded area, excluding widely scattered pits (see 2.3.2.2).

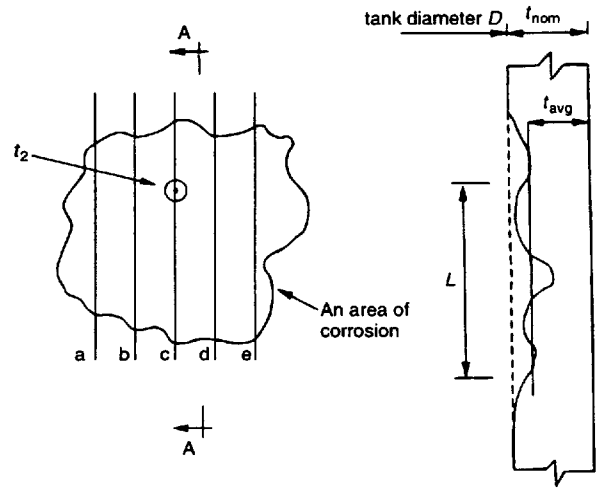
b. Then, calculate the critical length, L:

$$L = 3.7\sqrt{Dt_2}\text{, but not more than 40 inches.}$$

Where:

L = The maximum vertical length, in inches, over which hoop stresses are assumed to "average out" around local discontinuities.

Note: The actual vertical length of the corroded area may *exceed L*.



Legend:
a–e are inspection planes selected by inspector.

t_2 = least min. thickness in entire area, exclusive of pits.

SECTION A-A
Profile along plane "c", the plane having the lowest average thickness, t_1.

Procedure:
1. Determine t_2.
2. Calculate $L = 3.7\sqrt{Dt_2}$, but not more than 40".
3. Locate "L" to get minimum t_{avg}, which is t_1.

Figure 2-1—Inspection of Corrosion Areas

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607746 224 ■

D = Tank diameter, in feet.
t_2 = The least thickness, in inches, in an area of corrosion, exclusive of pits.

c. Then, the inspector shall visually or otherwise decide which vertical plane(s) in the area is likely to be the most affected by corrosion. Profile measurements shall be taken along each vertical plane for a distance, L. In the plane(s), determine the lowest average thickness, t_1, averaged over a length of L, using at least five equally spaced measurements over length L.
d. Refer to 2.3.3.1 for minimum permitted values for t_1 and t_2. The additional loads in 2.3.3.4 shall also be considered.

2.3.2.2 Widely scattered pits may be ignored provided that:

a. No pit depth results in the remaining shell thickness being less than one-half the minimum acceptable tank shell thickness exclusive of the corrosion allowance; and
b. The sum of their dimensions along any vertical line does not exceed 2 inches in an 8-inch length (see Figure 2-2).

2.3.3 MINIMUM THICKNESS CALCULATION FOR WELDED TANK SHELL

2.3.3.1 The minimum acceptable shell plate thickness for tanks with diameters equal to or less than 200 feet shall be calculated as follows:

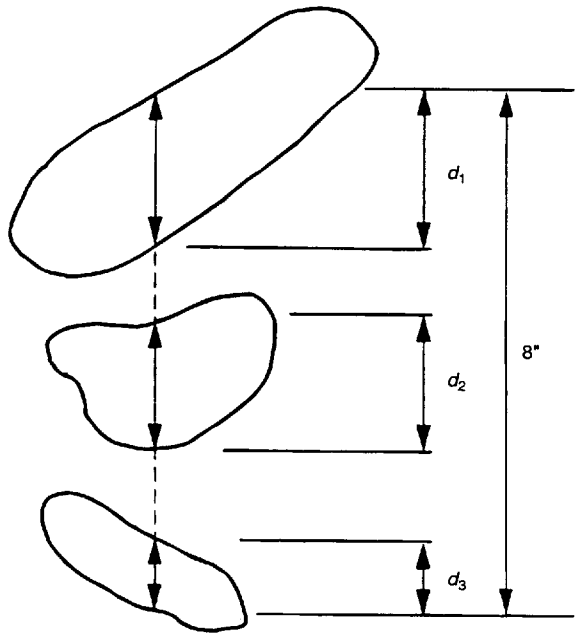


Figure 2-2—Pit Measurement

$$t_{min} = \frac{2.6D(H-1)G}{SE}$$

Where:

t_{min} = The minimum acceptable thickness, in inches, as calculated from the above formula; however, t_{min} shall not be less than 0.1 inch for any tank course.
D = Nominal diameter of tank, in feet.
H = Height, in feet, from the bottom of the length L (see 2.3.2.1) for the most severely corroded area in each shell course to the maximum design liquid level.
G = Highest specific gravity of the contents (including test water if tank will, or may, be tested in the future).
S = Maximum allowable stress in pounds per square inch; use the smaller of $0.80Y$ or $0.429T$ for bottom and second course; use the smaller of $0.88Y$ or $0.472T$ for all other courses. Note: For reconstructed tanks, S shall be in accordance with the current applicable standard.
Y = Specified minimum yield strength of the plate; use 30,000 pounds per square inch if not known.
T = The smaller of the specified minimum tensile strength of the plate or 80,000 pounds per square inch; use 55,000 pounds per square inch if not known.
E = Original joint efficiency for the tank. Use Table 2-1 if original E is unknown. $E = 1.0$ when evaluating the retirement thickness in a corroded plate, when away from welds or joints by at least the greater of one inch or twice the plate thickness.

The criteria for continued operation is:

a. The value t_1 (see 2.3.2.1.c) shall be greater than or equal to t_{min} (see above), subject to verification of all other loadings listed below; and
b. The value t_2 (see 2.3.2.1.b) shall be greater than or equal to 60 percent of t_{min}.

Any corrosion allowance for service until the time of the next tank inspection shall be added to the two required thicknesses (t_{min} and $0.60\ t_{min}$) in a. and b. above.

2.3.3.2 Alternatively, the minimum acceptable shell plate thickness for tanks with diameters equal to or less than 200 feet may be calculated in accordance with the variable design point method in API Standard 650, 3.6.4, substituting "S × E" for "S"; E and S may be defined as in 2.3.3.1.

2.3.3.3 The variable design point method shall be used for tanks greater than 200 feet in diameter, with all variables defined as in 2.3.3.1.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607747 160 ■

Table 2-1—Joint Efficiencies for Welded Joints

Standard	Edition & Year	Type of Joint	Joint Efficiency E	Applicability or Limits
API 650	7th & Later	Butt	1.00	Basic Standard
	(1980–Present)	Butt	0.85	Appendix A–Spot RT
		Butt	0.70	Appendix A–No RT
	1st–6th	Butt	0.85	Basic Standard
	(1961–1978)	Butt	1.00	Appendices D&G
API 12C	14th & 15th (1957–1958)	Butt	0.85	
	3rd–13th (1940–1956)	Lap[a]	0.75	3/8 inch max. *t*
		Butt[c]	0.85	
	1st & 2nd (1936–1939)	Lap[a]	0.70	7/16 inch max. *t*
		Lap[b]	0.50 + k/5	1/4 inch max. *t*
		Butt[c]	0.85	
Unknown		Lap[a]	0.70	7/16 inch max. *t*
		Lap[b]	0.50 + k/5	1/4 inch max. *t*
		Butt[c]	0.70	
		Lap[d]	0.35	

[a]Full double lap welded.
[b]Full fillet weld with at least 25 percent intermittent full fillet opposite side; k = percent of intermittent weld expressed in decimal form.
[c]Single butt-welded joints with a back-up bar were permitted from the years of 1936 to 1940 and 1948 to 1954.
[d]Single lap welded only.

2.3.3.4 The thickness determinations of 2.3.3.1, 2.3.3.2, and 2.3.3.3 consider liquid loading only. All other loads shall also be evaluated according to the original standard of construction; and engineering judgement shall be used to evaluate different conditions or new information. As applicable, the following loadings shall be taken into account:

a. Wind-induced buckling.
b. Seismic loads.
c. Operation at temperatures over 200°F.
d. Vacuum-induced external pressure.
e. External loads caused by piping, tank-mounted equipment, hold down lugs, etc.
f. Wind-induced overturning.
g. Loads due to settlement.

2.3.3.5 As an alternative to the procedures described above, any thinning of the tank shell below minimum required wall thickness due to corrosion or other wastage may be evaluated to determine the adequacy for continued service by employing the "design by analysis methods," defined in Section VIII, Division 2, Appendix 4 of the ASME Code. When using these criteria, the stress value used in the original tank design shall be substituted for the S_m value of Division 2, if the design stress is less than or equal to the lesser of $^2/_3Y$ (Specified Minimum Yield Strength) or $^1/_3T$ (Specified Minimum Tensile Strength). If the original design stress is greater than $^2/_3Y$ or $^1/_3T$, then the lesser of $^2/_3Y$ or $^1/_3T$ shall be substituted for S_m.

2.3.4 MINIMUM THICKNESS CALCULATION FOR RIVETED TANK SHELL

2.3.4.1 The minimum acceptable thickness for riveted tank shells shall be calculated using the formula of 2.3.3.1 except that the following allowable stress criteria and joint efficiencies shall be used:

S = 21,000 pounds per square inch

E = 1.0 for shell plate 6 inches or more away from rivets. See Table 2-2 for joint efficiencies for locations within 6 inches or rivets.

2.3.4.2 The rivet joint efficiencies given in Table 2-2 are conservative minimums for riveted tank construction details and are included to simplify riveted tank evaluations. However, in some cases it may be advantageous to calculate the actual rivet joint efficiencies using computational methods applicable to lap and butt type riveted joints. When this alternative of calculated joint efficiencies is used, the following maximum allowable stresses shall apply:

a. For the maximum tensile stress in net section of plate, use the lesser of $0.80Y$ or $0.426T$; use 21,000 pounds per square inch if T or Y is unknown.
b. For the maximum shear in net section of rivet, use 16,000 pounds per square inch.
c. For the maximum bearing stress on plates or rivets, use 32,000 pounds per square inch for rivets in single shear, and 35,000 pounds per square inch for rivets in double shear.

2.3.4.3 For tanks with riveted joints, consideration shall be given to whether, and to what extent, corrosion affects such joints. If calculations show that excess thickness exists, this excess may be taken as corrosion allowance.

2.3.4.4 Non-liquid loads (see 2.3.3.4) shall also be considered in the analysis of riveted tanks.

2.3.5 DISTORTIONS

2.3.5.1 Shell distortions include out-of-roundness, buckled areas, flat spots, and peaking and banding at welded joints.

Table 2-2—Joint Efficiencies for Riveted Joints

Type of Joint	Number of Rivet Rows	Joint Efficiency E
Lap	1	0.45
Lap	2	0.60
Lap	3	0.70
Lap	4	0.75
Butt[a]	2[b]	0.75
Butt	3[b]	0.85
Butt	4[b]	0.90
Butt	5[b]	0.91
Butt	6[b]	0.92

[a]All butt joints listed have butt straps both inside and outside.
[b]Number of row on each side of joint center line.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607748 0T7 ■

2.3.5.2 Shell distortions can be caused by many conditions such as foundation settlement, over- or under-pressuring, high wind, poor shell fabrication, or repair techniques, and so forth.

2.3.5.3 Shell distortions shall be evaluated on an individual basis to determine if specific conditions are considered acceptable for continuing tank service and/or the extent of corrective action.

2.3.6 FLAWS

Flaws such as cracks or laminations shall be thoroughly examined and evaluated to determine their nature and extent and need for repair. If a repair is needed, a repair procedure shall be developed and implemented. The requirement for repairing scars such as arc strikes, gouges, or tears from temporary attachment welds must be evaluated on a case-by-case basis. Cracks in the shell-to-bottom weld are critical and shall be removed and the weld repaired.

2.3.7 WIND GIRDERS AND SHELL STIFFENERS

The evaluation of an existing tank shell for suitability for service must also consider the details and condition of any wind girders or shell stiffeners. Degradation by corrosion of these structural elements or their attachment welds to the shell may render these elements inadequate for the design conditions.

2.3.8 SHELL WELDS

The condition of the tank shell welds shall be evaluated for suitability for service. Any deterioration of the existing welds that results from corrosion or pitting must be evaluated and appropriate repair procedures established or the tank rerated as necessary. Some typical shell butt-weld flaws and
96 | recommended procedures for repairs are given in 7.6.

2.3.9 SHELL PENETRATIONS

2.3.9.1 The condition and details of existing shell penetrations (nozzles, manways, cleanout openings, etc.) shall be reviewed when assessing the integrity of an existing tank shell. Details such as type and extent of reinforcement, weld spacing, and thickness of components (reinforcing plate, nozzle neck, bolting flange, and cover plate), are important considerations and shall be reviewed for structural adequacy and compliance with the applicable standard. Any noncompliance, or deterioration due to corrosion, must be assessed and repair procedures established where appropriate or the tank rerated, as necessary.

2.3.9.2 Nozzle wall thickness shall be evaluated for pressure and all other loads.

2.4 Tank Bottom Evaluation

2.4.1 GENERAL

Leaks in tank bottoms are not acceptable. Each aspect of corrosion phenomena, and other potential leak or failure mechanism must be examined. Periodic assessment of tank bottom integrity shall be performed in addition to the internal inspections specified in 4.4.2. The assessment period shall be less than or equal to the appropriate internal inspection interval given in 4.4.2. The use of leak detection tests or monitoring systems (such as double bottoms or liners under tank bottoms with leak detection pipes) will satisfy the requirement for periodic assessment between internal inspections.

Excessive foundation settlement of storage tanks can affect the integrity of tank shells and bottoms. Therefore, monitoring the settlement behavior of tanks is a recognized practice to assess the integrity of tank bottoms. Refer to Appendix B for techniques for evaluating tank bottom settlement.

2.4.2 CAUSES OF BOTTOM FAILURE

The following list gives some historical causes of tank bottom leakage or failure that shall be considered in the decision to line, repair, or replace a tank bottom:

a. Internal pitting and pitting rates in the anticipated service.
b. Corrosion of weld joints (weld and heat affected zone).
c. Weld joint cracking history.
d. Stresses placed on the bottom plates by roof support loads and shell settlement.
e. Underside corrosion (normally in the form of pitting).
f. Inadequate drainage resulting in surface water flowing under the tank bottom.
g. The lack of an annular plate ring when required.
h. Uneven settlement that results in high localized stresses in the bottom plates.
i. Roof support columns or other supports welded to the tank bottom where adequate allowance for movement was not made.
j. Rock or gravel foundation pads with inadequately filled-in surface voids.
k. Nonhomogeneous fill under the tank bottom (for example, a lump of clay in a sand foundation pad).
l. Inadequately supported sumps.

2.4.3 CATHODIC PROTECTION OF TANK BOTTOMS

A selection basis for cathodic protection systems for the underside of tank bottoms is covered by API RP 651.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607749 T33 ■

2.4.4 INTERNAL LINING PROTECTION OF TANK BOTTOMS

Applied linings for internal surfaces of tank bottoms are covered by API RP 652.

2.4.5 BOTTOM LEAK DETECTION

If a tank bottom is to be replaced, consideration should be given to installing a leak detection (tell-tale) system that will channel any leak in the bottom to a location where it can be readily observed from the outside of the tank.[1]

2.4.6 BOTTOM PLATE THICKNESS MEASUREMENTS

Various methods for determining tank bottom plate underside corrosion are available. The methods vary by the extent that they can reliably measure general corrosion and pitting. A combination of these methods may be required along with extrapolation techniques and analysis to establish the probable conditions of the entire tank bottom.

2.4.7 MINIMUM THICKNESS FOR TANK BOTTOM PLATE

Quantifying the remaining thickness of tank bottoms based on the results of measurement (see 2.4.6) can be done by either a deterministic method (see 2.4.7.1) or a probabilistic method (see 2.4.7.2) as follows:

2.4.7.1 An acceptable deterministic method for calculating the minimum remaining bottom thicknesses is as follows:

$$MRT_1 = T_o - GC_a - StP_a - UP_m - (StP_r + UP_r + GC_r)O_r$$

$$MRT_2 = T_o - GC_a - StP_m - UP_a - (StP_r + UP_r + GC_r)O_r$$

Where:

MRT_1, MRT_2 = minimum remaining thickness at the end of the in-service period of operation, in inches. MRT_1 represents minimum remaining thickness due to average internal pitting and maximum external pitting. MRT_2 represents minimum remaining thickness due to maximum internal pitting and average external pitting.

T_o = original plate thickness, in inches.

StP_a = average depth of internal pitting prior to repairs (in accordance with 2.4.7.5), in inches, measured from the original thickness. 96

StP_m = maximum depth of internal pitting remaining in the bottom plates after repairs in accordance with 2.4.7.5 are completed, in inches, measured from the original thickness. 96

UP_a = average depth of underside pitting, in inches.

UP_m = maximum depth of underside pitting, in inches.

StP_r = maximum internal pitting rate, in inches per year; $StP_r = 0$ if the tank bottom is internally lined (see API RP 652).

UP_r = maximum underside pitting rate, in inches per year; $UP_r = 0$ if tank bottom is cathodically protected (see API RP 651).

O_r = anticipated in-service period of operation (normally 10 years).

GC_a = average depth of generally corroded area, in inches.

GC_r = maximum rate of general corrosion, in inches per year.

2.4.7.2 For the probabilistic method, a statistical analysis is made of thickness data from measurements (see 2.4.6) projecting remaining thickness, based on sample scanning of the bottom.

2.4.7.3 If the minimum bottom thicknesses, at the end of the in-service period of operation, are calculated to be less than the minimum bottom renewal thicknesses given in Table 4-1, the bottom shall be lined, repaired, replaced or the interval to the next internal inspection shortened.

2.4.7.4 Unless a stress analysis is performed, the minimum bottom plate thickness in the critical zone of the tank bottom defined by paragraph 7.10.1.1 shall be the smaller of $^1/_2$ the original bottom plate thickness (not including the original corrosion allowance) or 50 percent of t_{min} of the lower shell course per paragraph 2.3.3.1 but not less than 0.1 inch. Isolated pitting will not appreciably affect the strength of the plate. 96

[1] For existing tanks, API supports the use of a Release Prevention System (RPS). The term RPS refers to the suite of API standards that are designed to maintain aboveground storage tank integrity and thus protect the environment. These are: the frequency of internal inspections; lining the bottom of the tank interior; fitting the tank with Release Prevention Barriers (RPBs); installing cathodic protections; or some combination of these measures, depending on the operating environment and service of the tank.

If a decision is made to replace an existing bottom, API supports the evaluation of installing an RPB or continued use of an RPS. The evaluation should consider the effectiveness of RPS controls, the product stored, the location of the tank, and environmental sensitivities. An RPB includes steel bottoms, synthetic materials, clay liners, and all other barriers or combination of barriers placed in the bottom of or under an aboveground storage tank, which have the functions of: 1) preventing the escape of contaminated material; and 2) containing or channeling released material for leak detection (covered in detail in the non-mandatory Appendix I of Standard 650).

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

2.4.7.5 The repair of internal pitting, when performed to extend the in-service period of operation, shall be by pit welding, overlay welding, or lap patching, followed by inspection and testing. The extent of weld repairs is limited in the critical zone in accordance with 7.10.1.1.

2.4.7.6 The treatment of bottom pitting by the use of non-welded repairs (for example, coatings, caulking) is not considered a repair in the context of how StP_m affects the remaining service life of a tank bottom. 96

2.4.7.7 The thickness of the projection of the bottom plate beyond the shell as measured at the toe of the outside bottom-to-shell fillet weld shall not be less than 0.1 inch. The projection of the bottom plate beyond the outside toe of the shell-to-bottom weld shell shall be at least $^3/_8$ inch.

2.4.8 MINIMUM THICKNESS FOR ANNULAR PLATE RING

2.4.8.1 Due to strength requirements, the minimum thickness of annular plate ring is usually greater than 0.10 inch. Isolated pitting will not appreciably affect the strength of the plate. Unless a stress analysis is performed, the annular plate thickness shall be in accordance with 2.4.8.2 or 2.4.8.3, as applicable.

2.4.8.2 For tanks in service with a product specific gravity less than 1.0, which require annular plates for other than seismic loading considering, the thickness of the annular plates shall be not less than the thicknesses given in Table 2-3, plus any specified corrosion allowance.

2.4.8.3 For tanks in service with a product specific gravity of 1.0 or greater, which require annular plates for other than seismic loading considerations, the thickness of the annular plates shall be in accordance with Table 3-1 of API Standard 650 plus any specified corrosion allowance.

2.4.8.4 For tanks that utilize thickened annular plates for seismic considerations, a seismic evaluation shall be performed in accordance with the requirements of the applicable standard, using the actual thickness of the existing annular plate.

2.4.8.5 For the thickness and projection of the annular plate beyond the shell refer to 2.4.7.7. 96

2.5 Tank Foundation Evaluation

2.5.1 GENERAL

2.5.1.1 The principal causes of foundation deterioration are settlement, erosion, cracking, and deterioration of concrete

Table 2-3—Annular Bottom Plate Thicknesses (Inches) (Product Specific Gravity < 1.0)

Plate Thickness[a] of First Shell Course (Inches)	Stress[b] in First Shell Course (pounds per square inch) <24,300	<27,000	<29,700	<32,400
$t \leq 0.75$	0.17	0.20	0.23	0.30
$0.75 < t \leq 1.00$	0.17	0.22	0.31	0.38
$1.00 < t \leq 1.25$	0.17	0.26	0.38	0.48
$1.25 < t \leq 1.50$	0.22	0.34	0.47	0.59
$t > 1.50$	0.27	0.40	0.53	0.68

Note: The thicknesses specified in the table are based on the foundation providing a uniform support under the full width of the annular plate. Unless the foundation is properly compacted, particularly at the inside of a concrete ringwall, settlement will produce additional stresses in the annular plate.
[a]Plate thickness refers to the tank shell as constructed.
[b]Stresses are calculated from $[2.34\ D\ (H\text{-}1)]/t$.

crete initiated by: calcining, attack by underground water, attack by frost, and attack by alkalies and acids. To ensure suitability for service, all tank foundations shall be inspected periodically (see 4.3).

2.5.1.2 Some mechanisms of concrete deterioration are briefly described below:

a. Calcining (loss of water of hydration) can occur when concrete has been exposed to sufficiently high temperature for a period of time. During intermediate cooling periods, the concrete can absorb moisture, swell, lose its strength, and crack.

b. Deterioration of concrete exposed to underground water can be caused by chemical attack, by cyclic changes in temperature, and by freezing moisture.

c. Expansion of freezing moisture in porous concrete, or in concrete with minor settlement cracks or temperature cracks, can result in spalling and/or the development of serious structural cracks.

d. Sulfate-type alkalies, and to a lesser extent, chlorides, can act corrosively to destroy the bond of the concrete.

e. Temperature cracks (hairline cracks of uniform width) do not seriously affect the strength of the concrete foundation structure; however, these cracks can be potential access points for moisture or water seepage that could eventually result in corrosion of the reinforcing steel.

2.5.2 FOUNDATION REPAIR OR REPLACEMENT

2.5.2.1 If there is a need for foundation repair or replacement, foundations shall be restored to the tolerance limits of 8.5.6.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607751 691 ■

2.5.2.2 Concrete pads, ringwalls, and piers, showing evidence of spalling, structural cracks, or general deterioration, shall be repaired to prevent water from entering the concrete structure and corroding the reinforcing steel.

2.5.3 ANCHOR BOLTS

Distortion of anchor bolts and excessive cracking of the concrete structures in which they are embedded may be indications of either serious foundation settlement or a tank overpressure uplift condition.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 4—INSPECTION

4.1 General

Periodic in-service inspection of tanks shall be performed as defined herein. The purpose of this inspection is to assure continued tank integrity. 96

4.2 Inspection Frequency Considerations

4.2.1 Several factors must be considered to determine inspection intervals for storage tanks. These include, but are not limited to, the following:

a. The nature of the product stored.
b. The results of visual maintenance checks.
c. Corrosion allowances and corrosion rates.
d. Corrosion prevention systems.
e. Conditions at previous inspections.
f. The methods and materials of construction and repair.
g. The location of tanks, such as those in isolated or high risk areas.
h. The potential risk of air or water pollution.
i. Leak detection systems.
j. Change in operating mode (for example: frequency of fill cycling, frequent grounding of floating roof support legs).
k. Jurisdictional requirements.
l. Changes in service (including changes in water bottoms).

4.2.2 The interval between inspections of a tank (both internal and external) should be determined by its service history unless special reasons indicate that an earlier inspection must be made. A history of the service of a given tank or a tank in similar service (preferably at the same site) should be available so that complete inspections can be scheduled with a frequency commensurate with the corrosion rate of the tank. On-stream, nondestructive methods of inspection shall be considered when establishing inspection frequencies.

4.2.3 Jurisdictional regulations, in some cases, control the frequency and interval of the inspections. These regulations may include vapor loss requirements, seal condition, leakage, proper diking, and repair procedures. Knowledge of such regulations is necessary to insure compliance with scheduling and inspection requirements.

4.3 Inspections from the Outside of the Tank 96

4.3.1 ROUTINE IN-SERVICE INSPECTIONS

4.3.1.1 The external condition of the tank shall be monitored by close visual inspection from the ground on a routine basis. This inspection may be done by owner/operator personnel, and can be done by other than inspectors described in 4.10. Personnel performing this inspection should be knowledgeable of the storage facility operations, the tank, and the characteristics of the product stored.

4.3.1.2 The interval of such inspections shall be consistent with conditions at the particular site, but shall not exceed one month.

4.3.1.3 This routine in-service inspection shall include a visual inspection of the tank's exterior surface checking for: leaks; shell distortions; signs of settlement; corrosion; and condition of the foundation, paint coatings, insulation systems, and appurtenances.

4.3.2 EXTERNAL INSPECTION

4.3.2.1 All tanks shall be given a visual external inspection by an Authorized Inspector. This inspection shall be called the external inspection and must be conducted at least every 5 years or at the quarter corrosion rate life of the shell, whichever is less. Tanks may be in operation during this inspection. 96

4.3.2.2 Insulated tanks need to have insulation removed only to the extent necessary to determine the condition of the exterior wall of the tank or the roof.

4.3.2.3 Tank grounding system components such as shunts or mechanical connections of cables shall be visually checked. Recommended practices dealing with the prevention of hydrocarbon ignition are covered by API Recommended Practice 2003. 96

4.3.3 ULTRASONIC THICKNESS INSPECTION 96

4.3.3.1 External, ultrasonic thickness measurements of the shell can be a means of determining a rate of uniform general corrosion while the tank is in service, and can provide an indication of the integrity of the shell. The extent of such measurements shall be determined by the owner/operator.

4.3.3.2 When used, the ultrasonic thickness measurements shall be made at intervals not to exceed the following:

a. When the corrosion rate is not known, the maximum interval shall be five years.

Note: Corrosion rates may be estimated from tanks in similar service based on thickness measurements taken at an interval not exceeding five years.

b. When the corrosion rate is known, the maximum interval shall be the smaller of $RCA/2N$ years (where RCA is the remaining corrosion allowance in mils and N is the shell corrosion rate in mils per year) of 15 years.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607753 464 ■

4.3.3.3 *Internal inspection of the tank shell*, when the tank is out of service, can be substituted for a program of external ultrasonic thickness measurement if the internal inspection interval is equal to or less than the interval required in 4.3.3.2.b.

4.3.4 CATHODIC PROTECTION SURVEYS

4.3.4.1 Where exterior tank bottom corrosion is controlled by a cathodic protection system, periodic surveys of the system shall be conducted in accordance with API RP 651. The owner/operator shall review the survey results.

4.3.4.2 The owner/operator shall assure competency of personnel performing surveys.

4.4 Internal Inspection

4.4.1 GENERAL

4.4.1.1 Internal inspection is primarily required to:

a. Ensure that the bottom is not severely corroded and leaking.
b. Gather the data necessary for the minimum bottom and shell thickness assessments detailed in Section 2. As applicable, these data shall also take into account external ultrasonic thickness measurements made during in-service inspections (see 4.3.3).
c. Identify and evaluate any tank bottom settlement.

4.4.1.2 All tanks shall be given a formal internal inspection at intervals defined by 4.4.2 below. The Authorized Inspector who is responsible for evaluation of a tank must conduct a visual inspection and overview of the NDE results.

4.4.2 INSPECTION INTERVALS

4.4.2.1 Intervals between internal inspections shall be determined by the corrosion rates measured during previous inspections or anticipated based on experience with tanks in similar service. Normally, bottom corrosion rates will control and the inspection interval will be governed by the measured or anticipated corrosion rates and the calculations for minimum required thickness of tank bottoms (see 2.4.7). The actual inspection interval shall be set to ensure that the bottom plate minimum thicknesses at the next inspection are not less than the values listed in Table 4-1. In no case, however, shall the internal inspection interval exceed 20 years.

4.4.2.2 When corrosion rates are not known and similar service experience is not available to determine the bottom plate minimum thickness at the next inspection, the actual bottom thickness shall be determined by inspection(s) within the next 10 years of tank operation to establish corrosion rates.

4.4.3 ALTERNATIVE INTERNAL INSPECTION INTERVAL

For unique combinations of service, environment, and construction, the owner/operator may establish the internal inspection interval using an alternative procedure. The alternative procedure shall include method for determining bottom plate thickness, consideration of environmental risk, consideration of quality of inspection, and analysis of corrosion measurement. This alternative procedure shall be documented and made part of the permanent record of the tank.

4.5 Alternative to Internal Inspection to Determine Bottom Thickness

In cases where construction, size, or other aspects allow external access to the tank bottom to determine bottom thickness, an external inspection in lieu of an internal inspection is allowed to meet the data requirements of Table 4-1. However, in these cases, consideration of other maintenance items may dictate internal inspection intervals. This alternative approach shall be documented and made part of the permanent record of the tank.

4.6 Preparatory Work for Internal Inspection

Specific work procedures shall be prepared and followed when conducting inspections that will assure personnel safety and health and prevent property damage in the workplace (see 1.4).

4.7 Inspection Checklists

Appendix C provides sample checklists of items for consideration when conducting in-service and out-of-service inspections.

4.8 Records

4.8.1 GENERAL

Inspection records form the basis of a scheduled inspection/maintenance program. (It is recognized that records may not exist for older tanks, and judgements must be based on experience with tanks in similar services.) The owner/

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607754 3T0 ■

Table 4-1—Bottom Plate Minimum Thickness

Minimum Bottom Plate Thickness[a] at Next Inspection (inches)	Tank Bottom/Foundation Design
0.10	Tank bottom/foundation design with no means for detection and containment of a bottom leak.
0.05	Tank bottom/foundation design with means to provide detection and containment of a bottom leak.
0.05	Applied tank bottom reinforced lining, >0.05 inch thick, in accordance with API RP 652.

[a] See 2.4.7

operator shall maintain complete record file consisting of three types of records, namely: construction records, inspection history, and repair/alteration history.

4.8.2 CONSTRUCTION RECORDS

Construction records may include nameplate information, drawings, specifications, construction completion report, and any results of material tests and analyses.

4.8.3 INSPECTION HISTORY

The inspection history includes all measurements taken, the condition of all parts inspected, and a record of all examinations and tests. A complete description of any unusual conditions with recommendations for correction of details which caused the conditions shall also be included. This file will also contain corrosion rate and inspection interval calculations.

4.8.4 REPAIR/ALTERATION HISTORY

The repair/alteration history includes all data accumulated on a tank from the time of its construction with regard to repairs, alterations, replacements, and service changes (recorded with service conditions such as stored product temperature and pressure). These records should include the results of any experiences with coatings and linings.

4.9 Reports

4.9.1 Reports recommending repairs shall include reasons for the repairs, and sketches showing location and extent.

4.9.2 General inspection reports shall include metal thickness measurement, conditions found, repairs, any settlement measurements, and recommendations.

4.10 Inspector Qualifications

4.10.1 Qualified inspectors shall have education and experience equal to at least one of the following:

a. A degree in engineering plus one year of experience in inspection of tanks, pressure vessels, or piping.
b. A two-year certificate in engineering or technology from a technical college, and two years of experience in construction, repair, operation, or inspection, of which one year must be in inspection of tanks, pressure vessels, or piping.
c. The equivalent of a high school education, and three years of experience in construction, repair, operation, or inspection, of which one year must be in inspection of tanks, pressure vessels, or piping.
d. Five years of experience in the inspection of aboveground storage tanks in the petroleum or chemical industries.

4.10.2 An owner/operator of tanks may designate tank inspectors qualified in accordance with 4.10.1. Such inspectors shall have the necessary authority and organizational freedom to perform their duties. Authorized Inspectors shall be certified by an agency as provided in this standard, in accordance with Appendix D. This requirement will become effective eighteen (18) months after the date of issuance of the requirement.

4.10.3 Personnel performing nondestructive examinations shall meet the qualifications identified in 10.1.1.2, but need not be certified in accordance with 4.10.2. The results of their work, however, must be considered in the evaluation of the tank by an inspector who is so certified.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 5—MATERIALS

5.1 General

This section provides general requirements for the selection of materials for the repair, alteration, and reconstruction of existing tanks. Specific requirements for repairs and alterations are covered in Section 7.

5.2 New Materials

All new materials used for repair, alterations, or reconstruction shall conform to current applicable tank standards.

5.3 Original Materials for Reconstructed Tanks

5.3.1 SHELL AND BOTTOM PLATES WELDED TO THE SHELL

5.3.1.1 All shell plate materials and bottom plates welded to the shell shall be identified. Materials identified by original contract drawings, API name plates, or other suitable documentation do not require further identification. Material not identified shall be tested and identified by the requirements as outlined in 5.3.1.2. After identification, determination shall be made as to suitability of the material for intended service.

5.3.1.2 Each individual plate for which adequate identification does not exist shall be subjected to chemical analysis and mechanical tests as required in ASTM A6 and A370 including Charpy V-Notch. Impact values shall satisfy the requirements of API Standard 650. When the direction of rolling is not definitely known, two tension specimens shall be taken at right angles to each other from a corner of each plate, and one of those test specimen must meet the specification requirements.

5.3.1.3 For known materials, all shell plates and bottom plates welded to the shell shall meet, as a minimum, the chemistry and mechanical properties of material specified for the application with regard to thickness and design metal temperature given in API Standard 650.

5.3.2 STRUCTURAL

Existing rolled structural shapes that are to be reused shall meet the requirement of ASTM A7 as a minimum. New structural material shall meet the requirements of ASTM A36 as a minimum.

Note: ASTM A7 was a steel specification that was discontinued in the Fourth Edition of API Standard 650, 1970.

5.3.3 FLANGES and FASTENERS

5.3.3.1 Flange material shall meet the minimum requirements of the material specifications in the original standard of construction.

5.3.3.2 Fasteners shall meet the material specifications of the current applicable standard.

5.3.4 ROOF, BOTTOM, AND PLATE WINDGIRDERS

If existing plates are to be used to reconstruct the tank, they shall be checked for excessive corrosion and pitting. (See Sections 2 and 4.)

5.4 Welding Consumables

Welding consumables shall conform to the American Welding Society (AWS) classification that is applicable to the intended use.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 6—DESIGN CONSIDERATIONS FOR RECONSTRUCTED TANKS

6.1 General

Any specific design considerations other than normal product loading shall be specified by the owner/operator. Refer to 2.4.5 for release prevention systems and release prevention barrier definition.

6.2 New Weld Joints

6.2.1 Weld joint details shall meet the welding requirements of the applicable standard.

6.2.2 All new shell joints shall be butt-welded joints with complete penetration and complete fusion.

6.3 Existing Weld Joints

Existing weld joints shall meet the requirements of the original standard of construction for the tank.

6.4 Shell Design

6.4.1 Thickness to be used for each shell course when checking tank design shall be based on measurements taken within 180 days prior to relocation. (See 2.3.2 for measuring procedure, number, and locations of measured thicknesses.)

6.4.2 The maximum design liquid level for product shall be determined by calculating the maximum design liquid level for each shell course based on the specific gravity of the product, the actual thickness measured for each shell course, the allowable stress for the material in each course, and the design method to be used. The allowable stress for the material shall be determined using Table 3-2 of API Standard 650. For material not listed in Table 3-2, an allowable stress value of the lesser of $^{2}/_{3}$ yield strength or $^{3}/_{8}$ tensile strength shall be used.

6.4.3 The maximum liquid level for hydrostatic test shall be determined by using the actual thickness measured for each shell course, the allowable stress for the material in each course, and the design method to be used. The allowable stress for the material shall be determined using Table 3-2 of API Standard 650. For material not listed in Table 3-2, an allowable stress value of the lesser of $^{3}/_{4}$ yield strength or $^{2}/_{5}$ tensile strength shall be used.

6.4.4 If a corrosion allowance is required for the reconstructed tank, the required corrosion allowance shall be deducted from the actual thickness before calculating the maximum liquid level. If the actual thickness is greater than that necessary to allow the liquid level required, the extra thickness can be considered as corrosion allowance.

6.4.5 The joint efficiency and allowable stress levels used for the design liquid level calculations shall be consistent with the design method used and with the degree and type of inspection made on welded joints. The joint efficiency and allowable stress levels for existing welded joints that are not to be removed and replaced shall be based on the original degree and type of inspection.

6.5 Shell Penetrations

6.5.1 Replacement and new penetrations shall be designed, detailed, welded, and examined to meet the requirements of the current applicable standard.

6.5.2 Existing penetrations shall be evaluated for compliance with the original standard of construction.

6.6 Windgirders and Shell Stability

6.6.1 Top and intermediate windgirders for open top tanks shall meet the requirements of the applicable standard.

6.6.2 Tanks to be reconstructed shall be checked for wind-induced buckling in accordance with the procedures of the applicable standard, using the wind requirements for the location where the tank will be reconstructed.

6.7 Roofs

6.7.1 Roof designs shall meet the requirements of the applicable standard.

6.7.2 If the new site location requires a larger design live load than the original site, the adequacy of the existing roof shall be evaluated.

6.8 Seismic Design

Tanks that will be reconstructed in Seismic Zone 2 or greater (see Table E-1 in Appendix E of API Standard 650) shall be checked for seismic stability based on the rules of the applicable standard using the dimensions and thicknesses of the reconstructed tank. Reconstructed tanks shall be built to meet the stability requirements of the applicable standard. Thickened bottom plates under the bottom shell course or anchoring of the tank may be required even if not used on the original tank.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 7—TANK REPAIR AND ALTERATION

7.1 General

7.1.1 The basis for repairs and alterations shall be an API Standard 650 equivalence.

7.1.2 Hydrostatic testing requirements, nondestructive examination requirements, acceptance criteria for the welds, and repairs to shell plate and existing welds are specified in Section 10.

7.2 Removal and Replacement of Shell Plate Material

7.2.1 MINIMUM THICKNESS OF REPLACEMENT SHELL PLATE

The minimum thickness of the replacement shell plate material shall be calculated in accordance with the applicable standard. The thickness of the replacement shell plate shall not be less than the greatest nominal thickness of any plate in the same course adjoining the replacement plate except where the adjoining plate is a thickened insert plate. Any changes from the original design conditions, such as specific gravity, design pressure, liquid level, and shell height, shall be considered.

7.2.2 MINIMUM DIMENSIONS OF REPLACEMENT SHELL PLATE

7.2.2.1 The minimum dimension for a replacement shell plate is 12 inches or 12 times the thickness of the replacement plate, whichever is greater. The replacement plate may be circular, oblong, square with rounded corners, or rectangular with rounded corners except when an entire shell plate is replaced. Refer to Figure 7-1 for typical details of acceptable replacement shell plates.

7.2.2.2 Where one or more entire shell plates or full height segments of shell plates are to be removed and replaced, the minimum spacing requirements specified in Figure 7-1 for vertical weld joints shall be maintained. It is acceptable to remove and replace entire shell plates or full height segments of shell plates by cutting and rewelding along the existing horizontal weld joints. Prior to welding the new vertical joints, the existing horizontal welds shall be cut for a minimum distance of 12 inches beyond the new vertical joints. The vertical joints shall be welded prior to welding the horizontal joints.

7.2.3 WELD JOINT DESIGN

7.2.3.1 Shell replacement plates shall be welded with butt joints with complete penetration and complete fusion, except as permitted for lapped patch shell repairs. 96

7.2.3.2 Weld joint design for replacement shell plates shall be in accordance with 3.1.5.1 through 3.2.5.3 of API Standard 650. Joints in lap-welded shell tanks may be repaired according to the original standard of construction. Lap welded joint design for lapped patch shell repairs shall meet the requirements of 7.3. Details of welding shall be in accordance with 5.2 of that standard, and Section 9 of this standard. 96

7.2.3.3 For existing shell plates over $^1/_2$ inch thick, the outer edge of the butt weld attaching the replacement shell plate shall be at least the greater of 8 times the weld thickness or 10 inches from the outer edge of any existing butt-welded shell joints. For existing shell plates $^1/_2$ inch thick and less, the spacing may be reduced to 6 inches from the outer edge of vertical joints or 3 inches from the outer edge of horizontal joints. See Figure 7-1 for minimum dimensions.

For existing shell plates over $^1/_2$ inch thick, the outer edge of the butt weld attaching the replacement shell plate shall be at least the greater of 8 times the weld size or 10 inches from the edge (toe) of the fillet weld attaching the bottom shell course to the bottom except when the replacement shell plate extends to and intersects the bottom-to-shell joint at approximately 90 degrees. For existing shell plates $^1/_2$ inch thick and less, this spacing may be reduced to 6 inches. The edge of any vertical weld joint attaching a replacement plate shall be at least 12 inches from the edge of a weld joint in the annular plate ring or sketch plate. Figure 7-1 has minimum dimensions.

7.2.3.4 To reduce the potential for distortion of an existing tank due to welding a replacement plate into an existing tank shell, fit-up, heat input, and welding sequence must be considered.

7.3 Shell Repairs Using Lap Welded Patch Plates

7.3.1 Lapped patch shell repairs are an acceptable form of repair for butt-welded, lap-welded, and riveted tank shells, under the conditions outlined in 7.3.2, 7.3.3, and 7.3.4; only when specified by the owner. In addition, the repair details shall comply with the requirements of 7.3.1.1 through 7.3.1.10. These repairs are permanent repairs subject to an ongoing inspection and maintenance program. These requirements may be used to evaluate existing lapped patch shell repairs; however, the plate thickness limits need not apply. 96

7.3.1.1 All repair material shall comply with the requirements of the applicable standard of construction and API Standard 653.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607758 T46 ■

7.3.1.2 Lapped patch shell repairs shall not be used on any shell course thickness (original construction) that exceeds $^1/_2$ inch, nor to replace doorsheets or shell plates.

7.3.1.3 Except as permitted in 7.3.3.2 and 7.3.4.3, the repair plate material shall be the smaller of $^1/_2$ inch or the thickness of the shell plate adjacent to the repairs, but not less than $^3/_{16}$ inch.

7.3.1.4 The shape of the repair plate may be circular, oblong, square, or rectangular. All corners, except at the shell-to-bottom joint, shall be rounded to a minimum radius of 2 inches. The nozzle reinforcing plate shapes of API Standard 650 are also acceptable.

7.3.1.5 The repair plate may cross any butt-welded vertical or horizontal shell seams that have been ground flush, but must overlap a minimum of 6 inches beyond the shell seam. The weld spacing requirements of Figure 7-1 shall be used as a basis for locating repair plates relative to butt-welded, fillet-welded, and riveted seams and other repair plates.

7.3.1.6 Repair plates may extend to and intersect with the external shell-to-bottom joint if the vertical sides intersect the tank bottom at a 90-degree angle and the shell-to-bottom weld is in conformance with Figure 7-2. Repair plates positioned on the shell interior shall be located such that the toe-to-toe weld clearances are a minimum of 6 inches to the shell-to-bottom weld.

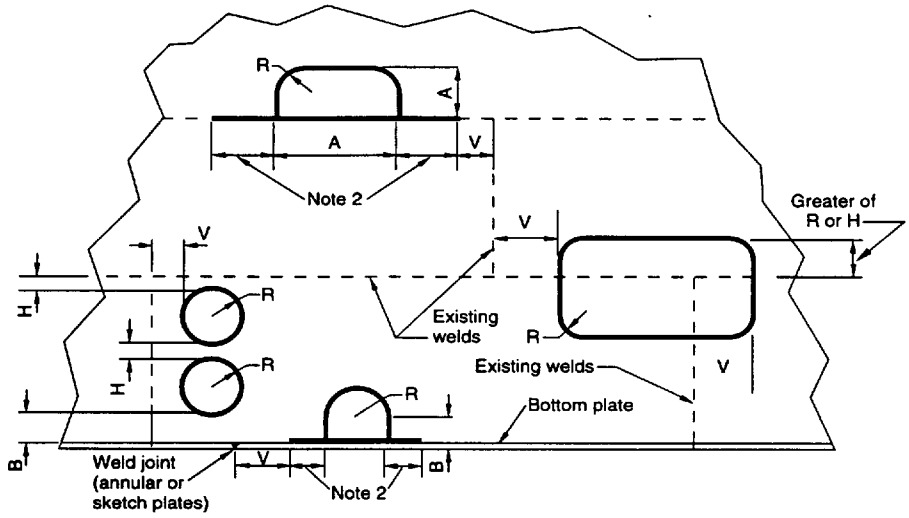
7.3.1.7 The maximum vertical and horizontal dimension of the repair plate is 48 inches and 72 inches, respectively. The minimum repair plate dimension is 6 inches. The repair plate shall be formed to the shell radius.

7.3.1.8 Shell openings and their reinforcements shall not be positioned within a lapped patch shell repair.

7.3.1.9 Prior to application of a lapped patch shell repair, the areas to be welded shall be ultrasonically inspected for plate defects and remaining thickness.

7.3.1.10 Repair plates shall not be lapped onto lap-welded shell seams, riveted shell seams, other lapped patch repair plates, distorted areas, or unrepaired cracks or defects.

7.3.2 Lapped patch repair plates may be used for the closure of holes caused by the removal of existing shell openings or the removal of severely corroded or eroded areas. In addition, the following requirements should be satisfied.



Dimension	Minimum weld spacing between edges (toes) of welds for thickness of replacement shell plate, *t*, (inches)	
	$t \leq 0.5$ inch	$t > 0.5$ inch
R	6"	Greater of 6" or 6*t*
B	6"	Greater of 10" or 8*t*
H	3"	Greater of 10" or 8*t*
V	6"	Greater of 10" or 8*t*
A	12"	Greater of 12" or 12*t*

Notes:
1. All weld intersections shall be at approximately 90°.
2. Prior to welding new vertical joints, cut existing horizontal weld for a minimum of 12" beyond the new vertical joints. Weld the horizontal joint last.

Figure 7-1—Acceptable Details for Replacement of Shell Plate Material

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

7.3.2.1 The welding shall be continuous on the outer perimeter of the repair plate and the inner perimeter of the hole in the shell plate. The minimum hole diameter is 2 inches. Shell openings due to plate removal shall have a minimum corner radius of 2 inches.

7.3.2.2 Nozzle necks and reinforcing plates shall be entirely removed prior to installation of a repair plate.

7.3.2.3 The repair plate thickness selection shall be based upon a design that conforms to the applicable standard of construction and API Standard 653, using a joint efficiency not exceeding 0.70. The welds of the repair plate shall be full fillet welds. The overlap of repair plate onto the shell shall not exceed 8 times the shell thickness. The minimum overlap is 1 inch.

7.3.2.4 The repair plate thickness shall not exceed the nominal thickness of the shell plate adjacent to the repair.

7.3.3 Lapped patch repair plates may be used to reinforce areas of severely deteriorated shell plates that are not able to resist the service loads to which the tank is to be subjected. Lapped patch repair plates may also be used for shell plates that are below the retirement thickness, providing the following additional requirements are satisfied.

96 **7.3.3.1** The selection of the repair plate thickness shall be based upon a design that conforms to the applicable standard of construction and API Standard 653, using a joint efficiency not exceeding 0.35. The perimeter weld shall be a full fillet weld.

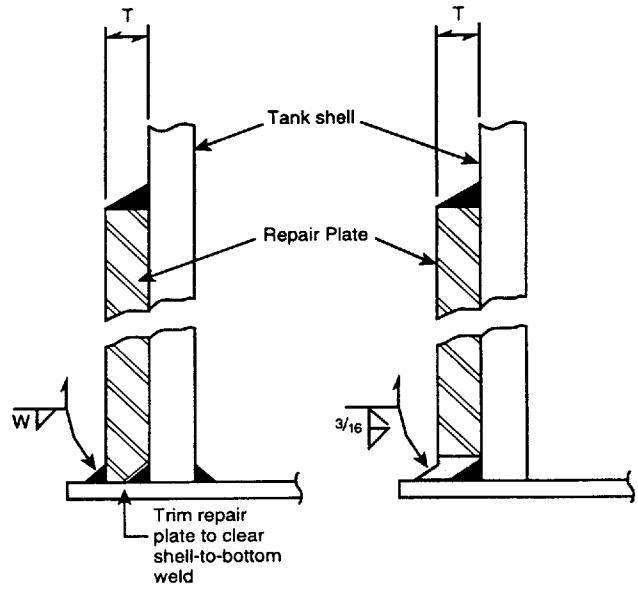
7.3.3.2 The repair plate thickness shall not exceed the shell plate thickness at the perimeter of the repair plate by more than one-third, but no more than $^1/_8$ inch. The repair plate thickness shall not exceed $^1/_2$ inch.

7.3.3.3 The remaining strength of the deteriorated areas under the repair plate shall not be considered as effective in carrying the calculated service or hydrotest loads.

7.3.4 Lapped patch repair plates may be used to repair small shell leaks, or minimize the potential from leaks from severely isolated or widely scattered pitting if the following requirements are satisfied.

7.3.4.1 The existing shell thickness, excluding the holes and pitting, meets the minimum acceptable shell thickness as determined by 2.3.2 and 2.3.3.

7.3.4.2 The repair plate is designed to withstand the hydrostatic pressure load between the repair plate and the shell assuming a hole exists in the shell using a joint efficiency of 0.35.



W = lesser of repair plate thickness or bottom plate thickness

Figure 7-2—Lapped Patch Repair Plates at the External Shell-to-Bottom Joint

7.3.4.3 The repair plate thickness shall not exceed the shell plate thickness at the perimeter of the repair plate by more than one-third, but no more than $^1/_8$ inch. The repair plate thickness shall be no thinner than $^3/_{16}$ inch nor thicker than $^1/_2$ inch. A full fillet perimeter weld is required.

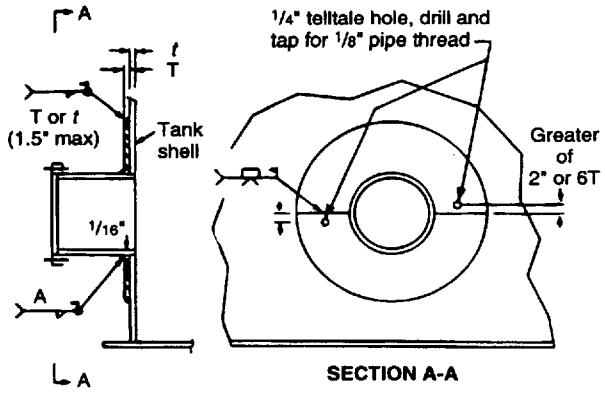
7.3.4.4 This repair method shall not be used if exposure of the fillet welds to the product will produce crevice corrosion or if a corrosion cell between the shell plate and repair plate is likely to occur.

96 **7.3.4.5** This repair method shall not be used to repair shell leaks if the presence of product between the shell plate and repair plate will prevent gas freeing from the tank to perform hot work.

7.3.4.6 The existing shell plate under the repair plate shall be evaluated at each future inspection to ensure it satisfies the requirements of 7.3.4.1. If the existing shell plate thickness does not satisfy 7.3.4.1 or the repair plate does not satisfy 7.3.3, the area is to be repaired in accordance with 7.2 or 7.3.2.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607760 6T4 ■



t and T (maximum)	A
9/16"	1/4"
3/4"	5/16"
15/16"	3/8"
1 1/8"	7/16"
1 5/16"	1/2"
1 9/16"	9/16"
1 3/4"	5/8"

Note: All details, dimensions, and weld spacing shall be in accordance with the requirements of API Standard 650.

Figure 7-2a—Typical Details for Addition of Reinforcing Plate to Existing Shell Penetration

7.4 Repair of Defects in Shell Plate Material

The need for repairing flaws such as cracks, gouges or tears (such as those often remaining after the removal of temporary attachments) discovered during an inspection of the tank shell shall be determined on an individual case basis. In areas of the shell where the plate thickness exceeds that required by design conditions, it is permissible to grind scars to a smooth contour so long as the remaining thickness is adequate for the design conditions. Where grinding to a smoothly contoured surface will result in unacceptable metal thickness, weld metal must be added to repair the scar using a qualified weld procedure.

7.5 Alteration of Tank Shells to Change Shell Height

Tank shells may be altered by adding new plate material to increase the height of the tank shell. The modified shell height shall be in accordance with the requirements of the applicable standard and shall take into consideration all anticipated loadings such as wind and seismic.

7.6 Repair of Defective Welds

Types of weld flaws and nonconformances that need repair are described in 7.6.1 through 7.6.4. 96

7.6.1 Cracks, lack of fusion, and rejectable slag and porosity that need repair shall be removed completely by gouging and/or grinding and the resulting cavity properly prepared for welding.

7.6.2 Generally, it is not necessary to remove existing weld reinforcement in excess of that allowed by API Standard 650 when discovered on an existing tank with a satisfactory service history. However, if operating conditions are such that the excessive weld reinforcement may be deleterious (such as for a floating roof with flexible seals), consideration shall be given to repairing the welds by grinding.

7.6.3 Existing weld undercut deemed unacceptable based upon suitability for service considerations shall be repaired by additional weld metal, or grinding, as appropriate.

7.6.4 Welded joints that have experienced loss of metal due to corrosion may be repaired by welding.

7.6.5 Arc strikes discovered in or adjacent to welded joints shall be repaired by grinding and/or welding. Arc strikes repaired by welding shall be ground flush with the plate.

7.7 Repair of Shell Penetrations

7.7.1 Repairs to existing shell penetrations shall be in compliance with API Standard 650.

7.7.2 Reinforcing plates may be added to existing unreinforced nozzles when deemed appropriate. The reinforcing plate shall meet all dimensional and weld spacing requirements of API Standard 650. See Figures 7-2a and 7-3 for acceptable details. 96

7.7.3 As a alternative, the reinforcing plates may be added to the inside of the tank provided that sufficient nozzle projection exists.

7.8 Addition or Replacement of Shell Penetrations

7.8.1 New shell penetrations (addition or replacement) shall be in accordance with all requirements of API Standard 650 or 7.8.2 of this standard. 96

7.8.2 Penetrations larger than 2 inches NPS shall be installed with the use of an insert plate if the shell plate thickness is greater than 1/2 inch and the shell plate material does

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

not meet the current design metal temperature criteria. In addition, the following requirement shall be met:

a. The minimum diameter of the insert plate shall be at least twice the diameter of the penetration or the diameter plus 12 inches, whichever is greater.
b. When reinforcing plates are used, the minimum diameter of the insert plate shall equal the diameter of the reinforcing plate plus 12 inches.
c. If an integral reinforcement design is used, the thickness of the insert plate–at its periphery and for a distance of $2t$ from the edge of the weld (where t is the shell plate thickness)–shall be the same thickness as the shell, or the insert plate shall have a 1:4 taper to match the shell plate thickness.
d. Spacing of welds shall be in accordance with Figure 7-1.
e. The joint between new material of the insert plate and old material of the existing shell plate shall be butt-welded.

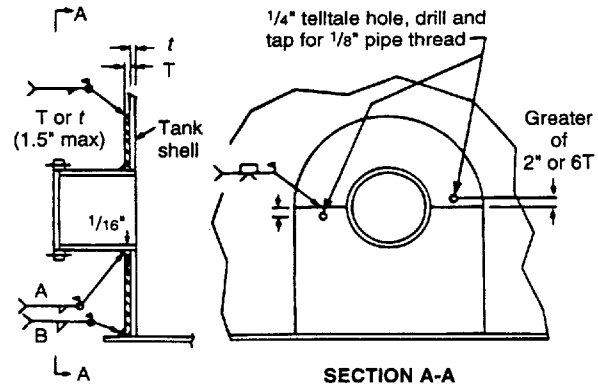
7.9 Alteration of Existing Shell Penetrations

7.9.1 Existing shell penetrations may be modified if the altered details comply with the requirements of API Standard 650, including the requirements for minimum reinforcing area and the requirements for spacing of welds around connections.

7.9.2 When installing a new tank bottom above the existing bottom, it may be necessary to alter existing shell penetrations in the bottom course of a tank shell. If the new bottom is "slotted" through the tank shell several inches above the existing bottom, the spacing between existing welds around penetrations and the new bottom-to-shell weld may not comply with API Standard 650 requirements. Options for altering the penetrations and/or reinforcing plates are given in 7.9.2.1 through 7.9.2.3.

7.9.2.1 The existing reinforcing plate may be trimmed to increase the spacing between the welds provided that the modified detail complies with the requirements of API Standard 650. Care must be exercised during the trimming operation to avoid damaging the shell material beneath the reinforcing plate. The existing weld attaching the portion of the reinforcing plate to be removed shall be completely removed by gouging and grinding.

7.9.2.2 The existing reinforcing plate may be removed and a new reinforcing plate added. Care must be exercised when removing the existing reinforcing plate to avoid damaging the shell plate beneath the reinforcing plate. The welds around the perimeter of the reinforcing plate and between the reinforcing plate and neck of the penetration shall be completely removed by gouging and grinding. The new reinforcing plate shall be in accordance with Figure 7-2a. If required to maintain weld spacing, a "tombstone" shaped reinforcing plate may be used. (See Figure 7-3).



SECTION A-A

t and T (maximum)	A	B
9/16"	1/4"	1/4"
3/4"	5/16"	1/4"
15/16"	3/8"	5/16"
1 1/8"	7/16"	5/16"
1 5/16"	1/2"	3/8"
1 9/16"	9/16"	3/8"
1 3/4"	5/8"	3/8"

Note: All details, dimensions, and weld spacing shall be in accordance with the requirements of API Standard 650.

Figure 7-3—Typical Details for Addition of "Tombstone" Shape Reinforcing Plate to Existing Shell Penetration

7.9.2.3 The existing penetration may be moved by cutting the section of the shell containing the fitting and reinforcing plate, and raising the entire assembly to the correct elevation (see Figure 7-4).

7.9.3 Any components of the penetration (neck, flange, and reinforcing plate) that are in serviceable condition after removal may be reused.

7.10 Repair of Tank Bottoms

7.10.1 REPAIRING A PORTION OF TANK BOTTOMS

7.10.1.1 The critical zone for repairs to tank bottoms is within the annular plate ring, within 12 inches of the shell, or within 12 inches of the inside edge of the annular plate ring. No welding, welded-on patch plates or weld overlays are permitted within the critical zone except for welding of: widely scattered pits (see 2.3.2.2), cracks in the bottom plates, the shell-to-bottom weld, or where the bottom or annular plate is being replaced.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607762 477 ■

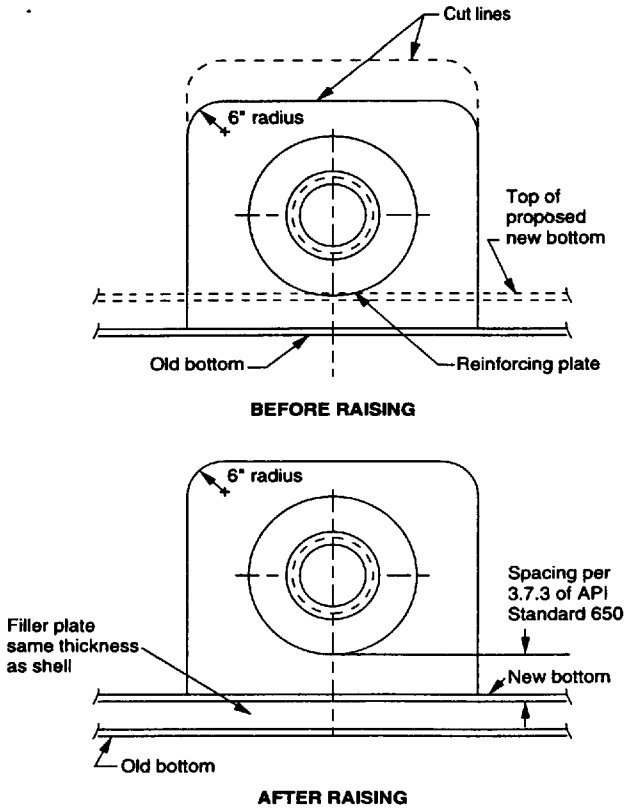


Figure 7-4—Method for Raising Shell Nozzles

7.10.1.2 If more extensive repairs are required within the critical zone than those listed in 7.10.1.1, the bottom plate under the bottom shell course shall be cut out and a new plate shall be installed. Weld spacing requirements shall be in accordance with 3.1.5.4 and 3.1.5.5 of API Standard 650. 96

7.10.1.3 The repair of sumps located within the critical zone shall be in accordance with 7.10.1.2.

7.10.2 REPLACEMENT OF ENTIRE TANK BOTTOM

7.10.2.1 Requirements governing the installation of a replacement bottom over an existing bottom are given in 7.10.2.1.1 through 7.10.2.1.6. 96

7.10.2.1.1 A suitable non-corrosive material cushion such as sand, gravel, or concrete shall be used between the old bottom and the new bottom. 96

7.10.2.1.2 The shell shall be "slotted" with a uniform cut made parallel to the tank bottom. The cut edges in the slot shall be ground to remove all slag and burrs from the cutting operation. The new bottom plate shall be extended outside the shell as required by API Standard 650. All rules for weld spacing shall be followed.

7.10.2.1.3 Voids in the foundation below the old bottom shall be filled with sand, crushed limestone, grout, or concrete.

7.10.2.1.4 Existing shell penetrations shall be raised if the elevation of the new bottom cuts through the reinforcing plate or if the weld space requirements given in 3.7.3 of API Standard 650 are not met.

7.10.2.1.5 For floating roof tanks, the new bottom profile must keep the roof level when it is resting on its support legs in the down position. The support legs can either be jacked up to maintain the original height or be shortened by the same amount as the thickness of the cushion and new bottom plate.

7.10.2.1.6 New bearing plates for floating roof support legs and for fixed roof support columns shall be installed.

7.10.2.2 When removing an existing tank bottom, the tank shell shall be separated from tank bottom either by:

a. Cutting the shell parallel to the tank bottom a minimum of $^1/_2$ inch above the bottom to shell weld (cut line B-B as shown in Figure 8-1), or
b. Removing the entire shell to bottom attachment weld, including any penetration and heat affected zone by suitable methods such as arc gouging and/or grinding.

If b. is elected, all arc gouged areas of the tank shell shall be magnetic particle examined, and defective areas repaired and re-examined.

7.10.2.3 Installation of a new tank bottom, after removal of the existing tank bottom, shall meet all requirements of API 650 including shell weld spacing.

7.10.2.4 When planning a bottom replacement, consideration shall be given to removing the old tank bottom or providing other means of preventing potential galvanic corrosion (refer to API RP 651). Also see 2.4.5 regarding bottom leak detection.

7.11 Repair of Fixed Roofs

7.11.1 SUPPORTED CONE ROOFS

7.11.1.1 The minimum thickness of new roof plates shall be $^3/_{16}$ inch plus any corrosion allowance as specified in the repair specifications. In the event roof live loads in excess of 25 pounds per square foot are specified (such as insulation, operating vacuum, high snow loads), the plate thickness shall be based on analysis using the allowable stresses in conformance with 3.10.3 of API Standard 650 (see 7.11.2.2). 96

7.11.1.2 The roof supports (rafters, girders, columns, and bases) shall be repaired or altered such that under design conditions the resulting stresses do not exceed the stress levels given in 3.10.3 of API Standard 650.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607763 303 ■

7.11.2 SELF-SUPPORTING ROOFS

7.11.2.1 The nominal thickness of new roof plate shall be $^{3}/_{16}$ inch or the required plate thickness given in 3.10.5 or 3.10.6 of API Standard 650, plus any specified corrosion allowance, whichever is greater.

7.11.2.2 The details of the roof-to-shell junction shall meet the requirements of 3.10.5 or 3.10.6 or Appendix F of API Standard 650, as applicable, for the intended service. Frangible joint design, or the need for emergency venting devices in accordance with API Standard 2000, shall meet the requirements of 3.10.2.5 of API Standard 650.

7.12 Floating Roofs

7.12.1 EXTERNAL FLOATING ROOFS

Any method of repair is acceptable that will restore the roof to a condition enabling it to perform as required.

7.12.2 INTERNAL FLOATING ROOFS

Repairs to internal floating roofs shall be made in accordance with the original construction drawings, if available. If the original construction drawings are not available, the roof repairs shall be in compliance with the requirements of Appendix H of API Standard 650.

7.12.3 REPAIR OF LEAKS IN PONTOONS

All leaks in pontoons or compartments of double deck floating roofs shall be repaired by rewelding the leaking joints and/or use of patch plates.

7.13 Repair or Replacement of Floating Roof Perimeter Seals

7.13.1 PRIMARY SEALS

Rim-mounted primary shoe seals and toroidal seal systems can be removed, repaired, or replaced. To minimize evaporation losses and reduce potential hazard to the workers, no more than one-fourth of the roof seal system should be out of an in-service tank at one time. Temporary spacers to keep the roof centered shall be used during the repairs. Primary seals systems mounted partly or fully below the bolting bar or top of the rim usually cannot be reached to allow removal in service. In this case, in-service repairs are limited to replacement of the primary seal fabric.

7.13.2 SECONDARY SEALS

Rim-mounted and shoe-mounted secondary seals may be readily installed, repaired, or replaced while the tank is in service.

7.13.3 SEAL-TO-SHELL GAP

Repair and other corrective actions to maintain seal-to-shell gap requirements, includes:

a. Adjusting the hanger system on primary shoe seals, and adding foam filler in toroidal seals.
b. Increasing the length of rim mounted secondary seals in the problem area.
c. Replacing all or part of the primary seal system along with possible installation of a rim extension for a secondary seal. This step shall be taken only after checking the annular space variation at several levels from low pump out to high liquid level.

7.13.4 MECHANICAL DAMAGE

Damaged parts shall be repaired or replaced. Before taking this action, the cause of the damage shall be identified and corrected. Buckled parts shall be replaced, not straightened. Torn seal fabric shall be replaced.

7.13.5 DETERIORATION OF SEAL MATERIAL

Material deterioration results from wear and corrosion on metallic elements, and chemical and weather deterioration of seal fabric. The service life and inspection information shall be used to determine whether a change of material is warranted.

7.13.6 INSTALLATION OF PRIMARY AND SECONDARY SEALS

7.13.6.1 The replacement or addition of primary and secondary seals shall be in accordance with the recommendations of the seal manufacturer. In addition, the final installation shall comply with all applicable jurisdictions.

7.13.6.2 If the roof rim thickness is less than 0.10 inch thick, it shall be replaced. The new roof rim shall be $^{3}/_{16}$ inch thickness, minimum.

7.14 Hot Taps

7.14.1 GENERAL

7.14.1.1 The requirements given herein cover the installation of radial hot tap connections on existing in-service tanks constructed with shell material that does not require postweld heat treatment.

a. For tank shell plates of recognized toughness, shell plates of unknown toughness having a maximum thickness less than or equal to $^{1}/_{2}$ inch, or shell plates of unknown toughness having a minimum shell design metal temperature at or above the curve given in Figure 7-5, the connection size and shell thickness limitations are shown in Table 7-1. 96

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607764 24T ■

Table 7-1—Hot Tap Connection Sizes and Shell Plate Thicknesses

Connection Size, NPS (inches)	Minimum Shell Plate Thickness (inches)
≤8	1/4
≤14	3/8
≤18	1/2

Recognized toughness is defined as meeting or exceeding the toughness requirements of API 650, Sixth Edition, Appendix D or G, API 650, Seventh Edition, or later, or other industry accepted toughness requirements.

b. For tank shell plates of unknown toughness with thickness greater than 1/2 inch and having a minimum shell design metal temperature below the curve given in Figure 7-5 the following limitations apply.

1. Nozzles shall be limited to a maximum diameter of 4 inches NPS.
2. The shell plate temperature shall be at or above the minimum shell design metal temperature for the entire hot tapping operation.
3. All nozzles shall be reinforced. The reinforcement shall be calculated per API 650, 3.7.2. The minimum thickness of the reinforcing plate shall be equal to the shell plate thickness, and the minimum reinforcing plate diameter shall not be less than the diameter of the shell cutout plus 2 inches. 96
4. The maximum height of tank liquid above the hot tap location during the hot tapping operation shall be such that the hydrostatic tank shell stress is less than 7,000 pounds per square inch at the elevation of the hot tap.

7.14.1.2 The minimum height of tank liquid above the hot tap location shall be at least 3 feet during the hot tapping operation.

7.14.1.3 Welding shall be done with low hydrogen electrodes.

7.14.1.4 Hot taps are not permitted on the roof of a tank or within the gas/vapor space of the tank.

7.14.1.5 Hot taps shall not be installed on laminated or severely pitted shell plate.

7.14.1.6 Hot taps are not permitted on tanks where the heat of welding may cause environmental cracking (such as caustic cracking or stress corrosion cracking).

7.14.2 HOT TAP PROCEDURES

A hot tap procedure specific to carrying out the work shall be developed and documented. The procedure shall include the practices given in API Publication 2201.

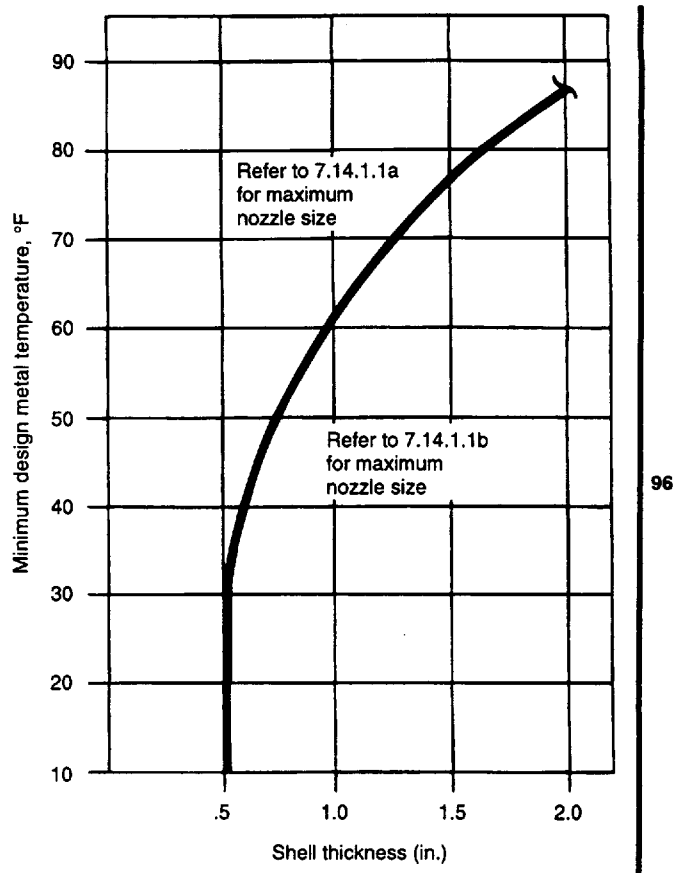


Figure 7-5—Curve for Hot Tapping Tanks With Steels of Unknown Toughness

7.14.3 PREPARATORY WORK

7.14.3.1 Minimum spacing in any direction (toe-to-toe of welds) between the hot tap and adjacent nozzles shall be equivalent to the square root of RT where R is the tank shell radius, in inches, and T is the shell plate thickness, in inches.

7.14.3.2 Shell plate thickness measurements shall be taken at a minimum of four places along the circumference of the proposed nozzle location.

7.14.4 MATERIAL LIMITATIONS

Hot tap only steels of known acceptable toughness or steels of unknown toughness that have a minimum shell design metal temperature at or above the exemption curve shown in Figure 7-5, unless the additional requirements of 7.14.1.1b are met. 96

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

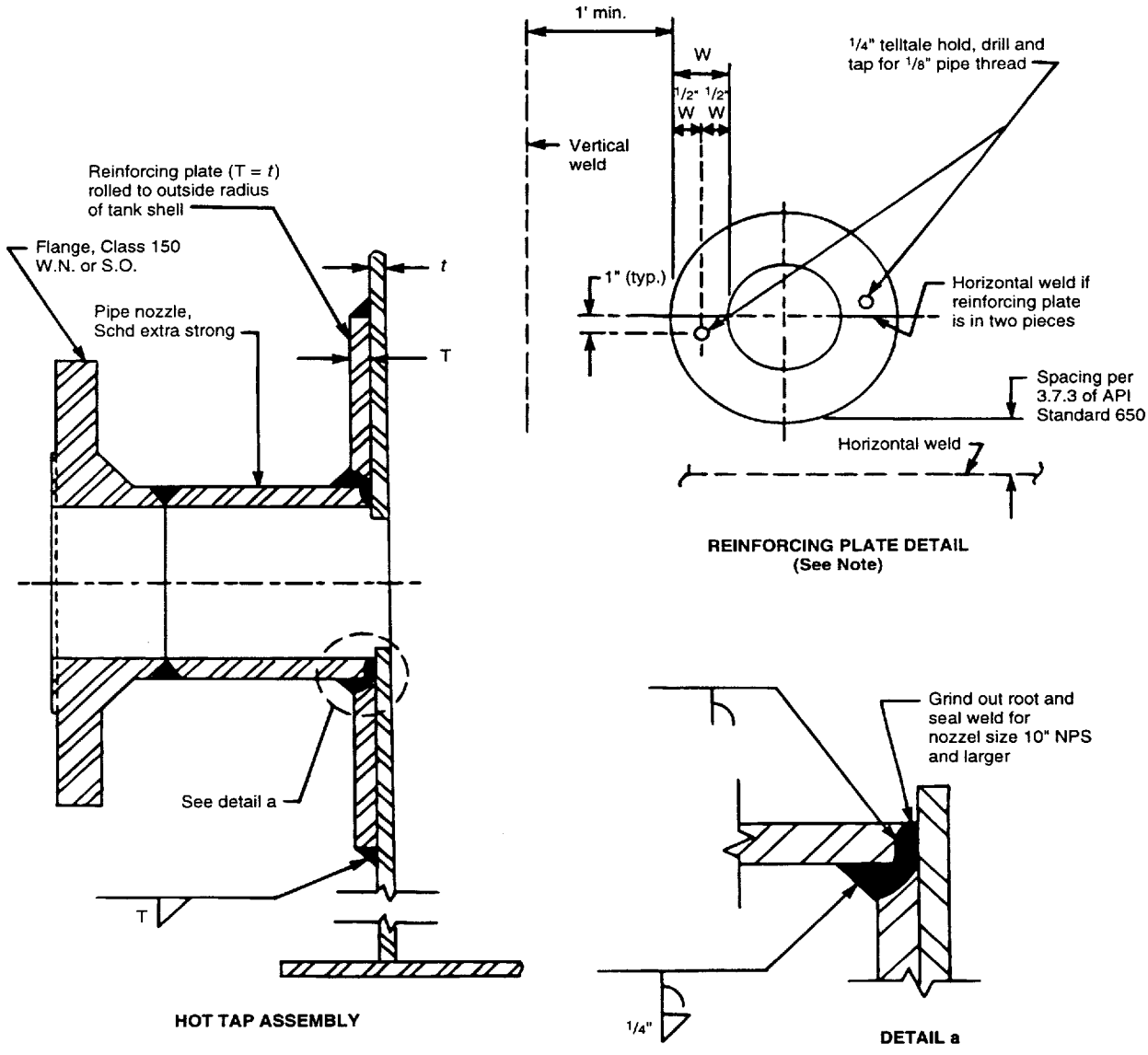
7.14.5 INSTALLATION PROCEDURE

7.14.5.1 Pipe nozzles shall be cut to the contour of the shell and beveled from the outside for a full penetration weld (see Figure 7-6).

7.14.5.2 After the pipe is welded, the reinforcing plate shall be installed either in one piece or two pieces with horizontal weld. The reinforcing plate to nozzle shall be installed with a full penetration weld. Care shall be taken to limit the heat input to the welds.

7.14.5.3 After the reinforcing plate has been welded to the shell and nondestructive examination carried out, the pad shall be pneumatically tested by the procedure described in API Standard 650. After the valve has been installed on the flange, a pressure test at least 1.5 times the hydrostatic head shall be performed on the nozzle prior to mounting the hot tap machine–which shall be bolted to the valve.

7.14.5.4 A qualified operator shall operate the hot tap machine and cut the hole in the tank following the hot tap machine manufacturer's procedures.



Note: Reinforcing plate diameters as given in Table 3-8 of API Standard 650.

Figure 7-6—Hot Tap for Tanks

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 8—DISMANTLING AND RECONSTRUCTION

8.1 General

8.1.1 This section provides procedures for dismantling and reconstructing existing welded tanks that are to be relocated from their original site.

8.1.2 Hydrostatic testing requirements, nondestructive examination requirements, and acceptance criteria for the quality of welds for reconstructed tanks are specified in Section 10.

8.2 Cleaning and Gas Freeing

The tank shall be cleaned and gas-freed prior to commencement of dismantling.

8.3 Dismantling Methods

8.3.1 GENERAL

Roof, shell, and bottom plates may be cut into any size pieces that are readily transportable to the new site for reconstruction.

8.3.2 BOTTOMS

8.3.2.1 Bottom plates that will be reused shall be cut by deseaming of lapwelds; or by cutting alongside of the remaining welds at a minimum of 2 inches away from existing welds, except where cuts cross existing weld seams.

8.3.2.2 If the bottom is to be reused, it shall be cut from the shell along line A-A as shown in Figure 8-1; except if the entire bottom is to be reused intact, the shell shall be cut along line C-C.

8.3.3 SHELLS

8.3.3.1 Shell plates shall be cut in accordance with one of the following methods or a combination thereof:

a. Cuts made to remove existing welds and the heat affected zones.
b. Cuts made a minimum of 6 inches away from existing welds except where cuts cross existing welds.

8.3.3.2 Shell stiffening rings, including wind girders and top angles, may be left attached to the shell plates or may be removed by cutting at the attachment welds. The area where temporary attachments are removed shall be ground flush with the shell plate.

8.3.3.3 The shell shall be cut from the bottom plate along line B-B as shown in Figure 8-1. The existing shell-to-bottom weld connection shall not be reused unless the entire bottom is to be reused intact.

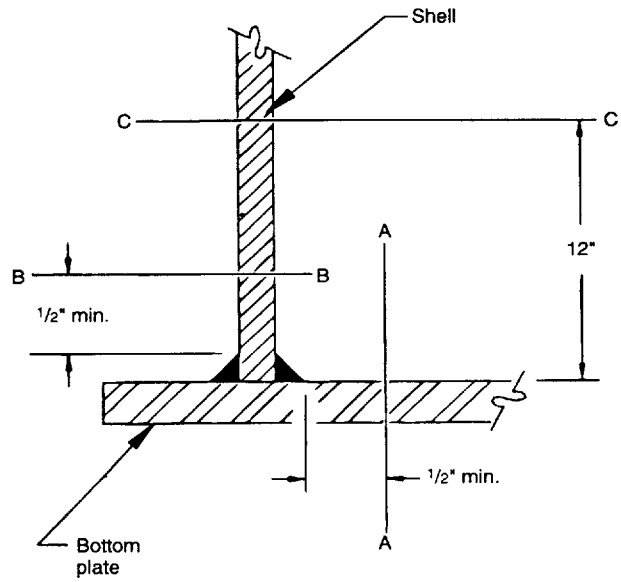


Figure 8-1—Tank Shell and Bottom Cut Locations

8.3.4 ROOFS

8.3.4.1 Roof plates shall be cut by deseaming of lap welds, or by cutting alongside of the remaining welds at a minimum of 2 inches away from existing welds, except where cuts cross existing welds.

8.3.4.2 Roof supporting structures shall be dismantled by removing bolts (if bolted) or deseaming the structural attachment welds.

8.3.5 PIECE MARKING

8.3.5.1 Shell, bottom, and roof plates shall be marked prior to dismantling for ready identification and placement when the tank is reconstructed. Marking material shall be of a durable type. Drawings showing piece mark locations are also a useful adjunct.

8.3.5.2 A minimum of two sets of matching center punch marks shall be located on the top and bottom edges of each shell plate to facilitate proper alignment during reconstruction.

8.4 Reconstruction

8.4.1 GENERAL

8.4.1.1 The foundation for the reconstructed tank shall meet the construction tolerances given in 8.5.6.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607767 T59 ■

8.4.1.2 Temporary attachments shall be removed, and the attachment area ground flush with the plate surface.

8.4.2 WELDING

8.4.2.1 Provisions shall be made during the reconstruction of a tank to ensure that:

a. Vertical weld joints in the shell are not aligned with the weld joints in the bottom plates.
b. There will be no welding over heat affected zones that remain from the original tank welds except where new joints cross original joints.
c. Weld spacing requirements of Figure 7-1 are maintained.

8.4.2.2 Tanks and their structural attachments shall be welded in accordance with the processes specified in API Standard 650 and the requirements of 8.4.2.3 through 8.4.2.11.

8.4.2.3 No welding of any kind shall be performed when the surfaces of the parts to be welded are wet from rain, snow, or ice; when rain or snow is falling on such surfaces; or during periods of high winds unless the welder and the work are properly shielded. No welding of any kind shall be performed when the temperature of the base metal is less than 0°F. When the temperature of the base metal is between 0°F and 32°F or the thickness is in excess of 1 inch, the base metal within 3 inches of the place where welding is to be started shall be heated to a temperature warm to the hand (approximately 140°F) before welding. (See 8.4.4.3 for preheat requirements for shell plates over $1^1/_2$ inches thick.)

8.4.2.4 Each layer of weld metal of multilayer welding shall be cleaned of slag and other deposits before the next layer is applied.

8.4.2.5 The edges of all welds shall merge with the surface of the plate without a sharp angle. The maximum acceptable undercutting is $^1/_{64}$ inch of the base metal for vertical butt joints. For horizontal butt joints undercutting not exceeding $^1/_{32}$ inch in depth is acceptable.

8.4.2.6 The reinforcement of the new welds on all butt joints on each side of the plate shall not exceed the thicknesses shown in Table 8-1.

Table 8-1—Maximum Thicknesses on New Welds

Plate Thickness (inches)	Maximum Reinforcement Thickness (inches)	
	Vertical Joints	Horizontal Joints
≤1/2	3/22	1/8
>1/2 through 1	1/8	3/16
>1	3/16	1/4

8.4.2.7 Tack welds used in the assembly of vertical joints of tank shells shall be removed and shall not remain in the finished joint when the joints are welded manually. When such joints are welded by the submerged-arc process, the tack welds shall be thoroughly cleaned of all welding slag but need not be removed provided that they are sound and are thoroughly fused into the subsequently applied weld beads. Tack welds shall be made using a fillet-weld or butt-weld procedure qualified in accordance with Section IX of the ASME Code. Tack welds to be left in place shall be made by qualified welders.

8.4.2.8 If weldable primer coatings have been applied on surfaces to be welded, they shall be included in welding procedure qualification tests for the brand, formulation, and maximum thickness of primer applied. All other coatings shall be completely removed from the weld area prior to welding.

8.4.2.9 Low-hydrogen electrodes shall be used for manual metal-arc welds, including the attachment of the first shell course to the bottom plates or annular plate ring, as follows:

a. For all welds in shell courses over 0.5 inch thick of API Standard 650 Group I-III materials.
b. For all welds in all shell courses of API Standard 650 Group IV-VI materials.

8.4.2.10 Low-hydrogen electrodes shall be used for welding temporary and new permanent attachments to shells of API Standard 650 Group IV, IVA, V, or VI materials. The welding procedure selected shall not cause underbead cracking; also, the need for preheat of thick plates and the effect of low atmospheric temperature during welding shall be considered.

8.4.2.11 If existing welds are found to be unsatisfactory by the original standard of construction, they shall be repaired in accordance with 7.6. | 96

8.4.3 BOTTOMS

8.4.3.1 After the bottom plates are laid out and tacked, they shall be joined by welding the joints in a sequence that results in the least distortion from shrinkage and provides as nearly as possible a plane surface.

8.4.3.2 The welding of the shell to the bottom (except for door sheets) shall be completed before the welding of bottom joints.

8.4.3.3 Plates shall be held in close contact at all lap joints during welding.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 10—EXAMINATION AND TESTING

10.1 Nondestructive Examinations

10.1.1 GENERAL

10.1.1.1 Nondestructive examination procedures, qualifications, and acceptance criteria shall be prepared for visual, magnetic particle, liquid penetrant, ultrasonic, and radiographic methods in accordance with API Standard 650 and the supplemental requirements given herein.

10.1.1.2 Personnel performing nondestructive examinations shall be qualified in accordance with API Standard 650 and the supplemental requirements given herein.

10.1.2 SHELL PENETRATIONS

10.1.2.1 Ultrasonic examination of shell plate for laminations shall be made in the immediate area affected when:

a. Adding a reinforcing plate to an existing unreinforced penetration.

b. Adding a hot tap connection.

10.1.2.2 Cavities resulting from gouging or grinding operations to remove attachment welds of existing reinforcing plates shall be examined by visual as well as by magnetic particle or liquid penetrant methods.

10.1.2.3 Completed welds attaching nozzle neck to shell, and reinforcing plate to shell and to nozzle neck, shall be examined by the magnetic particle or liquid penetrant methods.

10.1.2.4 Completed welds of stress relieved assemblies shall be examined by visual as well as by magnetic particle or liquid penetrant methods after stress relief, but before hydrostatic test.

10.1.2.5 For penetrations installed using insert plates, the completed butt welds between the insert plate and the shell plate shall be fully radiographed.

10.1.3 REPAIRED WELD FLAWS

10.1.3.1 Cavities resulting from gouging or grinding operations to remove weld defects shall be examined by visual as well as by magnetic particle or liquid penetrant methods.

10.1.3.2 Completed repairs of butt welds shall be examined over their full length by radiographic or ultrasonic methods.

10.1.3.3 Completed repairs of fillet welds shall be examined over their full length by the appropriate nondestructive examination method listed herein.

10.1.4 TEMPORARY AND PERMANENT ATTACHMENTS TO SHELL PLATES

10.1.4.1 Areas where weld projections (remaining after removal of temporary attachments) have been ground flush with the shell plate shall be examined by the appropriate nondestructive examination method listed herein.

10.1.4.2 Completed welds of new permanent attachments (not including shell-to-bottom welds) and areas where temporary attachments have been removed (API Standard 650 tank shell materials of Groups IV, IVA, V, or VI) shall be examined visually and by the magnetic particle method (or, at the option of the purchaser, by the liquid penetrant method). 96

10.1.5 SHELL PLATE TO SHELL PLATE WELDS

New welds attaching shell plate to shell plate shall be examined by visual and by radiographic methods (see 10.2). In addition, for plate thicknesses greater than one inch, the back-gouged surface of the root pass and final pass (each side) shall be examined for its complete length by magnetic particle or liquid penetrant methods.

10.1.6 SHELL-TO-BOTTOM WELD

New welding on the shell-to-bottom joint shall be inspected for its entire length by a right-angle vacuum box and a solution film, or by applying light diesel oil. Additionally, the first weld pass shall be inspected by applying light diesel oil to the side opposite the first weld pass made. The oil shall be allowed to stand at least four hours (preferably overnight) and then the weld inspected for "wicking" action. The oil shall be removed before the weld is completed.

10.1.7 BOTTOMS

Newly welded bottom joints shall be tested using a vacuum box and a solution film, or a tracer gas and detector. Every joint shall be tested for its entire length.

10.2 Radiographs

10.2.1 NUMBER AND LOCATION OF RADIOGRAPHS

The number and location of radiographs shall be in accordance with API Standard 650 and the following additional requirements:

10.2.1.1 For new or repaired shell vertical joints one radiograph shall be taken in every joint.

10.2.1.2 For new or repaired shell horizontal joints one radiograph shall be taken for each 50 feet of horizontal joint, not including those required at intersections of vertical and horizontal joints.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607769 821 ■

10.2.1.3 For each new or repaired intersection between shell vertical and horizontal joints one radiograph shall be taken.

10.2.1.4 For reconstructed tanks, each butt-welded annular plate joint shall be radiographed in accordance with API Standard 650.

10.2.1.5 New and replaced shell insert plate and door sheet welds shall be radiographed. For circular replacement plates, a minimum of one radiograph shall be taken regardless of thickness. For square or rectangular replacement plates, at least one radiograph shall be taken in a vertical joint, and at least one in a horizontal joint, and one in each corner. All junctions between repair and existing welds shall be radiographed. If defects are found, 100 percent radiography shall be performed on the repaired weld.

10.2.1.6 The minimum diagnostic length of each radiograph shall be 6 inches.

10.2.2 ACCEPTANCE CRITERIA FOR EXISTING SHELL PLATE TO SHELL PLATE WELDS

If the radiograph of an intersection between a new and old weld detects unacceptable welds by current standards, the existing welds may be evaluated according to the original standard of construction.

10.2.3 MARKING AND IDENTIFICATION OF RADIOGRAPHS

10.2.3.1 Each film shall show an identification of the welder(s) making the weld. A weld map showing location of welds, weld number, radiograph number, welder identification, and grading of each weld is an acceptable alternative to this requirement.

96 **10.2.3.2** Radiographs and radiograph records of all repaired welds shall be marked with the letter "R".

10.3 Hydrostatic Testing

10.3.1 WHEN HYDROSTATIC TESTING REQUIRED

10.3.1.1 A full hydrostatic test, held for 24 hours, shall be performed on:

a. A reconstructed tank.
b. Any tank that has undergone major repairs or major alterations (see 10.3.1.2) unless exempted by 10.3.2 for the applicable combination of materials, design, and construction features.

10.3.1.2 The terms major repair and major alteration refer to operations that require cutting, addition, removal and/or replacement of the annular plate ring, the shell-to-bottom weld, or a sizable portion of the shell. Within this context, *major repairs* and *major alterations* would include:

a. The installation of any shell penetration beneath the design liquid level larger than 12 inches NPS, or any bottom penetration located within 12 inches of the shell.
b. The removal and replacement or addition of any shell plate beneath the design liquid level, or any annular plate ring material where the longest dimension of the replacement plate exceeds 12 inches.
c. The complete or partial (more than one-half of the weld thickness) removal and replacement of more than 12 inches of vertical weld joining shell plates, or radial weld joining the annular plate ring.
d. The installation of a new bottom.
e. The removal and replacement of any part of the weld attaching the shell to the bottom or to the annular plate ring.
f. Partial or complete jacking of a tank shell.

10.3.2 WHEN HYDROSTATIC TESTING NOT REQUIRED

10.3.2.1 A full hydrostatic test of the tank is not required for major repairs and major alterations when any of the basic conditions listed in Column A of Table 10-1, in combination with the specific requirements of Column B, are satisfied. (See Figure 10-1.)

10.3.2.2 Specific requirements that must be satisfied in establishing an exemption basis from hydrostatic testing are identified for use in Column B of Table 10-1.

a. The repair method has been reviewed and approved by an engineer experienced in storage tank design in accordance with API Standard 650.
b. Materials used for the repair shall meet API Standard 650 requirements.
c. Vertical and horizontal shell joint welds shall have complete penetration and complete fusion. The root pass and final pass examination shall be in accordance with 10.1.5. In addition, the finished weld shall be fully radiographed.
d. Shell penetrations shall be installed with complete penetration and complete fusion weld for the reinforcing plate to nozzle neck and nozzle neck to shell joints. The root pass of the nozzle attachment weld shall be back-gouged and examined by magnetic particle or liquid penetrant methods; the completed weld shall be examined by the ultrasonic method. Examination and acceptance criteria for nondestructive examinations shall be in accordance with 10.1.1.

10.3.2.3 A full hydrostatic test is not required after replacement of a door sheet provided that both of the following conditions are met:

a. Door sheet is in accordance with the requirements of 7.3, except that it shall not extend to or intersect the bottom to the shell joint. 96
b. The tank is otherwise exempt from hydrostatic testing as permitted by 10.3.2.1.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607770 543 ■

10.4 Leak Tests

New or altered reinforcing plates of shell penetrations shall be given an air leak test in accordance with API Standard 650.

10.5 Measured Settlement During Hydrostatic Testing

10.5.1 INITIAL SURVEY

10.5.1.1 Where settlement is anticipated, a tank receiving a hydrostatic test shall have the foundation checked for settlement.

10.5.1.2 Tank settlement shall initially be surveyed with the tank empty using the number of bottom plate projection elevation measurement points, N, uniformly distributed around the circumference, as indicated by the following formula:

$$N = D/10$$

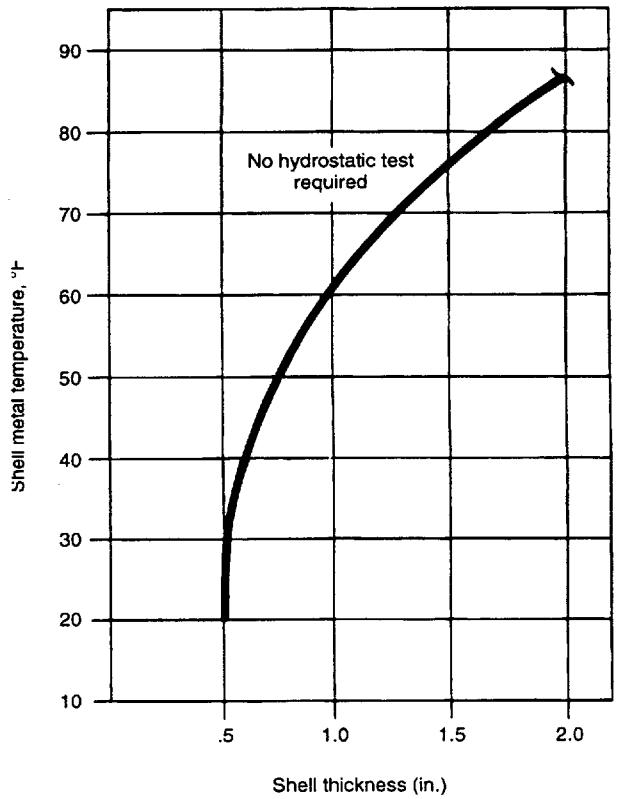
Where:

N = minimum required number of settlement measurement points, but no less than eight. All fractional values shall be rounded to the next higher whole number. The maximum spacing between settlement measurement points shall be 30 feet.

D = tank diameter, in feet.

Table 10-1—Conditions for Exemption From Tank Hydrostatic Test for Major Repairs and Alterations

A Basic Conditions	B Specific Requirements (see 10.3.2.2 for subitem identification)
1. Tank material of unknown toughness, but satisfies the exemption criteria of Table 10-1.	a, b, c, d
2. Tank material meets the toughness requirements of API Standard 650, seventh edition or later.	a, b, c, d
3. Tank repairs limited to bottom plates or annular plate ring (excluding shell-to-bottom weld).	a, b



Note: For unheated tanks, the shell metal temperature shall equal the design metal temperature as defined in 2.2.9.3 of API Standard 650. The original nominal thickness for the thickest tank shell plate shall be used for the assessment.

Figure 10-1—Exemption Curve for Hydrostatic Testing of Tanks

10.5.1.3 Tank settlement measurements in 10.5.1.2 shall be evaluated for acceptance in accordance with Appendix B.

10.5.2 SURVEY DURING HYDROSTATIC TESTING

Settlement shall be measured during filling and when water reaches 100 percent of test level. Excessive settlement in accordance with Appendix B shall be cause to stop the test for foundation investigation and/or repair.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 11—MARKING AND RECORD KEEPING

11.1 Nameplates

11.1.1 Tanks reconstructed in accordance with this standard shall be identified by a corrosion-resistant metal nameplate similar to that shown in Figure 11-1. Letters and numerals not less than 5/32 inch high shall be embossed, engraved, or stamped in the plate to indicate information as follows:

a. Reconstructed to API 653.
b. Edition and revision number.
c. Year reconstruction was completed.
d. If known, the original applicable standard and the year of original construction.
e. Nominal diameter.
f. Nominal shell height.
g. Design specific gravity.
h. Maximum permissible operating liquid level.
i. The name of the reconstruction contractor and the assigned serial number or contract number.
j. The owner/operator's tank number.
k. Shell material for each shell course.
l. Maximum operating temperature.
m. Allowable stress used in calculations of each shell course.

11.1.2 The new nameplate shall be attached to the tank shell adjacent to the existing nameplate, if any. An existing nameplate shall be left attached to the tank. Nameplates shall be attached as specified in API Standard 650.

11.2 Record Keeping

When a tank is evaluated, repaired, altered, or reconstructed in accordance with this standard, the following information, as applicable, shall be made a part of the owner/operator's records for the tank (see 4.8).

11.2.1 Calculations for:

a. Component evaluation for integrity, including brittle fracture considerations (Section 3).
b. Rerating (including liquid level).
c. Repair and alteration considerations.

11.2.2 Construction and repair drawings.

11.2.3 Additional support data including, but not limited to, information pertaining to:

a. Inspections (including thicknesses).
b. Material test reports/certifications.
c. Tests.
d. Radiographs (radiographs shall be retained for at least one year).
e. Brittle fracture considerations.
f. Original tank construction data (date, original standard, etc.).
g. Location and identification (owner/operator's number, serial number).
h. Description of the tank (diameter, height, service).
i. Design conditions (liquid level, specific gravity, allowable stress, unusual design loadings, etc.).
j. Shell material and thickness by course.
k. Tank perimeter elevations.
l. Construction completion record.

11.3 Certification

Tanks reconstructed in accordance with this standard shall require documentation of such reconstruction, and certification that the design, reconstruction, inspection, and testing was performed in compliance with this standard. The certification shall contain information as shown in Figure 11-2 for design and/or reconstruction as applicable.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607772 316 ■

RECONSTRUCTED TO API 653 EDITION _____

Reconstructed by:

_____

_____

Date Completed _____

Serial No. _____

Shell Course	Allowable Stress	Material
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

Original Standard _____

Tank No. _____

Tank Dia. _____ Height _____

Specific Gravity _____

Design Pressure _____

Orig. Const. Date _____

Year Reconstructed _____

Liquid Level Max. _____

Capacity _____

Max. Operating Temp. _____

Figure 11-1—Nameplate

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607773 252 ■

CERTIFICATION FOR TANK RECONSTRUCTED TO API 653

We hereby certify that the tank reconstructed at _____ and described as follows:
Location

_____	_____	_____	_____	_____
Serial No.	Owner's No.	Height	Capacity	Floating or Fixed Roof

was reconstructed, inspected, and tested in accordance with all applicable requirements of API Standard 653,

_____ Edition, _____ Revision, Dated _____ (including all material supplied by the reconstruction organization).

_____
Reconstruction Organization

_____
Authorized Representative

_____
Date

CERTIFICATION FOR TANK DESIGNED TO API 653

We hereby certify that the design evaluation of the tank reconstructed at _____
and described as follows:
Location

_____	_____	_____	_____	_____
Serial No.	Owner's No.	Height	Capacity	Floating or Fixed Roof

was performed by the undersigned organization in accordance with all design requirements of API Standard 653,

_____ Edition, _____ Revision, Dated _____.

_____
Reconstruction Organization

_____
Authorized Representative

_____
Date

Figure 11-2—Certification Forms

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX B—EVALUATION OF TANK BOTTOM SETTLEMENT

B.1 Introduction

B.1.1 In determining the effects of soil settlement on storage tanks, it is common practice to monitor settlement of the tank bottom. In most cases, such a monitoring program is initiated during the construction and continued during hydrotesting and operations. During operations, settlement measurements should be taken at a planned frequency, based upon an assessment of soil settlement predictions. For existing tanks which do not have initial settlement data, a program of settlement monitoring should be based on prior service history.

B.1.2 If at any time settlement is deemed excessive, the tanks should be emptied and releveled. Releveling of a sizable tank is expensive and rather difficult to achieve. Thus, a decision to relevel a tank is a crucial one, and relies very much on the proper interpretation and evaluation of the monitored settlement data.

B.1.3 Approaches used to correct tank shell and bottom settlement include techniques such as localized repairs of the bottom plates, partial releveling of the tank periphery, and major releveling of the entire tank bottom. Major releveling of the tank, involving total lifting of the tank shell and bottom at one time, can introduce high localized stresses in the structure and impair its integrity. Therefore, when choosing techniques for correcting settlement problems, an alternative to total lifting of the tank shell and bottom should be considered as a first choice. If it is decided to lift the entire tank shell and bottom at one time, it should be done by personnel with demonstrated experience in this technique.

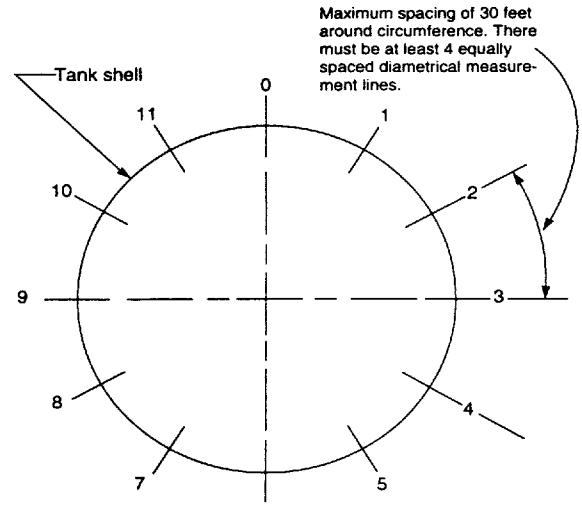
B.2 Types of Settlement

B.2.1 ELEVATION MEASUREMENTS

The principal types of tank settlement consist of settlements which relate to the tank shell and bottom plate. These settlements can be recorded by taking elevation measurements around the tank circumference and across the tank diameter. Figures B-1 and B-2 show recommended locations on a tank shell and bottom plate for settlement measurements. Data obtained from such measurements should be used to evaluate the tank structure. Additional settlement readings may be required to define local depressions.

B.2.2 SHELL SETTLEMENT EVALUATION

Settlement of a tank is considered to be the result of either one or a combination of the following three settlement components.



Notes:
1. There must be at least eight settlement points. The maximum spacing of settlement points is 30 feet around the circumference.
2. Points shall be equally spaced around the tank shell. See 10.5.1.2 for method of determining the number of measurement points.

Figure B-1—Measurements of Shell Settlement (External)

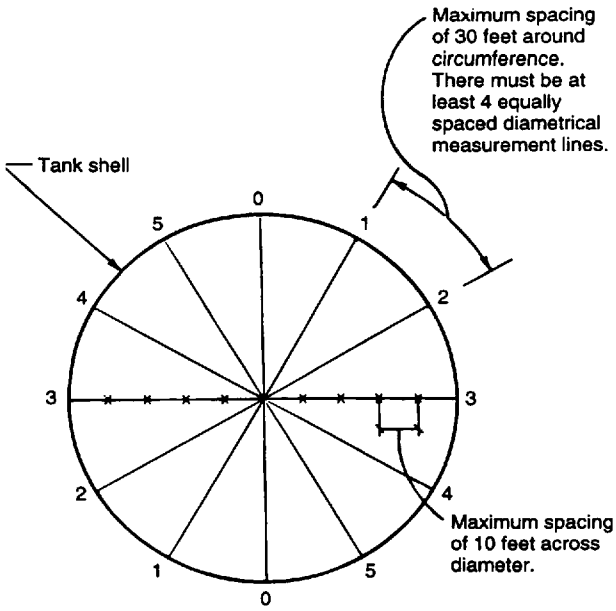
B.2.2.1 Uniform settlement. This component often can be predicted in advance, with sufficient accuracy from soil tests. It may vary in magnitude, depending on the soil characteristics. Uniform settlement of a tank does not induce stresses in the tank structure. However, piping and attachments must be given adequate consideration to prevent problems caused by such settlement.

B.2.2.2 Rigid body tilting of a tank (planar tilt). This component rotates the tank in a tilted plane. The tilt will cause an increase in the liquid level and, therefore, an increase in the hoop stress in the tank shell. Also, excessive tilting can cause binding of peripheral seals in a floating roof and inhibit roof travel. This type of settlement could affect tank nozzles that have piping attached to them. Figure B-3 shows that the settled location of the tank shell, after rigid body tilt, can be represented by either a cosine or sine wave with respect to its original position in a horizontal plane.

B.2.2.3 Out-of-plane settlement (differential settlement). Due to the fact that a tank is a rather flexible structure, chances are great that the tank shell will settle in a non-planar con-

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607775 025 ■



Note: See 10.5.1.2 for method of determining the number of measurement points.

Figure B-2—Measurements of Bottom Settlement (Internal) Tank Out-of-Service

figuration, inducing additional stresses in the tank shell. The out-of-plane settlements at the bottom edge lead to a lack of circularity at the top of the tank, and in the case of a floating roof tank, the extent of the induced ovality may impede the proper functioning of the floating roof in such a way that releveling is required. Also, such settlements may cause flat spots to develop in the tank shell. This type of settlement could affect tank nozzles that have piping attached to them.

B.2.2.4 While uniform settlement and rigid body tilt of a tank may cause problems as described above, the out-of-plane settlement is the important component to determine and evaluate in order to ensure the structural integrity of the shell and bottom. Based on this principle, a common approach is to determine the magnitudes of the uniform and rigid body tilt component (if any) for each data point on the tank periphery. Once this is carried out, the plane of rigid tilt is then important as a datum from which to measure the magnitudes of the out-of-plane settlements.

A graphical representation illustrating the effect of tank shell settlement components is shown in Figure B-3. The construction of this "settlement plot" has been developed in accordance with the following:

a. The actual edge settlement (in most cases an irregular curve) is plotted using points around the tank circumference as the abscissa.

b. The vertical distance between the abscissa and the lowest point on this curve (point 22) is the minimum settlement, and it is called the uniform settlement component. A line through this point, parallel to the abscissa, provides a new base or datum line for settlement measurements which are called "adjusted settlements."

c. The plane of rigid tilt settlement is represented by the "optimum cosine curve" between the maximum and minimum values of the actual edge settlements. Several methods exist for determining the optimum cosine curve. The least accurate method is by free hand drawing techniques, a kind of trial and error procedure to fit the best cosine curve through the data. A better method is to use the mathematical and graphical capabilities of a computer.

d. The vertical distances between the irregular curve and the cosine curve represent the magnitudes of the out-of-plane settlements (U_i at data point i).

B.2.2.5 A method of determining the acceptable settlement is given in B.3.2. As an alternative, a rigorous stress analysis could be carried out.

B.2.3 EDGE SETTLEMENT

B.2.3.1 Edge settlement occurs when the tank shell settles sharply around the periphery, resulting in deformation of the bottom plate near the shell-to-bottom corner junction. Figure B-4 illustrates this settlement.

B.2.3.2 The formula given in B.3.3 can be used to evaluate this criteria. Alternatively, a rigorous stress analysis can be carried out for the deformed profile.

B.2.4 BOTTOM SETTLEMENT NEAR TANK SHELL

B.2.4.1 Figure B-5 illustrates this type of settlement.

B.2.4.2 The formula given in B.3.3 can be used to evaluate this criteria. Alternatively, a rigorous stress analysis can be carried out for the deformed profile.

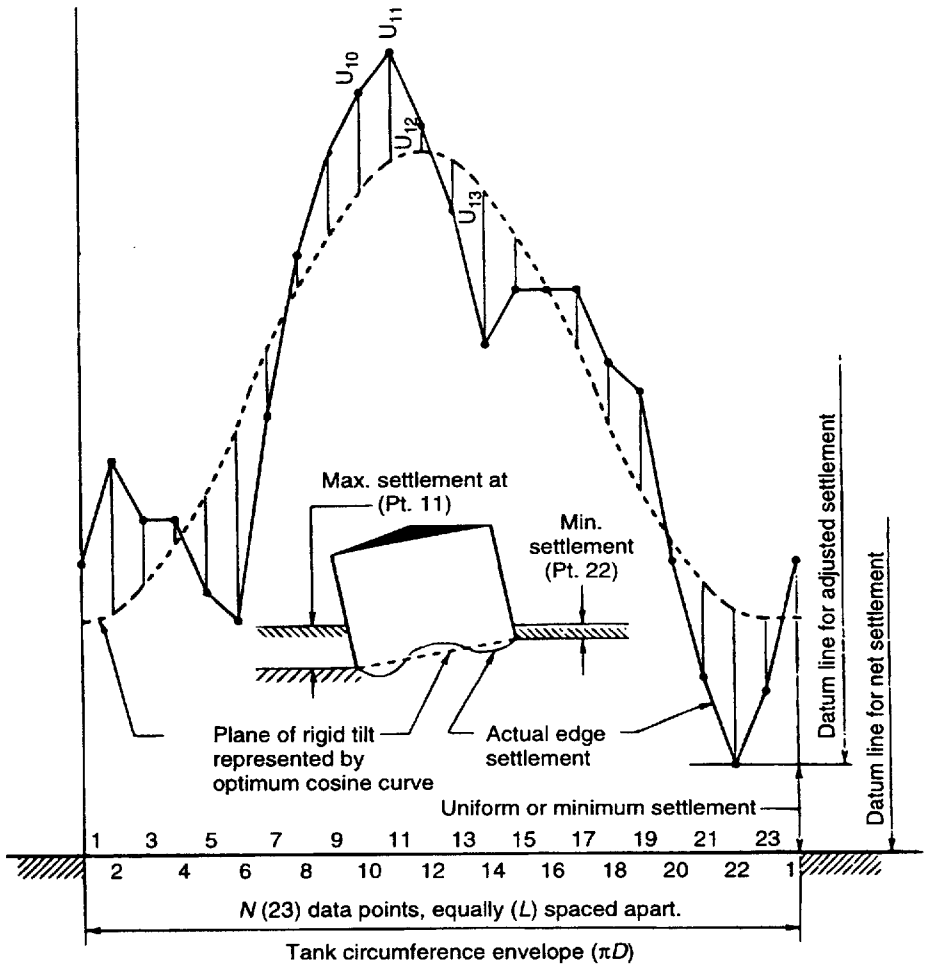
B.2.5 LOCALIZED BOTTOM SETTLEMENT REMOTE FROM TANK SHELL

B.2.5.1 These are depressions (or bulges) that occur in a random manner, remote from the shell (see Figure B-6).

B.2.5.2 Acceptability of these localized settlements is dependent upon localized stresses in the bottom plate, design and quality of the lap welds (single pass or multipass), and voids below the bottom plate. The formula given in B.3.3 can be used to evaluate this criteria. These limits are applicable to tank bottoms which have single pass lap welded joints.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607776 T61 ■



Out-of-plane deflection for point "i" is
$S_i = U_i - (1/2\ U_{i-1} + 1/2\ U_{i+1})$, for example
$S_{11} = U_{11} - (1/2 U_{10} + 1/2 U_{12})$

U_i = out-of-plane settlement of point *"i"*,
(+) when above cos. curve;
(–) when below cos. curve, for example:
U_{11} = (+)
U_{13} = (–)

Note: See 10.5.1.2 for definition of "N."

Figure B-3—Graphical Representation of Tank Shell Settlement

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607777 9T8 ■

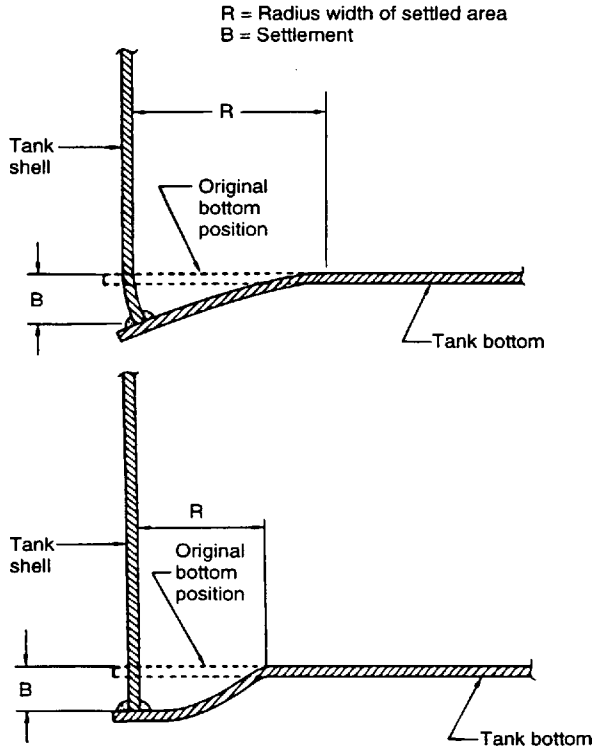


Figure B-4—Edge Settlement

B.3 Determination of Acceptable Settlement

B.3.1 GENERAL

For existing tanks with history of successful service, it may be possible to accept greater settlement and distortion of the foundation from a true plane than new tank construction standards allow. Each tank must be evaluated based on service conditions, materials of construction, soil characteristics, tank foundation design, and tank service history. The methods discussed in following sections are not mandatory and are an approximation of the maximum permissible settlement. However, experience has shown that if settlements exceed the following requirements, further assessment or repair is required.

B.3.2 SHELL SETTLEMENT

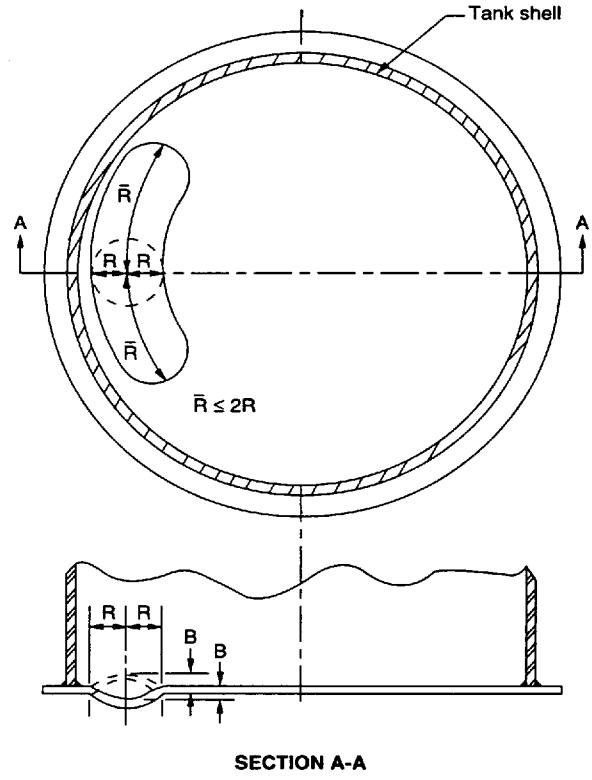
From the measurements described in B.2, determine the maximum out-of-plane deflection. Use the following formula to calculate the maximum permissible out-of-plane deflection (see note below.)

$$|S| \le \frac{(L^2 \times Y \times 11)}{2[(E \times H)]}$$

Where:

S = deflection, in feet (out of plane distortion)
L = arc length between measurement points, in feet
Y = yield strength, in pounds per square inch
E = Young's modulus, in pounds per square inch
H = tank height, in feet

Note: This formula is based on "Criteria for Settlement of Tanks," W. Allen Marr, M. ASCE, Jose A. Ramos, and T. William Lambe, F. ASCE, *Journal of Geotechnical Engineering Division, Proceedings of the American Society of Civil Engineers*, Vol. 108, August 1982.



R = Radius of inscribed circle in bulged or depressed area
B = Settlement or depression depth or height of bulge

Figure B-5—Bottom Settlement Near Shell

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607778 834 ■

B.3.3 INTERNAL BOTTOM SETTLEMENTS OR BULGES

Measure the bulge or depression. The permissible bulge or depression is given by the following formula (see note below).

$$B = 0.37R$$

Where:

B = maximum height of bulge or depth of local depression, inches

R = radius of inscribed circle in bulged area or local depression, in feet

Figure B-7 is a graphical representation of this formula.

Note: This formula is based on "Criteria for Settlement of Tanks," W. Allen Marr, M. ASCE, Jose A. Ramos, and T. William Lambe, F. ASCE, *Journal of Geotechnical Engineering Division. Proceedings of the American Society of Civil Engineers*, Vol. 108, August 1982.

B.4 Repairs

If it is determined that settlements have occurred which are beyond the permissible limits established in the previous sections, then consideration should be given to making repairs or a rigorous stress analysis should be performed to evaluate the deformed profile. Various repair techniques have been discussed above. The judgement on repairs should be tempered with knowledge of tank service history, previous repairs, previous inspections, tank foundation conditions, soil characteristics, the material of construction, and estimates of future settlement. See 7.10 for suggested repair details.

96

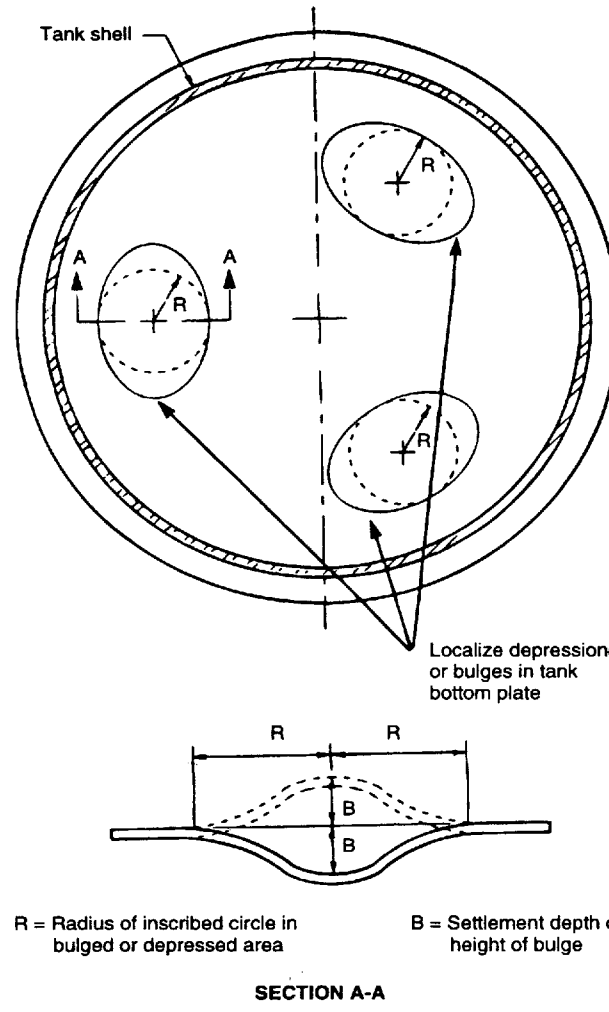


Figure B-6—Localized Bottom Depressions or Bulges Remote From Shell

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

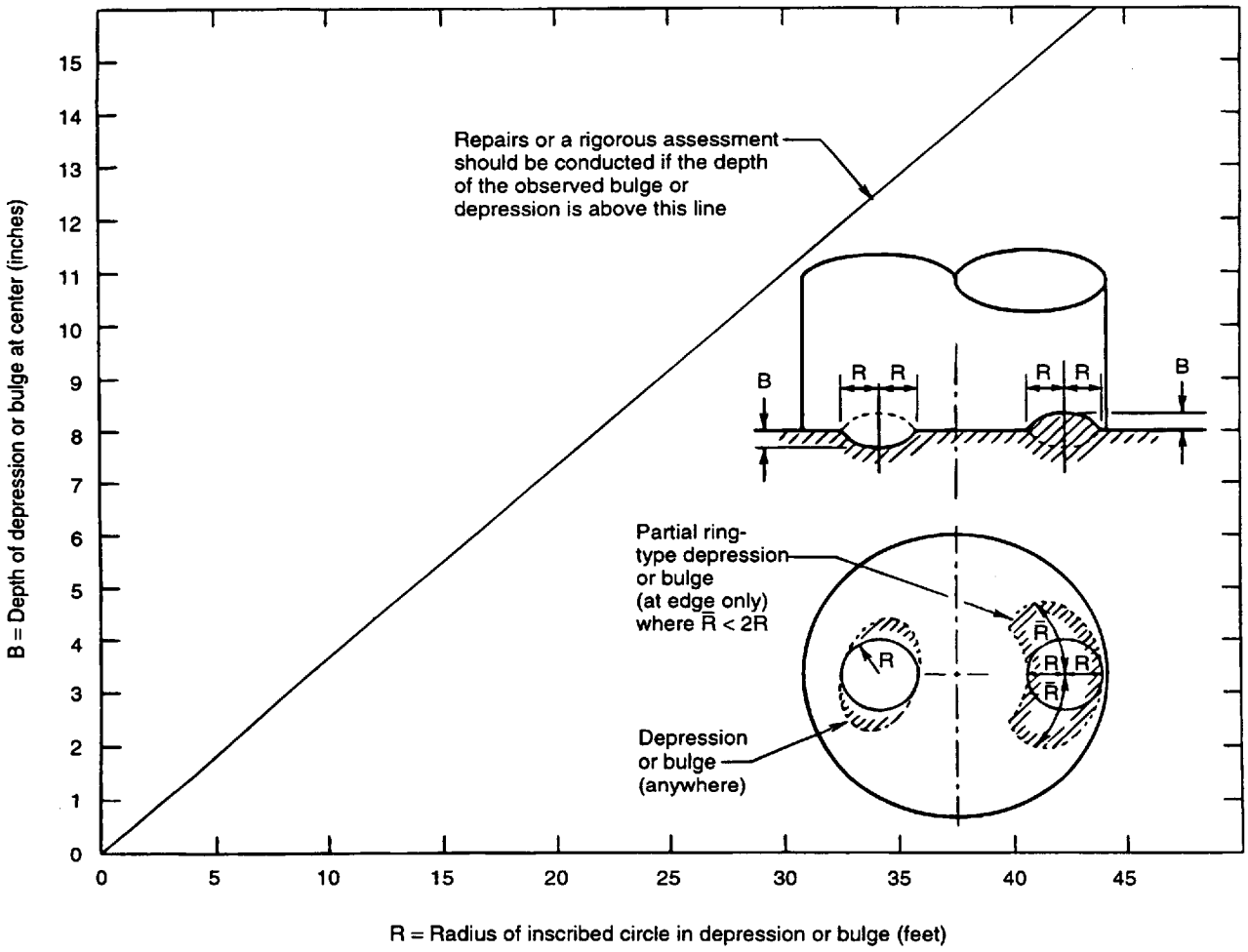


Figure B-7—Localized Tank Bottom Settlement Limits for Single Pass Welds

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Additional copies available from API Publications and Distribution:
(202) 682-8375

Information about API Publications, Programs and Services is available
on the World Wide Web at: http://www.api.org

American Petroleum Institute

1220 L Street, Northwest
Washington, D.C. 20005-4070
202-682-8000

Order No. C653A1

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34

Information Handling Services, 2000

Date of Issue: August 1996
Affected Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction,* Second Edition, December 1995 (1st printing)

CORRECTED PAGES FOR API STANDARD 653

As a followup to the Errata issued in May 1996 for API Standard 653, please substitute the attached corrected pages for those included in the first printing of this standard.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Date of Issue: May 1995
Affected Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction,* Second Edition, December 1995

ERRATA

Page 1-2, 1.6 Referenced Publications. Under API publications, add the following:

Publ 2207 Preparing Tank Bottoms for Hot Work

Page 2-1, Paragraph 2.2.3.1. Add the following sentence to the end of the paragraph:

Holes through roof plates shall be repaired or replaced.

Page 2-2, Paragraph 2.3.1.5. In the last sentence add the word "not" to change the phrase "shall be exceeded" to read:

shall not be exceeded.

Paragraph 2.3.2.1.b. The square root symbol is missing from the equation to calculate L. *Add the square root symbol to correct the equation to:*

$L = 3.7\sqrt{Dt_2}$, but not more than 40 inches.

Page 2-3, Paragraph 2.3.3.1. In the list beginning "Where:" S = *Maximum allowable stress..." change the number 0.426*T *to:*

0.429*T*

Page 2-6, Paragraph 2.4.7.3. Change "Table 4" to:

Table 4.1

Page 2-7, Paragraph 2.4.8.1. Correct the minimum thickness of annular plate ring from "1.10 inch" to:

0.10 inch

Paragraph 2.4.8.2. Delete the word "project" in "project specific gravity," and replace it with:

product specific gravity

Page 5-1, Paragraph 5.3.4. In the paragraph head, change WIND GIRDERS to one word:

WINDGIRDERS

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page 6-1, 6.4 Shell Design. The original Paragraph 6.4.3 from API Standard 653, First Edition, was omitted. Replace it and renumber the next paragraphs as 6.4.4 and 6.4.5. Text for 6.4.3 is as follows:

6.4.3 The maximum liquid level for hydrostatic test shall be determined by using the actual thickness measured for each shell course, the allowable stress for the material in each course, and the design method to be used. The allowable stress for the material shall be determined using Table 3-2 of API Standard 650. For material not listed in Table 3-2, an allowable stress value of the lesser of $^{3}/_{4}$ yield strength or $^{2}/_{5}$ tensile strength shall be used.

Page 6-1, 6.6 Change "Wind Girders" to one word:

Windgirders

Paragraph 6.6.1. Change "wind girders" to one word:

windgirders

Page 7-1, Paragraph 7.2.3.1. In the second sentence, replace the word "tapped" with:

lapped

Page 7-2, Figure 7-1. The dimension arrow for V should be extended all the way to the weld joint described in the figure as "Weld joint (annular or sketch plates)".

*Page 7-3, Figure 7-2. In the column of dimensions under the heading "*t *and* T *(maximum)" on the third line down $1^{5}/_{16}$" is incorrect. Replace it with the correct dimension:*

$^{15}/_{16}$"

*Page 7-4, Figure 7-3. In the column of dimensions under the heading "*t *and* T *(maximum)" on the third line down $1^{5}/_{16}$" is incorrect. Replace it with the correct dimension:*

$^{15}/_{16}$"

Paragraph 7.8.2.2. In the second to last sentence, change the referenced figure number from "Figure 6" to:

Figure 7-2

Page 10-2, Paragraph 10.3.2.2.d. At the end of the first sentence, after "... plate to nozzle neck..." add: "and nozzle neck" so the sentence reads:

Shell penetrations shall be installed with complete penetration and complete fusion weld for the reinforcing plate to nozzle neck and nozzle neck to shell joints.

Paragraph 10.3.2.3.a. Delete the word "of" in the phrase "the bottom of the shell joint" and replace it with "to." Also, delete the second sentence that starts with "Note...."

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Table 10-1, Column A, Basic Conditions Item 1. Correct the Figure 8 reference to read:

Figure 10-1

Page 11-1, Paragraph 11.3 Certification. After the Paragraph 11.3, delete list items e. through l., and delete duplicate Paragraph 11.3.

Page 11-2, Figure 11-1. In the heading on the sample form, delete the words "REPAIRED/ALTERED" and replace with:

RECONSTRUCTED

Figure 11-1. Three columns are labeled "Shell Course." Only the first column on the left should remain as Shell Course. Rename the middle column "Allowable Stress," and rename the next column "Material."

Page 11-3, Figure 11-2—Certification Forms. In the title of the second form, change the word "RECONSTRUCTED" to:

DESIGNED

Appendix B, Page B-1, Figure B-1. Add the following description to the arc dimension from points 2 to 3:

Maximum spacing of 30 feet around circumference. There must be at least 4 equally spaced diametrical measurement lines.

Figure B-1. Delete the dimension arrow from point 3 to point 4.

Section B.1.3, second sentence. Delete the fraction "$^1/_2$" from the sentence.

Page B-1, Section B.2.2.2. In the first sentence, correct "(planar tilts)" to read:

(planar tilt)

Page B-2, Paragraph B.2.2.4.c. Correct the second word in the first sentence from "plate" to:

plane

Page B-3, Figure B-3. The formulas shown under Out-of-plane deflection for point "i" are incorrect. Replace them with the following:

$$S_i = U_i - \left(\tfrac{1}{2}U_{i-1} + \tfrac{1}{2}U_{i+1}\right), \text{ for example}$$

$$S_{11} = U_{11} - \left(\tfrac{1}{2}U_{10} + \tfrac{1}{2}U_{12}\right)$$

Page B- 4, Paragraph B.3.2. Correct the equation given to calculate the maximum permissible out-of-plane deflection to read:

$$|S| \leq \frac{\left(L^2 \times Y \times 11\right)}{2[(E \times H)]}$$

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Date of Issue: February 1996
Affected Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*, Second Edition, December 1995 (1st printing)

ERRATA

Please substitute the attached corrected Cover and Pages 10-1 and 10-2.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Tank Inspection, Repair, Alteration, and Reconstruction

API STANDARD 653
SECOND EDITION, DECEMBER 1995





American
Petroleum
Institute

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34
Information Handling Services, 2000



One of the most significant long-term trends affecting the future vitality of the petroleum industry is the public's concerns about the environment. Recognizing this trend, API member companies have developed a positive, forward looking strategy called STEP: Strategies for Today's Environmental Partnership. This program aims to address public concerns by improving our industry's environmental, health and safety performance; documenting performance improvements; and communicating them to the public. The foundation of STEP is the API Environmental Mission and Guiding Environmental Principles.

API ENVIRONMENTAL MISSION AND GUIDING ENVIRONMENTAL PRINCIPLES

The members of the American Petroleum Institute are dedicated to continuous efforts to improve the compatibility of our operations with the environment while economically developing energy resources and supplying high quality products and services to consumers. The members recognize the importance of efficiently meeting society's needs and our responsibility to work with the public, the government, and others to develop and to use natural resources in an environmentally sound manner while protecting the health and safety of our employees and the public. To meet these responsibilities, API members pledge to manage our businesses according to these principles:

- To recognize and to respond to community concerns about our raw materials, products and operations.
- To operate our plants and facilities, and to handle our raw materials and products in a manner that protects the environment, and the safety and health of our employees and the public.
- To make safety, health and environmental considerations a priority in our planning, and our development of new products and processes.
- To advise promptly, appropriate officials, employees, customers and the public of information on significant industry-related safety, health and environmental hazards, and to recommend protective measures.
- To counsel customers, transporters and others in the safe use, transportation, and disposal of our raw materials, products, and waste materials.
- To economically develop and produce natural resources and to conserve those resources by using energy efficiently.
- To extend knowledge by conducting or supporting research on the safety, health, and environmental effects of our raw materials, products, processes, and waste materials.
- To commit to reduce overall emission and waste generation.
- To work with others to resolve problems created by handling and disposal of hazardous substances from our operations.
- To participate with government and others in creating responsible laws, regulations, and standards to safeguard the community, workplace, and environment.
- To promote these principles and practices by sharing experiences and offering assistance to others who produce, handle, use, transport, or dispose of similar raw materials, petroleum products and wastes.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607787 847 ■

Tank Inspection, Repair, Alteration, and Reconstruction

Manufacturing, Distribution and Marketing Department

API STANDARD 653
SECOND EDITION, DECEMBER 1995



American Petroleum Institute

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34
Information Handling Services, 2000

SPECIAL NOTES

1. API PUBLICATIONS NECESSARILY ADDRESS PROBLEMS OF A GENERAL NATURE. WITH RESPECT TO PARTICULAR CIRCUMSTANCES, LOCAL, STATE, AND FEDERAL LAWS AND REGULATIONS SHOULD BE REVIEWED.

2. API IS NOT UNDERTAKING TO MEET THE DUTIES OF EMPLOYERS, MANUFACTURERS, OR SUPPLIERS TO WARN AND PROPERLY TRAIN AND EQUIP THEIR EMPLOYEES, AND OTHERS EXPOSED, CONCERNING HEALTH AND SAFETY RISKS AND PRECAUTIONS, NOR UNDERTAKING THEIR OBLIGATIONS UNDER LOCAL, STATE, OR FEDERAL LAWS.

3. INFORMATION CONCERNING SAFETY AND HEALTH RISKS AND PROPER PRECAUTIONS WITH RESPECT TO PARTICULAR MATERIALS AND CONDITIONS SHOULD BE OBTAINED FROM THE EMPLOYER, THE MANUFACTURER OR SUPPLIER OF THAT MATERIAL, OR THE MATERIAL SAFETY DATA SHEET.

4. NOTHING CONTAINED IN ANY API PUBLICATION IS TO BE CONSTRUED AS GRANTING ANY RIGHT, BY IMPLICATION OR OTHERWISE, FOR THE MANUFACTURE, SALE, OR USE OF ANY METHOD, APPARATUS, OR PRODUCT COVERED BY LETTERS PATENT. NEITHER SHOULD ANYTHING CONTAINED IN THE PUBLICATION BE CONSTRUED AS INSURING ANYONE AGAINST LIABILITY FOR INFRINGEMENT OF LETTERS PATENT.

5. GENERALLY, API STANDARDS ARE REVIEWED AND REVISED, REAFFIRMED, OR WITHDRAWN AT LEAST EVERY FIVE YEARS. SOMETIMES A ONE-TIME EXTENSION OF UP TO TWO YEARS WILL BE ADDED TO THIS REVIEW CYCLE. THIS PUBLICATION WILL NO LONGER BE IN EFFECT FIVE YEARS AFTER ITS PUBLICATION DATE AS AN OPERATIVE API STANDARD OR, WHERE AN EXTENSION HAS BEEN GRANTED, UPON REPUBLICATION. STATUS OF THE PUBLICATION CAN BE ASCERTAINED FROM THE API AUTHORING DEPARTMENT [TELEPHONE (202) 682-8000]. A CATALOG OF API PUBLICATIONS AND MATERIALS IS PUBLISHED ANNUALLY AND UPDATED QUARTERLY BY API, 1220 L STREET, N.W., WASHINGTON, D.C. 20005.

All rights reserved. No part of this work may be reproduced, stored in a retrieval system, or transmitted by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior written permission from the publisher. Contact API Publications Manager, 1220 L Street, N.W., Washington, D.C. 20005.

Copyright © 1995 American Petroleum Institute

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

FOREWORD

This standard is based on the accumulated knowledge and experience of owners, operators, manufacturers, and repairers of steel storage tanks. The object of this publication is to provide guidance in the inspection, repair, alteration, and reconstruction of steel storage tanks used in the petroleum and chemical industries.

If tanks are inspected, repaired, altered, or reconstructed in accordance with this standard, the owner/operator may elect to modify, delete, or amplify sections of this standard. It is strongly recommended that such modifications, deletions, or amplifications be made by supplementing this standard rather than by rewriting or incorporating sections into another complete standard.

The rules given in this standard are minimum requirements. This standard is not to be interpreted as approving, recommending, or endorsing any specific design nor limiting the methods of inspection, repair, alteration, or reconstruction.

Each edition, revision, or addenda to this API standard may be used beginning with the date of issuance shown on the cover page for that edition, revision, or addenda. Each edition, revision, or addenda to this API standard becomes effective six months after the date of issuance for equipment that is certified as being rerated, reconstructed, relocated, repaired, modified (altered), inspected, and tested per this standard. During the six-month time between the date of issuance of the edition, revision, or addenda and the effective date, the purchaser and manufacturer shall specify to which edition, revision, or addenda the equipment is to be rated, reconstructed, relocated, repaired, modified (altered), inspected, and tested.

API standards are published as an aid to procurement of standardized equipment and materials. API standards are not intended to inhibit purchasers or producers from purchasing or producing products made to specifications other than those of API.

API standards may be used by anyone desiring to do so. Every effort has been made by the Institute to assure the accuracy and reliability of the data contained in them; however, the Institute makes no representation, warranty, or guarantee in connection with the publication of API standards and hereby expressly disclaims any liability or responsibility for loss or damage resulting from their use; for any violation of any federal, state, or municipal regulation with which an API standard may conflict; or for the infringement of any patent resulting from the use of an API standard.

Suggested revisions are invited and should be submitted to the director of the Manufacturing, Distribution and Marketing Department, American Petroleum Institute, 1220 L Street, N.W., Washington, D.C. 20005.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

CONTENTS

Page

SECTION 1—INTRODUCTION
1.1 General 1-1
1.2 Compliance With This Standard 1-1
1.3 Jurisdiction 1-1
1.4 Safe Working Practices 1-1
1.5 Definitions 1-1
1.6 Referenced Publications 1-2

SECTION 2—SUITABILITY FOR SERVICE
2.1 General 2-1
2.2 Tank Roof Evaluation 2-1
2.2.1 General 2-1
2.2.2 Fixed Roofs 2-1
2.2.3 Floating Roofs 2-1
2.2.4 Change of Service 2-1
2.3 Tank Shell Evaluation 2-2
2.3.1 General 2-2
2.3.2 Actual Thickness Determination 2-2
2.3.3 Minimum Thickness Calculation for Welded Tank Shell 2-3
2.3.4 Minimum Thickness Calculation for Riveted Tank Shell 2-4
2.3.5 Distortions 2-4
2.3.6 Flaws 2-5
2.3.7 Wind Girders and Shell Stiffeners 2-5
2.3.8 Shell Welds 2-5
2.3.9 Shell Penetrations 2-5
2.4 Tank Bottom Evaluation 2-5
2.4.1 General 2-5
2.4.2 Causes of Bottom Failure 2-5
2.4.3 Cathodic Protection of Tank Bottoms 2-5
2.4.4 Internal Lining Protection of Tank Bottoms 2-6
2.4.5 Bottom Leak Detection 2-6
2.4.6 Bottom Plate Thickness Measurements 2-6
2.4.7 Minimum Thickness for Tank Bottom Plate 2-6
2.4.8 Minimum Thickness for Annular Plate Ring 2-7
2.5 Tank Foundation Evaluation 2-7
2.5.1 General 2-7
2.5.2 Foundation Repair or Replacement 2-7
2.5.3 Anchor Bolts 2-7

SECTION 3—BRITTLE FRACTURE CONSIDERATIONS
3.1 General 3-1
3.2 Basic Considerations 3-1
3.3 Notes for Figure 3 3-1

SECTION 4—INSPECTION
4.1 General 4-1
4.2 Inspection Frequency Considerations 4-1
4.3 External Inspection 4-1

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

4.3.1 Routine In-Service Inspections 4-1
4.3.2 Scheduled Inspections 4-1
4.3.3 In-Service Ultrasonic Thickness Measurements of the Shell 4-1
4.4 Internal Inspection 4-2
4.4.1 General 4-2
4.4.2 Inspection Intervals 4-2
4.4.3 Alternative Internal Inspection Interval 4-2
4.5 Alternative to Internal Inspection to Determine Bottom Thickness 4-2
4.6 Preparatory Work for Internal Inspection 4-2
4.7 Inspection Checklists 4-2
4.8 Records 4-2
4.8.1 General 4-2
4.8.2 Construction Records 4-3
4.8.3 Inspection History 4-3
4.8.4 Repair/Alteration History 4-3
4.9 Reports 4-3
4.10 Inspector Qualifications 4-3

SECTION 5—MATERIALS
5.1 General 5-1
5.2 New Materials 5-1
5.3 Original Materials for Reconstructed Tanks 5-1
5.3.1 Shell and Bottom Plates Welded to the Shell 5-1
5.3.2 Structural 5-1
5.3.3 Flanges and Fasteners 5-1
5.3.4 Roof, Bottom, and Plate Wind Girders 5-1
5.4 Welding Consumables 5-1

SECTION 6—DESIGN CONSIDERATIONS FOR RECONSTRUCTED TANKS
6.1 General 6-1
6.2 New Weld Joints 6-1
6.3 Existing Weld Joints 6-1
6.4 Shell Design 6-1
6.5 Shell Penetrations 6-1
6.6 Wind Girders and Shell Stability 6-1
6.7 Roofs 6-1
6.8 Seismic Design 6-1

SECTION 7—TANK REPAIR AND ALTERATION
7.1 General 7-1
7.2 Removal and Replacement of Shell Plate Material 7-1
7.2.1 Minimum Thickness of Replacement Shell Plate 7-1
7.2.2 Minimum Dimensions of Replacement Shell Plate 7-1
7.2.3 Weld Joint Design 7-1
7.3 Repair of Defects in Shell Plate Material 7-1
7.4 Alteration of Tank Shells to Change Shell Height 7-2
7.5 Repair of Defective Welds 7-2
7.6 Repair of Shell Penetrations 7-2
7.7 Addition or Replacement of Shell Penetrations 7-3

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

7.8 Alteration of Existing Shell Penetrations 7-3
7.9 Repair of Tank Bottoms 7-4
7.9.1 Repairing a Portion of Tank Bottoms 7-4
7.9.2 Replacement of Entire Tank Bottom 7-4
7.10 Repair of Fixed Roofs 7-5
7.10.1 Supported Cone Roofs 7-5
7.10.2 Self-Supporting Roofs 7-5
7.11 Floating Roofs 7-5
7.11.1 External Floating Roofs 7-5
7.11.2 Internal Floating Roofs 7-5
7.11.3 Repair of Leaks in Pontoons 7-5
7.12 Repair or Replacement of Floating Roof Perimeter Seals 7-5
7.12.1 Primary Seals 7-5
7.12.2 Secondary Seals 7-6
7.12.3 Seal-to-Shell Gap 7-6
7.12.4 Mechanical Damage 7-6
7.12.5 Deterioration of Seal Material 7-6
7.12.6 Installation of Primary and Secondary Seals 7-6
7.13 Hot Taps 7-6
7.13.1 General 7-6
7.13.2 Hot Tap Procedures 7-6
7.13.3 Preparatory Work 7-6
7.13.4 Material Limitations 7-6
7.13.5 Installation Procedure 7-6

SECTION 8—DISMANTLING AND RECONSTRUCTION
8.1 General 8-1
8.2 Cleaning and Gas Freeing 8-1
8.3 Dismantling Methods 8-1
8.3.1 General 8-1
8.3.2 Bottoms 8-1
8.3.3 Shells 8-1
8.3.4 Roofs 8-1
8.3.5 Piece Marking 8-1
8.4 Reconstruction 8-1
8.4.1 General 8-1
8.4.2 Welding 8-2
8.4.3 Bottoms 8-2
8.4.4 Shells 8-3
8.4.5 Roofs 8-3
8.5 Dimensional Tolerances 8-3
8.5.1 General 8-3
8.5.2 Plumbness 8-3
8.5.3 Roundness 8-3
8.5.4 Peaking 8-3
8.5.5 Banding 8-3
8.5.6 Foundations 8-3

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

SECTION 9—WELDING
9.1 Welding Qualifications 9-1
9.2 Identification and Records 9-1

SECTION 10—EXAMINATION AND TESTING
10.1 Nondestructive Examinations 10-1
10.1.1 General 10-1
10.1.2 Shell Penetrations 10-1
10.1.3 Repaired Weld Flaws 10-1
10.1.4 Temporary and Permanent Attachments to Shell Plates 10-1
10.1.5 Shell Plate to Shell Plate Welds 10-1
10.1.6 Shell-to-Bottom Weld 10-1
10.1.7 Bottoms 10-1
10.2 Radiographs 10-1
10.2.1 Number and Location of Radiographs 10-1
10.2.2 Acceptance Criteria for Existing Shell Plate to Shell Plate Welds 10-2
10.2.3 Marking and Identification of Radiographs 10-2
10.3 Hydrostatic Testing 10-2
10.3.1 When Hydrostatic Testing Required 10-2
10.3.2 When Hydrostatic Testing Not Required 10-2
10.4 Leak Tests 10-3
10.5 Measured Settlement During Hydrostatic Testing 10-3
10.5.1 Initial Survey 10-3
10.5.2 Survey During Hydrostatic Testing 10-3

SECTION 11—MARKING AND RECORDKEEPING
11.1 Nameplates 11-1
11.2 Recordkeeping 11-1
11.3 Certification 11-1

APPENDIX A—BACKGROUND ON PAST EDITIONS OF API WELDED STORAGE TANK STANDARDS A-1
APPENDIX B—EVALUATION OF TANK BOTTOM SETTLEMENT B-1
APPENDIX C—CHECKLISTS FOR TANK INSPECTION C-1
APPENDIX D—AUTHORIZED INSPECTOR CERTIFICATION D-1
APPENDIX E—TECHNICAL INQUIRIES E-1

Figures
2-1—Inspection of Corrosion Areas 2-2
2-2—Pit Measurement 2-3
3-1—Brittle Fracture Considerations 3-2
3-2—Exemption Curve for Tanks Constructed From Carbon Steel of Unknown Toughness 3-3
7-1—Acceptable Details for Replacement of Shell Plate Material 7-2
7-2—Typical Details for Addition of Reinforcing Plate to Existing Shell Penetration 7-3
7-3—Typical Details for Addition of "Tombstone" Shape Reinforcing Plate to Existing Shell Penetration 7-4
7-4—Method for Raising Shell Nozzles 7-5
7-5—Exemption Curve for Hot Tapping Tanks With Steels of Unknown Toughness 7-7

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Page

Figures

7-6—Hot Tap for Tanks 7-8
8-1—Tank Shell and Bottom Cut Locations 8-1
10-1—Exemption Curve for Hydrostatic Testing of Tanks 10-3
11-1—Nameplate 11-2
11-2—Certification Forms 11-3
B-1—Measurements of Shell Settlement (External) B-1
B-2—Measurements of Bottom Settlement (Internal) Tank Out-of-Service B-2
B-3—Graphical Representation of Tank Shell Settlement B-3
B-4—Edge Settlement B-4
B-5—Bottom Settlement Near Shell B-4
B-6—Localized Bottom Depressions or Bulges Remote From Shell B-5
B-7—Localized Tank Bottom Settlement Limits for Single Pass Welds B-6

Tables

2-1—Joint Efficiencies for Welded Joints 2-4
2-2—Joint Efficiencies for Riveted Joints 2-4
2-3—Annular Bottom Plate Thicknesses (Inches)
(Product Specific Gravity <1.0) 2-7
4-1—Bottom Plate Minimum Thickness 4-2
7-1—Hot Tap Connection Sizes and Shell Plate Thicknesses 7-6
8-1—Maximum Thicknesses on New Welds 8-2
8-2—Radii Tolerances 8-3
10-1—Conditions for Exemption From Tank Hydrostatic Test for
Major Repairs and Alterations 10-3
A-1—Editions of API Standard 650 and Its Precursor, Standard 12C A-1

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

Tank Inspection, Repair, Alteration, and Reconstruction

SECTION 1—INTRODUCTION

1.1 General

1.1.1 This standard covers carbon and low alloy steel tanks built to API Standard 650 and its predecessor 12C. This standard provides minimum requirements for maintaining the integrity of welded or riveted, non-refrigerated, atmospheric pressure, aboveground storage tanks after they have been placed in service. It covers the maintenance inspection, repair, alteration, relocation, and reconstruction of such tanks.

1.1.2 The scope is limited to the tank foundation, bottom, shell, structure, roof, attached appurtenances, and nozzles to the face of the first flange, first threaded joint, or first welding-end connection. Many of the design, welding, examination, and material requirements of API Standard 650 can be applied in the maintenance inspection, rating, repair, and alteration of in-service tanks. In the case of apparent conflicts between the requirements of this standard and API Standard 650 or its predecessor API 12C, this standard shall govern for tanks that have been placed in service.

1.1.3 This standard employs the principles of API Standard 650; however, storage tank owner/operators, based on consideration of specific construction and operating details, may apply this standard to any steel tank constructed in accordance with a tank specification.

1.1.4 This standard is intended for use by organizations that maintain or have access to engineering and inspection personnel technically trained and experienced in tank design, fabrication, repair, construction, and inspection.

1.1.5 This standard does not contain rules or guidelines to cover all the varied conditions which may occur in an existing tank. When design and construction details are not given, and are not available in the standard to which the tank was originally constructed, then details that will provide a level of integrity equal to the level provided by the current edition of API Standard 650 must be used.

1.2 Compliance With This Standard

The owner/operator has ultimate responsibility for complying with the provisions of this standard. Should a party other than the owner/operator be assigned certain tasks, such as relocating and reconstructing a tank, the limits of responsibility for each party shall be defined by the owner/operator prior to commencing work.

1.3 Jurisdiction

If any provision of this standards presents a direct or implied conflict with any statutory regulation, the regulation shall govern. However, if the requirements of this standard are more stringent than the requirements of the regulation, then the requirements of this standard shall govern.

1.4 Safe Working Practices

An assessment shall be made of the potential hazards to which personnel may be exposed when conducting internal tank inspections, making repairs, or dismantling tanks. Procedures shall be developed according to the guidelines given in API Publications 2015, 2015A, 2015B, 2207, and 2217 that will include safeguard for personnel health and safety, prevention of accidental fires and explosions, and the prevention of property damage.

Special procedures may need to be developed for certain activities described in this standard that are not fully covered by the referenced API publications; for example, safety precautions for personnel accessing floating roof tanks that are in service, or gas freeing the bottom side of a tank. Finally, procedures must comply with any federal or state safety regulations pertaining to "confined spaces" or any other relevant provisions.

1.5 Definitions

1.5.1 *Alteration* means any work on a tank involving cutting, burning, welding, or heating operations that changes the physical dimensions and/or configuration of a tank. Typical examples of alterations include:

a. The addition of manways and nozzles greater than 12-inch nominal pipe size (NPS).
b. An increase or decrease in tank shell height.

1.5.2 *Applicable standard* refers to the original standard of construction, such as API standards or specifications or Underwriter Laboratories (UL) standards, unless the original standard of construction has been superseded or withdrawn from publication; in this event, applicable standard means the current edition of the appropriate standard. See Appendix A for background on editions of API welded storage tank standards.

1.5.3 *Atmospheric pressure* is used to describe tanks designed to withstand an internal pressure up to but not exceeding 2½ pounds per square inch gauge.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607796 85T ■

1.5.4 An *Authorized Inspection Agency* is one of the following organizations or individuals.

a. The inspection organization of the jurisdiction in which the aboveground storage tank is operated.
b. The inspection organization of an insurance company which is licensed or registered to and does write aboveground storage tank insurance.
c. An owner or operator of one or more aboveground storage tank(s) who maintains an inspection organization for activities relating only to his equipment, and not for aboveground storage tanks intended for sale or resale.
d. An independent organization or individual under contract to and under the direction of an owner or operator and recognized or otherwise not prohibited by the jurisdiction in which the aboveground storage tank is operated. The owner or operator's inspection program shall provide the controls necessary for use by Authorized Inspectors contracted to inspect aboveground storage tanks.

1.5.5 *Change in service* means a change from previous operating conditions involving different properties of the stored product such as specific gravity or corrosivity and/or different service conditions of temperature and/or pressure.

1.5.6 *Hot tap* identifies a procedure for installing a nozzle in the shell of a tank that is in service.

1.5.7 An *Inspector* as used in this document is an Authorized Inspector and is an employee of an Authorized Inspection Agency who is qualified and certified to perform tank inspections under this standard.

1.5.8 *Owner/operator* refers to the legal entity having both control of and/or responsibility for the operation and maintenance of an existing storage tank.

1.5.9 *Reconstruction* means the work necessary to reassemble a tank that has been dismantled and relocated to a new site.

1.5.10 *Reconstruction organization* refers to that organization having assigned responsibility by the owner/operator to design and/or reconstruct a tank.

1.5.11 *Repair* means any work necessary to maintain or restore a tank to a condition suitable for safe operation. Typical examples of repairs includes:

a. Removal and replacement of material (such as roof, shell, or bottom material, including weld metal) to maintain tank integrity.
b. Re-leveling and/or jacking of a tank shell, bottom, or roof.
c. Addition of reinforcing plates to existing shell penetrations.
d. Repair of flaws, such as tears or gouges, by grinding and/or gouging followed by welding.

1.6 Referenced Publications

The following standards, codes, publications, and specifications are cited in this standard. The latest edition or revision shall be used unless otherwise noted.

API

Std 620	*Design and Construction of Large, Welded Low-Pressure Storage Tanks*
Std 650	*Welded Steel Tanks for Oil Storage*
RP 651	*Cathodic Protection of Aboveground Petroleum Storage Tanks*
RP 652	*Lining of Aboveground Petroleum Storage Tank Bottoms*
Std 2000	*Venting Atmospheric and Low-Pressure Storage Tanks (Non-refrigerated and Refrigerated)*
RP 2003	*Protection Against Ignitions Arising Out of Static, Lightning, and Stray Currents*
Publ 2015	*Cleaning Petroleum Storage Tanks*
Publ 2015A	*Guide for Controlling the Lead Hazard Associated with Tank Entry and Cleaning*
Publ 2015B	*Cleaning Open-Top and Covered Floating-Roof Tanks*
Publ 2201	*Procedure for Welding or Hot Tapping on Equipment Containing Flammables*
Publ 2217	*Guidelines for Confined Space Work in the Petroleum Industry*

ASME[1]

Boiler and Pressure Vessel Code, Section V, "Nondestructive Examination"; Section VIII, "Pressure Vessels" Alternative Rules, Division 2; Section IX, "Welding and Brazing Qualifications"

ASTM[2]

A6	*General Requirements for Rolled Steel Plates, Shapes, Sheet Piling, and Bars for Structural Use*
A20	*General Requirements for Steel Plates for Pressure Vessels*
A36	*Structural Steel*
A370	*Standard Test Methods and Definitions for Mechanical Testing of Steel Products*

[1] American Society for Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[2] American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 2—SUITABILITY FOR SERVICE

2.1 General

2.1.1 When the results of a tank inspection show that a change has occurred from the original physical condition of that tank, an evaluation shall be made to determine its suitability for continued use.

2.1.2 This section provides an evaluation of the suitability of an existing tank for continued service, or for a change of service, or when making decisions involving repairs, alterations, dismantling, relocating, or reconstructing an existing tank.

2.1.3 The following list of factors for consideration is not all-inclusive for all situations, nor is it intended to be a substitute for the engineering analysis and judgements required for each situation:

a. Internal corrosion due to the product stored or water bottoms.
b. External corrosion due to environmental exposure.
c. Stress levels and allowable stress levels.
d. Properties of the stored product such as specific gravity, temperature, and corrosivity.
e. Metal design temperatures at the service location of the tank.
f. External roof live load, wind, and seismic loadings.
g. Tank foundation, soil, and settlement conditions.
h. Chemical analysis and mechanical properties of the materials of construction.
i. Distortions of the existing tank.
j. Operating conditions such as filling/emptying rates and frequency.

2.2 Tank Roof Evaluation

2.2.1 GENERAL

2.2.1.1 The structural integrity of the roof and roof support system shall be verified.

2.2.1.2 Roof plates corroded to an average thickness of less than 0.09 inch in any 100 square inch area or roof plates with any holes through the roof plate shall be repaired or replaced.

2.2.2 FIXED ROOFS

Roof support members (rafters, girders, columns, and bases) shall be inspected for soundness by a method acceptable to the responsible inspector; corroded and damaged members shall be evaluated and repaired or replaced if necessary. Particular attention must be given to the possibility of severe internal corrosion of pipe columns (corrosion may not be evidenced by external visual inspection).

2.2.3 FLOATING ROOFS

2.2.3.1 Areas of roof plates and pontoons exhibiting cracks or punctures shall be repaired or the affected sections replaced.

2.2.3.2 Areas that are pitted shall be evaluated to determine the likelihood of through-pitting occurring before the next scheduled internal inspection. If so, the affected areas shall be repaired or replaced.

2.2.3.3 Roof support systems, perimeter seal systems, appurtenances such as a roof rolling ladder, anti-rotation devices, water drain systems, and venting systems shall be evaluated for needed repairs or replacements.

2.2.3.4 Guidance for the evaluation of existing floating roofs shall be based on the criteria of API Standard 650, Appendix C for external floating roofs, and Appendix H for internal floating roofs. However, upgrading to meet this standard is not mandatory.

2.2.4 CHANGE OF SERVICE

2.2.4.1 Internal Pressure

All requirements of the applicable standard (for example, API 650, Appendix F) shall be considered in the evaluation and subsequent modifications to the tank roof and roof-to-shell junction.

2.2.4.2 External Pressure

As applicable, the roof support structure (if any), and the roof-to-shell junction shall be evaluated for the effects of a design partial vacuum. The criteria outlined in API Standard 620 shall be used.

2.2.4.3 Operation at Elevated Temperature

All requirements of API Standard 650, Appendix M, shall be considered before changing the service of a tank to operation at temperatures above 200°F.

2.2.4.4 Operation at Lower Temperature Than Original Design

If the operating temperature is changed to a lower temperature than the original design, the requirements of the

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607798 622 ■

applicable standard or API Standard 650 for the lower temperature shall be met.

2.2.4.5 Normal and Emergency Venting

Effects of change of service on normal and emergency venting shall be considered.

2.3 Tank Shell Evaluation

2.3.1 GENERAL

2.3.1.1 Flaws, deterioration, or other conditions (for example, change of service, relocation, corrosion greater than the original corrosion allowance) that might adversely affect the performance or structural integrity of the shell of an existing tank must be evaluated and a determination made regarding suitability for intended service.

2.3.1.2 The evaluation of the existing tank shell shall be conducted by personnel experienced in tank design and shall include an analysis of the shell for the intended design conditions, based on existing shell plate thickness and material. The analysis shall take into consideration all anticipated loading conditions and combinations, including pressure due to fluid static head, internal and external pressure, wind loads, seismic loads, roof live loads, nozzle loads, settlement, and attachment loads.

2.3.1.3 Shell corrosion occurs in many forms and varying degrees of severity and may result in a generally uniform loss of metal over a large surface area or in localized areas. Pitting may also occur. Each case must be treated as a unique situation and a thorough inspection conducted to determine the nature and extent of corrosion prior to developing a repair procedure. Pitting does not normally represent a significant threat to the overall structural integrity of a shell unless present in a severe form with pits in close proximity to one another. Criteria for evaluating both general corrosion and pitting are defined below.

2.3.1.4 Methods for determining the minimum shell thickness suitable for continued operation are given in 2.3.2, 2.3.3, and 2.3.4. (See Section 4 for frequency of inspection.)

2.3.1.5 If the requirements of 2.3.3 (welded) or 2.3.4 (riveted) cannot be satisfied, the corroded or damaged areas shall be repaired, or the allowable liquid level of the tank reduced, or the tank retired. The allowable liquid level for the continued use of a tank may be established by using the formulas for a minimum acceptable thickness (see 2.3.3.1 and 2.3.4.1) and solving for height H. The actual thickness, as determined by inspection, minus the corrosion allowance shall be used to establish the liquid level limit. The maximum design liquid level shall be exceeded.

2.3.2 ACTUAL THICKNESS DETERMINATION

2.3.2.1 For determining the controlling thicknesses in each shell course when there are corroded areas of considerable size, measured thicknesses shall be averaged in accordance with the following procedure (see Figure 2-1):

a. For each area, the inspector shall determine the minimum thickness, t_2, at any point in the corroded area, excluding widely scattered pits (see 2.3.2.2).

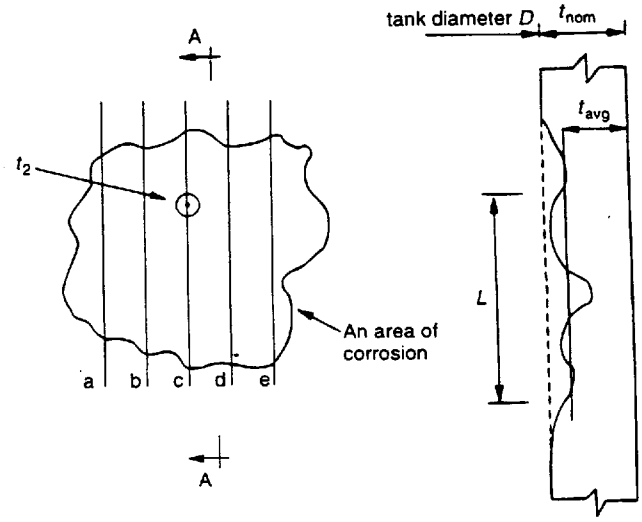
b. Then, calculate the critical length, L:

$$L = 3.7\,Dt_2, \text{ but not more than 40 inches.}$$

Where:

L = The maximum vertical length, in inches, over which hoop stresses are assumed to "average out" around local discontinuities.

Note: The actual vertical length of the corroded area may *exceed L*.



Legend:
a–e are inspection planes selected by inspector.

t_2 = least min. thickness in entire area, exclusive of pits.

SECTION A-A
Profile along plane "c", the plane having the lowest average thickness, t_1.

Procedure:
1. Determine t_2.
2. Calculate $L = 3.7\sqrt{Dt_2}$, but not more than 40".
3. Locate "L" to get minimum t_{avg}, which is t_1.

Figure 2-1—Inspection of Corrosion Areas

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607799 569 ■

D = Tank diameter, in feet.
t_2 = The least thickness, in inches, in an area of corrosion, exclusive of pits.

c. Then, the inspector shall visually or otherwise decide which vertical plane(s) in the area is likely to be the most affected by corrosion. Profile measurements shall be taken along each vertical plane for a distance, L. In the plane(s), determine the lowest average thickness, t_1, averaged over a length of L, using at least five equally spaced measurements over length L.
d. Refer to 2.3.3.1 for minimum permitted values for t_1 and t_2. The additional loads in 2.3.3.4 shall also be considered.

2.3.2.2 Widely scattered pits may be ignored provided that:

a. No pit depth results in the remaining shell thickness being less than one-half the minimum acceptable tank shell thickness exclusive of the corrosion allowance; and
b. The sum of their dimensions along any vertical line does not exceed 2 inches in an 8-inch length (see Figure 2-2).

2.3.3 MINIMUM THICKNESS CALCULATION FOR WELDED TANK SHELL

2.3.3.1 The minimum acceptable shell plate thickness for tanks with diameters equal to or less than 200 feet shall be calculated as follows:

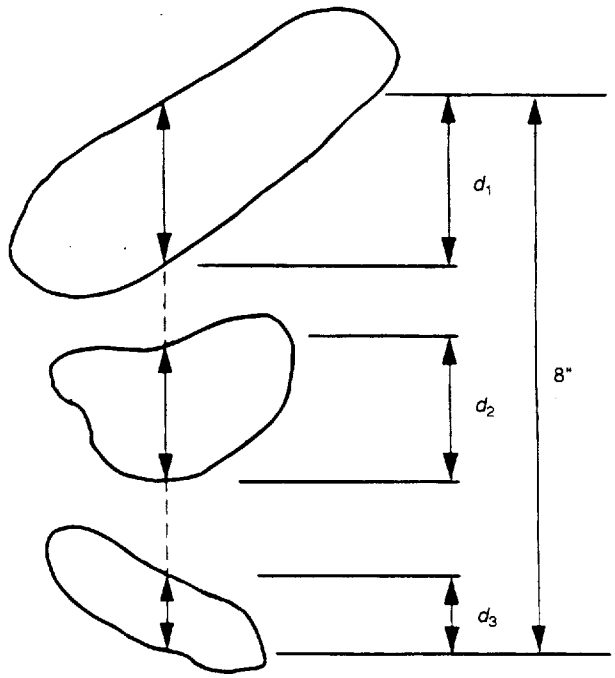


Figure 2-2—Pit Measurement

$$t_{min} = \frac{2.6D(H-1)G}{SE}$$

Where:

t_{min} = The minimum acceptable thickness, in inches, as calculated from the above formula; however, t_{min} shall not be less than 0.1 inch for any tank course.
D = Nominal diameter of tank, in feet.
H = Height, in feet, from the bottom of the length L (see 2.3.2.1) for the most severely corroded area in each shell course to the maximum design liquid level.
G = Highest specific gravity of the contents (including test water if tank will, or may, be tested in the future).
S = Maximum allowable stress in pounds per square inch; use the smaller of $0.80Y$ or $0.426T$ for bottom and second course; use the smaller of $0.88Y$ or $0.472T$ for all other courses. Note: For reconstructed tanks, S shall be in accordance with the current applicable standard.
Y = Specified minimum yield strength of the plate; use 30,000 pounds per square inch if not known.
T = The smaller of the specified minimum tensile strength of the plate or 80,000 pounds per square inch; use 55,000 pounds per square inch if not known.
E = Original joint efficiency for the tank. Use Table 2-1 if original E is unknown. $E = 1.0$ when evaluating the retirement thickness in a corroded plate, when away from welds or joints by at least the greater of one inch or twice the plate thickness.

The criteria for continued operation is:

a. The value t_1 (see 2.3.2.1.c) shall be greater than or equal to t_{min} (see above), subject to verification of all other loadings listed below; and
b. The value t_2 (see 2.3.2.1.b) shall be greater than or equal to 60 percent of t_{min}.

Any corrosion allowance for service until the time of the next tank inspection shall be added to the two required thicknesses (t_{min} and $0.60\ t_{min}$) in a. and b. above.

2.3.3.2 Alternatively, the minimum acceptable shell plate thickness for tanks with diameters equal to or less than 200 feet may be calculated in accordance with the variable design point method in API Standard 650, 3.6.4, substituting "S × E" for "S"; E and S may be defined as in 2.3.3.1.

2.3.3.3 The variable design point method shall be used for tanks greater than 200 feet in diameter, with all variables defined as in 2.3.3.1.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607800 000 ■

Table 2-1—Joint Efficiencies for Welded Joints

Standard	Edition & Year	Type of Joint	Joint Efficiency E	Applicability or Limits
API 650	7th & Later	Butt	1.00	Basic Standard
	(1980–Present)	Butt	0.85	Appendix A–Spot RT
		Butt	0.70	Appendix A–No RT
	1st–6th	Butt	0.85	Basic Standard
	(1961–1978)	Butt	1.00	Appendices D&G
API 12C	14th & 15th (1957–1958)	Butt	0.85	
	3rd–13th (1940–1956)	Lap[a]	0.75	3/8 inch max. *t*
		Butt[c]	0.85	
	1st & 2nd (1936–1939)	Lap[a]	0.70	7/16 inch max. *t*
		Lap[b]	0.50 + k/5	1/4 inch max. *t*
		Butt[c]	0.85	
Unknown		Lap[a]	0.70	7/16 inch max. *t*
		Lap[b]	0.50 + k/5	1/4 inch max. *t*
		Butt[c]	0.70	
		Lap[d]	0.35	

[a]Full double lap welded.
[b]Full fillet weld with at least 25 percent intermittent full fillet opposite side; k = percent of intermittent weld expressed in decimal form.
[c]Single butt-welded joints with a back-up bar were permitted from the years of 1936 to 1940 and 1948 to 1954.
[d]Single lap welded only.

2.3.3.4 The thickness determinations of 2.3.3.1, 2.3.3.2, and 2.3.3.3 consider liquid loading only. All other loads shall also be evaluated according to the original standard of construction; and engineering judgement shall be used to evaluate different conditions or new information. As applicable, the following loadings shall be taken into account:

a. Wind-induced buckling.
b. Seismic loads.
c. Operation at temperatures over 200°F.
d. Vacuum-induced external pressure.
e. External loads caused by piping, tank-mounted equipment, hold down lugs, etc.
f. Wind-induced overturning.
g. Loads due to settlement.

2.3.3.5 As an alternative to the procedures described above, any thinning of the tank shell below minimum required wall thickness due to corrosion or other wastage may be evaluated to determine the adequacy for continued service by employing the "design by analysis methods," defined in Section VIII, Division 2, Appendix 4 of the ASME Code. When using these criteria, the stress value used in the original tank design shall be substituted for the S_m value of Division 2, if the design stress is less than or equal to the lesser of $\frac{2}{3}Y$ (Specified Minimum Yield Strength) or $\frac{1}{3}T$ (Specified Minimum Tensile Strength). If the original design stress is greater than $\frac{2}{3}Y$ or $\frac{1}{3}T$, then the lesser of $\frac{2}{3}Y$ or $\frac{1}{3}T$ shall be substituted for S_m.

2.3.4 MINIMUM THICKNESS CALCULATION FOR RIVETED TANK SHELL

2.3.4.1 The minimum acceptable thickness for riveted tank shells shall be calculated using the formula of 2.3.3.1 except that the following allowable stress criteria and joint efficiencies shall be used:

S = 21,000 pounds per square inch

E = 1.0 for shell plate 6 inches or more away from rivets. See Table 2-2 for joint efficiencies for locations within 6 inches or rivets.

2.3.4.2 The rivet joint efficiencies given in Table 2-2 are conservative minimums for riveted tank construction details and are included to simplify riveted tank evaluations. However, in some cases it may be advantageous to calculate the actual rivet joint efficiencies using computational methods applicable to lap and butt type riveted joints. When this alternative of calculated joint efficiencies is used, the following maximum allowable stresses shall apply:

a. For the maximum tensile stress in net section of plate, use the lesser of $0.80Y$ or $0.426T$; use 21,000 pounds per square inch if T or Y is unknown.
b. For the maximum shear in net section of rivet, use 16,000 pounds per square inch.
c. For the maximum bearing stress on plates or rivets, use 32,000 pounds per square inch for rivets in single shear, and 35,000 pounds per square inch for rivets in double shear.

2.3.4.3 For tanks with riveted joints, consideration shall be given to whether, and to what extent, corrosion affects such joints. If calculations show that excess thickness exists, this excess may be taken as corrosion allowance.

2.3.4.4 Non-liquid loads (see 2.3.3.4) shall also be considered in the analysis of riveted tanks.

2.3.5 DISTORTIONS

2.3.5.1 Shell distortions include out-of-roundness, buckled areas, flat spots, and peaking and banding at welded joints.

Table 2-2—Joint Efficiencies for Riveted Joints

Type of Joint	Number of Rivet Rows	Joint Efficiency E
Lap	1	0.45
Lap	2	0.60
Lap	3	0.70
Lap	4	0.75
Butt[a]	2[b]	0.75
Butt	3[b]	0.85
Butt	4[b]	0.90
Butt	5[b]	0.91
Butt	6[b]	0.92

[a]All butt joints listed have butt straps both inside and outside.
[b]Number of row on each side of joint center line.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607801 T47 ■

2.3.5.2 Shell distortions can be caused by many conditions such as foundation settlement, over- or under-pressuring, high wind, poor shell fabrication, or repair techniques, and so forth.

2.3.5.3 Shell distortions shall be evaluated on an individual basis to determine if specific conditions are considered acceptable for continuing tank service and/or the extent of corrective action.

2.3.6 FLAWS

Flaws such as cracks or laminations shall be thoroughly examined and evaluated to determine their nature and extent and need for repair. If a repair is needed, a repair procedure shall be developed and implemented. The requirement for repairing scars such as arc strikes, gouges, or tears from temporary attachment welds must be evaluated on a case-by-case basis. Cracks in the shell-to-bottom weld are critical and shall be removed and the weld repaired.

2.3.7 WIND GIRDERS AND SHELL STIFFENERS

The evaluation of an existing tank shell for suitability for service must also consider the details and condition of any wind girders or shell stiffeners. Degradation by corrosion of these structural elements or their attachment welds to the shell may render these elements inadequate for the design conditions.

2.3.8 SHELL WELDS

The condition of the tank shell welds shall be evaluated for suitability for service. Any deterioration of the existing welds that results from corrosion or pitting must be evaluated and appropriate repair procedures established or the tank rerated as necessary. Some typical shell butt-weld flaws and recommended procedures for repairs are given in 7.5.

2.3.9 SHELL PENETRATIONS

2.3.9.1 The condition and details of existing shell penetrations (nozzles, manways, cleanout openings, etc.) shall be reviewed when assessing the integrity of an existing tank shell. Details such as type and extent of reinforcement, weld spacing, and thickness of components (reinforcing plate, nozzle neck, bolting flange, and cover plate), are important considerations and shall be reviewed for structural adequacy and compliance with the applicable standard. Any noncompliance, or deterioration due to corrosion, must be assessed and repair procedures established where appropriate or the tank rerated, as necessary.

2.3.9.2 Nozzle wall thickness shall be evaluated for pressure and all other loads.

2.4 Tank Bottom Evaluation

2.4.1 GENERAL

Leaks in tank bottoms are not acceptable. Each aspect of corrosion phenomena, and other potential leak or failure mechanism must be examined. Periodic assessment of tank bottom integrity shall be performed in addition to the internal inspections specified in 4.4.2. The assessment period shall be less than or equal to the appropriate internal inspection interval given in 4.4.2. The use of leak detection tests or monitoring systems (such as double bottoms or liners under tank bottoms with leak detection pipes) will satisfy the requirement for periodic assessment between internal inspections.

Excessive foundation settlement of storage tanks can affect the integrity of tank shells and bottoms. Therefore, monitoring the settlement behavior of tanks is a recognized practice to assess the integrity of tank bottoms. Refer to Appendix B for techniques for evaluating tank bottom settlement.

2.4.2 CAUSES OF BOTTOM FAILURE

The following list gives some historical causes of tank bottom leakage or failure that shall be considered in the decision to line, repair, or replace a tank bottom:

a. Internal pitting and pitting rates in the anticipated service.
b. Corrosion of weld joints (weld and heat affected zone).
c. Weld joint cracking history.
d. Stresses placed on the bottom plates by roof support loads and shell settlement.
e. Underside corrosion (normally in the form of pitting).
f. Inadequate drainage resulting in surface water flowing under the tank bottom.
g. The lack of an annular plate ring when required.
h. Uneven settlement that results in high localized stresses in the bottom plates.
i. Roof support columns or other supports welded to the tank bottom where adequate allowance for movement was not made.
j. Rock or gravel foundation pads with inadequately filled-in surface voids.
k. Nonhomogeneous fill under the tank bottom (for example, a lump of clay in a sand foundation pad).
l. Inadequately supported sumps.

2.4.3 CATHODIC PROTECTION OF TANK BOTTOMS

A selection basis for cathodic protection systems for the underside of tank bottoms is covered by API RP 651.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607802 983 ■

2.4.4 INTERNAL LINING PROTECTION OF TANK BOTTOMS

Applied linings for internal surfaces of tank bottoms are covered by API RP 652.

2.4.5 BOTTOM LEAK DETECTION

If a tank bottom is to be replaced, consideration should be given to installing a leak detection (tell-tale) system that will channel any leak in the bottom to a location where it can be readily observed from the outside of the tank.[1]

2.4.6 BOTTOM PLATE THICKNESS MEASUREMENTS

Various methods for determining tank bottom plate underside corrosion are available. The methods vary by the extent that they can reliably measure general corrosion and pitting. A combination of these methods may be required along with extrapolation techniques and analysis to establish the probable conditions of the entire tank bottom.

2.4.7 MINIMUM THICKNESS FOR TANK BOTTOM PLATE

Quantifying the remaining thickness of tank bottoms based on the results of measurement (see 2.4.6) can be done by either a deterministic method (see 2.4.7.1) or a probabilistic method (see 2.4.7.2) as follows:

2.4.7.1 An acceptable deterministic method for calculating the minimum remaining bottom thicknesses is as follows:

$$MRT_1 = T_o - GC_a - StP_a - UP_m - (StP_r + UP_r + GC_r)O_r$$

$$MRT_2 = T_o - GC_a - StP_m - UP_a - (StP_r + UP_r + GC_r)O_r$$

Where:

MRT_1, MRT_2 = minimum remaining thickness at the end of the in-service period of operation, in inches. MRT_1 represents minimum remaining thickness due to average internal pitting and maximum external pitting. MRT_2 represents minimum remaining thickness due to maximum internal pitting and average external pitting.

T_o = original plate thickness, in inches.

StP_a = average depth of internal pitting, in inches, measured from the original thickness.

StP_m = maximum depth of internal pitting remaining in the bottom plates after repairs are completed, in inches, measured from the original thickness.

UP_a = average depth of underside pitting, in inches.

UP_m = maximum depth of underside pitting, in inches.

StP_r = maximum internal pitting rate, in inches per year; $StP_r = 0$ if the tank bottom is internally lined (see API RP 652).

UP_r = maximum underside pitting rate, in inches per year; $UP_r = 0$ if tank bottom is cathodically protected (see API RP 651).

O_r = anticipated in-service period of operation (normally 10 years).

GC_a = average depth of generally corroded area, in inches.

GC_r = maximum rate of general corrosion, in inches per year.

2.4.7.2 For the probabilistic method, a statistical analysis is made of thickness data from measurements (see 2.4.6) projecting remaining thickness, based on sample scanning of the bottom.

2.4.7.3 If the minimum bottom thicknesses, at the end of the in-service period of operation, are calculated to be less than the minimum bottom renewal thicknesses given in Table 4, the bottom shall be lined, repaired, replaced or the interval to the next internal inspection shortened.

2.4.7.4 Unless a stress analysis is performed, the minimum bottom plate thickness in the critical zone of the tank bottom defined by paragraph 7.9.1.1 shall be the smaller of ½ the original bottom plate thickness (not including the original corrosion allowance) or 50 percent of t_{min} of the lower shell course per paragraph 2.3.3.1 but not less than 0.1 inch. Isolated pitting will not appreciably affect the strength of the plate.

[1] For existing tanks, API supports the use of a Release Prevention System (RPS). The term RPS refers to the suite of API standards that are designed to maintain aboveground storage tank integrity and thus protect the environment. These are: the frequency of internal inspections; lining the bottom of the tank interior; fitting the tank with Release Prevention Barriers (RPBs); installing cathodic protections; or some combination of these measures, depending on the operating environment and service of the tank.

If a decision is made to replace an existing bottom, API supports the evaluation of installing an RPB or continued use of an RPS. The evaluation should consider the effectiveness of RPS controls, the product stored, the location of the tank, and environmental sensitivities. An RPB includes steel bottoms, synthetic materials, clay liners, and all other barriers or combination of barriers placed in the bottom of or under an aboveground storage tank, which have the functions of: 1) preventing the escape of contaminated material; and 2) containing or channeling released material for leak detection (covered in detail in the non-mandatory Appendix I of Standard 650).

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607803 81T ■

2.4.8 MINIMUM THICKNESS FOR ANNULAR PLATE RING

2.4.8.1 Due to strength requirements, the minimum thickness of annular plate ring is usually greater than 1.10 inch. Isolated pitting will not appreciably affect the strength of the plate. Unless a stress analysis is performed, the annular plate thickness shall be in accordance with 2.4.8.2 or 2.4.8.3, as applicable.

2.4.8.2 For tanks in service with a project specific gravity less than 1.0, which require annular plates for other than seismic loading considering, the thickness of the annular plates shall be not less than the thicknesses given in Table 2-3, plus any specified corrosion allowance.

2.4.8.3 For tanks in service with a product specific gravity of 1.0 or greater, which require annular plates for other than seismic loading considerations, the thickness of the annular plates shall be in accordance with Table 3-1 of API Standard 650 plus any specified corrosion allowance.

2.4.8.4 For tanks that utilize thickened annular plates for seismic considerations, a seismic evaluation shall be performed in accordance with the requirements of the applicable standard, using the actual thickness of the existing annular plate.

2.5 Tank Foundation Evaluation

2.5.1 GENERAL

2.5.1.1 The principal causes of foundation deterioration are settlement, erosion, cracking, and deterioration of concrete initiated by: calcining, attack by underground water, attack by frost, and attack by alkalies and acids. To ensure suitability for service, all tank foundations shall be inspected periodically (see 4.3).

Table 2-3—Annular Bottom Plate Thicknesses (Inches) (Product Specific Gravity < 1.0)

Plate Thickness[a] of First Shell Course (Inches)	Stress[b] in First Shell Course (pounds per square inch) <24,300	<27,000	<29,700	<32,400
$t \leq 0.75$	0.17	0.20	0.23	0.30
$0.75 < t \leq 1.00$	0.17	0.22	0.31	0.38
$1.00 < t \leq 1.25$	0.17	0.26	0.38	0.48
$1.25 < t \leq 1.50$	0.22	0.34	0.47	0.59
$t > 1.50$	0.27	0.40	0.53	0.68

Note: The thicknesses specified in the table are based on the foundation providing a uniform support under the full width of the annular plate. Unless the foundation is properly compacted, particularly at the inside of a concrete ringwall, settlement will produce additional stresses in the annular plate.
[a]Plate thickness refers to the tank shell as constructed.
[b]Stresses are calculated from $[2.34\ D\ (H\text{-}1)]/t$.

2.5.1.2 Some mechanisms of concrete deterioration are briefly described below:

a. Calcining (loss of water of hydration) can occur when concrete has been exposed to sufficiently high temperature for a period of time. During intermediate cooling periods, the concrete can absorb moisture, swell, lose its strength, and crack.

b. Deterioration of concrete exposed to underground water can be caused by chemical attack, by cyclic changes in temperature, and by freezing moisture.

c. Expansion of freezing moisture in porous concrete, or in concrete with minor settlement cracks or temperature cracks, can result in spalling and/or the development of serious structural cracks.

d. Sulfate-type alkalies, and to a lesser extent, chlorides, can act corrosively to destroy the bond of the concrete.

e. Temperature cracks (hairline cracks of uniform width) do not seriously affect the strength of the concrete foundation structure; however, these cracks can be potential access points for moisture or water seepage that could eventually result in corrosion of the reinforcing steel.

2.5.2 FOUNDATION REPAIR OR REPLACEMENT

2.5.2.1 If there is a need for foundation repair or replacement, foundations shall be restored to the tolerance limits of 8.5.6.

2.5.2.2 Concrete pads, ringwalls, and piers, showing evidence of spalling, structural cracks, or general deterioration, shall be repaired to prevent water from entering the concrete structure and corroding the reinforcing steel.

2.5.3 ANCHOR BOLTS

Distortion of anchor bolts and excessive cracking of the concrete structures in which they are embedded may be indications of either serious foundation settlement or a tank overpressure uplift condition.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 3—BRITTLE FRACTURE CONSIDERATIONS

3.1 General

This section provides a procedure for the assessment of existing tanks for suitability for continued operation or change of service with respect to the risk of brittle fracture and does not supplement or replace the requirements of Section 10 for the hydrotesting of repaired, modified, or reconstructed tanks. The procedure applies to both welded and riveted tanks; however, the procedure is based primarily on experience and data obtained from welded tanks.

3.2 Basic Considerations

A decision tree, Figure 3-1, is used to present the assessment procedure for failure due to brittle fracture. The decision tree is based on the following principles:

3.2.1 In all reported incidents of tank failure due to brittle fracture, failure occurred either shortly after erection during hydrostatic testing or on the first filling in cold weather, after a change to lower temperature service, or after a repair/alteration. This experience shows that once a tank has demonstrated the ability to withstand the combined effects of maximum liquid level (highest stresses) and lowest operating temperature without failing, the risk of failure due to brittle fracture with continued service is minimal.

3.2.2 Any change in service must be evaluated to determine if it increases the risk of failure due to brittle fracture. In the event of a change to a more severe service (such as operating at a lower temperature or handling product at a higher specific gravity) it is necessary to consider the need for a hydrostatic test to demonstrate fitness for a new more severe service. The following aspects should be considered:

a. The likelihood of repairs/alterations since the original hydrostatic test not meeting requirements of this standard.
b. Deterioration of the tank since original hydrostatic test.

3.3 Assessment Procedure

The assessment procedure as illustrated in Figure 3-1 shall be used for evaluating aboveground atmospheric storage tanks in petroleum and chemical services. Each of the key steps, numbered 1 through 11 on the decision tree, correspond sequentially to the explanations provided below.

3.3.1 These tanks meet the API Standard 650 (seventh edition or later) requirements to minimize the risk of failure due to brittle fracture. Tanks may also be shown to meet the toughness requirements of API Standard 650 (seventh edition or later) by impact testing coupon samples from a representative number of shell plates.

3.3.2 Many tanks that continue to operate successfully in the same service were not built to the requirements of API Standard 650 (seventh edition or later). These tanks are potentially susceptible to failure due to brittle fracture and require an assessment as illustrated by the decision tree.

3.3.3 For the purpose of this assessment, hydrostatic testing demonstrates that an aboveground atmospheric storage tank in a petroleum or chemical service is fit for continued service and at minimal risk of failure due to brittle fracture, provided that all governing requirements for repairs, alterations, reconstruction, or change in service are in accordance with this standard (including a need for hydrostatic testing after major repairs, modifications or reconstruction). The effectiveness of the hydrostatic test in demonstrating fitness for continued service is shown by industry experience.

3.3.4 If a tank shell thickness is no greater than 0.5 inch, the risk of failure due to brittle fracture is minimal, provided that an evaluation for suitability of service in accordance with Section 2 has been performed. The original nominal thickness for the thickest tank shell plate shall be used for this assessment.

3.3.5 No known tank failures due to brittle fracture have occurred at shell metal temperatures of 60°F or above. Similar assurance against brittle fracture can be gained by increasing the metal temperature by heating the tank contents.

3.3.6 Industry experience and laboratory tests have shown that a membrane stress in tank shell plates of at least 7 ksi is required to cause failure due to brittle fracture.

3.3.7 Tanks constructed from steels listed in Figure 2-1 of API Standard 650 can be used in accordance with their exemption curves, provided that an evaluate for suitability of service in conformance with Section 2 of this standard has been performed. Tanks fabricated from steels of unknown toughness thicker than ½ inch and operating at a shell metal temperature below 60°F can be used if the tank meets the requirements of Figure 3-2. The original nominal thickness for thickest tank shell plate shall be used for the assessment. For unheated tanks, the shell metal temperature shall be the design metal temperature as defined in 2.2.9.3 of API Standard 650.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

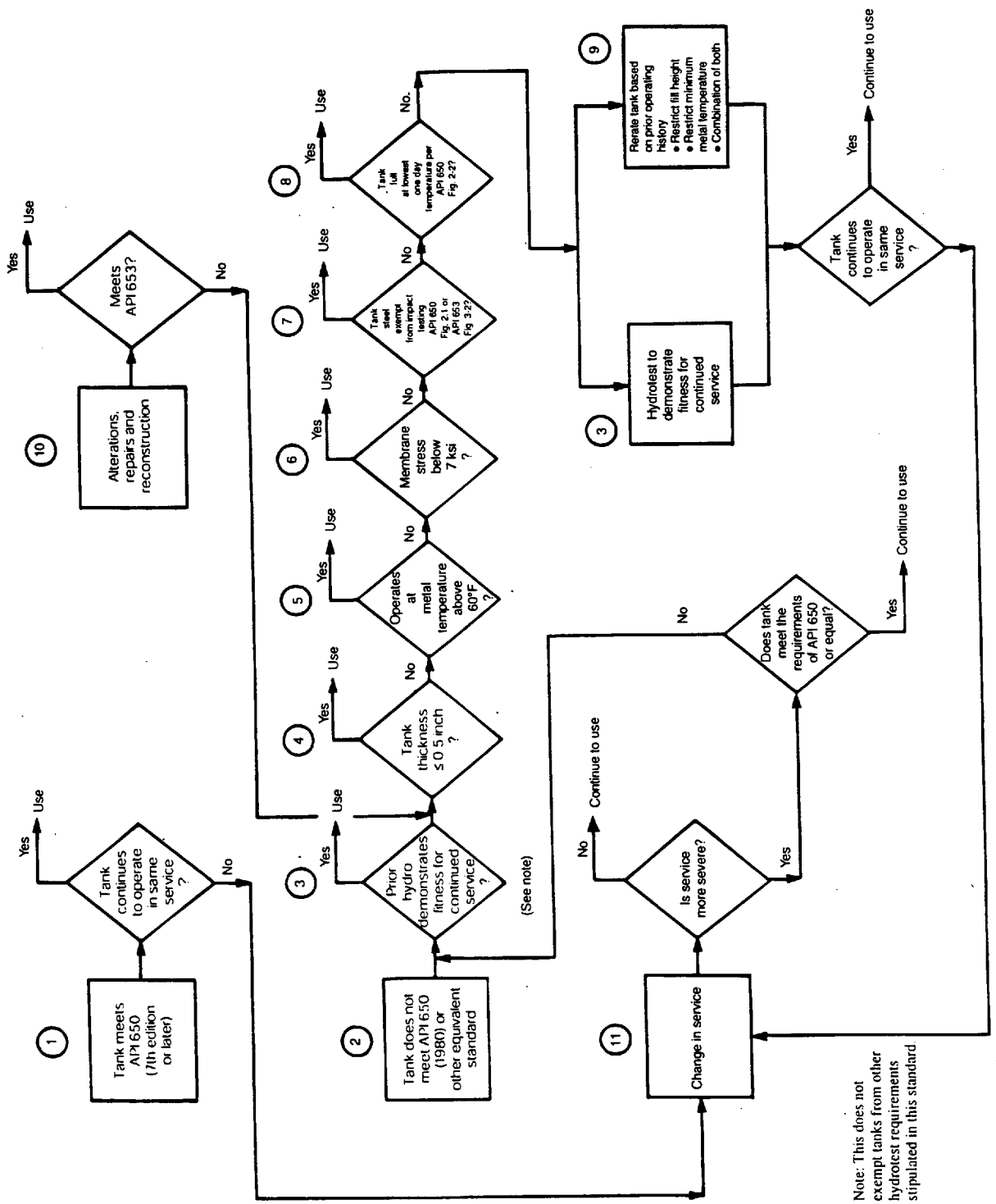


Figure 3-1—Brittle Fracture Considerations

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607806 529 ■

3.3.8 The risk of failure due to brittle fracture is minimal once a tank has demonstrated that it can operate at a specified maximum liquid level at the lowest expected temperature without failing. For the purpose of this assessment, the lowest expected temperature is defined as the lowest one-day mean temperature as shown in Figure 2-2 of API Standard 650 for the continental United States. It is necessary to check tank log records and meteorological records to ensure that the tank had operated at the specified maximum liquid level when the one-day mean temperature was as low as shown in Figure 2-2 of API Standard 650.

3.3.9 An evaluation can be performed to establish a safe operating envelope for a tank based on the past operating history. This evaluation shall be based on the most severe combination of temperature and liquid level experienced by the tank during its life. The evaluation may show that the tank needs to be rerated or operated differently; several options exist:

a. Restrict the liquid level.
b. Restrict the minimum metal temperature.
c. Change the service to a stored product with a lower specific gravity.
d. Combinations of a., b., and c., above.

The owner/operator can also make a more rigorous analysis to determine the risk of failure due to brittle fracture by performing a fracture mechanics analysis based upon established principles and practices. The procedures and acceptance criteria for conducting an alternative analysis are not included in this standard.

3.3.10 All repairs, alterations, and relocations shall be made in compliance with this standard.

3.3.11 An assessment shall be made to determine if the change in service places the tank at greater risk of failure due to brittle fracture. The service can be considered more severe and create a greater risk of brittle fracture if the service temperature is reduced (for example, changing from heated oil service to ambient temperature product), or the product is changed to one with a greater specific gravity and thus increase stresses.

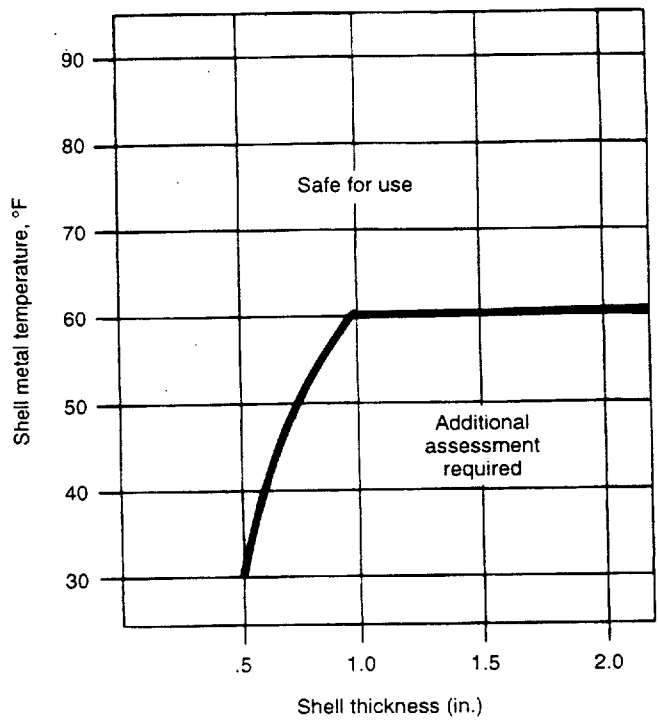


Figure 3-2—Exemption Curve for Tanks Constructed From Carbon Steel of Unknown Toughness

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 4—INSPECTION

4.1 General

Periodic in-service inspection of tanks shall be performed by an Authorized Inspector as defined herein (see 4.10), unless otherwise noted.

4.2 Inspection Frequency Considerations

4.2.1 Several factors must be considered to determine inspection intervals for storage tanks. These include, but are not limited to, the following:

a. The nature of the product stored.
b. The results of visual maintenance checks.
c. Corrosion allowances and corrosion rates.
d. Corrosion prevention systems.
e. Conditions at previous inspections.
f. The methods and materials of construction and repair.
g. The location of tanks, such as those in isolated or high risk areas.
h. The potential risk of air or water pollution.
i. Leak detection systems.
j. Change in operating mode (for example: frequency of fill cycling, frequent grounding of floating roof support legs).
k. Jurisdictional requirements.
l. Changes in service (including changes in water bottoms).

4.2.2 The interval between inspections of a tank (both internal and external) should be determined by its service history unless special reasons indicate that an earlier inspection must be made. A history of the service of a given tank or a tank in similar service (preferably at the same site) should be available so that complete inspections can be scheduled with a frequency commensurate with the corrosion rate of the tank. On-stream, nondestructive methods of inspection shall be considered when establishing inspection frequencies.

4.2.3 Jurisdictional regulations, in some cases, control the frequency and interval of the inspections. These regulations may include vapor loss requirements, seal condition, leakage, proper diking, and repair procedures. Knowledge of such regulations is necessary to insure compliance with scheduling and inspection requirements.

4.3 External Inspection

4.3.1 ROUTINE IN-SERVICE INSPECTIONS

4.3.1.1 The external condition of the tank shall be monitored by close visual inspection from the ground on a routine basis. This inspection may be done by owner/operator personnel, and can be done by other than inspectors described in 4.10. Personnel performing this inspection should be knowledgeable of the storage facility operations, the tank, and the characteristics of the product stored.

4.3.1.2 The interval of such inspections shall be consistent with conditions at the particular site, but shall not exceed one month.

4.3.1.3 This routine in-service inspection shall include a visual inspection of the tank's exterior surface checking for: leaks; shell distortions; signs of settlement; corrosion; and condition of the foundation, paint coatings, insulation systems, and appurtenances.

4.3.2 SCHEDULED INSPECTIONS

4.3.2.1 All tanks shall be given a formal visual external inspection by an inspector qualified in accordance with 4.10 at least every five years or at the quarter corrosion-rate life of the shell, whichever is less. Tanks may be in operation during this inspection.

4.3.2.2 Insulated tanks need to have insulation removed only to the extent necessary to determine the condition of the exterior wall of the tank or the roof.

4.3.2.3 Where exterior tank bottom corrosion is controlled by a cathodic protection system, periodic surveys of the system shall be conducted in accordance with API Recommended Practice 651.

4.3.2.4 Tank grounding system components such as shunts or mechanical connections of cables shall be visually checked. Recommended practices dealing with the prevention of hydrocarbon ignition are covered by API Recommended Practice 2003.

4.3.3 IN-SERVICE ULTRASONIC THICKNESS MEASUREMENTS OF THE SHELL

4.3.3.1 External, ultrasonic thickness measurements of the shell can be a means of determining a rate of uniform general corrosion while the tank is in service, and can provide an indication of the integrity of the shell. The extent of such measurements shall be determined by the owner/operator.

4.3.3.2 When used, the ultrasonic thickness measurements shall be made at intervals not to exceed the following:

a. When the corrosion rate is not known, the maximum interval shall be five years.

Note: Corrosion rates may be estimated from tanks in similar service based on thickness measurements taken at an interval not exceeding five years.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607808 3T1 ■

b. When the corrosion rate is known, the maximum interval shall be the smaller of *RCA/2N* years (where *RCA* is the remaining corrosion allowance in mils and *N* is the shell corrosion rate in mils per year) of 15 years.

4.3.3.3 Internal inspection of the tank shell, when the tank is out of service, can be substituted for a program of external ultrasonic thickness measurement if the internal inspection interval is equal to or less than the interval required in 4.3.3.2.b.

4.4 Internal Inspection

4.4.1 GENERAL

4.4.1.1 Internal inspection is primarily required to:

a. Ensure that the bottom is not severely corroded and leaking.
b. Gather the data necessary for the minimum bottom and shell thickness assessments detailed in Section 2. As applicable, these data shall also take into account external ultrasonic thickness measurements made during in-service inspections (see 4.3.3).
c. Identify and evaluate any tank bottom settlement.

4.4.1.2 All tanks shall be given a formal internal inspection at intervals defined by 4.4.2 below. The Authorized Inspector who is responsible for evaluation of a tank must visually examine each tank.

4.4.2 INSPECTION INTERVALS

4.4.2.1 Intervals between internal inspections shall be determined by the corrosion rates measured during previous inspections or anticipated based on experience with tanks in similar service. Normally, bottom corrosion rates will control and the inspection interval will be governed by the measured or anticipated corrosion rates and the calculations for minimum required thickness of tank bottoms (see 2.4.7). The actual inspection interval shall be set to ensure that the bottom plate minimum thicknesses at the next inspection are not less than the values listed in Table 4-1. In no case, however, shall the internal inspection interval exceed 20 years.

4.4.2.2 When corrosion rates are not known and similar service experience is not available to determine the bottom plate minimum thickness at the next inspection, the actual bottom thickness shall be determined by inspection(s) within the next 10 years of tank operation to establish corrosion rates.

4.4.3 ALTERNATIVE INTERNAL INSPECTION INTERVAL

For unique combinations of service, environment, and construction, the owner/operator may establish the internal inspection interval using an alternative procedure. The alternative procedure shall include method for determining bottom plate thickness, consideration of environmental risk, consideration of quality of inspection, and analysis of corrosion measurement. This alternative procedure shall be documented and made part of the permanent record of the tank.

4.5 Alternative to Internal Inspection to Determine Bottom Thickness

In cases where construction, size, or other aspects allow external access to the tank bottom to determine bottom thickness, an external inspection in lieu of an internal inspection is allowed to meet the data requirements of Table 4-1. However, in these cases, consideration of other maintenance items may dictate internal inspection intervals. This alternative approach shall be documented and made part of the permanent record of the tank.

4.6 Preparatory Work for Internal Inspection

Specific work procedures shall be prepared and followed when conducting inspections that will assure personnel safety and health and prevent property damage in the workplace (see 1.4).

4.7 Inspection Checklists

Appendix C provides sample checklists of items for consideration when conducting in-service and out-of-service inspections.

4.8 Records

4.8.1 GENERAL

Inspection records form the basis of a scheduled inspection/maintenance program. (It is recognized that records may not exist for older tanks, and judgements must be based on experience with tanks in similar services.) The owner/

Table 4-1—Bottom Plate Minimum Thickness

Minimum Bottom Plate Thickness[a] at Next Inspection (inches)	Tank Bottom/Foundation Design
0.10	Tank bottom/foundation design with no means for detection and containment of a bottom leak.
0.05	Tank bottom/foundation design with means to provide detection and containment of a bottom leak.
0.05	Applied tank bottom reinforced lining, >0.05 inch thick, in accordance with API RP 652.

[a] See 2.4.7

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607809 238 ■

operator shall maintain complete record file consisting of three types of records, namely: construction records, inspection history, and repair/alteration history.

4.8.2 CONSTRUCTION RECORDS

Construction records may include nameplate information, drawings, specifications, construction completion report, and any results of material tests and analyses.

4.8.3 INSPECTION HISTORY

The inspection history includes all measurements taken, the condition of all parts inspected, and a record of all examinations and tests. A complete description of any unusual conditions with recommendations for correction of details which caused the conditions shall also be included. This file will also contain corrosion rate and inspection interval calculations.

4.8.4 REPAIR/ALTERATION HISTORY

The repair/alteration history includes all data accumulated on a tank from the time of its construction with regard to repairs, alterations, replacements, and service changes (recorded with service conditions such as stored product temperature and pressure). These records should include the results of any experiences with coatings and linings.

4.9 Reports

4.9.1 Reports recommending repairs shall include reasons for the repairs, and sketches showing location and extent.

4.9.2 General inspection reports shall include metal thickness measurement, conditions found, repairs, any settlement measurements, and recommendations.

4.10 Inspector Qualifications

4.10.1 Qualified inspectors shall have education and experience equal to at least one of the following:

a. A degree in engineering plus one year of experience in inspection of tanks, pressure vessels, or piping.

b. A two-year certificate in engineering or technology from a technical college, and two years of experience in construction, repair, operation, or inspection, of which one year must be in inspection of tanks, pressure vessels, or piping.

c. The equivalent of a high school education, and three years of experience in construction, repair, operation, or inspection, of which one year must be in inspection of tanks, pressure vessels, or piping.

d. Five years of experience in the inspection of aboveground storage tanks in the petroleum or chemical industries.

4.10.2 An owner/operator of tanks may designate tank inspectors qualified in accordance with 4.10.1. Such inspectors shall have the necessary authority and organizational freedom to perform their duties. Authorized Inspectors shall be certified by an agency as provided in this standard, in accordance with Appendix D. This requirement will become effective eighteen (18) months after the date of issuance of the requirement.

4.10.3 Personnel performing nondestructive examinations shall meet the qualifications identified in 10.1.1.2, but need not be certified in accordance with 4.10.2. The results of their work, however, must be considered in the evaluation of the tank by an inspector who is so certified.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 5—MATERIALS

5.1 General

This section provides general requirements for the selection of materials for the repair, alteration, and reconstruction of existing tanks. Specific requirements for repairs and alterations are covered in Section 7.

5.2 New Materials

All new materials used for repair, alterations, or reconstruction shall conform to current applicable tank standards.

5.3 Original Materials for Reconstructed Tanks

5.3.1 SHELL AND BOTTOM PLATES WELDED TO THE SHELL

5.3.1.1 All shell plate materials and bottom plates welded to the shell shall be identified. Materials identified by original contract drawings, API name plates, or other suitable documentation do not require further identification. Material not identified shall be tested and identified by the requirements as outlined in 5.3.1.2. After identification, determination shall be made as to suitability of the material for intended service.

5.3.1.2 Each individual plate for which adequate identification does not exist shall be subjected to chemical analysis and mechanical tests as required in ASTM A6 and A370 including Charpy V-Notch. Impact values shall satisfy the requirements of API Standard 650. When the direction of rolling is not definitely known, two tension specimens shall be taken at right angles to each other from a corner of each plate, and one of those test specimen must meet the specification requirements.

5.3.1.3 For known materials, all shell plates and bottom plates welded to the shell shall meet, as a minimum, the chemistry and mechanical properties of material specified for the application with regard to thickness and design metal temperature given in API Standard 650.

5.3.2 STRUCTURAL

Existing rolled structural shapes that are to be reused shall meet the requirement of ASTM A7 as a minimum. New structural material shall meet the requirements of ASTM A36 as a minimum.

Note: ASTM A7 was a steel specification that was discontinued in the Fourth Edition of API Standard 650, 1970.

5.3.3 FLANGES and FASTENERS

5.3.3.1 Flange material shall meet the minimum requirements of the material specifications in the original standard of construction.

5.3.3.2 Fasteners shall meet the material specifications of the current applicable standard.

5.3.4 ROOF, BOTTOM, AND PLATE WIND GIRDERS

If existing plates are to be used to reconstruct the tank, they shall be checked for excessive corrosion and pitting. (See Sections 2 and 4.)

5.4 Welding Consumables

Welding consumables shall conform to the American Welding Society (AWS) classification that is applicable to the intended use.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 6—DESIGN CONSIDERATIONS FOR RECONSTRUCTED TANKS

6.1 General

Any specific design considerations other than normal product loading shall be specified by the owner/operator. Refer to 2.4.5 for release prevention systems and release prevention barrier definition.

6.2 New Weld Joints

6.2.1 Weld joint details shall meet the welding requirements of the applicable standard.

6.2.2 All new shell joints shall be butt-welded joints with complete penetration and complete fusion.

6.3 Existing Weld Joints

Existing weld joints shall meet the requirements of the original standard of construction for the tank.

6.4 Shell Design

6.4.1 Thickness to be used for each shell course when checking tank design shall be based on measurements taken within 180 days prior to relocation. (See 2.3.2 for measuring procedure, number, and locations of measured thicknesses.)

6.4.2 The maximum design liquid level for product shall be determined by calculating the maximum design liquid level for each shell course based on the specific gravity of the product, the actual thickness measured for each shell course, the allowable stress for the material in each course, and the design method to be used. The allowable stress for the material shall be determined using Table 3-2 of API Standard 650. For material not listed in Table 3-2, an allowable stress value of the lesser of ⅔ yield strength or ⅜ tensile strength shall be used.

6.4.3 If a corrosion allowance is required for the reconstructed tank, the required corrosion allowance shall be deducted from the actual thickness before calculating the maximum liquid level. If the actual thickness is greater than that necessary to allow the liquid level required, the extra thickness can be considered as corrosion allowance.

6.4.4 The joint efficiency and allowable stress levels used for the design liquid level calculations shall be consistent with the design method used and with the degree and type of inspection made on welded joints. The joint efficiency and allowable stress levels for existing welded joints that are not to be removed and replaced shall be based on the original degree and type of inspection.

6.5 Shell Penetrations

6.5.1 Replacement and new penetrations shall be designed, detailed, welded, and examined to meet the requirements of the current applicable standard.

6.5.2 Existing penetrations shall be evaluated for compliance with the original standard of construction.

6.6 Wind Girders and Shell Stability

6.6.1 Top and intermediate wind girders for open top tanks shall meet the requirements of the applicable standard.

6.6.2 Tanks to be reconstructed shall be checked for wind-induced buckling in accordance with the procedures of the applicable standard, using the wind requirements for the location where the tank will be reconstructed.

6.7 Roofs

6.7.1 Roof designs shall meet the requirements of the applicable standard.

6.7.2 If the new site location requires a larger design live load than the original site, the adequacy of the existing roof shall be evaluated.

6.8 Seismic Design

Tanks that will be reconstructed in Seismic Zone 2 or greater (see Table E-1 in Appendix E of API Standard 650) shall be checked for seismic stability based on the rules of the applicable standard using the dimensions and thicknesses of the reconstructed tank. Reconstructed tanks shall be built to meet the stability requirements of the applicable standard. Thickened bottom plates under the bottom shell course or anchoring of the tank may be required even if not used on the original tank.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 7—TANK REPAIR AND ALTERATION

7.1 General

7.1.1 The basis for repairs and alterations shall be an API Standard 650 equivalence.

7.1.2 Hydrostatic testing requirements, nondestructive examination requirements, acceptance criteria for the welds, and repairs to shell plate and existing welds are specified in Section 10.

7.2 Removal and Replacement of Shell Plate Material

7.2.1 MINIMUM THICKNESS OF REPLACEMENT SHELL PLATE

The minimum thickness of the replacement shell plate material shall be calculated in accordance with the applicable standard. The thickness of the replacement shell plate shall not be less than the greatest nominal thickness of any plate in the same course adjoining the replacement plate except where the adjoining plate is a thickened insert plate. Any changes from the original design conditions, such as specific gravity, design pressure, liquid level, and shell height, shall be considered.

7.2.2 MINIMUM DIMENSIONS OF REPLACEMENT SHELL PLATE

7.2.2.1 The minimum dimension for a replacement shell plate is 12 inches or 12 times the thickness of the replacement plate, whichever is greater. The replacement plate may be circular, oblong, square with rounded corners, or rectangular with rounded corners except when an entire shell plate is replaced. Refer to Figure 7-1 for typical details of acceptable replacement shell plates.

7.2.2.2 Where one or more entire shell plates or full height segments of shell plates are to be removed and replaced, the minimum spacing requirements specified in Figure 7-1 for vertical weld joints shall be maintained. It is acceptable to remove and replace entire shell plates or full height segments of shell plates by cutting and rewelding along the existing horizontal weld joints. Prior to welding the new vertical joints, the existing horizontal welds shall be cut for a minimum distance of 12 inches beyond the new vertical joints. The vertical joints shall be welded prior to welding the horizontal joints.

7.2.3 WELD JOINT DESIGN

7.2.3.1 Shell replacement plates shall be welded with butt joints with complete penetration and complete fusion. Fillet welded tapped patch plates are not permitted.

7.2.3.2 Weld joint design for replacement shell plates shall be in accordance with 3.1.5.1 through 3.2.5.3 of API Standard 650. Joints in lap-welded shell tanks may be repaired according to the original standard of construction; however, lapped patch plates are not permitted. Details of welding shall be in accordance with 5.2 of that standard, and Section 9 of this standard.

7.2.3.3 For existing shell plates over ½ inch thick, the outer edge of the butt weld attaching the replacement shell plate shall be at least the greater of 8 times the weld thickness or 10 inches from the outer edge of any existing butt-welded shell joints. For existing shell plates ½ inch thick and less, the spacing may be reduced to 6 inches from the outer edge of vertical joints or 3 inches from the outer edge of horizontal joints. See Figure 7-1 for minimum dimensions.

For existing shell plates over ½ inch thick, the outer edge of the butt weld attaching the replacement shell plate shall be at least the greater of 8 times the weld size or 10 inches from the edge (toe) of the fillet weld attaching the bottom shell course to the bottom except when the replacement shell plate extends to and intersects the bottom-to-shell joint at approximately 90 degrees. For existing shell plates ½ inch thick and less, this spacing may be reduced to 6 inches. The edge of any vertical weld joint attaching a replacement plate shall be at least 12 inches from the edge of a weld joint in the annular plate ring or sketch plate. Figure 7-1 has minimum dimensions.

7.2.3.4 To reduce the potential for distortion of an existing tank due to welding a replacement plate into an existing tank shell, fit-up, heat input, and welding sequence must be considered.

7.3 Repair of Defects in Shell Plate Material

The need for repairing flaws such as cracks, gouges or tears (such as those often remaining after the removal of temporary attachments) discovered during an inspection of the tank shell shall be determined on an individual case basis. In areas of the shell where the plate thickness exceeds that required by design conditions, it is permissible to grind scars to a smooth contour so long as the remaining thickness is adequate for the design conditions. Where grinding to a smoothly contoured surface will result in unacceptable metal thickness, weld metal must be added to repair the scar using a qualified weld procedure.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607813 769 ■

7.4 Alteration of Tank Shells to Change Shell Height

Tank shells may be altered by adding new plate material to increase the height of the tank shell. The modified shell height shall be in accordance with the requirements of the applicable standard and shall take into consideration all anticipated loadings such as wind and seismic.

7.5 Repair of Defective Welds

Types of weld flaws and nonconformances that need repair are described in 7.5.1 through 7.5.4.

7.5.1 Cracks, lack of fusion, and rejectable slag and porosity that need repair shall be removed completely by gouging and/or grinding and the resulting cavity properly prepared for welding.

7.5.2 Generally, it is not necessary to remove existing weld reinforcement in excess of that allowed by API Standard 650 when discovered on an existing tank with a satisfactory service history. However, if operating conditions are such that the excessive weld reinforcement may be deleterious (such as for a floating roof with flexible seals), consideration shall be given to repairing the welds by grinding.

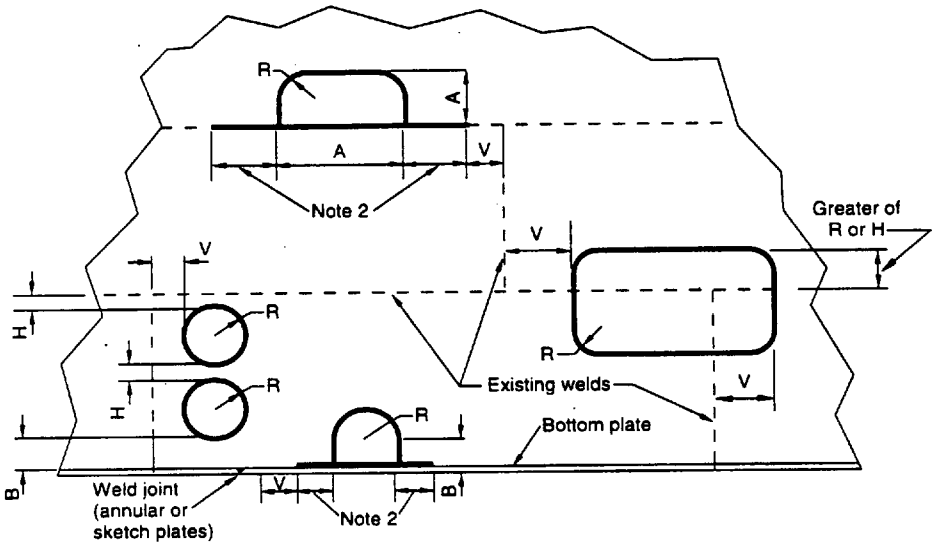
7.5.3 Existing weld undercut deemed unacceptable based upon suitability for service considerations shall be repaired by additional weld metal, or grinding, as appropriate.

7.5.4 Welded joints that have experienced loss of metal due to corrosion may be repaired by welding.

7.5.5 Arc strikes discovered in or adjacent to welded joints shall be repaired by grinding and/or welding. Arc strikes repaired by welding shall be ground flush with the plate.

7.6 Repair of Shell Penetrations

7.6.1 Repairs to existing shell penetrations shall be in compliance with API Standard 650.



Dimension	Minimum weld spacing between edges (toes) of welds for thickness of replacement shell plate, t, (inches)	
	$t \leq 0.5$ inch	$t > 0.5$ inch
R	6"	Greater of 6" or 6t
B	6"	Greater of 10" or 8t
H	3"	Greater of 10" or 8t
V	6"	Greater of 10" or 8t
A	12"	Greater of 12" or 12t

Notes:
1. All weld intersections shall be at approximately 90°.
2. Prior to welding new vertical joints, cut existing horizontal weld for a minimum of 12" beyond the new vertical joints. Weld the horizontal joint last.

Figure 7-1—Acceptable Details for Replacement of Shell Plate Material

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

7.6.2 Reinforcing plates may be added to existing unreinforced nozzles when deemed appropriate. The reinforcing plate shall meet all dimensional and weld spacing requirements of API Standard 650. See Figures 7-2 and 7-3 for acceptable details.

7.6.3 As a alternative, the reinforcing plates may be added to the inside of the tank provided that sufficient nozzle projection exists.

7.7 Addition or Replacement of Shell Penetrations

7.7.1 New shell penetrations (addition or replacement) shall be in accordance with all requirements of API Standard 650 or 7.7.2 of this standard.

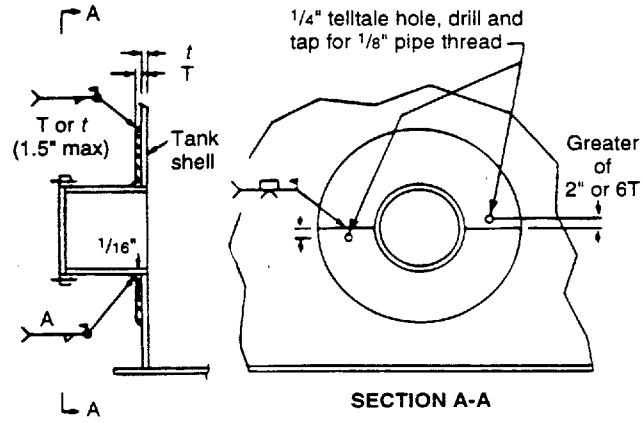
7.7.2 Penetrations larger than 2 inches NPS shall be installed with the use of an insert plate if the shell plate thickness is greater than ½ inch and the shell plate material does not meet the current design metal temperature criteria. In addition, the following requirement shall be met:

a. The minimum diameter of the insert plate shall be at least twice the diameter of the penetration or the diameter plus 12 inches, whichever is greater.
b. When reinforcing plates are used, the minimum diameter of the insert plate shall equal the diameter of the reinforcing plate plus 12 inches.
c. If an integral reinforcement design is used, the thickness of the insert plate–at its periphery and for a distance of $2t$ from the edge of the weld (where t is the shell plate thickness)–shall be the same thickness as the shell, or the insert plate shall have a 1:4 taper to match the shell plate thickness.
d. Spacing of welds shall be in accordance with Figure 7-1.
e. The joint between new material of the insert plate and old material of the existing shell plate shall be butt-welded.

7.8 Alteration of Existing Shell Penetrations

7.8.1 Existing shell penetrations may be modified if the altered details comply with the requirements of API Standard 650, including the requirements for minimum reinforcing area and the requirements for spacing of welds around connections.

7.8.2 When installing a new tank bottom above the existing bottom, it may be necessary to alter existing shell penetrations in the bottom course of a tank shell. If the new bottom is "slotted" through the tank shell several inches above the existing bottom, the spacing between existing welds around penetrations and the new bottom-to-shell weld may not comply with API Standard 650 requirements. Options for altering the penetrations and/or reinforcing plates are given in 7.8.2.1 through 7.8.2.3.



t and T (maximum)	A
9/16"	1/4"
3/4"	5/16"
15/16"	3/8"
1 1/8"	7/16"
1 5/16"	1/2"
1 9/16"	9/16"
1 3/4"	5/8"

Note: All details, dimensions, and weld spacing shall be in accordance with the requirements of API Standard 650.

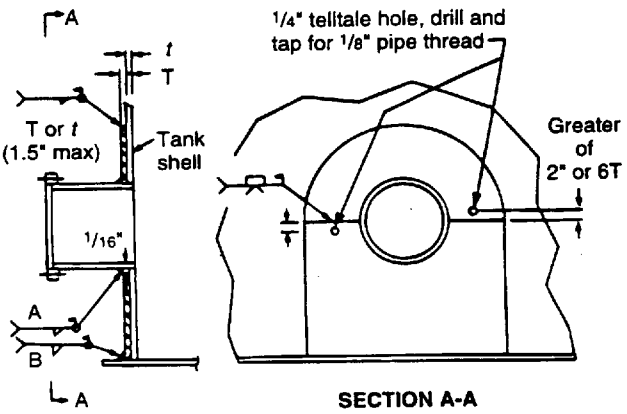
Figure 7-2—Typical Details for Addition of Reinforcing Plate to Existing Shell Penetration

7.8.2.1 The existing reinforcing plate may be trimmed to increase the spacing between the welds provided that the modified detail complies with the requirements of API Standard 650. Care must be exercised during the trimming operation to avoid damaging the shell material beneath the reinforcing plate. The existing weld attaching the portion of the reinforcing plate to be removed shall be completely removed by gouging and grinding.

7.8.2.2 The existing reinforcing plate may be removed and a new reinforcing plate added. Care must be exercised when removing the existing reinforcing plate to avoid damaging the shell plate beneath the reinforcing plate. The welds around the perimeter of the reinforcing plate and between the reinforcing plate and neck of the penetration shall be completely removed by gouging and grinding. The new reinforcing plate shall be in accordance with Figure 6. If required to maintain weld spacing, a "tombstone" shaped reinforcing plate may be used. (See Figure 7-3).

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607815 531 ■



t and T (maximum)	A	B
9/16"	1/4"	1/4"
3/4"	5/16"	1/4"
15/16"	3/8"	5/16"
1 1/8"	7/16"	5/16"
1 5/16"	1/2"	3/8"
1 9/16"	9/16"	3/8"
1 3/4"	5/8"	3/8"

Note: All details, dimensions, and weld spacing shall be in accordance with the requirements of API Standard 650.

Figure 7-3—Typical Details for Addition of "Tombstone" Shape Reinforcing Plate to Existing Shell Penetration

7.8.2.3 The existing penetration may be moved by cutting the section of the shell containing the fitting and reinforcing plate, and raising the entire assembly to the correct elevation (see Figure 7-4).

7.8.3 Any components of the penetration (neck, flange, and reinforcing plate) that are in serviceable condition after removal may be reused.

7.9 Repair of Tank Bottoms

7.9.1 REPAIRING A PORTION OF TANK BOTTOMS

7.9.1.1 The critical zone for repairs to tank bottoms is within the annular plate ring, within 12 inches of the shell, or within 12 inches of the inside edge of the annular plate ring. No welding, welded-on patch plates or weld overlays are permitted within the critical zone except for welding of: widely scattered pits (see 2.3.2.2), cracks in the bottom plates, the shell-to-bottom weld, or where the bottom or annular plate is being replaced.

7.9.1.2 If more extensive repairs are required within the critical zone than those listed in 7.9.1.1, the bottom plate under the bottom shell course shall be cut out and a new plate shall be installed. Weld spacing requirements shall be in accordance with 3.1.5.4 and 3.1.5.5 of API Standard 650.

7.9.1.3 The repair of sumps located within the critical zone shall be in accordance with 7.9.1.2.

7.9.2 REPLACEMENT OF ENTIRE TANK BOTTOM

7.9.2.1 Requirements governing the installation of a replacement bottom over an existing bottom are given in 7.9.2.1.1 through 7.9.2.1.6.

7.9.2.1.1 A suitable non-corrosive material cushion such as sand, gravel, or concrete, 3 to 4 inches thick, shall be used between the old bottom and the new bottom.

7.9.2.1.2 The shell shall be "slotted" with a uniform cut made parallel to the tank bottom. The cut edges in the slot shall be ground to remove all slag and burrs from the cutting operation. The new bottom plate shall be extended outside the shell as required by API Standard 650. All rules for weld spacing shall be followed.

7.9.2.1.3 Voids in the foundation below the old bottom shall be filled with sand, crushed limestone, grout, or concrete.

7.9.2.1.4 Existing shell penetrations shall be raised if the elevation of the new bottom cuts through the reinforcing plate or if the weld space requirements given in 3.7.3 of API Standard 650 are not met.

7.9.2.1.5 For floating roof tanks, the new bottom profile must keep the roof level when it is resting on its support legs in the down position. The support legs can either be jacked up to maintain the original height or be shortened by the same amount as the thickness of the cushion and new bottom plate.

7.9.2.1.6 New bearing plates for floating roof support legs and for fixed roof support columns shall be installed.

7.9.2.2 When removing an existing tank bottom, the tank shell shall be separated from tank bottom either by:

a. Cutting the shell parallel to the tank bottom a minimum of ½ inch above the bottom to shell weld (cut line B-B as shown in Figure 8-1), or
b. Removing the entire shell to bottom attachment weld, including any penetration and heat affected zone by suitable methods such as arc gouging and/or grinding.

If b. is elected, all arc gouged areas of the tank shell shall be magnetic particle examined, and defective areas repaired and re-examined.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607816 478 ■

7.9.2.3 Installation of a new tank bottom, after removal of the existing tank bottom, shall meet all requirements of API 650 including shell weld spacing.

7.9.2.4 When planning a bottom replacement, consideration shall be given to removing the old tank bottom or providing other means of preventing potential galvanic corrosion (refer to API RP 651). Also see 2.4.5 regarding bottom leak detection.

7.10 Repair of Fixed Roofs

7.10.1 SUPPORTED CONE ROOFS

7.10.1.1 The minimum thickness of new roof plates shall be 3/16 inch plus any corrosion allowance as specified in the repair specifications. In the event roof live loads in excess of 25 pounds per square foot are specified (such as insulation, operating vacuum, high snow loads), the plate thickness shall be based on analysis using the allowable stresses in conformance with 3.10.3 of API Standard 650 (see 7.10.2.2).

7.10.1.2 The roof supports (rafters, girders, columns, and bases) shall be repaired or altered such that under design conditions the resulting stresses do not exceed the stress levels given in 3.10.3 of API Standard 650.

7.10.2 SELF-SUPPORTING ROOFS

7.10.2.1 The nominal thickness of new roof plate shall be 3/16 inch or the required plate thickness given in 3.10.5 or 3.10.6 of API Standard 650, plus any specified corrosion allowance, whichever is greater.

7.10.2.2 The details of the roof-to-shell junction shall meet the requirements of 3.10.5 or 3.10.6 or Appendix F of API Standard 650, as applicable, for the intended service. Frangible joint design, or the need for emergency venting devices in accordance with API Standard 2000, shall meet the requirements of 3.10.2.5 of API Standard 650.

7.11 Floating Roofs

7.11.1 EXTERNAL FLOATING ROOFS

Any method of repair is acceptable that will restore the roof to a condition enabling it to perform as required.

7.11.2 INTERNAL FLOATING ROOFS

Repairs to internal floating roofs shall be made in accordance with the original construction drawings, if available. If the original construction drawings are not available, the roof repairs shall be in compliance with the requirements of Appendix H of API Standard 650.

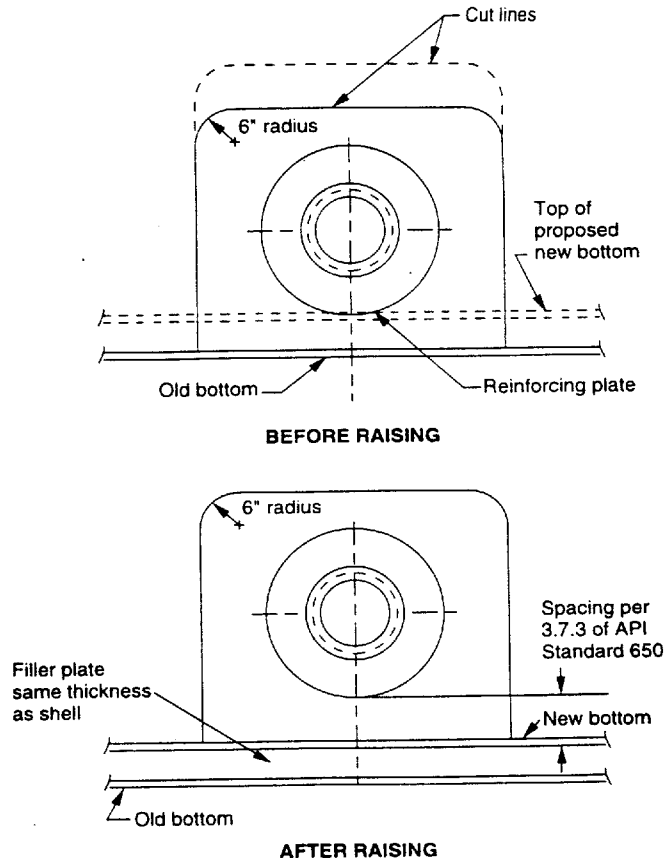


Figure 7-4—Method for Raising Shell Nozzles

7.11.3 REPAIR OF LEAKS IN PONTOONS

All leaks in pontoons or compartments of double deck floating roofs shall be repaired by rewelding the leaking joints and/or use of patch plates.

7.12 Repair or Replacement of Floating Roof Perimeter Seals

7.12.1 PRIMARY SEALS

Rim-mounted primary shoe seals and toroidal seal systems can be removed, repaired, or replaced. To minimize evaporation losses and reduce potential hazard to the workers, no more than one-fourth of the roof seal system should be out of an in-service tank at one time. Temporary spacers to keep the roof centered shall be used during the repairs. Primary seals systems mounted partly or fully below the bolting bar or top of the rim usually cannot be reached to allow removal in service. In this case, in-service repairs are limited to replacement of the primary seal fabric.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607817 304 ■

7.12.2 SECONDARY SEALS

Rim-mounted and shoe-mounted secondary seals may be readily installed, repaired, or replaced while the tank is in service.

7.12.3 SEAL-TO-SHELL GAP

Repair and other corrective actions to maintain seal-to-shell gap requirements, includes:

a. Adjusting the hanger system on primary shoe seals, and adding foam filler in toroidal seals.
b. Increasing the length of rim mounted secondary seals in the problem area.
c. Replacing all or part of the primary seal system along with possible installation of a rim extension for a secondary seal. This step shall be taken only after checking the annular space variation at several levels from low pump out to high liquid level.

7.12.4 MECHANICAL DAMAGE

Damaged parts shall be repaired or replaced. Before taking this action, the cause of the damage shall be identified and corrected. Buckled parts shall be replaced, not straightened. Torn seal fabric shall be replaced.

7.12.5 DETERIORATION OF SEAL MATERIAL

Material deterioration results from wear and corrosion on metallic elements, and chemical and weather deterioration of seal fabric. The service life and inspection information shall be used to determine whether a change of material is warranted.

7.12.6 INSTALLATION OF PRIMARY AND SECONDARY SEALS

7.12.6.1 The replacement or addition of primary and secondary seals shall be in accordance with the recommendations of the seal manufacturer. In addition, the final installation shall comply with all applicable jurisdictions.

7.12.6.2 If the roof rim thickness is less than 0.10 inch thick, it shall be replaced. The new roof rim shall be 3/16 inch thickness, minimum.

7.13 Hot Taps

7.13.1 GENERAL

7.13.1.1 The requirements given herein cover the installation of radial hot tap connections on existing in-service tanks constructed with shell material that does not require postweld heat treatment. Connection size and shell thickness limitations are shown in Table 7-1.

Table 7-1—Hot Tap Connection Sizes and Shell Plate Thicknesses

Connection Size, NPS (inches)	Minimum Shell Plate Thickness (inches)
≤8	1/4
≤14	3/8
≤18	1/2

7.13.1.2 Welding shall be done with low hydrogen electrodes.

7.13.1.3 Hot taps are not permitted on the roof of a tank or within the gas/vapor space of the tank.

7.13.1.4 Hot taps shall not be installed on laminated or severely pitted shell plate.

7.13.1.5 Hot taps are not permitted on tanks where the heat of welding may cause environmental cracking (such as caustic cracking or stress corrosion cracking).

7.13.2 HOT TAP PROCEDURES

A hot tap procedure specific to carrying out the work shall be developed and documented. The procedure shall include the practices given in API Publication 2201.

7.13.3 PREPARATORY WORK

7.13.3.1 Minimum spacing in any direction (toe-to-toe of welds) between the hot tap and adjacent nozzles shall be equivalent to the square root of RT where R is the tank shell radius, in inches, and T is the shell plate thickness, in inches.

7.13.3.2 Shell plate thickness measurements shall be taken at a minimum of four places along the circumference of the proposed nozzle location.

7.13.4 MATERIAL LIMITATIONS

Material limitations include hot tap only steels of known acceptable toughness or steels of unknown toughness that have a minimum shell metal temperature at or above the exemption curve shown in Figure 7-5.

7.13.5 INSTALLATION PROCEDURE

7.13.5.1 Pipe nozzles shall be cut to the contour of the shell and beveled from the outside for a full penetration weld (see Figure 7-6).

7.13.5.2 After the pipe is welded, the reinforcing plate shall be installed either in one piece or two pieces with horizontal weld. The reinforcing plate to nozzle shall be installed with a full penetration weld. Care shall be taken to limit the heat input to the welds.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607818 240 ■

7.13.5.3 After the reinforcing plate has been welded to the shell and nondestructive examination carried out, the pad shall be pneumatically tested by the procedure described in API Standard 650. After the valve has been installed on the flange, a pressure test at least 1.5 times the hydrostatic head shall be performed on the nozzle prior to mounting the hot tap machine–which shall be bolted to the valve.

7.13.5.4 A qualified operator shall operate the hot tap machine and cut the hole in the tank following the hot tap machine manufacturer's procedures.

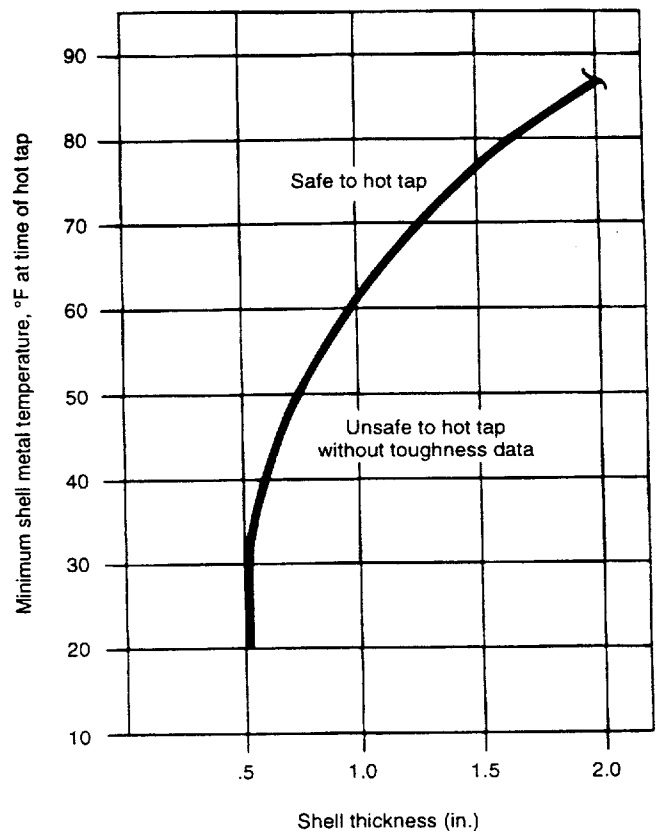
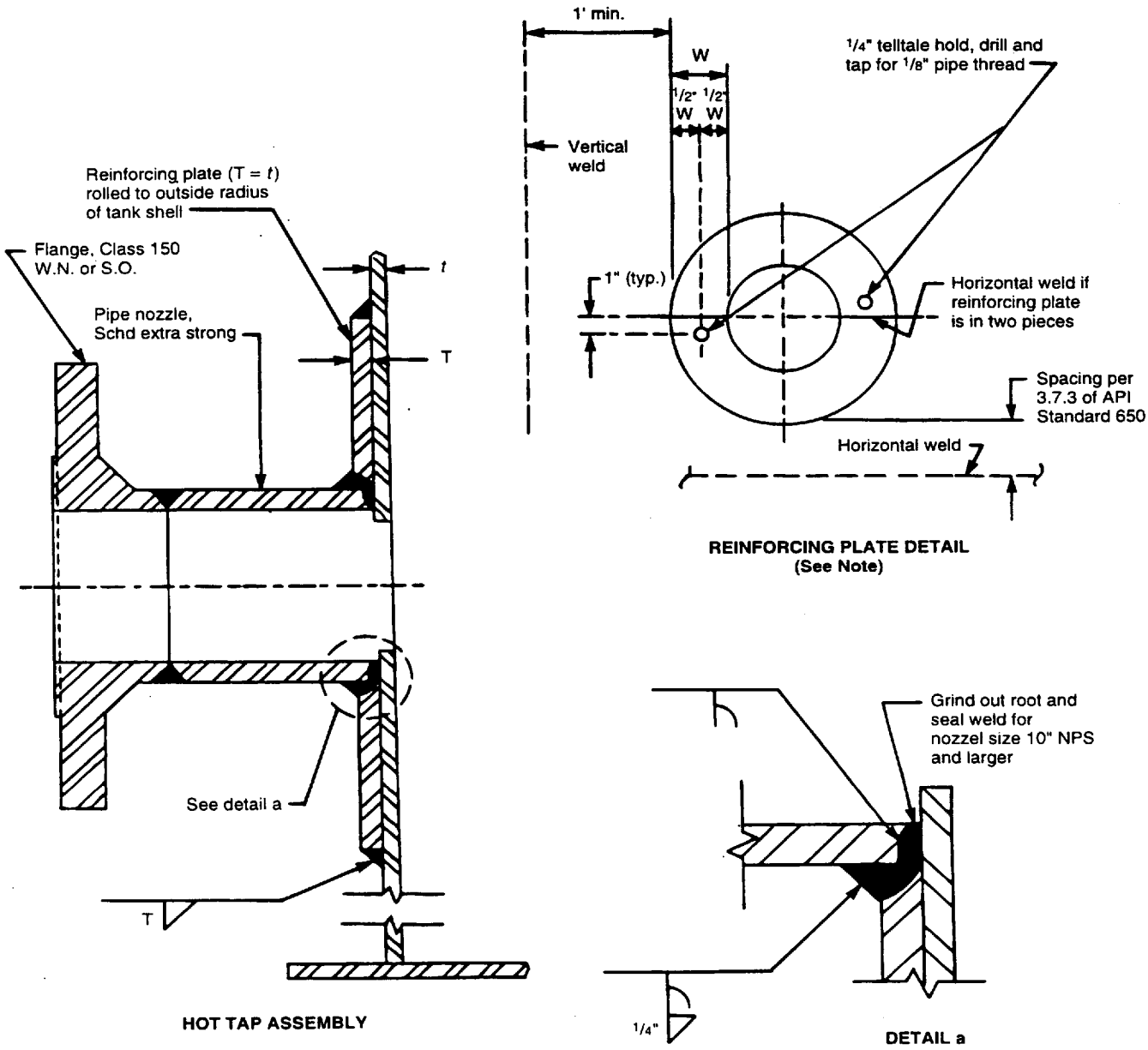


Figure 7-5—Exemption Curve for Hot Tapping Tanks With Steels of Unknown Toughness

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34



Note: Reinforcing plate diameters as given in Table 3-8 of API Standard 650.

Figure 7-6—Hot Tap for Tanks

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 8—DISMANTLING AND RECONSTRUCTION

8.1 General

8.1.1 This section provides procedures for dismantling and reconstructing existing welded tanks that are to be relocated from their original site.

8.1.2 Hydrostatic testing requirements, nondestructive examination requirements, and acceptance criteria for the quality of welds for reconstructed tanks are specified in Section 10.

8.2 Cleaning and Gas Freeing

The tank shall be cleaned and gas-freed prior to commencement of dismantling.

8.3 Dismantling Methods

8.3.1 GENERAL

Roof, shell, and bottom plates may be cut into any size pieces that are readily transportable to the new site for reconstruction.

8.3.2 BOTTOMS

8.3.2.1 Bottom plates that will be reused shall be cut by descaming of lapwelds; or by cutting alongside of the remaining welds at a minimum of 2 inches away from existing welds, except where cuts cross existing weld seams.

8.3.2.2 If the bottom is to be reused, it shall be cut from the shell along line A-A as shown in Figure 8-1; except if the entire bottom is to be reused intact, the shell shall be cut along line C-C.

8.3.3 SHELLS

8.3.3.1 Shell plates shall be cut in accordance with one of the following methods or a combination thereof:

a. Cuts made to remove existing welds and the heat affected zones.
b. Cuts made a minimum of 6 inches away from existing welds except where cuts cross existing welds.

8.3.3.2 Shell stiffening rings, including wind girders and top angles, may be left attached to the shell plates or may be removed by cutting at the attachment welds. The area where temporary attachments are removed shall be ground flush with the shell plate.

8.3.3.3 The shell shall be cut from the bottom plate along line B-B as shown in Figure 8-1. The existing shell-to-bottom weld connection shall not be reused unless the entire bottom is to be reused intact.

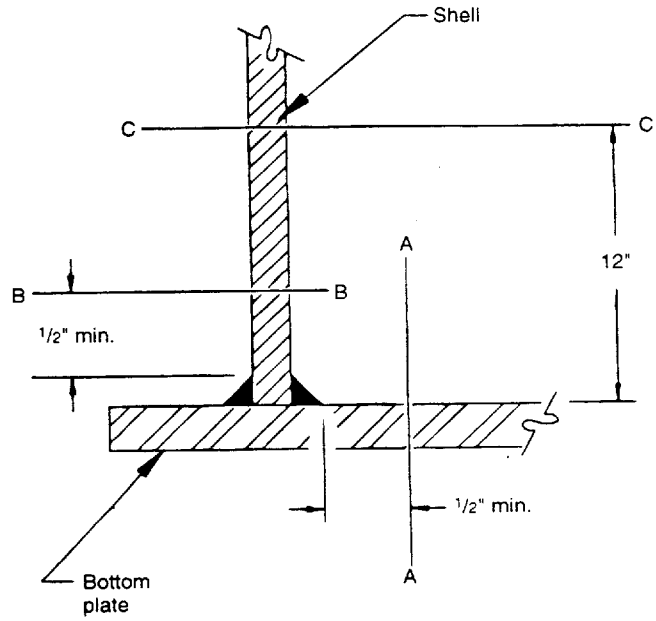


Figure 8-1—Tank Shell and Bottom Cut Locations

8.3.4 ROOFS

8.3.4.1 Roof plates shall be cut by descaming of lap welds, or by cutting alongside of the remaining welds at a minimum of 2 inches away from existing welds, except where cuts cross existing welds.

8.3.4.2 Roof supporting structures shall be dismantled by removing bolts (if bolted) or descaming the structural attachment welds.

8.3.5 PIECE MARKING

8.3.5.1 Shell, bottom, and roof plates shall be marked prior to dismantling for ready identification and placement when the tank is reconstructed. Marking material shall be of a durable type. Drawings showing piece mark locations are also a useful adjunct.

8.3.5.2 A minimum of two sets of matching center punch marks shall be located on the top and bottom edges of each shell plate to facilitate proper alignment during reconstruction.

8.4 Reconstruction

8.4.1 GENERAL

8.4.1.1 The foundation for the reconstructed tank shall meet the construction tolerances given in 8.5.6.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607821 835 ■

8.4.1.2 Temporary attachments shall be removed, and the attachment area ground flush with the plate surface.

8.4.2 WELDING

8.4.2.1 Provisions shall be made during the reconstruction of a tank to ensure that:

a. Vertical weld joints in the shell are not aligned with the weld joints in the bottom plates.
b. There will be no welding over heat affected zones that remain from the original tank welds except where new joints cross original joints.
c. Weld spacing requirements of Figure 7-1 are maintained.

8.4.2.2 Tanks and their structural attachments shall be welded in accordance with the processes specified in API Standard 650 and the requirements of 8.4.2.3 through 8.4.2.11.

8.4.2.3 No welding of any kind shall be performed when the surfaces of the parts to be welded are wet from rain, snow, or ice; when rain or snow is falling on such surfaces; or during periods of high winds unless the welder and the work are properly shielded. No welding of any kind shall be performed when the temperature of the base metal is less than 0°F. When the temperature of the base metal is between 0°F and 32°F or the thickness is in excess of 1 inch, the base metal within 3 inches of the place where welding is to be started shall be heated to a temperature warm to the hand (approximately 140°F) before welding. (See 8.4.4.3 for preheat requirements for shell plates over 1½ inches thick.)

8.4.2.4 Each layer of weld metal of multilayer welding shall be cleaned of slag and other deposits before the next layer is applied.

8.4.2.5 The edges of all welds shall merge with the surface of the plate without a sharp angle. The maximum acceptable undercutting is 1/64 inch of the base metal for vertical butt joints. For horizontal butt joints undercutting not exceeding 1/32 inch in depth is acceptable.

8.4.2.6 The reinforcement of the new welds on all butt joints on each side of the plate shall not exceed the thicknesses shown in Table 8-1.

Table 8-1—Maximum Thicknesses on New Welds

Plate Thickness (inches)	Maximum Reinforcement Thickness (inches)	
	Vertical Joints	Horizontal Joints
≤1/2	3/32	1/8
>1/2 through 1	1/8	3/16
>1	3/16	1/4

8.4.2.7 Tack welds used in the assembly of vertical joints of tank shells shall be removed and shall not remain in the finished joint when the joints are welded manually. When such joints are welded by the submerged-arc process, the tack welds shall be thoroughly cleaned of all welding slag but need not be removed provided that they are sound and are thoroughly fused into the subsequently applied weld beads. Tack welds shall be made using a fillet-weld or butt-weld procedure qualified in accordance with Section IX of the ASME Code. Tack welds to be left in place shall be made by qualified welders.

8.4.2.8 If weldable primer coatings have been applied on surfaces to be welded, they shall be included in welding procedure qualification tests for the brand, formulation, and maximum thickness of primer applied. All other coatings shall be completely removed from the weld area prior to welding.

8.4.2.9 Low-hydrogen electrodes shall be used for manual metal-arc welds, including the attachment of the first shell course to the bottom plates or annular plate ring, as follows:

a. For all welds in shell courses over 0.5 inch thick of API Standard 650 Group I-III materials.
b. For all welds in all shell courses of API Standard 650 Group IV-VI materials.

8.4.2.10 Low-hydrogen electrodes shall be used for welding temporary and new permanent attachments to shells of API Standard 650 Group IV, IVA, V, or VI materials. The welding procedure selected shall not cause underbead cracking; also, the need for preheat of thick plates and the effect of low atmospheric temperature during welding shall be considered.

8.4.2.11 If existing welds are found to be unsatisfactory by the original standard of construction, they shall be repaired in accordance with 7.5.

8.4.3 BOTTOMS

8.4.3.1 After the bottom plates are laid out and tacked, they shall be joined by welding the joints in a sequence that results in the least distortion from shrinkage and provides as nearly as possible a plane surface.

8.4.3.2 The welding of the shell to the bottom (except for door sheets) shall be completed before the welding of bottom joints.

8.4.3.3 Plates shall be held in close contact at all lap joints during welding.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607822 771 ■

8.4.4 SHELLS

8.4.4.1 Plates to be joined by butt welding shall be matched accurately and retained in position during welding. Misalignment in completed vertical joints over 5/8 inch thick shall not exceed 10 percent of the plate thickness, with a maximum of 1/8 inch. Misalignment in joint 5/8 inch thick or less shall not exceed 1/16 inch. Vertical joints shall be completed before the lower horizontal weld is made.

8.4.4.2 In completed horizontal butt joints, the upper plate shall not project beyond the face of the lower plate at any point by more than 20 percent of the thickness of the upper plate, with a maximum projection of 1/8 inch, except that a projection of 1/16 inch is acceptable for upper plates less than 5/16 inch thick.

8.4.4.3 For horizontal and vertical joints in tank shell courses constructed of material over 1½ inches thick (based on the thickness of the thicker plate at the joint), multipass weld procedures are required, with no pass more than ¾ inch thick permitted. A minimum preheat of 200°F is required of these welds.

8.4.5 ROOFS

This standard does not include special stipulations for reconstruction thereof, except that the structural framing (such as rafters and girders) must be reasonably true to line and surface. Other requirements shall be in accordance with the applicable standard.

8.5 Dimensional Tolerances

8.5.1 GENERAL

8.5.1.1 The tolerances given in this section have been established to produce a reconstructed tank of acceptable appearance and structural integrity and to permit proper functioning of floating roofs and seals.

8.5.1.2 Measurements to verify these tolerances shall be taken before the hydrostatic test of the reconstructed tank.

8.5.2 PLUMBNESS

8.5.2.1 The maximum out-of-plumbness of the top of the shell relative to the bottom of the shell shall not exceed 1/100 of the total tank height, with a maximum of 5 inches.

8.5.2.2 The out-of-plumbness in one shell course shall not exceed the values specified for mill tolerances in ASTM A6 or A20, whichever is applicable.

8.5.3 ROUNDNESS

Radii measured at 1 foot above the shell-to-bottom weld shall not exceed the tolerances shown in Table 8-2.

Table 8-2—Radii Tolerances

Tank Diameter (feet)	Radius Tolerances (inches)
< 40	± ½
40 to < 150	± ¾
150 to < 250	± 1
≥ 250	± 1¼

Radius tolerances measured higher than one foot above the shell-to-bottom weld shall not exceed three times the tolerances given above.

8.5.4 PEAKING

With a horizontal sweep board 36 inches long, peaking shall not exceed ½ inch. The sweep board shall be made to the true outside radius of the tank.

8.5.5 BANDING

With a vertical sweep board 36 inches long, banding shall not exceed one inch.

8.5.6 FOUNDATIONS

8.5.6.1 To achieve the tolerances specified in 8.5.1 through 8.5.5, it is essential that foundations true to a plane be provided for the tank reconstruction. The foundation shall have adequate bearing capacity to maintain the trueness of the foundation.

8.5.6.2 Where foundations true to a horizontal plane are specified, tolerances shall be as follows:

a. Where concrete ringwalls are provided under the shell, the top of the ringwall shall be level within ±1/8 inch in any 30 feet of the circumference and within ±¼ inch in the total circumference measured from the average elevation.
b. Where concrete ringwalls are not provided, the foundation under the shell shall be level within ±1/8 inch in any 10 feet of circumference and within ±½ inch in the total circumference measured from the average elevation.

8.5.6.3 For foundations specified to be sloped from a horizontal plan, elevation differences about the circumference shall be calculated from the specified high point. Actual elevation differences about the circumference shall be determined from the actual elevation of the specified high point. The actual elevation differences shall not deviate from the calculated differences by more than the following tolerances:

a. Where concrete ringwalls are provided ±1/8 inch in any 30 feet of circumference and ±¼ inch in the total circumference.
b. Where concrete ringwalls are not provided, ±1/8 inch in any 10 feet of circumference and ±½ inch in the total circumference.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 9—WELDING

9.1 Welding Qualifications

9.1.1 Welding procedure specifications (WPS) and welders and welding operators shall be qualified in accordance with Section IX of the ASME Code.

9.1.2 Weldability of steel from existing tanks shall be verified. If the material specification for the steel from an existing tank is unknown or obsolete, test coupons for the welding procedure qualification shall be taken from the actual plate to be used.

9.2 Identification and Records

9.2.1 Each welder and welding operator shall be assigned an identifying number, letter, or symbol. Records of this identification, along with the date and results of the welder's qualification tests shall be accessible to the inspector.

9.2.2 The welder or welding operator's identification mark shall be hand- or machine-stamped adjacent to and at intervals not exceeding 3 feet along the completed welds. In lieu of stamping, a record may be kept that identifies the welder or welding operator employed for each welded joint; these records shall be accessible to the inspector. Roof plate welds and flange-to-nozzle-neck welds do not require welder identification.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 10—EXAMINATION AND TESTING

10.1 Nondestructive Examinations

10.1.1 GENERAL

10.1.1.1 Nondestructive examination procedures, qualifications, and acceptance criteria shall be prepared for visual, magnetic particle, liquid penetrant, ultrasonic, and radiographic methods in accordance with API Standard 650 and the supplemental requirements given herein.

10.1.1.2 Personnel performing nondestructive examinations shall be qualified in accordance with API Standard 650 and the supplemental requirements given herein.

10.1.2 SHELL PENETRATIONS

10.1.2.1 Ultrasonic examination of shell plate for laminations shall be made in the immediate area affected when:

a. Adding a reinforcing plate to an existing unreinforced penetration.

b. Adding a hot tap connection.

10.1.2.2 Cavities resulting from gouging or grinding operations to remove attachment welds of existing reinforcing plates shall be examined by visual as well as by magnetic particle or liquid penetrant methods.

10.1.2.3 Completed welds attaching nozzle neck to shell, and reinforcing plate to shell and to nozzle neck, shall be examined by the magnetic particle or liquid penetrant methods.

10.1.2.4 Completed welds of stress relieved assemblies shall be examined by visual as well as by magnetic particle or liquid penetrant methods after stress relief, but before hydrostatic test.

10.1.2.5 For penetrations installed using insert plates, the completed butt welds between the insert plate and the shell plate shall be fully radiographed.

10.1.3 REPAIRED WELD FLAWS

10.1.3.1 Cavities resulting from gouging or grinding operations to remove weld defects shall be examined by visual as well as by magnetic particle or liquid penetrant methods.

10.1.3.2 Completed repairs of butt welds shall be examined over their full length by radiographic or ultrasonic methods.

10.1.3.3 Completed repairs of fillet welds shall be examined over their full length by the appropriate nondestructive examination method listed herein.

10.1.4 TEMPORARY AND PERMANENT ATTACHMENTS TO SHELL PLATES

10.1.4.1 Areas where weld projections (remaining after removal of temporary attachments) have been ground flush with the shell plate shall be examined by the appropriate nondestructive examination method listed herein.

10.1.4.2 Completed welds of temporary or new permanent attachments made to shells of API Standard 650 Group IV, IVA, V, or VI materials shall be examined by visual as well as by magnetic particle or liquid penetrant methods.

10.1.5 SHELL PLATE TO SHELL PLATE WELDS

New welds attaching shell plate to shell plate shall be examined by visual and by radiographic methods (see 10.2). In addition, for plate thicknesses greater than one inch, the back-gouged surface of the root pass and final pass (each side) shall be examined for its complete length by magnetic particle or liquid penetrant methods.

10.1.6 SHELL-TO-BOTTOM WELD

New welding on the shell-to-bottom joint shall be inspected for its entire length by a right-angle vacuum box and a solution film, or by applying light diesel oil. Additionally, the first weld pass shall be inspected by applying light diesel oil to the side opposite the first weld pass made. The oil shall be allowed to stand at least four hours (preferably overnight) and then the weld inspected for "wicking" action. The oil shall be removed before the weld is completed.

10.1.7 BOTTOMS

Newly welded bottom joints shall be tested using a vacuum box and a solution film, or a tracer gas and detector. Every joint shall be tested for its entire length.

10.2 Radiographs

10.2.1 NUMBER AND LOCATION OF RADIOGRAPHS

The number and location of radiographs shall be in accordance with API Standard 650 and the following additional requirements:

10.2.1.1 For new or repaired shell vertical joints one radiograph shall be taken in every joint.

10.2.1.2 For new or repaired shell horizontal joints one radiograph shall be taken for each 50 feet of horizontal joint, not including those required at intersections of vertical and horizontal joints.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607825 480 ■

10.2.1.3 For each new or repaired intersection between shell vertical and horizontal joints one radiograph shall be taken.

10.2.1.4 For reconstructed tanks, each butt-welded annular plate joint shall be radiographed in accordance with API Standard 650.

10.2.1.5 New and replaced shell insert plate and door sheet welds shall be radiographed. For circular replacement plates, a minimum of one radiograph shall be taken regardless of thickness. For square or rectangular replacement plates, at least one radiograph shall be taken in a vertical joint, and at least one in a horizontal joint, and one in each corner. All junctions between repair and existing welds shall be radiographed. If defects are found, 100 percent radiography shall be performed on the repaired weld.

10.2.1.6 The minimum diagnostic length of each radiograph shall be 6 inches.

10.2.2 ACCEPTANCE CRITERIA FOR EXISTING SHELL PLATE TO SHELL PLATE WELDS

If the radiograph of an intersection between a new and old weld detects unacceptable welds by current standards, the existing welds may be evaluated according to the original standard of construction.

10.2.3 MARKING AND IDENTIFICATION OF RADIOGRAPHS

10.2.3.1 Each film shall show an identification of the welder(s) making the weld. A weld map showing location of welds, weld number, radiograph number, welder identification, and grading of each weld is an acceptable alternative to this requirement.

10.2.3.2 Radiographs of repaired work and records of radiographs of repaired work shall be marked with the letter "R."

10.3 Hydrostatic Testing

10.3.1 WHEN HYDROSTATIC TESTING REQUIRED

10.3.1.1 A full hydrostatic test, held for 24 hours, shall be performed on:

a. A reconstructed tank.
b. Any tank that has undergone major repairs or major alterations (see 10.3.1.2) unless exempted by 10.3.2 for the applicable combination of materials, design, and construction features.

10.3.1.2 The terms major repair and major alteration refer to operations that require cutting, addition, removal and/or replacement of the annular plate ring, the shell-to-bottom weld, or a sizable portion of the shell. Within this context, *major repairs* and *major alterations* would include:

a. The installation of any shell penetration beneath the design liquid level larger than 12 inches NPS, or any bottom penetration located within 12 inches of the shell.
b. The removal and replacement or addition of any shell plate beneath the design liquid level, or any annular plate ring material where the longest dimension of the replacement plate exceeds 12 inches.
c. The complete or partial (more than one-half of the weld thickness) removal and replacement of more than 12 inches of vertical weld joining shell plates, or radial weld joining the annular plate ring.
d. The installation of a new bottom.
e. The removal and replacement of any part of the weld attaching the shell to the bottom or to the annular plate ring.
f. Partial or complete jacking of a tank shell.

10.3.2 WHEN HYDROSTATIC TESTING NOT REQUIRED

10.3.2.1 A full hydrostatic test of the tank is not required for major repairs and major alterations when any of the basic conditions listed in Column A of Table 10-1, in combination with the specific requirements of Column B, are satisfied. (See Figure 10-1.)

10.3.2.2 Specific requirements that must be satisfied in establishing an exemption basis from hydrostatic testing are identified for use in Column B of Table 10-1.

a. The repair method has been reviewed and approved by an engineer experienced in storage tank design in accordance with API Standard 650.
b. Materials used for the repair shall meet API Standard 650 requirements.
c. Vertical and horizontal shell joint welds shall have complete penetration and complete fusion. The root pass and final pass examination shall be in accordance with 10.1.5. In addition, the finished weld shall be fully radiographed.
d. Shell penetrations shall be installed with complete penetration and complete fusion weld for the reinforcing plate to nozzle neck to shell joints. The root pass of the nozzle attachment weld shall be back-gouged and examined by magnetic particle or liquid penetrant methods; the completed weld shall be examined by the ultrasonic method. Examination and acceptance criteria for nondestructive examinations shall be in accordance with 10.1.1.

10.3.2.3 A full hydrostatic test is not required after replacement of a door sheet provided that both of the following conditions are met:

a. Door sheet is in accordance with the requirements of 7.2, except that it shall not extend to or intersect the bottom of the shell joint. **Note to Rick Basile: please check this info.**
b. The tank is otherwise exempt from hydrostatic testing as permitted by 10.3.2.1.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607826 317 ■

10.4 Leak Tests

New or altered reinforcing plates of shell penetrations shall be given an air leak test in accordance with API Standard 650.

10.5 Measured Settlement During Hydrostatic Testing

10.5.1 INITIAL SURVEY

10.5.1.1 Where settlement is anticipated, a tank receiving a hydrostatic test shall have the foundation checked for settlement.

10.5.1.2 Tank settlement shall initially be surveyed with the tank empty using the number of bottom plate projection elevation measurement points, N, uniformly distributed around the circumference, as indicated by the following formula:

$$N = D/10$$

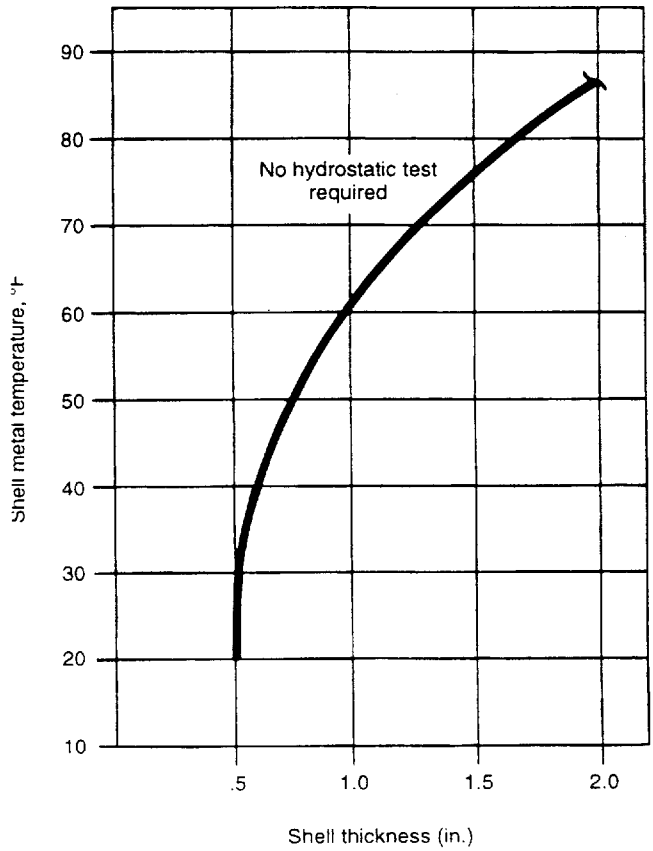
Where:

N = minimum required number of settlement measurement points, but no less than eight. All fractional values shall be rounded to the next higher whole number. The maximum spacing between settlement measurement points shall be 30 feet.

D = tank diameter, in feet.

Table 10-1—Conditions for Exemption From Tank Hydrostatic Test for Major Repairs and Alterations

A Basic Conditions	B Specific Requirements (see 10.3.2.2 for subitem identification)
1. Tank material of unknown toughness, but satisfies the exemption criteria of Figure 8.	a, b, c, d
2. Tank material meets the toughness requirements of API Standard 650, seventh edition or later.	a, b, c, d
3. Tank repairs limited to bottom plates or annular plate ring (excluding shell-to-bottom weld).	a, b



Note: For unheated tanks, the shell metal temperature shall equal the design metal temperature as defined in 2.2.9.3 of API Standard 650. The original nominal thickness for the thickest tank shell plate shall be used for the assessment.

Figure 10-1—Exemption Curve for Hydrostatic Testing of Tanks

10.5.1.3 Tank settlement measurements in 10.5.1.2 shall be evaluated for acceptance in accordance with Appendix B.

10.5.2 SURVEY DURING HYDROSTATIC TESTING

Settlement shall be measured during filling and when water reaches 100 percent of test level. Excessive settlement in accordance with Appendix B shall be cause to stop the test for foundation investigation and/or repair.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

SECTION 11—MARKING AND RECORD KEEPING

11.1 Nameplates

11.1.1 Tanks reconstructed in accordance with this standard shall be identified by a corrosion-resistant metal nameplate similar to that shown in Figure 11-1. Letters and numerals not less than 5/32 inch high shall be embossed, engraved, or stamped in the plate to indicate information as follows:

a. Reconstructed to API 653.
b. Edition and revision number.
c. Year reconstruction was completed.
d. If known, the original applicable standard and the year of original construction.
e. Nominal diameter.
f. Nominal shell height.
g. Design specific gravity.
h. Maximum permissible operating liquid level.
i. The name of the reconstruction contractor and the assigned serial number or contract number.
j. The owner/operator's tank number.
k. Shell material for each shell course.
l. Maximum operating temperature.
m. Allowable stress used in calculations of each shell course.

11.1.2 The new nameplate shall be attached to the tank shell adjacent to the existing nameplate, if any. An existing nameplate shall be left attached to the tank. Nameplates shall be attached as specified in API Standard 650.

11.2 Record Keeping

When a tank is evaluated, repaired, altered, or reconstructed in accordance with this standard, the following information, as applicable, shall be made a part of the owner/operator's records for the tank (see 4.8).

11.2.1 Calculations for:

a. Component evaluation for integrity, including brittle fracture considerations (Section 3).
b. Rerating (including liquid level).
c. Repair and alteration considerations.

11.2.2 Construction and repair drawings.

11.2.3 Additional support data including, but not limited to, information pertaining to:

a. Inspections (including thicknesses).
b. Material test reports/certifications.
c. Tests.
d. Radiographs (radiographs shall be retained for at least one year).
e. Brittle fracture considerations.
f. Original tank construction data (date, original standard, etc.).
g. Location and identification (owner/operator's number, serial number).
h. Description of the tank (diameter, height, service).
i. Design conditions (liquid level, specific gravity, allowable stress, unusual design loadings, etc.).
j. Shell material and thickness by course.
k. Tank perimeter elevations.
l. Construction completion record.

11.3 Certification

Tanks reconstructed in accordance with this standard shall require documentation of such reconstruction, and certification that the design, reconstruction, inspection, and testing was performed in compliance with this standard. The certification shall contain information as shown in Figure 11-2 for design and/or reconstruction as applicable.

e. Brittle fracture considerations.
f. Original tank construction data (date, original standard, etc.).
g. Location and identification (owner/operator's Number, serial Number).
h. Description of the tank (diameter, height, service).
i. Design conditions (liquid level, specific gravity, allowable stress, unusual design loadings, etc.).
j. Shell material and thickness by course.
k. Tank perimeter elevations.
l. Construction completion record.

11.3 Certification

Tanks reconstructed in accordance with this standard shall require documentation of such reconstruction, and certification that the design, reconstruction, inspection, and testing was performed in compliance with this standard. The certification shall contain information as shown in Figure 11-2 for design and/or reconstruction as applicable.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607828 19T ■

REPAIRED/ALTERED TO API 653 EDITION _____

Reconstructed by:

_____

_____

Date Completed _____

Serial No. _____

Shell Course	Shell Course	Shell Course
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

Original Standard _____

Tank No. _____

Tank Dia. _____ Height _____

Specific Gravity _____

Design Pressure _____

Orig. Const. Date _____

Year Reconstructed _____

Liquid Level Max. _____

Capacity _____

Max. Operating Temp. _____

Figure 11-1—Nameplate

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607829 026 ■

CERTIFICATION FOR TANK RECONSTRUCTED TO API 653

We hereby certify that the tank reconstructed at _____ and described as follows:
Location

_____ Serial No. _____ Owner's No. _____ Height _____ Capacity _____ Floating or Fixed Roof

was reconstructed, inspected, and tested in accordance with all applicable requirements of API Standard 653,

_____ Edition, _____ Revision, Dated _____ (including all material supplied by the reconstruction organization).

_____
Reconstruction Organization

_____
Authorized Representative

_____
Date

CERTIFICATION FOR TANK RECONSTRUCTED TO API 653

We hereby certify that the design evaluation of the tank reconstructed at _____ and described as follows:
Location

_____ Serial No. _____ Owner's No. _____ Height _____ Capacity _____ Floating or Fixed Roof

was performed by the undersigned organization in accordance with all design requirements of API Standard 653,

_____ Edition, _____ Revision, Dated _____.

_____
Reconstruction Organization

_____
Authorized Representative

_____
Date

Figure 11-2—Certification Forms

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX A—BACKGROUND ON PAST EDITIONS OF API WELDED STORAGE TANK STANDARDS

API published a specification for welded steel storage tanks in 1936 entitled Standard 12C, *API Specification for Welded Oil Storage Tanks.* Fifteen editions of and seven supplements to Standard 12C were published between 1936 and 1961. Standard 12C was replaced by API Standard 650, *Welded Steel Tanks for Oil Storage;* nine editions of and eleven supplements or revision packages to API Standard 650 have been issued. The current edition of API Standard 650 is the Ninth Edition, published in July 1993.

The table below provides a list of editions, supplements, and revisions to Standard 12C and API Standard 650.

Table A-1—Editions of API Standard 650 and Its Precursor, Standard 12C

Standard 12C, *API Specification for Welded Oil Storage Tanks*

Edition	Date
First	July 1936
Second	October 1937
Supplement 1	April 1938
Supplement 2	September 1938
Supplement 3	April 1939
Third	April 1940
Fourth	March 1941
Fifth	May 1942
Sixth	August 1944
Seventh	August 1946
Supplement 1	September 1947
Eighth	September 1948
Supplement 1	December 1949
Ninth	October 1950
Tenth	September 1951
Eleventh	September 1952
Supplement 1	September 1953
Twelfth	October 1954
Thirteenth	September 1955
Supplement 1	October 1956
Fourteenth	October 1957
Fifteenth	1958

Standard 650, *Welded Steel Tanks for Oil Storage*

Edition	Date
First	1961
Supplement	1963
Second	1964
Third	1966
Supplement 1	December 1967
Fourth	June 1970
Supplement 1	April 1971
Fifth	July 1973
Supplement 1	October 1973
Supplement 2	April 1974
Supplement 3	March 1975
Sixth	April 1977
Revision 1	May 1978
Revision 2	December 1978
Revision 3	October 1979
Seventh	November 1980
Revision 1	February 1984
Eighth	November 1988
Ninth	July 1993
Addendum 1	December 1994

Ordering information for past editions of API Standard 12C and API Standard 650 is available by calling the API Library at (202) 682-8042.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX B—EVALUATION OF TANK BOTTOM SETTLEMENT

B.1 Introduction

B.1.1 In determining the effects of soil settlement on storage tanks, it is common practice to monitor settlement of the tank bottom. In most cases, such a monitoring program is initiated during the construction and continued during hydrotesting and operations. During operations, settlement measurements should be taken at a planned frequency, based upon an assessment of soil settlement predictions. For existing tanks which do not have initial settlement data, a program of settlement monitoring should be based on prior service history.

B.1.2 If at any time settlement is deemed excessive, the tanks should be emptied and releveled. Releveling of a sizable tank is expensive and rather difficult to achieve. Thus, a decision to relevel a tank is a crucial one, and relies very much on the proper interpretation and evaluation of the monitored settlement data.

B.1.3 Approaches used to correct tank shell and bottom settlement include techniques such as localized repairs of the bottom plates, partial releveling of the tank periphery, and major releveling of the entire tank bottom. Major releveling of the tank, involving total lifting of the ½ tank shell and bottom at one time, can introduce high localized stresses in the structure and impair its integrity. Therefore, when choosing techniques for correcting settlement problems, an alternative to total lifting of the tank shell and bottom should be considered as a first choice. If it is decided to lift the entire tank shell and bottom at one time, it should be done by personnel with demonstrated experience in this technique.

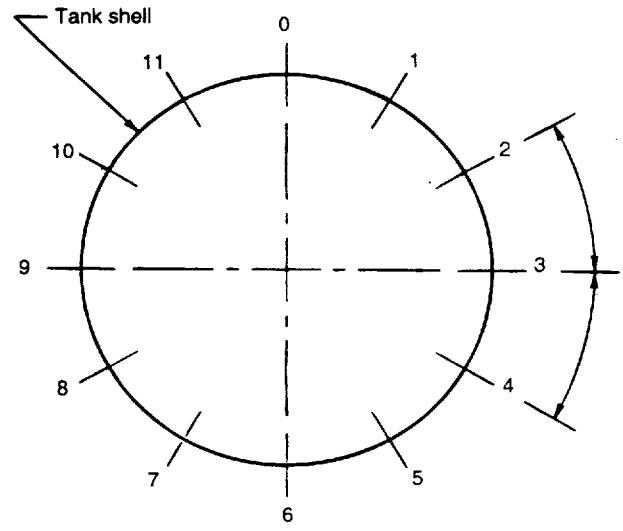
B.2 Types of Settlement

B.2.1 ELEVATION MEASUREMENTS

The principal types of tank settlement consist of settlements which relate to the tank shell and bottom plate. These settlements can be recorded by taking elevation measurements around the tank circumference and across the tank diameter. Figures B-1 and B-2 show recommended locations on a tank shell and bottom plate for settlement measurements. Data obtained from such measurements should be used to evaluate the tank structure. Additional settlement readings may be required to define local depressions.

B.2.2 SHELL SETTLEMENT EVALUATION

Settlement of a tank is considered to be the result of either one or a combination of the following three settlement components.



Notes:
1. There must be at least eight settlement points. The maximum spacing of settlement points is 30 feet around the circumference.
2. Points shall be equally spaced around the tank shell. See 10.5.1.2 for method of determining the number of measurement points.

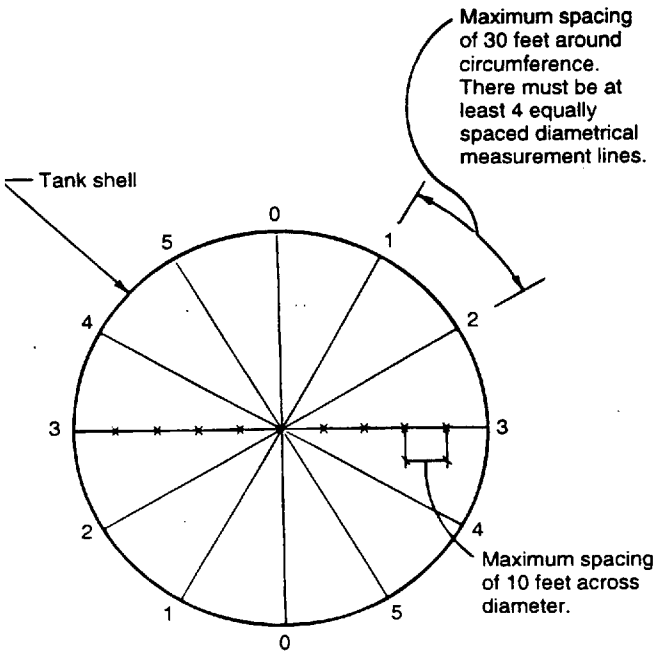
Figure B-1—Measurements of Shell Settlement (External)

B.2.2.1 Uniform settlement. This component often can be predicted in advance, with sufficient accuracy from soil tests. It may vary in magnitude, depending on the soil characteristics. Uniform settlement of a tank does not induce stresses in the tank structure. However, piping and attachments must be given adequate consideration to prevent problems caused by such settlement.

B.2.2.2 Rigid body tilting of a tank (planar tilts). This component rotates the tank in a tilted plane. The tilt will cause an increase in the liquid level and, therefore, an increase in the hoop stress in the tank shell. Also, excessive tilting can cause binding of peripheral seals in a floating roof and inhibit roof travel. This type of settlement could affect tank nozzles that have piping attached to them. Figure B-3 shows that the settled location of the tank shell, after rigid body tilt, can be represented by either a cosine or sine wave with respect to its original position in a horizontal plane.

B.2.2.3 Out-of-plane settlement (differential settlement). Due to the fact that a tank is a rather flexible structure, chances are great that the tank shell will settle in a non-planar con-

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34



Note: See 10.5.1.2 for method of determining the number of measurement points.

Figure B-2—Measurements of Bottom Settlement (Internal) Tank Out-of-Service

figuration, inducing additional stresses in the tank shell. The out-of-plane settlements at the bottom edge lead to a lack of circularity at the top of the tank, and in the case of a floating roof tank, the extent of the induced ovality may impede the proper functioning of the floating roof in such a way that releveling is required. Also, such settlements may cause flat spots to develop in the tank shell. This type of settlement could affect tank nozzles that have piping attached to them.

B.2.2.4 While uniform settlement and rigid body tilt of a tank may cause problems as described above, the out-of-plane settlement is the important component to determine and evaluate in order to ensure the structural integrity of the shell and bottom. Based on this principle, a common approach is to determine the magnitudes of the uniform and rigid body tilt component (if any) for each data point on the tank periphery. Once this is carried out, the plane of rigid tilt is then important as a datum from which to measure the magnitudes of the out-of-plane settlements.

A graphical representation illustrating the effect of tank shell settlement components is shown in Figure B-3. The construction of this "settlement plot" has been developed in accordance with the following:

a. The actual edge settlement (in most cases an irregular curve) is plotted using points around the tank circumference as the abscissa.

b. The vertical distance between the abscissa and the lowest point on this curve (point 22) is the minimum settlement, and it is called the uniform settlement component. A line through this point, parallel to the abscissa, provides a new base or datum line for settlement measurements which are called "adjusted settlements."

c. The plate of rigid tilt settlement is represented by the "optimum cosine curve" between the maximum and minimum values of the actual edge settlements. Several methods exist for determining the optimum cosine curve. The least accurate method is by free hand drawing techniques, a kind of trial and error procedure to fit the best cosine curve through the data. A better method is to use the mathematical and graphical capabilities of a computer.

d. The vertical distances between the irregular curve and the cosine curve represent the magnitudes of the out-of-plane settlements (U_i at data point i).

B.2.2.5 A method of determining the acceptable settlement is given in B.3.2. As an alternative, a rigorous stress analysis could be carried out.

B.2.3 EDGE SETTLEMENT

B.2.3.1 Edge settlement occurs when the tank shell settles sharply around the periphery, resulting in deformation of the bottom plate near the shell-to-bottom corner junction. Figure B-4 illustrates this settlement.

B.2.3.2 The formula given in B.3.3 can be used to evaluate this criteria. Alternatively, a rigorous stress analysis can be carried out for the deformed profile.

B.2.4 BOTTOM SETTLEMENT NEAR TANK SHELL

B.2.4.1 Figure B-5 illustrates this type of settlement.

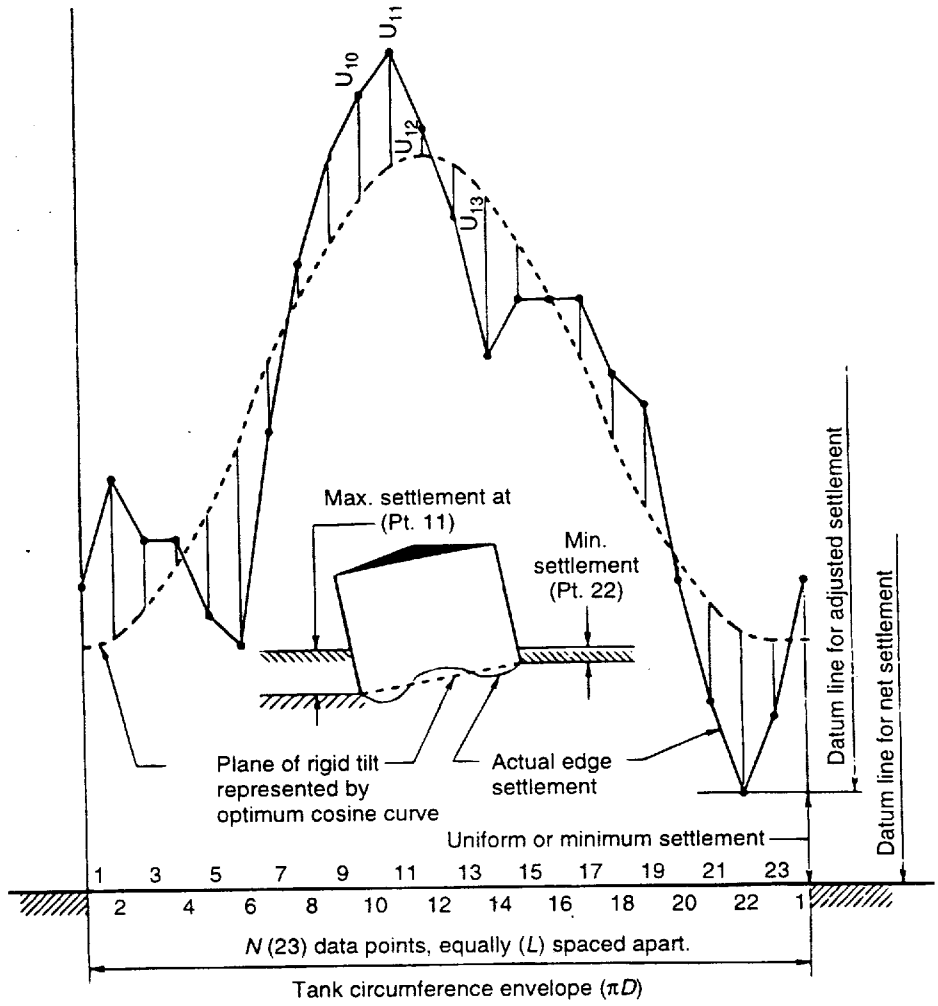
B.2.4.2 The formula given in B.3.3 can be used to evaluate this criteria. Alternatively, a rigorous stress analysis can be carried out for the deformed profile.

B.2.5 LOCALIZED BOTTOM SETTLEMENT REMOTE FROM TANK SHELL

B.2.5.1 These are depressions (or bulges) that occur in a random manner, remote from the shell (see Figure B-6).

B.2.5.2 Acceptability of these localized settlements is dependent upon localized stresses in the bottom plate, design and quality of the lap welds (single pass or multipass), and voids below the bottom plate. The formula given in B.3.3 can be used to evaluate this criteria. These limits are applicable to tank bottoms which have single pass lap welded joints.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34



Out-of-plane deflection for point "i" is
$S_i = U_i - (1/2\ U_i + 1/2\ U_{i+1})$, for example
$S_{11} = U_{11}\ (1/2_{U10} + 1/2_{U12})$

U_i = out-of-plane settlement of point *"i"*,
(+) when above cos. curve;
(–) when below cos. curve, for example:
U_{11} = (+)
U_{13} = (–)

Note: See 10.5.1.2 for definition of "N."

Figure B-3—Graphical Representation of Tank Shell Settlement

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607834 493 ■

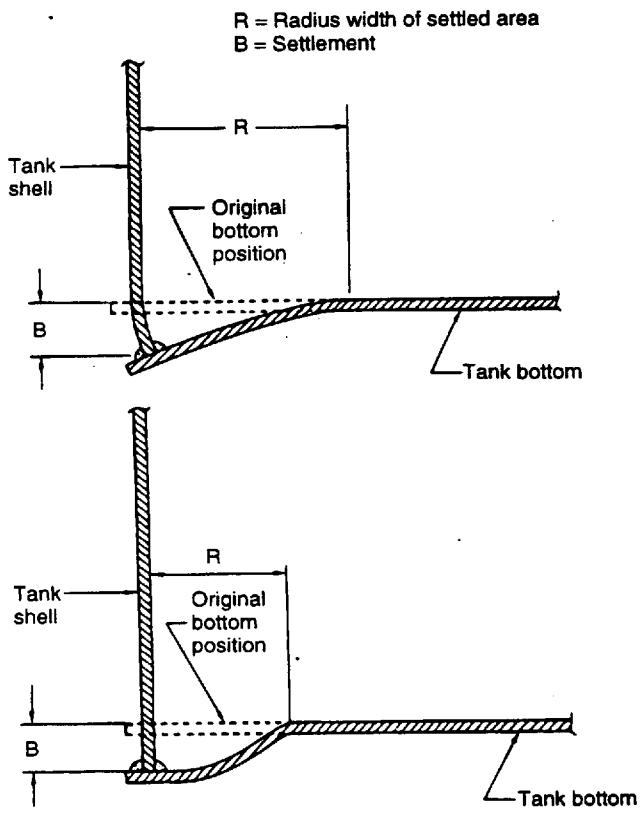


Figure B-4—Edge Settlement

B.3 Determination of Acceptable Settlement

B.3.1 GENERAL

For existing tanks with history of successful service, it may be possible to accept greater settlement and distortion of the foundation from a true plane than new tank construction standards allow. Each tank must be evaluated based on service conditions, materials of construction, soil characteristics, tank foundation design, and tank service history. The methods discussed in following sections are not mandatory and are an approximation of the maximum permissible settlement. However, experience has shown that if settlements exceed the following requirements, further assessment or repair is required.

B.3.2 SHELL SETTLEMENT

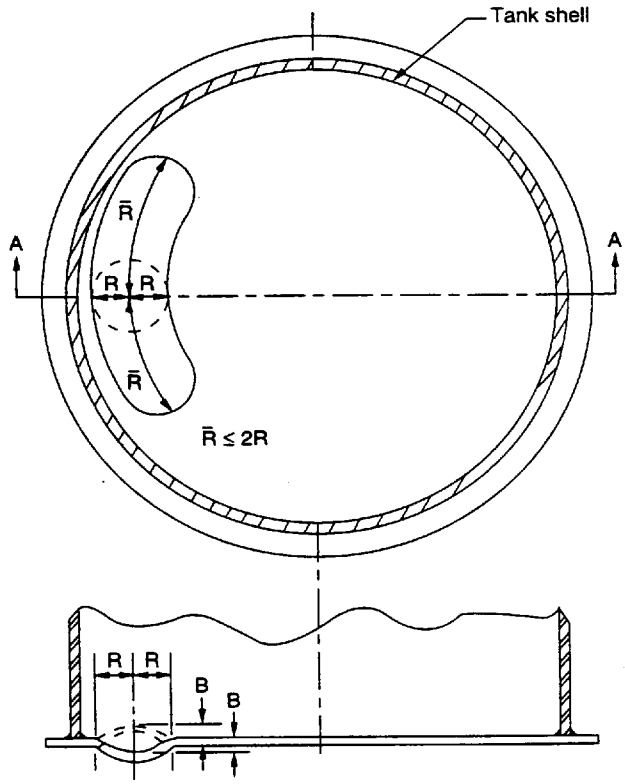
From the measurements described in B.2, determine the maximum out-of-plane deflection. Use the following formula to calculate the maximum permissible out-of-plane deflection (see note below.)

$$|S| \le \frac{(L^2 \times Y \times 11)}{[2(E \times H)]}$$

Where:

S = deflection, in feet (out of plane distortion)
L = arc length between measurement points, in feet
Y = yield strength, in pounds per square inch
E = Young's modulus, in pounds per square inch
H = tank height, in feet

Note: This formula is based on "Criteria for Settlement of Tanks," W. Allen Marr, M. ASCE, Jose A. Ramos, and T. William Lambe, F. ASCE, *Journal of Geotechnical Engineering Division, Proceedings of the American Society of Civil Engineers*, Vol. 108, August 1982.



SECTION A-A

R = Radius of inscribed circle in bulged or depressed area
B = Settlement or depression depth or height of bulge

Figure B-5—Bottom Settlement Near Shell

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607835 32T ■

B.3.3 INTERNAL BOTTOM SETTLEMENTS OR BULGES

Measure the bulge or depression. The permissible bulge or depression is given by the following formula (see note below).

$$B = 0.37R$$

Where:

B = maximum height of bulge or depth of local depression, inches

R = radius of inscribed circle in bulged area or local depression, in feet

Figure B-7 is a graphical representation of this formula.

Note: This formula is based on "Criteria for Settlement of Tanks," W. Allen Marr, M. ASCE, Jose A. Ramos, and T. William Lambe, F. ASCE, *Journal of Geotechnical Engineering Division, Proceedings of the American Society of Civil Engineers*, Vol. 108, August 1982.

B.4 Repairs

If it is determined that settlements have occurred which are beyond the permissible limits established in the previous sections, then consideration should be given to making repairs or a rigorous stress analysis should be performed to evaluate the deformed profile. Various repair techniques have been discussed above. The judgement on repairs should be tempered with knowledge of tank service history, previous repairs, previous inspections, tank foundation conditions, soil characteristics, the material of construction, and estimates of future settlement. See 7.9 for suggested repair details.

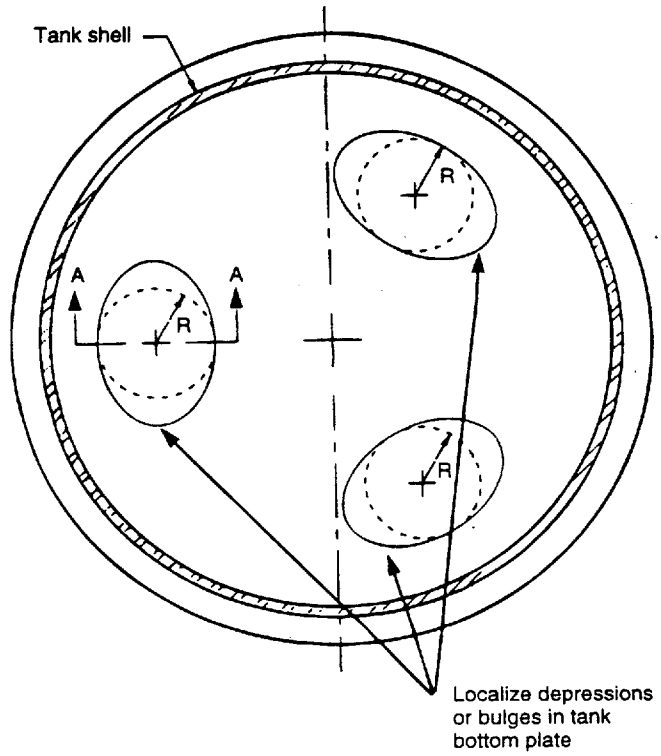


Figure B-6—Localized Bottom Depressions or Bulges Remote From Shell

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

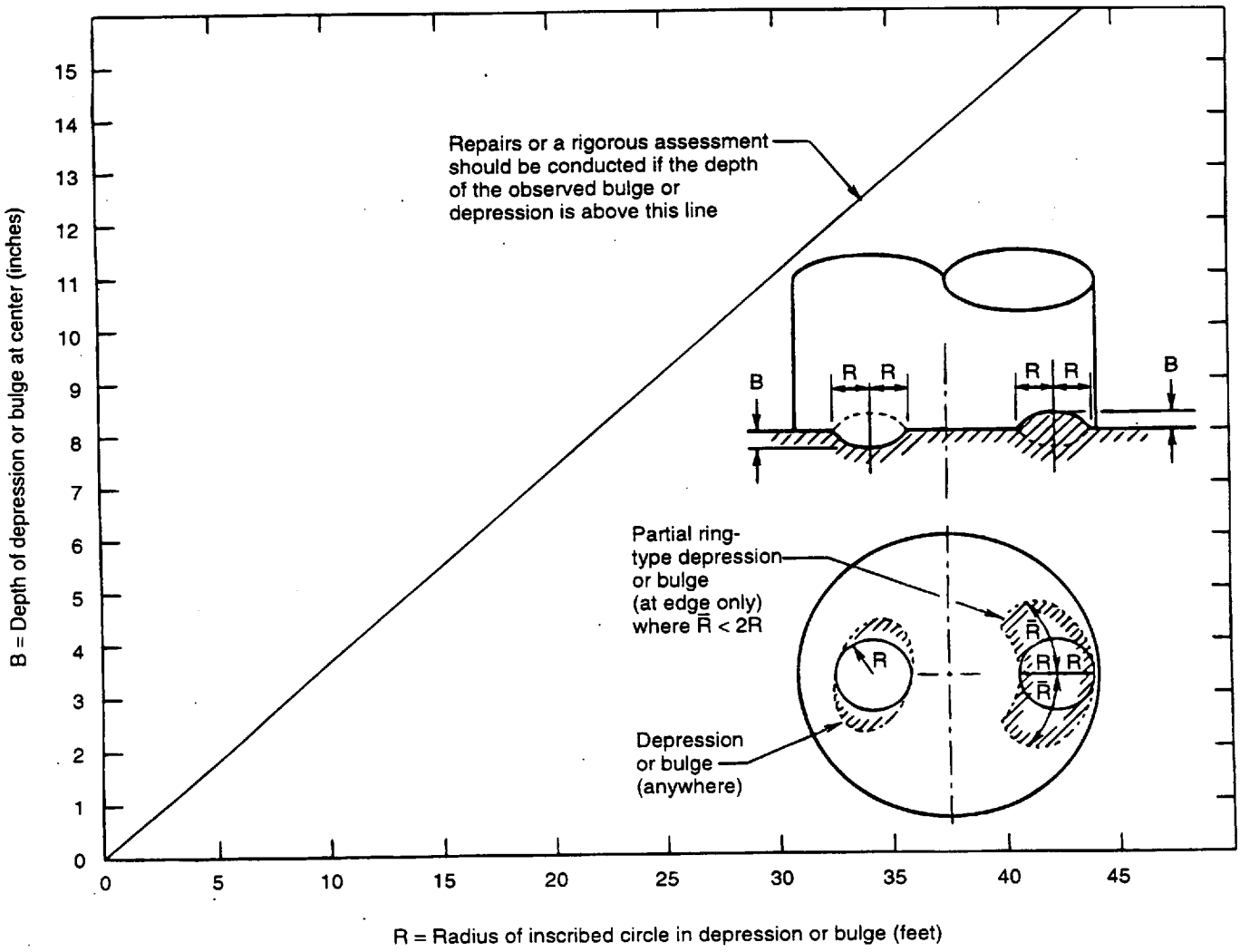


Figure B-7—Localized Tank Bottom Settlement Limits for Single Pass Welds

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX C—CHECKLISTS FOR TANK INSPECTION

Appendix C contains sample checklists illustrating tank components and auxiliary items that should be considered for internal and external inspection of tanks. This information is provided as guidance to the owner/operator for developing an inspection assessment schedule for any specific tank installation. The checklist format facilitates the recording of inspection findings.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607838 039 ■

TANK IN-SERVICE INSPECTION CHECKLIST

	Item	Completed ✔	Comments
C.1.1	**FOUNDATION**		
	Measure foundation levelness and bottom elevations (see Appendix B for extent of measurements).		
C.1.1.1	**Concrete Ring**		
	a. Inspect for broken concrete, spalling, and cracks, particularly under backup bars used in welding butt welded annular rings under the shell.		
	b. Inspect drain openings in ring, back of waterdraw basins and top surface of ring for indications of bottom leakage.		
	c. Inspect for cavities under foundation and vegetation against bottom of tank.		
	d. Check that runoff rainwater from the shell drains away from tank.		
	e. Check for settlement around perimeter of tank.		
C.1.1.2	**Asphalt**		
	a. Check for settling of tank into asphalt base which would direct runoff rain water under the tank instead of away from it.		
	b. Look for areas where leaching of oil has left rock filler exposed, which indicates hydrocarbon leakage.		
C.1.1.3	**Oiled Dirt or Sand**		
	Check for settlement into the base which would direct runoff rain water under the tank rather than away from it.		
C.1.1.4	**Rock**		
	Presence of crushed rock under the steel bottom usually results in severe underside corrosion. Make a note to do additional bottom plate examination (ultrasonic, hammer testing, or turning of coupons) when the tank is out of service.		
C.1.1.5	**Site Drainage**		
	a. Check site for drainage away from the tank and associated piping and manifolds.		
	b. Check operating condition of the dike drains.		
C.1.1.6	**Housekeeping**		
	Inspect the area for buildup of trash, vegetation, and other inflammables buildup.		
C.1.2	**SHELLS**		
C.1.2.1	**External Visual Inspection**		
	a. Visually inspect for paint failures, pitting, and corrosion.		
	b. Clean off the bottom angle area and inspect for corrosion and thinning on plate and weld.		
	c. Inspect the bottom-to-foundation seal, if any.		
C.1.2.2	**Internal (Floating Roof Tank)**		
	Visually inspect for grooving, corrosion, pitting, and coating failures.		
C.1.2.3	**Riveted Shell Inspection**		
	a. Inspect external surface for rivet and seam leaks.		
	b. Locate leaks by sketch or photo (location will be lost when shell is abrasive cleaned for painting).		
	c. Inspect rivets for corrosion loss and wear.		
	d. Inspect vertical seams to see if they have been full fillet lap welded to increase joint efficiency.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607839 T75 ■

TANK IN-SERVICE INSPECTION CHECKLIST—Continued			
	Item	Completed ✔	Comments
	e. If no record exists of vertical riveted seams, dimension and sketch (or photograph) the rivet pattern: number of rows, rivet size, pitch length, and note whether the joint is butt riveted or lap riveted.		
C.1.2.4	**Wind Girder (Floating Roof Tanks)**		
	a. Inspect wind girder and handrail for corrosion damage (paint failure, pitting, corrosion product buildup), especially where it occurs at tack welded junction, and for broken welds.		
	b. Check support welds to shell for pitting, especially on shell plates.		
	c. Note whether supports have reinforcing pads welded to shell.		
C.1.3	**SHELL APPURTENANCES**		
C.1.3.1	**Manways and Nozzles**		
	a. Inspect for cracks or signs of leakage on weld joint at nozzles, manways, and reinforcing plates.		
	b. Inspect for shell plate dimpling around nozzles, caused by excessive pipe deflection.		
	c. Inspect for flange leaks and leaks around bolting.		
	d. Inspect sealing of insulation around manways and nozzles.		
	e. Check for inadequate manway flange and cover thickness on mixer manways.		
C.1.3.2	**Tank Piping Manifolds**		
	a. Inspect manifold piping, flanges, and valves for leaks.		
	b. Inspect fire fighting system components.		
	c. Check for anchored piping which would be hazardous to the tank shell or bottom connections during earth movement.		
	d. Check for adequate thermal pressure relief of piping to the tank.		
	e. Check operation of regulators for tanks with purge gas systems.		
	f. Check sample connections for leaks and for proper valve operation.		
	g. Check for damage and test the accuracy of temperature indicators.		
	h. Check welds on shell-mounted davit clips above valves 6 inches and larger.		
C.1.3.3	**Autogauge System**		
	a. Inspect autogauge tape guide and lower sheave housing (floating swings) for leaks.		
	b. Inspect autogauge head for damage.		
	c. Bump the checker on autogauge head for proper movement of tape.		
	d. Identify size and construction material of autogauge tape guide (floating roof tanks).		
	e. Ask operator if tape tends to hang up during tank roof movement (floating roof tanks).		
	f. Compare actual product level to the reading on the autogauge (maximum variation is 2 inches).		
	g. On floating roof tanks, when the roof is in the lowest position, check that no more than two feet of tape are exposed at the end of the tape guide.		
	h. Inspect condition of board and legibility of board-type autogauges.		
	i. Test freedom of movement of marker and float.		
C.1.3.4	**Shell-Mounted Sample Station**		
	a. Inspect sample lines for function of valves and plugging of lines, including drain or return-to-tank line.		
	b. Check circulation pump for leaks and operating problems.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607840 797 ■

TANK IN-SERVICE INSPECTION CHECKLIST—Continued			
	Item	**Completed ✔**	**Comments**
	c. Test bracing and supports for sample lines and equipment.		
C.1.3.5	**Heater (Shell Manway Mounted)**		
	Inspect condensate drain for presence of oil indicating leakage.		
C.1.3.6	**Mixer**		
	a. Inspect for proper mounting flange and support.		
	b. Inspect for leakage.		
	c. Inspect condition of power lines and connections to mixer.		
C.1.3.7	**Swing Lines: Winch Operation**		
	a. Nonfloating. Raise, then lower the swing line with the winch, and check for cable tightness to confirm that swing line lowered properly.		
	b. Floating. With tank half full or more, lower the swing line, then let out cable and check if swing has pulled cable tight, indicating that the winch is operating properly.		
	c. Indicator. Check that the indicator moves in the proper direction: Floating swing line indicators show a lower level as cable is wound up on the winch. Non-floating swing line indicators show the opposite.		
C.1.3.8	**Swing Lines: External Guide System**		
	Check for leaks at threaded and flanged joints.		
C.1.3.9	**Swing Lines: Identify Ballast Varying Need**		
	Check for significant difference in stock specific gravity.		
C.1.3.10	**Swing Lines: Cable Material and Condition**		
	a. For non-stainless steel cable, check for corrosion over entire length.		
	b. All cable: check for wear or fraying.		
C.1.3.11	**Swing lines: Product Sample Comparison**		
	Check for water or gravity differences that would indicate a leaking swing joint.		
C.1.3.12	**Swing Lines: Target**		
	Target should indicate direction of swing opening (up or down) and height above bottom where suction will be lost with swing on bottom support.		
C.1.4	**ROOFS**		
C.1.4.1	**Deck Plate Internal Corrosion**		
	For safety, before accessing the roof, check with ultrasonic instrument or lightly use a ball peen hammer to test the deck plate near the edge of the roof for thinning. (Corrosion normally attacks the deck plate at the edge of a fixed roof and at the rafters in the center of the roof first.)		
C.1.4.2	**Deck Plate External Corrosion**		
	Visually inspect for paint failure, holes, pitting, and corrosion product on the roof deck.		
C.1.4.3	**Roof Deck Drainage**		
	Look for indication of standing water. (Significant sagging of fixed roof deck indicates potential rafter failure. Large standing water areas on a floating roof indicate inadequate drainage design or, if to one side, a nonlevel roof with possible leaking pontoons.)		
C.1.4.4	**Level of Floating Roof**		
	At several locations, measure distance from roof rim to a horizontal weld seam above the roof. A variance in the readings indicates a nonlevel roof with possible shell out-of-round, out-of-plumb, leaking pontoons, or hang-up. On small diameter tanks, an unlevel condition can indicate unequal loading at that level.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

TANK IN-SERVICE INSPECTION CHECKLIST—Continued			
	Item	**Completed ✔**	**Comments**
C.1.4.5	**Gas Test Internal Floating Roof**		
	Test for explosive gas on top of the internal floating roof. Readings could indicate a leaking roof, leaking seal system, or inadequate ventilation of the area above the internal floating roof.		
C.1.4.6	**Roof Insulation**		
	a. Visually inspect for cracks or leaks in the insulation weather coat where runoff rain water could penetrate the insulation.		
	b. Inspect for wet insulation under the weather coat.		
	c. Remove small test sections of insulation and check roof deck for corrosion and holes near the edge of the insulated area.		
C.1.4.7	**Floating Roof Seal Systems**		
	a. Measure and record maximum seal-to-shell gaps at:		
	1. Low pump out.		
	2. Mid-shell.		
	3. High liquid level.		
	b. Measure and record annular space at 30 foot spacing (minimum of four quadrants) around roof and record. Measurements should be taken in directly opposite pairs.		
	1. _____ _____ Opposite pair 1.		
	2. _____ _____ Opposite pair 2.		
	c. Check if seal fabric on primary shoe seals is pulling shoes away from shell (fabric not wide enough).		
	d. Inspect fabric for deterioration, holes, tears, and cracks.		
	e. Inspect visible metallic parts for corrosion and wear.		
	f. Inspect for openings in seals that would permit vapor emissions.		
	g. Inspect for protruding bolt or rivet heads against the shell.		
	h. Pull both primary and secondary seal systems back all around the shell to check their operation.		
	i. Inspect secondary seals for signs of buckling or indications that their angle with the shell is too shallow.		
	j. Inspect wedge-type wiper seals for flexibility, resilience, cracks, and tears.		
C.1.5	**ROOF APPURTENANCES**		
C.1.5.1	**Sample Hatch**		
	a. Inspect condition and functioning of sample hatch cover.		
	b. On tanks governed by Air Quality Monitoring District rules, check for the condition of seal inside hatch cover.		
	c. Check for corrosion and plugging on thief and gauge hatch cover.		
	d. Where sample hatch is used to reel gauge stock level, check for marker and tab stating hold off distance.		
	e. Check for reinforcing pad where sample hatch pipe penetrates the roof deck.		
	f. On floating roof sample hatch and recoil systems, inspect operation of recoil reel and condition of rope.		
	g. Test operation of system.		
	h. On ultra clean stocks such as JP4, check for presence and condition of protective coating or liner inside sample hatch (preventing rust from pipe getting into sample).		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607842 56T ■

TANK IN-SERVICE INSPECTION CHECKLIST—Continued			
	Item	**Completed ✔**	**Comments**
C.1.5.2	**Gauge Well**		
	a. Inspect visible portion of the gauge well for thinning, size of slots, and cover condition.		
	b. Check for a hold off distance marker and tab with hold off distance (legible).		
	c. On floating roofs, inspect condition of roof guide for gauge well, particularly the condition of the rollers for grooving.		
	d. If accessible, check the distance from the gauge well pipe to the tank shell at different levels.		
	e. If tank has a gauge well washer, check valve for leakage and for presence of a bull plug or blind flange.		
C.1.5.3	**Fixed Roof Scaffold Support**		
	Inspect scaffold support for corrosion, wear, and structural soundness.		
C.1.5.4	**Autogauge: Inspection Hatch and Guides (Fixed Roof)**		
	a. Check the hatch for corrosion and missing bolts.		
	b. Look for corrosion on the tape guide's and float guide's wire anchors.		
C.1.5.5	**Autogauge: Float Well Cover**		
	a. Inspect for corrosion.		
	b. Check tape cable for wear or fraying caused by rubbing on the cover.		
C.1.5.6	**Sample Hatch (Internal Floating Roof)**		
	a. Check overall conditions.		
	b. When equipped with a fabric seal, check for automatic sealing after sampling.		
	c. When equipped with a recoil reel opening device, check for proper operations.		
C.1.5.7	**Roof-Mounted Vents (Internal Floating Roof)**		
	Check condition of screens, locking and pivot pins.		
C.1.5.8	**Gauging Platform Drip Ring**		
	On fixed roof tanks with drip rings under the gauging platform or sampling area, inspect for plugged drain return to the tank.		
C.1.5.9	**Emergency Roof Drains**		
	Inspect vapor plugs for emergency drain: that seal fabric discs are slightly smaller than the pipe ID and that fabric seal is above the liquid level.		
C.1.5.10	**Removable Roof Leg Racks**		
	Check for leg racks on roof.		
C.1.5.11	**Vacuum Breakers**		
	Report size, number, and type of vacuum breakers. Inspect vacuum breakers. If high legs are set, check for setting of mechanical breaker in high leg position.		
C.1.5.12	**Rim Vents**		
	a. Check condition of the screen on the rim vent cover.		
	b. Check for plating off or removal of rim vents where jurisdictional rules do not permit removal.		
C.1.5.13	**Pontoon Inspection Hatches**		
	a. Open pontoon inspection hatch covers and visually check inside for pontoon leakage.		
	b. Test for explosive gas (an indicator of vapor space leaks).		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

TANK IN-SERVICE INSPECTION CHECKLIST—Continued

Item	Completed ✔	Comments
c. If pontoon hatches are equipped with locked down coves, check for vent tubes. Check that vent tubes are not plugged up. Inspect lock down devices for condition and operation.		
C.1.6 Accessways		
See Tank Out-of-Service Inspection Checklist, Item C.2.12.		

Notes:

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

TANK OUT-OF-SERVICE INSPECTION CHECKLIST			
	Item	**Completed ✔**	**Comments**
C.2.1	**OVERVIEW**		
	a. Check that tank has been cleaned, is gas free, and safe for entry.		
	b. Check that the tank is completely isolated from product lines, all electrical power, and steam lines.		
	c. Check that roof is adequately supported, including fixed roof structure and floating roof legs.		
	d. Check for presence of falling object hazards, such as corroded-through roof rafters, asphalt stalactites, and trapped hydrocarbons in unopened or plugged equipment or appurtenances, ledges, etc.		
	e. Inspect for slipping hazards on the bottom and roof decks.		
	f. Inspect structural welds on accessways and clips.		
	g. Check surfaces needing inspection for a heavy-scale buildup and check weld seams and oily surfaces where welding is to be done. Note areas needing more cleaning, including blasting.		
C.2.2	**TANK EXTERIOR**		
	a. Inspect appurtenances opened during cleaning such as lower floating swing sheave assemblies, nozzle interiors (after removal of valves).		
	b. Hammer test or ultrasonically test the roof.		
	c. Enter and inspect the floating roof pontoon compartments.		
C.2.3	**BOTTOM INTERIOR SURFACE**		
	a. Using a flashlight held close to and parallel to the bottom plates, and using the bottom plate layout as a guide, visually inspect and hammer test the entire bottom.		
	b. Measure the depth of pitting and describe the pitting appearance (sharp edged, lake type, dense, scattered, etc.)		
	c. Mark areas requiring patching or further inspection.		
	d. Mark locations for turning coupons for inspection.		
	e. Inspect all welds for corrosion and leaks, particularly the shell-to-bottom weld.		
	f. Inspect sketch plates for corrosion.		
	g. Locate and mark voids under the bottom.		
	h. Record bottom data on a layout sketch using the existing bottom plates as a grid. List the number and sizes of patches required.		
	i. Vacuum test the bottom lap welds.		
	j. Hammer test or ultrasonically examine any slightly discolored spots or damp areas.		
	k. Check for reinforcing pads under all bottom attached clips, brackets, and supports.		
	l. Inspect floating roof leg pads for pitting or cutting, and excessive dimpling (indicating excessive loading).		
	m. Check the column bases of fixed roof supports for adequate pads and restraining clips.		
	n. In earthquake zones 3 and 4, check that roof supports are not welded down to the tank bottom, but are only restrained from horizontal movement.		
	o. Check area beneath swing line cable for indications of cable cutting or dragging.		
	p. Mark old oil and air test connection for removal and patching.		
	q. Identify and report low areas on the bottom that do not drain adequately.		
	r. Inspect coating for holes, disbonding, deterioration, and discolorization.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607845 279 ■

TANK OUT-OF-SERVICE INSPECTION CHECKLIST—Continued			
	Item	Completed ✔	Comments
C.2.4	**SHELL SEAMS AND PLATE**		
	a. On cone up bottoms, closely inspect and gauge the depth of metal loss on the lower 2 to 4 inches of the shell (area of standing water).		
	b. Measure the depth of pitting on each course.		
	c. Inspect and estimate the amount of metal loss on the heads of rivets and bolts.		
	d. Inspect shell-to-bottom riveted lap joints.		
	e. Inspect for vertical grooving damage from seal assembly protrusions.		
	f. Inspect existing protective coatings for damage, deterioration, and disbonding.		
	g. Check for areas of rubbing (indicating too much pressure by the seal assembly shoes or inadequate annular space).		
	h. Visually inspect the shell plates and seams for indications of leakage.		
	i. If the shell has riveted or bolted seams, record the leak locations by film or chart in case the locations are lost during surface preparation for painting.		
	j. Measure annular space at 40-foot intervals.		
	k. Survey the shell to check for roundness and plumb.		
C.2.5	**SHELL-MOUNTED OVERFLOWS**		
	a. Inspect overflow for corrosion and adequate screening.		
	b. Check location of overflow that it is not above any tank valves or equipment.		
C.2.6	**ROOF INTERIOR SURFACE**		
C.2.6.1	**General**		
	a. Visually inspect the underside surface of the roof plates for holes, scale buildup, and pitting.		
	b. Hammer test or ultrasonically examine to check for thin areas, particularly in the vapor space of floating roofs and at edge of roof on cone roof tank.		
	c. Check all clips, brackets, braces, etc., welded to the roof deck plate for welded reinforcing pads and see that they have not broken free.		
	d. If no pad is present, penetrant test for cracking of the weld or deck plate.		
	e. Inspect for protective coating for breaks, disbondment, and deterioration.		
	f. Spark test the interior surface coating if recoating is not planned.		
C.2.6.2	**Fixed Roof Support Structure**		
	a. Inspect the support columns for thinning in the upper two feet.		
	b. On API columns (two channels welded together) check for corrosion scale breaking the tack welds, unless the joint between the channels is completely seal welded.		
	c. Check that the reinforcing pad on the bottom is seal welded to the tank bottom with horizontal movement restraining clips welded to the pad.		
	d. Determine if pipe column supports are concrete filled or open pipe. If open pipe, check for a drain opening in the bottom of the pipe.		
	e. Inspect and gauge rafters for thinning, particularly near the center of the roof. Report metal loss.		
	f. Check for loose or twisted rafters.		
	g. Inspect girders for thinning and check that they are attached securely to the top of the columns.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607846 105 ■

TANK OUT-OF-SERVICE INSPECTION CHECKLIST—Continued

Item	Completed ✔	Comments
h. Report if the columns have cross bracing in the area between the low pump out of the top of the shell (for future internal floating roof installation).		
i. Inspect and report presence of any roof-mounted swing line bumpers.		
j. Photograph the roof structure if no rafter layout drawing exists.		
C.2.7 FIXED ROOF APPURTENANCES		
C.2.7.1 Inspection and Light Hatches		
a. Inspect the hatches for corrosion, paint and coating failures, holes, and cover sealing.		
b. On loose covers, check for a safety chain in good condition.		
c. On light hatches over 30 inches across, check for safety rods.		
d. Inspect the condition of the gaskets on bold or latched down hatch covers.		
C.2.7.2 Staging Support Connection		
Inspect the condition of the staging support for corrosion.		
C.2.7.3 Breathers and Vents		
a. Inspect and service the breather.		
b. Inspect screens on vents and breathers.		
C.2.7.4 Emergency P/V Hatches		
a. Inspect and service pressure/vacuum hatches. (Setting should be high enough to prevent chattering of breather during normal operation. See breather manufacturer's guide.)		
b. Inspect liquid seal hatches for corrosion and proper liquid level in the seal.		
C.2.7.5 Sample Hatch		
a. Inspect sample hatch for corrosion.		
b. Check that the cover operates properly.		
c. If the tank has no gauge well, check for a hold off distance marker and check measurement.		
C.2.8 FLOATING ROOF		
C.2.8.1 Roof Deck		
a. Hammer test the area between roof rim and shell. (If access for hammer testing is inadequate, measure the distance from the bottom edge of the roof to the corroded area and then hammer test from inside the pontoon.)		
b. In sour water service, clean and test all deck plate weld seams for cracking unless the lower laps have been seal welded.		
c. Check that either the roof drain is open or the drain plug in the roof is open in case of unexpected rain.		
d. On flat bottomed and cone bottom roof decks, check for a vapor dam around the periphery of the roof. The dam should be continuous without break to prevent escape of vapors to the seal area from under the center of the roof.		
C.2.8.2 Floating Roof Pontoons		
a. Visually inspect each pontoon for liquid leakage.		
b. Run a light wire through the gooseneck vents on locked down inspection hatch covers to make sure they are open.		
c. Inspect lockdown latches on each cover.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

TANK OUT-OF-SERVICE INSPECTION CHECKLIST—Continued			
	Item	Completed ✔	Comments
	d. Check and report if each pontoon is:		
	1. Vapor tight (bulkhead seal welded on one side on bottom, sides, and top),		
	2. Liquid tight seal welded on bottom and sides only), or		
	3. Unacceptable (minimum acceptable condition is liquid tight).		
C.2.8.3	**Floating Roof Cutouts**		
	a. Inspect underside of cutouts for mechanical damage.		
	b. Inspect welds for cracks.		
	c. Inspect plate for thinning, pitting, and erosion.		
	d. Measure mixer cutouts and record plate thickness for future mixer installation or replacement. Plate thickness _____ .		
C.2.8.4	**Floating Roof Supports**		
	a. Inspect fixed low and removable high floating roof legs for thinning.		
	b. Inspect for notching at bottom of legs for drainage.		
	c. Inspect for leg buckling or felling at bottom.		
	d. Inspect pin hole in roof guide for tears.		
	e. Check plumb of all legs.		
	f. Inspect for adequate reinforcing gussets on all legs through a single portion of the roof.		
	g. Inspect the area around the roof legs for cracking if there is no internal reinforcing pad or if the topside pad is not welded to the deck plate on the underside.		
	h. Inspect the sealing system on the two-position legs and the vapor plugs in the fixed low leg for deterioration of the gaskets.		
	i. On shell mounted roof supports, check for adequate clearance based on the maximum floating roof movement as determined by the position of the roof relative to the gauge well and/or counter rotational device.		
C.2.9	**FLOATING ROOF SEAL ASSEMBLIES**		
C.2.9.1	**Primary Shoe Assembly**		
	a. Remove four sections of foam log (foam filled seals) for inspection on 90° locations.		
	b. Inspect hanger attachment to roof rim for thinning, bending, broken welds, and wear of pin holes.		
	c. Inspect clips welded to roof rim for thinning.		
	d. Shoes–inspect for thinning and holes in shoes.		
	e. Inspect for bit-metal bolts, clips, and attachments.		
	f. Seal fabric–inspect for deterioration, stiffening, holes, and tears in fabric.		
	g. Measure length of fabric from top of shoe to roof rim, and check against maximum anticipated annular space as roof operates.		
	h. Inspect any modification of shoes over shell nozzles, mixers, etc., for clearance.		
	i. Inspect shoes for damage caused by striking shell nozzles, mixers, etc.		
C.2.9.2	**Primary Toroidal Assembly**		
	a. Inspect seal fabric for wear, deterioration, holes, and tears.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607848 T88 ■

TANK OUT-OF-SERVICE INSPECTION CHECKLIST—Continued			
	Item	Completed ✔	Comments
	b. Inspect hold down system for buckling or bending.		
	c. Inspect foam for liquid absorption and deterioration.		
C.2.9.3	**Rim Mounted Secondaries**		
	a. Inspect the rim-mounted bolting bar for corrosion and broken welds.		
	b. Measure and chart seal-to-shell gaps.		
	c. Visually inspect seam from below, looking for holes as evident by light.		
	d. Inspect fabric for deterioration and stiffness.		
	e. Inspect for mechanical damage, corrosion, and wear on tip in contact with shell.		
	f. Inspect for contact with obstructions above top of shell.		
C.2.10	**FLOATING ROOF APPURTENANCES**		
C.2.10.1	**Roof Manways**		
	a. Inspect walls of manways for pitting and thinning.		
	b. On tanks with interface autogauges, check seal around gauge tape cable and guide wires through manway cover.		
	c. Inspect cover gasket and bolts.		
C.2.10.2	**Rim Vent**		
	a. Check rim vent for pitting and holes.		
	b. Check vent for condition of screen.		
	c. On floating roof tanks where the environmental rules require closing off the vent, check the vent pipe for corrosion at the pipe-to-rim joint and check that the blinding is adequate.		
C.2.10.3	**Vacuum Breaker, Breather Type**		
	a. Service and check operation of breather valve.		
	b. Check that nozzle pipe projects no more than ½ inch below roof deck.		
C.2.10.4	**Vacuum Breaker, Mechanical Type**		
	Inspect the stem for thinning. Measure how far the vacuum breaker cover is raised off the pipe when the roof is resting on high or low legs.		
	a. On high legs: _____ .		
	b. On low legs: _____ .		
C.2.10.5	**Roof Drains: Open Systems, Including Emergency Drains**		
	a. Check liquid level inside open roof drains for adequate freeboard. Report if there is insufficient distance between liquid level and top of drain.		
	b. If tank comes under Air Quality Monitoring District rules, inspect the roof drain vapor plug.		
	c. If emergency drain is not at the center of the roof, check that there are at least three emergency drains.		
C.2.10.6	**Closed Drain Systems: Drain Basins**		
	a. Inspect for thinning and pitting.		
	b. Inspect protective coating (topside).		
	c. Inspect basin cover or screen for corrosion.		
	d. Test operation of check valve.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD·API/PETRO STD 653-ENGL 1995 ■ 0732290 0607849 914 ■

TANK OUT-OF-SERVICE INSPECTION CHECKLIST—Continued			
	Item	Completed ✔	Comments
	e. Check for presence of check valve where bottom of basin is below product level.		
	f. Inspect drain basin(s) to roof deck welds for cracking.		
	g. Check drain basin(s) outlet pipe for adequate reinforcement to roof deck (including reinforcing pad).		
C.2.10.7	**Closed Drain Systems: Fixed Drain Line on Tank Bottom**		
	a. Hammer test fixed drain line on tank bottom for thinning and scale/debris plugging.		
	b. Inspect supports and reinforcing pads for weld failures and corrosion.		
	c. Check that pipe is guided, not rigidly locked to support, to avoid tearing of tank bottom plate.		
C.2.10.8	**Closed Drain Systems: Flexible Pipe Drain**		
	a. Inspect for damage to exterior of pipe.		
	b. Check for obstructions that pipe could catch on.		
	c. Inspect shields to protect pipe from snagging.		
	d. Inspect results of hydrotest on flexible roof drain system.		
C.2.10.9	**Closed Drain Systems: Articulated Joint Drain**		
	a. Hammer test rigid pipe in flexible joint systems for thinning and scale/debris plugging.		
	b. Inspect system for signs of bending or strain.		
	c. Inspect results of system hydrotest.		
	d. Inspect landing leg and pad.		
C.2.10.10	**Autogauge System and Alarms**		
	a. Check freedom of movement of tape through autogauge tape guide.		
	b. Inspect sheaves for freedom of movement.		
	c. Test operation checker.		
	d. Inspect tape and tape cable for twisting and fraying.		
	e. Test the tape's freedom of movement through guide sheaves and tape guide pipe.		
	f. On open-top tanks, check that gate tapes with cables have no more than one foot of tape exposed with float at lowest point.		
	g. Check float for leakage.		
	h. Test float guide wire anchors for spring action by pulling on wire and releasing.		
	i. Inspect floatwells in floating roofs for thinning and pitting of walls just above the liquid level.		
	j. Check that the autogauge tape is firmly attached to the float.		
	k. Inspect the tape cable and float guide wire fabric seals through the float well cover.		
	l. Inspect the bottom guide wire attachment clip: inspect for a temporary weighted bar instead of a permanent welded down clip.		
	m. Inspect board-type autogauge indicators for legibility and freedom of movement of indicator.		
	n. Measure and record these distances to determine if seal damage will occur if tank is run over from:		
	1. Shell top angle to underside of tape guide system.		
	2. Liquid level on floating top to top of secondary seal.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607850 636 ■

TANK OUT-OF-SERVICE INSPECTION CHECKLIST—Continued			
	Item	**Completed ✔**	**Comments**
	o. Identify floating roofs where the tape is connected directly to the roof.		
	p. Overfill alarm: Inspect tank overfill prevention alarm switches for proper operation.		
C.2.11	**COMMON TANK APPURTENANCES**		
C.2.11.1	**Gauge Well**		
	a. Inspect gate well pipe for thinning at about two-thirds distance above the bottom: look for thinning at the edge of the slots.		
	b. Check for corrosion on the pipe joint. Check that sample cords, weights, thermometers, etc., have been removed from the pipe.		
	c. Check for cone at bottom end of pipe about one foot above the bottom.		
	d. Check condition of well washer pipe and that its flared end is directed at the near side of the hold off pad.		
	e. Check that supports for gauge well are welded to pad or to shell and not directly to bottom plate.		
	f. Check operation of gauge well cover.		
	g. Check presence of a hold-off distance marker in well pipe and record hold-off distance. Hold-off distance _____ .		
	h. Identify and report size and pipe schedule, and whether pipe is solid or slotted. Report slot size.		
	i. Check that the hold-off distance plate is seal welded to the bottom and that any gauge well supports are welded to the plate and not directly to the bottom.		
	j. Inspect vapor control float and cable.		
	k. Check for presence and condition of gauge well washer.		
	l. Check for bull plug or plate blind on gauge well washer valve.		
	m. Inspect gauge well guide in floating roof for pitting and thinning.		
	n. Inspect the guide rollers and sliding plates for freedom of movement.		
	o. Inspect condition of gauge well pipe seal system.		
	p. On black oil and diesel services: if gauge well is also used for sampling, check for presence of a thief- and gauge-type hatch to avoid spillage.		
	q. Visually inspect inside of pipe for pipe weld protrusions which could catch or damage vapor control float.		
C.2.11.2	**Sampling Systems: Roof Sample Hatches**		
	a. Inspect roof mounted sample hatches for reinforcing pads and cracking.		
	b. Inspect cover for operation.		
	c. For tanks complying with Air Quality Monitoring District rules, inspect sample hatch covers for adequate sealing.		
	d. Check horizontal alignment of internal floating roof sample hatches under fixed roof hatches.		
	e. Inspect the sealing system on the internal floating roof sample hatch cover.		
	f. Inspect floating roof sample hatch cover recoil reel and rope.		
C.2.11.3	**Shell Nozzles**		
	a. Inspect shell nozzles for thinning and pitting.		
	b. Inspect hot tap nozzles for trimming of holes.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

TANK OUT-OF-SERVICE INSPECTION CHECKLIST—Continued			
	Item	Completed ✔	Comments
	c. Identify type of shell nozzles.		
	d. Identify and describe internal piping, including elbow up and elbow down types.		
C.2.11.4	**For Nozzles Extended Into the Tank**		
	a. Inspect pipe support pads welded to tank bottom.		
	b. Inspect to see that pipe is free to move along support without strain or tearing action on bottom plate.		
	c. Inspect nozzle valves for packing leaks and damaged flange faces.		
	d. Inspect heater stream nozzle flanges and valves for wire cutting.		
	e. Report which nozzles have thermal pressure relief bosses and valves.		
	f. In internal elbow-down fill line nozzles, inspect the wear plate on the tank bottom.		
	g. On elbow-up fill lines in floating roof tanks, check that opening is directed against underside of roof, not against vapor space. Inspect impact are for erosion.		
C.2.11.5	**Diffusers and Air Rolling Systems**		
	a. Inspect diffuser pipe for erosion and thinning.		
	b. Check holes in diffuser for excessive wear and enlargement.		
	c. Inspect diffuser supports for damage and corrosion.		
	d. Check that diffuser supports restrain, not anchor, longitudinal line movement.		
	e. Inspect air spiders on bottom of lube oil tanks for plugging and damaged or broken threaded joints.		
C.2.11.6	**Swing Lines**		
	a. Inspect flexible joint for cracks and leaks.		
	b. Scribe the flexible joint across the two moving faces and raise end of swing line to check the joint's freedom of movement, indicated by separation of scribe marks.		
	c. Check that flexible joints over six inches are supported.		
	d. Inspect the swing pipe for deep pitting and weld corrosion.		
	e. Loosen the vent plugs in the pontoons and listen for a vacuum. Lack of a vacuum indicates a leaking pontoon.		
	f. Check the results of air test on pontoons during repairs.		
	g. Inspect the pontoons for pitting.		
	h. Inspect the pull-down cable connections to the swing.		
	i. Inspect the condition of the bottom-mounted support, fixed roof limiting bumper, or shell mounted limiting bumper for wood condition, weld and bolt corrosion, and seal welding to bottom or shell.		
	j. Inspect safety hold-down chain for corrosion and weak links.		
	k. Check that there is a welded reinforcing pad where the chain connects to the bottom.		
	l. If the floating swing in a floating or internal floating roof tank does not have a limiting device preventing the swing from exceeding 60 degrees, measure and calculate the maximum angle possible with the roof on overflow. Max. angle on overflow _____ . (If the calculated angle exceeds 65 degrees, recommended installation of a limiting bracket.)		
	m. Inspect pull down cable for fraying.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

STD.API/PETRO STD 653-ENGL 1995 ■ 0732290 0607852 409 ■

TANK OUT-OF-SERVICE INSPECTION CHECKLIST—Continued			
	Item	**Completed ✔**	**Comments**
	n. Inspect for three cable clamps where cable attaches to end of swing line (single-reeved) or to roof assembly (double-reeved). Inspect sheaves for freedom of movement.		
	o. Inspect winch operation and check the height indicator for legibility and accuracy.		
	p. Inspect bottom-mounted sheave assembly at end of pontoon for freedom of rotation of sheave.		
	q. Inspect shell-mounted lower sheave assembly for freedom of rotation of sheave, corrosion thinning, and pitting of sheave housing.		
	r. Inspect upper sheave assembly for freedom of movement of sheave.		
	s. Inspect the cable counterbalance assembly for corrosion and freedom of operation.		
C.2.11.7	**Manway Heater Racks**		
	a. Inspect the manway heater racks for broken welds and bending of the sliding rails.		
	b. Measure and record the length of the heater and length of the track.		
C.2.11.8	**Mixer Wear Plates and Deflector Stands**		
	a. Inspect bottom and shell plates and deflector stands.		
	b. Inspect for erosion and corrosion on the wear plates. Inspect for rigidity, structural soundness, corrosion, and erosion of deck plates and reinforcing pads that are seal welded to the bottom under the deflector stand legs.		
	c. Measure for propeller clearance between the bottom of deflector stand and roof when the roof is on low legs.		
C.2.12	**ACCESS STRUCTURES**		
C.2.12.1	**Handrails**		
	a. Identify and report type (steel pipe, galvanized pipe, square tube, angle) and size of handrails.		
	b. Inspect for pitting and holes, paint failure.		
	c. Inspect attachment welds.		
	d. Identify cold joints and sharp edges. Inspect the handrails and midrails.		
	e. Inspect safety drop bar (or safety chain) for corrosion, functioning, and length.		
	f. Inspect the handrail between the rolling ladder and the gaging platform for a hazardous opening when the floating roof is at its lowest level.		
C.2.12.2	**Platform Frame**		
	a. Inspect frame for corrosion and paint failure.		
	b. Inspect the attachment of frame to supports and supports to tank for corrosion and weld failure.		
	c. Check reinforcing pads where supports are attached to shell or roof.		
	d. Inspect the surface that deck plate or grating rests on, for thinning and holes.		
	e. Check that flat-surface to flat-surface junctures reseal welded.		
C.2.12.3	**Deck Plate and Grating**		
	a. Inspect deck plate for corrosion-caused thinning or holes (not drain holes) and paint failure.		
	b. Inspect plate-to-frame weld for rust scale buildup.		
	c. Inspect grating for corrosion-caused thinning of bars and failure of welds.		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

TANK OUT-OF-SERVICE INSPECTION CHECKLIST—Continued		
Item	**Completed ✔**	**Comments**
d. Check grating tie down clips. Where grating has been retrofitted to replace plate, measure the rise of the step below and above the grating surface and compare with other risers on the stairway.		
C.2.12.4 Stairway Stringers		
a. Inspect spiral stairway stringers for corrosion, paint failure, and weld failure. Inspect attachment of stairway treads to stringer.		
b. Inspect stairway supports to shell welds and reinforcing pads.		
c. Inspect steel support attachment to concrete base for corrosion.		
C.2.12.5 Rolling Ladder		
a. Inspect rolling ladder stringers for corrosion.		
b. Identify and inspect ladder fixed rungs (square bar, round bar, angles) for weld attachment to stringers and corrosion, particularly where angle rungs are welded to stringers.		
c. Check for wear and corrosion where rolling ladder attaches to gaging platform.		
d. Inspect pivot bar for wear and secureness.		
e. Inspect operation of self-leveling stairway treads.		
f. Inspect for corrosion and wear on moving parts.		
g. Inspect rolling ladder wheels for freedom of movement, flat spots, and wear on axle.		
h. Inspect alignment of rolling ladder with roof rack.		
i. Inspect top surface of rolling ladder track for wear by wheels to assure at least 18 inches of unworn track (track long enough).		
j. Inspect rolling ladder track welds for corrosion.		
k. Inspect track supports on roof for reinforcing pads seal welded to deck plate.		
l. Check by dimensioning, the maximum angle of the rolling ladder when the roof is on low legs. Max. angle _____ .		
m. If rolling ladder track extends to within five feet of the edge of the roof on the far side, check for a handrail on the top of the shell on that side.		
Notes:		

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX D—AUTHORIZED INSPECTOR CERTIFICATION

D.1 Examination/Grading/Validation

D.1.1 EXAMINATION

A written examination to certify inspector candidates within the scope of this standard shall be administered by a third party designated by the American Petroleum Institute. The examination shall be based on the content of the latest edition of this standard. API IRE Chapter XIII[1], ASME Section V and IX, and the applicable section of API Standard 650, API RP651, API RP652, API IRE Chapter II[2], and API IRE Chapter XVI[3].

D.1.2 GRADING

The third party testing agency shall administer and grade the examinations using an equating system that statistically adjusts the passing score of each individual examination administration to compensate for any potential differences in the difficulty levels of the exams. Applicants may appeal any examination grievance to the American Petroleum Institute within ninety days of receipt of their grades.

D.1.3 VALIDATION

The examination shall be validated as to its suitability for determining the applicant's level of knowledge on how to inspect aboveground storage tanks.

[1] API IRE Chapter XIII is under revision and will be reissued as API Recommended Practice 575, *Inspection of Atmospheric and Low Pressure Storage Tanks.*

[2] API IRE Chapter II is under revision and will be reissued as API Recommended Practice 571, *Recognition of Conditions Causing Deterioration or Failure.*

[3] API IRE Chapter XVI is under revision and will be reissued as API Recommended Practice 576, *Inspection of Pressure-Relieving Devices.*

D.2 Examination Methods/Types

The examination shall be written.

D.3 Certification

D.3.1 An API certificate will be issued upon compliance with the provisions of 4.10 and D.1, or D.3.2.

D.3.2 During the first twelve months after the date of issuance of this appendix, Inspectors whose regular duties routinely have included atmospheric storage tank inspection for at least the previous five years in the petroleum or chemical industries will be eligible for a special, one-time, three-year certification. Notarized employment documentation verifying years of experience and satisfactory performance will be required.

D.4 Certification Agency

The American Petroleum Institute shall be the certifying agency.

D.5 Recertification

D.5.1 All Authorized Inspectors will be required to provide notarized documentation of employment as an API 653 Inspector whose regular duties routinely have included atmospheric storage tank inspection during the previous three-year certification period; otherwise, re-examination is required.

D.5.2 Recertification by written examination is required for Authorized Inspectors who qualified under D.3.2.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

APPENDIX E—TECHNICAL INQUIRIES

E.1 Introduction

API will consider written requests for interpretations of Standard 653. API staff will make such interpretations in writing after consulting, if necessary, with the appropriate committee officers and committee members. The API committee responsible for maintaining Standard 653 meets regularly to consider written requests for interpretations and revisions and to develop new criteria dictated by technological development. The committee's activities in this regard are limited strictly to interpretations of the standard and to the consideration of revisions to the present standard on the basis of new data or technology. As a matter of policy, API does not approve, certify, rate, or endorse any item, construction, proprietary device, or activity, and accordingly, inquiries that require such consideration will be returned. Moreover, API does not act as a consultant on specific engineering problems or on the general understanding or application of the standard. If it is the opinion of the committee, based on the inquiry information submitted, that the inquirer should seek other assistance, the inquiry will be returned with the recommendation that such assistance be obtained. All inquiries that do not provide the information needed for the committee's full understanding will be returned.

E.2 Inquiry Format

E.2.1 Inquiries shall be limited strictly to requests for interpretation of the standard or to the consideration of revisions to the standard on the basis of new data or technology. Inquiries shall be submitted in the format described in E.2.2 through E.2.5.

E.2.2 The scope of an inquiry shall be limited to a single subject or a group of closely related subjects. An inquiry concerning two or more unrelated subjects will be returned.

E.2.3 An inquiry shall start with a background section that states the purpose of the inquiry, which would be either to obtain an interpretation of the standard or to propose a revision to the standard. The background section shall concisely provide the information needed for the committee's understanding of the inquiry (with sketches as necessary) and shall cite the applicable edition, revision, paragraphs, figures, and tables.

E.2.4 After the background section, an inquiry's main section shall state the inquiry as a condensed, precise question, omitting superfluous background information and, where appropriate, posing the question so that the reply could take the form of "yes" or "no" (perhaps with provisos). This inquiry statement should be technically and editorially correct. The inquirer shall state what he believes the standard requires. If he believes a revision to the standard is needed, he shall provide recommended wording.

E.2.5 The inquirer shall include his name and mailing address. The inquiry should be typed; however, legible handwritten inquiries will be considered. Inquiries should be submitted to the director of the Manufacturing, Distribution and Marketing Department, American Petroleum Institute, 1220 L Street, N.W., Washington, D.C. 20005.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 2000
April 10, 2000 04:35:34

ADDITIONAL COPIES AVAILABLE FROM
PUBLICATIONS AND DISTRIBUTION
(202) 682-8375

American Petroleum Institute

1220 L Street, Northwest
Washington, D.C. 20005-4070
202-682-8000

Order No. C65302

COPYRIGHT 2000 American Petroleum Institute
April 10, 2000 04:35:34

Information Handling Services, 2000